



Dahua DSS Platform

API Document



Overview

VMS is a digital and intelligent video management platform comprised of seven core modules: Monitoring Center, Event Center, Intelligent Search, Access Control, Parking Management, Intelligent Analysis, and Operation and Maintenance Center, along with two subsystems for Retail and Energy. The system employs a Client/Server (C/S) architecture based on a Service-Oriented Architecture (SOA), allowing for clear functionality of service modules and maintaining loose coupling between services, which facilitates module reuse and flexible expansion. The system supports All-In-One deployment for all services and also allows for distributed deployment to achieve elastic scaling for medium to large project scenarios. Additionally, VMS is a fully open northbound system that defines a comprehensive set of open APIs. These open northbound APIs enable rapid integration between third-party applications and the VMS platform.

Target Audience

This document is mainly for developers of third-party manufacturers who want to develop applications based on DSS open capabilities.

To better understand the content of this document, the following technical skills are required:

- Proficiency in programming languages such as Java and C#.
- Familiarity with the security industry, including understanding terms related to video surveillance, alarms, and PTZ (Pan-Tilt-Zoom) cameras.
- A good understanding of video-related protocols, including knowledge of the usage of RTSP (Real Time Streaming Protocol) and HTTP (Hypertext Transfer Protocol).

Terms

Terms/ Abbreviation	Description
VMS	Video Management Platform
TLS	Transport Layer Security
HTTPS	The HTTP channel aimed at security ensures the integrity of the transmission process through encryption and authentication based on HTTP.
RTSP	Real-time streaming protocol is specially used to control the transmission of multimedia streams, which can play, pause and stop the video stream through a series of commands.
RTP	Real-time Transport Protocol
FLV	Flash Video
HLS	HTTP Live Streaming
Token	After a user successfully logs in to the VMS platform, the platform will automatically assign a token as the unique credential of the user
Organization	The platform defaults to a root organization, which can encompass sub-organizations as well as devices.
Device	Referring to a physical device, such as an NVR, IPC, access control or alarm hub
Unit	The platform categorizes devices into multiple units based on their capabilities. Unit types include encoding units, decoding units, alarm input units, alarm output units, among others. A single device may contain multiple units.
Channel	Each unit contains multiple channels, with channels being the smallest units that provide data capabilities. For example, an encoding channel can output video stream data, while an alarm channel can detect and output alarms.

Revision History

Version	Release Time	Revision Content
01	2020.04.10	First release.
02	2021.02.20	1. Updated the content of personnel management, access control and facial recognition business. 2. Focuses on the rapid integration process and the open service capabilities. 3. Updated the API protocols.

Version	Release Time	Revision Content
03	2021.07.18	1. Added the module of intelligent analysis. 2. Update the protocols on device details and visitor business.
04	2021.10.25	New protocols for MPT record searching, alarm hub operations, and access control group authorizations.
05	2022.04.20	1. Added the protocols on searching for device status and notifications. 2. Opened the protocols of PTZ control 3. Opened smart retrieval remaining query export protocol. 4. Updated the face database sending control protocol. 5. Added parking record query export protocol.
06	2022.11.14	1. Added alarm correlation information query protocol. 2. Added alarm hub query operation protocol. 3. Added time zone list query protocol. 4. Added time template list query protocol. 5. Added the elevator control information for the protocols of devices, personnel and visitors.
07	2023.06.10	1. Reconstruction of access control services to add features for zones and passage rules. 2. Remove the protocols of exporting, MPT and POS.
08	2023.12.21	1. Added the interface for searching for roles. 2. Isolation of authentication tokens for interfaces and image resources. 3. Added# and Python code examples for invocation.
09	2024.07.16	1. Added the interface for audio. 2. The parking lot business changes, along with the addition of personnel, may result in some agreements being incompatible with the previous version. 3. The alarm subscription feature now includes image information delivery, which may lead to incompatibility with the previous logic. 4. Added the interface for searching for MPT records. 5. The method of accessing static resources, such as images on the platform, is not compatible. Please refer chapter 3.3.6 for details.

Version	Release Time	Revision Content
10	2025.01.01	1. Replace the old interface with the new hierarchical loading tree interface. 2. Added an interface to retrieve streaming media addresses via FLV/HLS methods. 3. Added an interface to retrieve the door status. 4. Added an interface to retrieve visual intercom call records. 5. Added new interfaces for entrance and exit operations, including parking lot details, vehicle details, alarm details, and vehicle monitoring lists. 6. Cancel the interfaces for stream key retrieval and fingerprint collection. The services will still be operational.
11	2025.07.30	1. Added an interface to capture picture from real-time video. 2. Added an interface to lock or unlock person of access control. 3. Cancel the interfaces for alarm associated with traffic and entrance/exit information. The services will still be operational.

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1 Overview

1.1 Instruction

To quickly understand the API capabilities offered by the VMS platform and expedite development, please follow the document content in the order provided below.

[Forward]: Explain the business capabilities and architecture of the VMS platform, and provide a preliminary understanding of the technical terms that will appear in the document.

[Overview]: Describe the open capabilities of the VMS platform and introduce integration solutions for different scenarios, providing a preliminary understanding of the integration process.

[Interface Specification]: Introduce the calling specifications for the VMS Open API and the MQ message push specifications.

[API Interface]: Categorize the APIs for easy search, allowing developers to understand specific functions and usage methods through API descriptions and usage examples. Developers can then select the required API interfaces.

[MQ Notification Message]: Developers should pay attention to this section when they need to receive real-time notifications or alerts from the platform. Notifications can be categorized for easy searching, allowing developers to understand specific functions and usage methods through descriptions and examples. They can then subscribe to the necessary notification messages.

[Best Practice]: This section provides detailed integration methods for commonly used platform features through several classic integration cases, enabling developers to gain a deeper understanding of how to use the platform API effectively.

[Appendix]: The data dictionary, error code and code example DEMO of the platform are provided for the developer's reference. There are API login authentication example DEMO, MQ docking example DEMO, and encryption and decryption example DEMO.

[FAQ]: Provide solutions for common issues encountered during the integration process.

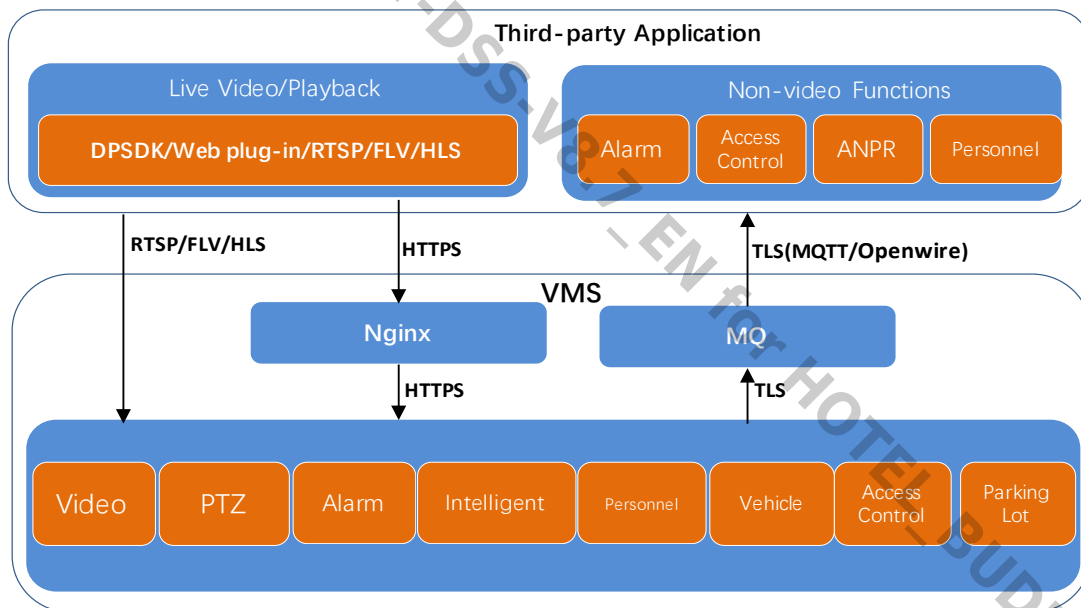
1.2 Architecture

The VMS (Video Management System) platform provides comprehensive management capabilities for video, alarms, access control, attendance, personnel, vehicles, parking lots, facial recognition, and structured data. It defines standard REST API interfaces for open access, allowing third-party developers to interact with the platform via HTTPS.

Additionally, developers can receive real-time notifications from the platform through subscribed MQ messages.

For video services, the platform supports pulling streams directly via RTSP, FLV, and HLS protocols. To address the complexity of video services and facilitate third-party integration, it provides a packaged DPSDK.dll for client and app integration, as well as a JS package for Web plug-in integration. Third-party developers can choose the appropriate integration method based on their specific requirements.

The architecture is as follows:



1.3 Preparation

Step 1: Prepare the VMS environment

✓ Install the VMS platform

Get the installation package, install the platform according to the platform user's manual, and then enable all the services.

- ✓ **Install the VMS client**

Get the client installation package and install it. Configure the login account. The default user is system.

- ✓ **Import the License**

Get the valid activation code of the License, and import the License to the platform to authenticate the platform.

- ✓ **Adding Devices**

Add front-end equipment to the platform to ensure that the platform has accessible front-end codec resources.

- ✓ **Select the Language**

Select a suitable programming language that facilitates interaction with protocols such as HTTP, MQ, and RTSP. If video playback functionality is required, incorporate the implementation of the playback component as well. The platform offers two access methods: HTTP (unencrypted) and HTTPS (encrypted). The default access port for HTTP is port 80, while the default port for HTTPS is 443. During the development and debugging phase, HTTP can be used for easier packet capturing and troubleshooting. In actual usage, HTTPS must be employed to ensure the security of information transmission. To enable access via HTTP on port 80, the client must configure the HTTP allowlist in the [System Parameters -> Security Configuration] page, adding the IP addresses of third-party developer machines to the allowlist.

Step 2 Select the integration method based on the scenario.

VMS provides various integration methods for different scenarios. Third-party developers can choose the appropriate integration method based on their business requirements and development languages.

- ✓ Directly integrating with the video service using DPSDK.dll allows you to utilize the DPSDK integration method without needing to focus on the API documentation. Please refer to the DPSDK usage instructions for guidance. The download link for DPSDK is:
<https://depp.dahuasecurity.com/integration/guide/download/DSS%20SDK>.

- ✓ Integrate with the video service using the API along with a Web plugin.
Please contact DSS marketing personnel to obtain the Web plugin.
- ✓ Integrate with the video service using the API combined with FLV/HLS. The default baseline installation package of DSS does not support FLV/HLS streaming. Please contact DSS marketing personnel to obtain the installation package for FLV/HLS services and install it in the DSS environment.
- ✓ Integrate with the video service using the API with the RTSP method. This is supported by the default DSS baseline installation.
- ✓ Non-video services need to be integrated using the API method.

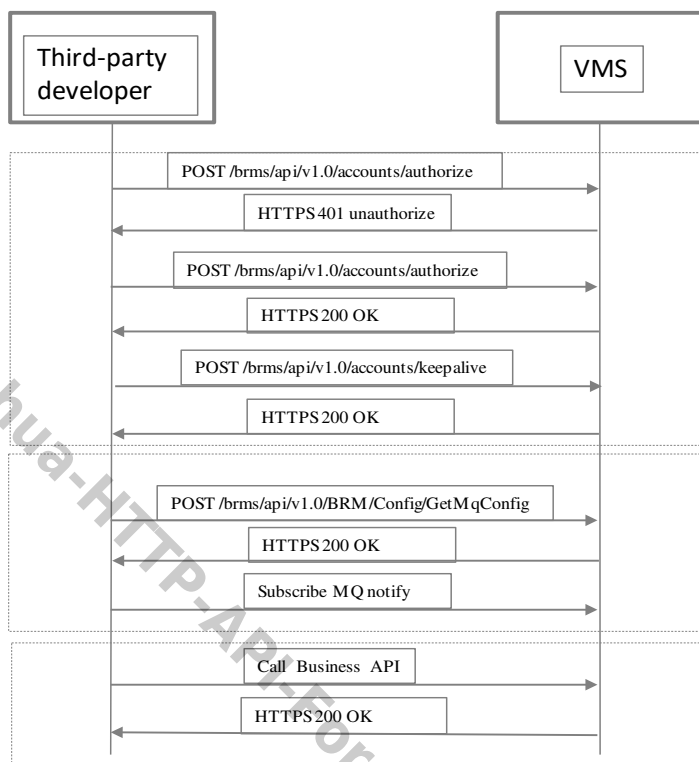
Step 3 Integrated with VMS

- ✓ After the third-party developers obtain the username and password for logging into the VMS and confirm the integration method, they can start integrating with the VMS.

1.4 Integration Process

The API integration with VMS involves 3 steps:

Log in and authenticate through the HTTP Digest method-> Obtain MQ configuration and subscribe to MQ notification message-> Call the business interface to complete business docking. The process is as follows:



2 Interface Specifications

2.1 HTTP Interface Protocol Specification

All APIs provided by VMS are based on the HTTP/1.1 protocol and follow RESTful principles. Data exchanged is standardized in JSON format. VMS defaults to open HTTPS ports and does not recommend using HTTP for calls. To enable HTTP access, the allowlist for port 80 must be configured in VMS (System Parameters -> Security Configuration -> HTTP Allowlist).

Item	Description
Transmission Method	To ensure the security of transmission, uses HTTPS transmission.
URL format	http(s)://[platform ip]:[platform port]/[subsystem]/api/[version]/[module]/[interface]/... For detail, see the rules of URL below.
Request Method	GET / POST / PUT / DELETE
Data Format	All in JSON format for the request and response data.
Character Encoding	Using UTF-8 encoding
Safety Authentication	Access to the API is secured through username/password authentication; only users who successfully authenticate and obtain a Token can access the API.
Permission Control	Most APIs are controlled by VMS menu permissions. Calling an unauthorized API will result in an error message.
Certificate Requirements	None
Calling Result	See the instruction on HTTP returned parameters.

2.1.1 Request Instructions

✓ Request Methods:

Request Method	Operation
POST	Obtain resources, mostly used for business data query and data acquisition.

Request Method	Operation
GET	Resource creation and data submission, primarily used for sending data to the backend or inserting it into a database.
PUT	Resource updating, primarily used for refreshing backend caches, databases, etc.
DELETE	Resource deletion, primarily used for removing data records.

✓ URL Instructions:

The HTTP request URL format is as follows: `http(s)://[platform ip]:[platform port]/[subsystem]/api/[version]/[module]/[interface]/...`

Item	Instruction
ip	VMS IP address
port	Port Number
subsystem	Sub-system name, including brms, obms, ipms, iams and more
version	API versioning is indicated by "V" followed by the major version and minor version, such as v1.0 or v1.1. The major version number changes when there are breaking changes, while the minor version number changes when new features are added.
module	Module name, can be omitted.
interface	Port name

In the format above, the ellipsis (...) may represent multiple hierarchical nodes in the resource description (e.g., node1/node2), depending on the specific context.

The completed example is as follows:

```
# new resources
POST /brms/api/v1.1/user
# resource acquisition
GET /brms/api/v1.1/user/{id}
# resource update
PUT /brms/api/v1.1/user/{id}
# resource update (an attribute)
PUT /brms/api/v1.1/user/{id}/password
# resource operation
PUT /brms/api/v1.1/user/{id}/lock
# resource deletion
DELETE /brms/api/v1.1/user/{id}
# delete in batches (Parameters are simple and do not exceed the Uri length limit)
DELETE /brms/api/v1.1/user?ids={ids}
# delete in batches (Parameter is complex or may exceed Uri length limit)
POST /brms/api/v1.1/user/delete/batch
# search for (list)
GET /brms/api/v1.1/user/list
# search for (page)
GET /brms/api/v1.1/user/page
```

✓ **Head of HTTP public message:**

The HTTP request header needs to contain the following fields: Accept-Language, Content-Type, X-Subject-Token, not including login interface.

Parameter	Required	Type	Description	Example
Accept-Language	Yes	string	Language environment, is primarily used for internationalization translation; in correct entries may affect the language of returned data.	Accept-language: E N. For details, please refer to the internationalization guidelines.
Content-Type	Yes	string	Data Type: Unless specifically stated otherwise, the format for all APIs in this document is application/json.	application/json; charset=UTF-8
X-Subject-Token	Yes	string	Authentication Token: Used for API authorization and authentication. The Token is returned upon successful login to the platform.	d725f34e4f034038bf1771a8ea43c931

Example of request head:

```
Accept-Language: zh
Content-Type: application/json; charset=UTF-8
X-Subject-Token: d725f34e4f034038bf1771a8ea43c931
```

✓ **International:**

The internationalization parameters for the interface should be included in the header information under the Accept-Language field.

```
PUT /brms/api/v1.0/accounts/keepalive HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: zh
Content-Type: application/json; charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84

{
}
```

The list of language code abbreviations corresponding to the internationalization local values is as follows:

Language Code	Country
ar	Arabic
bg	Bulgarian
cs	Czech

Language Code	Country
da	Danish
de	German
en	English
es	Spanish
es-LA	Latin American Spanish
fi	Finnish
fr	French
he	Hebrew
hu	Hungarian
it	Italian
ja	Japanese
ko	Korean
mk	Macedonian
pl	Polish
pt-BR	Brazilian Portuguese
ru	Russian
sk	Slovak
th	Thai
tr	Turkish
uk	Ukrainian
vi	Vietnamese
zh	Chinese
zh-TW	Traditional Chinese

2.1.2 Response Instructions

The status code of HTTP interface call success is uniformly 200 (401 will be returned for individual interface authentication failure). Various exceptions at the business level are uniformly reflected by returning the code field and desc in the body.

Parameter	Type	Description
code	int	Error code: 1000 indicates success, while any code other than 1000 indicates a failure. Common error codes can be found in Appendix 6.2.

Parameter	Type	Description
desc	string	Description of the code
data	object	The result data: the data content for different business interfaces is not fixed.

Example of a successful response body:

Example 1: (no data)

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

Example 2: (has data)

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "doorGroupName": "b007",
    "timeTemplateId": "3",
    "channelIds": [
      "1000001$7$0$0"
    ],
    "holidayPlanId": "0",
    "remark": "test"
  }
}
```

Example of a failed response body:

Example 1: (no data)

```
{
  "code": 1004,
  "desc": "The parameter is illegal.",
  "data": null
}
```

Example 2: (has data)

```
{
  "code": 10004,
  "desc": "The carNo is alredy exists or repeat.",
  "data": {
    "repeatCarNos": [],
    "existingCarNos": ["JA2563D8"]
  }
}
```

2.2 Specification of MQ Message Subscription and Push

Notifications from the VMS platform are pushed in real-time to clients or third parties via ActiveMQ. Therefore, developers must subscribe to the MQ message queue as needed according to VMS specifications to receive real-time notifications or alerts from the VMS. The protocol body uses JSON format. The message body format is consistent for both the sending and receiving queues.

Request Parameter:

Parameter	Type	Description
id	string	SN
method	string	Method
info	object	Message content

Example of message body:

```
{
  "id": "112456",
  "method": "...",
  "info":
  {
    ...
  }
}
```

Currently, the VMS platform has 4 topics that require subscription:

The `userId` and `userGroupId` represent the ID of the logged-in user and the user group ID, respectively. The login authentication interface returns the current user's `userId` and `userGroupId`. For example, if the current user's `userId` is 1 and the `userGroupId` is 2, the topics to subscribe to are:

Parameter	Type	Example
Alarm topic	<code>mq.alarm.msg.topic.{userId}</code> <code>mq.alarm.msg.group.topic.{userGroupId}</code>	<code>mq.alarm.msg.topic.1</code> <code>mq.alarm.msg.group.topic.2</code>
Event topic	<code>mq.event.msg.topic.{userId}</code>	<code>mq.event.msg.topic.1</code>
Public topic	<code>mq.common.msg.topic</code>	<code>mq.common.msg.topic</code>

The messages pushed by the MQ are in JSON format. The specific format can be referenced in the MQ message.

2.3 Specifications of Accessing Images

Obtain platform image resources. For security reasons, append the resource access credential to the image URL in the following format: `Url?token={credential}`. This credential is solely used for downloading platform files, while uploading files to the platform requires a login token. The credential is derived from the response during platform login; refer to section [3.1.2 Submit Credentials] for the returned credential. An example of a complete download image URL is as follows:

```
https://192.168.1.1:9901/56602311-65f5-11eb-94b1-4cedfb428a36/20200101/1/dfs_cdaaefb1-6698-11eb-988d-4cedfb428a36_53486430_53797182.jpg?token=3caf4abd6e584013973602345c565072
```

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3 API Interface

All API interfaces of the system are based on the HTTP/1.1 protocol and follow the RESTful style. All the interaction data is in JSON format. The version of the specific interface protocol is reflected by the interface URL.

3.1 Login Authentication

The VMS API can only be accessed after successful login authentication. The login authentication uses the secondary authentication process of Http Digest. The protocol is as follows.

3.1.1 Trigger Authentication

Brief Description

- First login. Initiate the authentication process, the server will return the status code 401 Unauthorized and will provide random authentication information such as realm and randomKey in the response.

Request URL

- /brms/api/v1.0/accounts/authorize

Request Method

- POST

Request Parameter

- Body Parameters

Parameter Required Type			Description
userName	Yes	string	Username
ipAddress	No	string	Client computer IP address
clientType	No	string	Client type, write WINPC_V2 directly

Request Example

```
POST /brms/api/v1.0/accounts/authorize HTTP/1.1
```

```
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
```

```
{
  "userName": "system",
  "ipAddress": "",
  "clientType": "WINPC_V2"
}
```

Return Parameter

Parameter	Type	Description
realm	string	Encrypted field. Used to generate a signature for the second login.
randomKey	string	Random secret key. Can be used as a parameter for the second login and will take effect after 10 seconds.
encryptType	string	Encryption mode. Can be used as a parameter for the second login.
publickey	string	Platform RSA-encrypted public key, base64 coded. Can be used to encrypt the secret key and vector of the AES encryption algorithm.

Return Example

```
{
  "realm": "f689ed8c40030d68007d6e95002f7f87",
  "randomKey": "48c008d0026a49c4",
  "encryptType": "MD5",
  "publickey": "MIGf000GCS...iH+FFwu9wID00AB"
}
```

3.1.2 Submit Credentials

Brief description

- Second login. For the procedure, see [5.1 Login Authentication]. A signature will be generated based on the authentication information returned from triggering the authentication process. You need to enter it into the request. The server will respond with a 200 OK status code and an access token. The random authentication information returned from triggering authentication is valid for 10 seconds. Please submit the credentials within 10 seconds.

Request Method

- POST

Request URL

- /brms/api/v1.0/accounts/authorize

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
mac	No	string	MAC address of client device, maximum length 50, required by some businesses. Null transmission is not recommended.
deviceSN	No	string	Device SN code, used for mobile device login
signature	Yes	string	Authentication data. The signature is calculated as follows: temp1 = md5(password) temp2 = md5(userName + temp1) temp3 = md5(temp2) temp4 = md5(userName + ":" + realm + ":" + temp3) signature = md5(temp4 + ":" + randomKey)
userName	Yes	string	Username
randomKey	Yes	string	Random value, returned from the interface in the first login and becoming invalid after 10 seconds
publicKey	No	string	Terminal RSA public key, note not the platform's public key. Compatible with previous versions, it can now be directly transferred to empty.
ipAddress	No	string	IP address
clientType	No	string	Client type, write WINPC_V2 directly.
userType	No	string	User type: 0: System user; 1: Domain user. 0 by default.
secretKey	No	string	User encryption and decryption key, ciphertext = encryption(secret key, platform public key). Use 32-bit random string in plaintext before encryption.
secretVector	No	string	User encryption and decryption vector, ciphertext = encryption(secret vector, platform public key). Use 16-bit random string in plaintext before encryption
loginType	否	string	Login type: 1 = normal login, 2 = multi-site login, default 1

Request Example

```
POST /brms/api/v1.0/accounts/authorize HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8

{
  "mac": "C8:D9:D2:16:AF:F6",
  "deviceSN": "",
  "signature": "44ec11d2efe26009375c403706ed1678",
  "userName": "system",
  "randomKey": "bb12f5aa90e64a73",
  "publicKey": "",
  "ipAddress": "",
  "clientType": "WINPC_V2",
  "userType": "0",
  "secretKey": "iYBQqQBCoq...N84A==",
  "secretVector": "AblgO+BAVca3...eW3GFPg==",
  "loginType": "1"
}
```

Return Parameter

Parameter	Type	Description
duration	int	Keep-alive interval (seconds). The [3.1.3 Heartbeat Keep-alive] interface needs to be called within this time
tokenRate	int	Session token update frequency (seconds). The [3.1.4 Update Token] interface needs to be called within this time
token	string	Used for interface authentication. For usage, see [5.1 Login Authentication]
credential	string	Session credentials for downloading static resources such as images and files. For usage, see [2.3 Specifications of Accessing Images]
userId	string	User ID
userName	string	Username
userGroupId	string	User group ID
userLevel	string	User level: 1=Super administrator; 2=Administrator; 3=Custom role
reused	string	Reused or not: 0=No, 1=Yes
loginDateUtc	long	Login UTC time
loginIpChanged	string	Has the IP address of the current user's login changed from the previous login: 0=No, 1=Yes
lastLoginIp	string	The IP address of the current user's last login

Parameter	Type	Description
verification	string	The session credential used to reset the password for the first login, valid for a single time and valid for 5 minutes
versionInfo	object	Version information
- lastVersion	string	Latest version on client
- updateUrl	string	Client update path
- fileSize	string	Size of installation package (unit: byte)
- patchVersion	string	Latest version on client patch
- patchUrl	string	Client patch update path
personId	string	The ID of the person bound to the app
houseHolder	int	Whether the personnel is the owner, APP judges whether the personnel management is displayed
sipNum	string	Calling number, for video intercom service
sipPassword	string	SIP password, encrypted with terminal public key, for video intercom service
pocId	string	Cluster intercom registration number
pocPassword	string	The cluster intercom registers the password and uses the terminal public key for encryption
secretKey	string	Ciphertext of secret key = RSA encryption (secret key, public key of terminal). Compatible with previous versions, when the publicKey parameter is empty, this field also returns empty.
secretVector	string	Vector ciphertext = RSA encryption (vector, public key of terminal). Compatible with previous versions, when the publicKey parameter is empty, this field also returns empty.
forceResetPassword	boolean	Forcibly modify password
enablePasswordValidityPeriod	string	Enable password validity verification: 0=No, 1=Yes
passwordRemainValidityPeriod	string	The remaining validity period of the user's current password, in days
passwdLeaveChangedDay	string	The remaining validity period of the user's current password, in days. The MPT220 device uses.
serviceAbility	array	Capability set, deprecated
emapUrl	string	E-map path, deprecated

Return Example

```
{
```



```
"duration": 30,
"token": "a15ee202efab4a54af01c5c0a4ed9bac",
"credential": "ff435202efab4a54af01c5c0a4e65abe",
"userId": "1",
"userName": "system",
"userGroupId": "1",
"verification": "69607e74f...03f1ef",
"serviceAbility": null,
"versionInfo": {
  "lastVersion": "1578852",
  "updateUrl": "/client/x86/xxxxx1.exe;/client/x64/xxxxxx2.exe",
  "fileSize": "",
  "patchVersion": "",
  "patchUrl": "",
},
"emapUrl": null,
"sipNum": "8888881000",
"sipPassword": "4CFC...EE8",
"pocId": "6666661000",
"pocPassword": "3ABF3...B78",
"tokenRate": 1800,
"secretKey": "",
"secretVector": "",
"reused": "1",
"userLevel": "1",
"personId": "1000001",
"houseHolder": 1,
"forceResetPassword": 0,
"loginDateUtc": 1726127578,
"enablePasswordValidityPeriod": "1",
"passwordRemainValidityPeriod": "90",
"passwdLeaveChangedDay": "90",
"loginIpChanged": "0",
"lastLoginIp": ""
}
```

3.1.3 Heartbeat Keep-alive

Brief Description

- Heartbeat keep-alive

Request Method

- PUT

Request URL

- /brms/api/v1.0/accounts/keepalive

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
token	Yes	string	Token

Request Example

```
PUT /brms/api/v1.0/accounts/keepalive HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84

{
  "token": "841e473bacef4b31abc5f53272afeb84"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- token	string	Token
- credential	string	Session credentials for downloading static resources such as images and files
- duration	int	Keep-alive interval (seconds)
- tokenRate	int	Update token interval (seconds)

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "token": "62cc3c253050429f9fdc839f833ffb2b",
    "credential": null,
    "duration": 30,
    "tokenRate": null
  }
}
```

```
}
```

3.1.4 Update Token

Brief Description

- Update token information

Request Method

- POST

Request URL

- /brms/api/v1.0/accounts/updateToken

Request Parameter

None

Request Example

```
POST /brms/api/v1.0/accounts/updateToken HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84

{
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- token	string	Token
- credential	string	Session credentials for downloading static resources such as images and files
- duration	int	Keep-alive interval (seconds)
- tokenRate	int	Update token interval (seconds)

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "token": "62cc3c253050429f9fdc839f833ffb2b",
    "credential": "ff435202efab4a54af01c5c0a4e65abe",
    "duration": 30,
    "tokenRate": 1800
  }
}
```

3.1.5 Logout

Brief Description

- Request to destroy a session authenticated by a user

Request Method

- POST

Request URL

- /brms/api/v1.0/accounts/unauthorize

Request Parameter

None

Request Example

```
POST /brms/api/v1.0/accounts/unauthorize HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 30c5f6bea5a643eaa761580cb762d787
```

```
{
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code

Parameter	Type	Description
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.2 Basic Data

3.2.1 Get MQ Configuration

Brief Description

- Get ActiveMQ configuration, judge intranet or Internet according to the request IP on client, and return corresponding configuration. At present, addresses of 5 protocols are returned and can be used on demand.

Request Method

- POST

Request URL

- /brms/api/v1.0/BRM/Config/GetMqConfig

Request Parameter

None

Request Example

```
POST /brms/api/v1.0/BRM/Config/GetMqConfig HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84

{
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- enableTls	string	Whether to enable TLS: 0: No; 1: Yes
- userName	string	Username
- mqtt	string	Mqtt protocol address, used on APP
- amqp	string	Amqp protocol address
- stomp	string	Stomp protocol address
- wss	string	Wss protocol address
- addr	string	Openwire protocol address, used on PC
- password	string	The password, after AES encryption, needs to be decrypted using the secret key secretKey and vector secretVector returned by [3.1.2 Submit Credentials]. For the decryption method, refer to the decryptWithAES7 method in [6.4 Encryption and Decryption Example]. Among them, the secretKey and the secretVector are not clear text, but are encrypted by RSA and can only be used after decryption. For the decryption method, refer to the decryptRSAByPrivateKey method in [6.4 Encryption and Decryption Example].

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "enableTls": "1",
    "mqtt": "192.168.1.1:1883",
    "amqp": "192.168.1.1:5672",
    "stomp": "192.168.1.1:61613",
    "wss": "192.168.1.1:61615",
    "addr": "192.168.1.1:61616",
    "userName": "consumer",
    "password":
      "954dd70aef4e85924ed3868bc295d6a59f646f1c262ce1bbc87b9cc44300355e"
  }
}
```

3.2.2 Get Time Template List

Brief Description

- Get time template list

Request Method

- GET

Request URL

- /brms/api/v1.1/time-template/list

Request Parameters

- Body Parameters

Parameter	Required	Type	Description
keyword	No	string	Keywords, supports fuzzy search, matches time template names

Request Example

```
GET /brms/api/v1.1/time-template/list?keyword=abc HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Example of a Response

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "id": "10",
        "timeTemplateName": "test",
        "type": "2",
        "monPeriods": [
          {
            "beginTime": "10:00:00",
            "endTime": "18:00:00"
          }
        ],
        "tuePeriods": [
          {
```

```

        "beginTime": "10:00:00",
        "endTime": "18:00:00"
      }
    ],
    "wedPeriods": [
      {
        "beginTime": "10:00:00",
        "endTime": "18:00:00"
      }
    ],
    "thuPeriods": [
      {
        "beginTime": "10:00:00",
        "endTime": "18:00:00"
      }
    ],
    "friPeriods": [
      {
        "beginTime": "10:00:00",
        "endTime": "18:00:00"
      }
    ],
    "satPeriods": [
      {
        "beginTime": "10:00:00",
        "endTime": "18:00:00"
      }
    ],
    "sunPeriods": [
      {
        "beginTime": "10:00:00",
        "endTime": "18:00:00"
      },
      {
        "beginTime": "19:00:00",
        "endTime": "23:00:00"
      }
    ]
  ]
}

```

Return Parameter

Parameter	Type	Description
code	int	The error code.

Parameter	Type	Description
desc	string	Result description
data	object	Null
- results	object	Result
- - id	string	Time template ID
- - timeTemplateName	string	Time template name
- - type	string	Template type: 1=default template, cannot be edited; 2=general template type
- - monPeriods	object	The period of Monday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss
- - tuePeriods	object	The period of Tuesday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss
- - wedPeriods	object	The period of Wednesday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss
- - thuPeriods	object	The period of Thursday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss
- - friPeriods	object	The period of Friday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss
- - satPeriods	object	The period of Saturday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss
- - sunPeriods	object	The period of Sunday, which is the value between begin time to end time
- - - beginTime	string	Begin time, format: HH:mm:ss
- - - endTime	string	End time, format: HH:mm:ss

3.2.3 Get Time Zone List

Brief Description

- Get system zone list

Request Method

- GET

Request URL

- /brms/api/v1.1/config/timezone/list

Request Parameters

None

Request Example

```
GET /brms/api/v1.1/config/timezone/list HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: e98cfd44f0e14a409e2d5ac7111b4547
```

Example of a Response

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "timezoneId": "1",
        "timezoneName": "(UTC-12:00) International Date Line West",
        "timezoneOffset": "+08:00",
        "enableDst": "1",
        "dstType": "1",
        "dstBeginTime": {
          "month": "4",
          "week": "0",
          "day": "2",
          "time": "02:40"
        },
        "dstEndTime": {
          "month": "10",
```

```

    "week": "0",
    "day": "6",
    "time": "03:50"
  }
}
]
}
}

```

Return Parameter

Parameter	Type	Description
code	int	The error code
desc	string	Result description
data	object	Null
- results	object	Result
- - timeZoneId	string	Time zone ID
- - timeZoneName	string	Time zone name
- - timeZoneOffset	string	Time zone offset, for example: +08:00
- - enableDst	string	Enable DST 0: Close 1: Open
- - dstType	string	DST type: 1=date mode, 2=week mode
- - - dstBeginTime	object	Start time of DST
- - - month	string	Month
- - - week	string	Week (the field value is 0 for the date mode); -1 means the final week
- - - day	string	Date (meaning a weekday in the week mode); 0 means Sunday
- - - time	string	Time, such as 09:30
- - dstEndTime	object	End time of DST
- - - month	string	Month
- - - week	string	Week (the field value is 0 for the date mode); -1 means the final week
- - - day	string	Date (meaning a weekday in the week mode); 0 means Sunday
- - - time	string	Time, such as 09:30

3.2.4 Get All Role List

Brief Description

- Get all role list

Request Method

- GET

Request URL

- /brms/api/v1.1/role/all/list

Request Parameters

None

Request Example

```
GET /brms/api/v1.1/role/all/list HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676462c
```

Return Parameter

Parameter	Type	Description
code	int	The error code
desc	string	Result description
data	object	Data
- results	array	Results
- - id	string	Role id
- - roleName	string	Role name
- - memo	string	Description
- - manageable	string	Whether the current user can manage the role: 0 = No, 1 = Yes
- - roleType	string	Role type: 1 = Super Administrator, 2 = Administrator (default), 3 = Custom Operator, 4 = Custom Administrator
- - owning	string	Whether the current user owns the role: 0 = No, 1 = Yes

Example of a Response

```
{
  "code": 1000,
  "desc": "Success",
```

```
{
  "data": {
    "results": [
      {
        "id": "1",
        "roleName": "Super Administrator",
        "memo": "",
        "manageable": "0",
        "roleType": "1",
        "owning": "0"
      },
      {
        "id": "2",
        "roleName": "Administrator",
        "memo": "",
        "manageable": "0",
        "roleType": "2",
        "owning": "0"
      },
      {
        "id": "3",
        "roleName": "test",
        "memo": "just for test",
        "manageable": "1",
        "roleType": "3",
        "owning": "1"
      }
    ]
  }
}
```

3.2.5 Get a List of Data Dictionaries

Brief Description

- Get a list of data dictionaries

Request Method

- GET

Request URL

- `/brms/api/v1.1/dictionary/list?types={types}`

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
types	Yes	string	Dictionary type, separated by commas

Request Example

```
GET /brms/api/v1.1/dictionary/list?types=9046 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ca19ae28474040f192b9fc8cc7b9c69a
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- results	array	Result
- - type	string	Dictionary type
- - name	string	Dictionary type name
- - items	array	Dictionary item
- - - code	int	Dictionary code
- - - name	string	Dictionary name
- - - param1	string	Parameter 1, separated by commas
- - - param2	string	Parameter 2, separated by commas
- - - sort	string	Sort

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "type": "8",
        "name": "alarm type",
        "items": [
          {
            "code": "41",
            "name": "Duress Alarm",
            "param1": "16",
```

```
{
  "param2": "0",
  "sort": "1"
}
```

3.3 Device

3.3.1 Device Information

3.3.1.1 Get Device Details

Brief Description

- Get device details. Calling interfaces requires administrator privileges.

Request Method

- GET

Request URL

- /brms/api/v1.1/device/{deviceCode}

Request Parameters

- Path Parameters

Parameter	Required	Type	Description
deviceCode	Yes	string	Device code

Request Example

```
GET /brms/api/v1.1/device/1000000 HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- deviceCode	string	Device code
- addType	string	Device adding type: 1 = IP address; 2 = IP segment; 3 = Domain name; 4 = Auto registration; 5 = P2P adding device
- deviceIp	string	IP address
- devicePort	string	Port
- loginName	string	Device account
- loginPass	string	Device password, AES encryption
- protocolType	string	Protocol type: 1=VMS, 2=Hikvision, 98=SNMP, 99=Onvif, 100=SIA, 185=JIUZHOU
- deviceCategory	string	Device category
- deviceType	string	Device sub-category
- deviceSeries	string	Device series
- deviceModel	string	Device model
- mac	string	Device mac address
- deviceName	string	Device name
- deviceSn	string	Device serial number
- capability	string	Device capability
- registerId	string	Registration ID
- manufacturer	string	Manufacturer: 1 = VMS; 2 = Hikvision; 3 = Other; 185 = Jiuzhou
- manufacturerName	string	Customized manufacturer name, which does not correspond to the manufacturer field one-to-one
- serverCode	string	Server code
- orgCode	string	Organization code
- sipId	string	Video call SIP-ID
- sipPwd	string	SIP password of video call, AES encryption
- enableBuilding	string	Enable building for talk devices
- enableUnit	string	Enable unit for talk devices
- enablePhase	string	Enable phase for talk devices

Parameter	Type	Description
- softwareVersion	string	Software version
- hardwareVersion	string	Hardware version
- protocolVersion	string	Supported protocol versions: 1=first generation, 2=second generation
- faceAlgorithmVersion	string	Face algorithm version
- domainId	string	Domain ID
- isVideoNexus	string	Whether large baseline for video communication nexus: 1 = Yes, 0 = No
- proxyIp	string	Proxy IP (Auto registration /P2P/ONVIF/Hikvision/GB28281 devices not connected by IP)
- proxyPort	string	Proxy port (Auto registration /P2P/ONVIF/Hikvision/GB28281 devices not connected by IP)
- snmpReadCredential	string	SNMP protocol read credential
- snmpWriteCredential	string	SNMP protocol write credential
- snmpAuthVersion	string	SNMP authentication version: 1=v1, 2=v2, 3=v3
- snmpAuthType	string	SNMP authentication encryption type: 1=MD5, 2=SHA
- snmpUser	string	SNMP authentication username
- snmpPassword	string	SNMP authentication password
- snmpEncryptType	string	SNMP message encryption type: 1=CBC-DES, 2=CFB-AES
- snmpEncryptSecretKey	string	SNMP message encryption key
- devicePosition	string	Device location
- palmVeinAlgorithmVersion	string	Palmar vein algorithm version
- deviceContacts	array	Device contact
- - contactName	string	Device contact name
- - contactPhone	string	Device contact phone
- - contactEmail	string	Device contact email
- - contactNote	string	Remark
- timeZoneId	string	Time zone ID
- timeZoneName	string	Time zone name
- timeZoneOffset	string	Time zone offset, such as +08:00

Parameter	Type	Description
- enableDst	string	Enable DST: 0 = Disabled; 1 = Enabled
- dstType	string	DST type: 1 = Date mode; 2 = Week mode
- dstBeginTime	object	Start time of DST
- - month	string	Month
- - week	string	Week (the field value is 0 for the date mode); -1 means the final week
- - day	string	Date (meaning a weekday in the week mode); 0 means Sunday
- - time	string	Time, such as 09:30
- dstEndTime	object	End time of DST
- - month	string	Month
- - week	string	Week (the field value is 0 for the date mode); -1 means the final week
- - day	string	Date (meaning a weekday in the week mode); 0 means Sunday
- - time	string	Time, such as 09:30
- timeSyncStatus	string	Syncing status: 0 = Failed; 1 = Successful; 2 = In distribution; 3 = Not supported
- controllers	array	Controller
- - controllerSeq	string	Controller number
- - name	string	Name
- - property	string	Unidirection/Bidirection information
- encoderUnits	array	Encoding unit information
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Total number of channels
- - streamType	string	Stream type
- - zeroChannelEncode	string	Zero-channel encoding or not
- - transcoder	string	Whether to use Hikvision private stream: 0 = No; 1 = Yes
- - channels	array	Channel information
- - - channelSeq	string	Channel number
- - - channelSn	string	Channel serial number

Parameter	Type	Description
- - - channelType	string	Channel type, such as video channel, picture channel, video/picture mixed channel
- - - channelSubtype	string	Channel sub-type: 0 = Normal video; 1 = Visible light; 2 = Thermal imaging
- - - remoteChannelType	string	Remote channel type, including: local encoding channel, remote channel, cascading channel, and analog matrix channel
- - - channelName	string	Channel name
- - - cameraType	string	Camera type, such as bullet camera, PTZ camera, dome camera, etc.
- - - keyCode	string	Keyboard code
- - - streamType	string	Channel stream type: 1: Main stream, 2: Sub stream 1, 3: Sub stream 2, default 1
- - - capability	string	Capability set, a decimal system value
- - - enableWeb	string	Support for access channels web:0 = No, 1 = Yes
- - - channelMac	string	Channel mac address
- - - encoderChannelType	string	Video Channel Type: 1 = Local channel, 2 = Remote channel
- - - gpsX	string	Longitude
- - - gpsY	string	Latitude
- - - mapId	string	Map ID
- - - rtspUrls	array	RTSP address
- - - - streamType	string	Channel stream type: 1: Main stream, 2: Sub stream 1, 3: Sub stream 2, default 1
- - - - rtspUrl	string	RTSP address, AES encryption
- - - channelId	string	Channel IP
- - - channelExt	object	Channel extension information
- - - - roadway	string	Lane number
- - - - cartMaxSpeed	string	Maximum speed of large vehicle
- - - - cartMinSpeed	string	Minimum speed of large vehicle
- - - - dollyMaxSpeed	string	Maximum speed of small vehicle
- - - - dollyMinSpeed	string	Minimum speed of small vehicle

Parameter	Type	Description
- - - direction	string	Driving direction
- decoderUnits	array	Decoding unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Number of channels
- - decodeMode	string	Decoding mode
- - supportCombine	string	Whether to support combination: 0 = No; 1 = Yes
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelSn	string	Channel serial number
- - - channelName	string	Channel name
- - - maxSplitCount	string	Maximum number of splits
- - - channelType	string	Channel type: 1 = Output channel; 2 = Decoding channel
- alarmInputUnits	array	Alarm input channel
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - alarmType	string	Alarm type
- - - alarmLevel	string	Alarm level
- - - parentTargetSeq	string	The target serial number. If it is empty, it indicates the device itself; otherwise, it indicates the serial number of the remote front-end channel
- - - parentTargetName	string	Target name, channel name or device name
- - - associateId	string	External ID of the external connection, which is the unique ID of the three-party system
- - - subsystemSeq	string	Subsystem serial number
- - - supportPartial	string	Partition of defense area

Parameter	Type	Description
- - - deviceIdentification	string	Attribute of accessory device category
- - - extendZone	string	Whether to extend the defense zone: 0 = No 1 = Yes
- - - supportSnapshot	string	Whether to support snapshot: 0 = No, 1 = Yes
- - - accessorySn	string	Defense area accessory serial number
- alarmOutputUnits	array	Alarm output unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - parentTargetSeq	string	The target serial number. If it is empty, it indicates the device itself; otherwise, it indicates the serial number of the remote front-end channel
- - - parentTargetName	string	Target name, channel name or device name
- - - activeStatus	string	Channel enable status: 0=disabled, 1=enabled, 2=unknown
- accessUnits	array	Access control unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - fingerPrintAuth	string	Whether to support fingerprint authentication: 0 = Unknown (compatible with the previous), 1 = Not supported; 2 = Supported
- - cardAuth	string	Whether to support card authentication: 1 = Yes; 0 = No
- - faceAuth	string	Whether to support face recognition authentication: 1 = Yes; 0 = No
- - palmVeinAuth	string	Whether to support palm vein authentication: 1 = Yes; 0 = No
- - multiFace	string	Whether to support multi-face recognition: 1 = Yes; 0 = No

Parameter	Type	Description
- - userIsolate	string	Whether to support human-card separation: 1 = Yes; 0 = No
- - unlockModes	string	Combination of the unlocking modes supported by the device; separated by commas for multiple modes, for example: 0, 1, 2, 3
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - controllerSeq	string	Controller number
- - - readIds	array	Read head ID list
- virtualUnits	object	Virtual unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- posUnits	array	POS unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- ledUnits	array	LED unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - direction	string	Display direction: 1 = Go straight; 2 = Left turn; 3 = Right turn; 4 = Go straight and left turn; 5 = Go straight and right turn; 13 = Pregnant; 14 = Vehicle

Parameter	Type	Description
- - - defaultText	string	Preset text
- - - showContents	object	Displayed content
- - - - id	string	Number
- - - - content	string	Displayed content
- - - - userDefined	string	Custom content or not: 0 = No; 1 = Yes
- - - fixedContentType	string	Fixed content in idle mode: 0 = Empty; 1 = Empty parking spots in field; 2 = Custom content
- - - fixedContent	string	Arrow direction: 1 = Left; 2 = Right
- - - row	string	Number of displayed rows
- - - height	string	LED screen height
- - - width	string	LED screen width
- - - color	string	LED display color: 1 = Red; 2 = Green; 3 = Yellow
- - - zeroDisplayType	string	LED display color: 1 = Red; 2 = Green; 3 = Yellow
- - - rollType	string	Content rolling type: 0 = Fixed; 1 = Rolling
- - - rollSpeed	string	Rolling speed
- cviAlarmUnits	array	CVI alarm unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - encoderChannelSeq	string	Video channel number
- - - encoderChannelName	string	Video channel name
- - - alarmType	string	Alarm type
- boxAlarmUnits	array	Alarm box unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number

Parameter	Type	Description
- - - channelName	string	Channel name
- - - boxSeq	string	Alarm box number, value range [0, 3]
- - - alarmType	string	Alarm type
- boxAlarmOutUnits	array	Alarm box output unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - boxSeq	string	Alarm box number, value range [0, 3]
- - - boxOutIndex	string	Alarm box output channel address
- audioLightAlarmUnits	object	Audio light alarm unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - parentTargetSeq	string	Target serial number, if it is empty, it means the device itself, otherwise it means the serial number of the remote front-end channel
- - - parentTargetName	string	Target name, channel name or device name
- - - channelType	string	Channel type: 1=Audio, 2=Light
- enableLift	string	Lift control enable
- liftUnits	array	Lift unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Quantity of channels
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - floorNum	string	Floor number

Parameter	Type	Description
- - - gloabControl	string	Global lift control
- - - floors	array	Floor
- - - - floorSerialNo	string	Floor serial no.
- - - - floorName	string	Floor name
- - fingerPrintAuth	string	Whether fingerprint authentication is supported: 0 = No, 1 = Yes
- easUnits	array	Eas unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - unitName	string	Unit name
- - channelCount	string	Channel count
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- alarmSignalUnits	array	Alarm signal unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Channel count
- - channels	object	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- uvssUnits	array	UVSS unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Channel count
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- parkingSpaceUnits	array	Parking space unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Channel count
- - parkingSpaceMode	string	Vehicle detector mode: 0 = parking space area mode or unknown, 1 = license plate mode, 2 = parking

Parameter	Type	Description
		space statistics mode (mainly outdoor vehicle detector)
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- - - encoderChannelSeq	string	Serial number of the video channel
- - - encoderChannelName	string	Name of the video channel
- ledPartitionUnits	array	Led partitionUnits unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Channel count
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- gateUnits	array	Gate unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Channel count
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- audioUnits	array	Audio unit
- - unitType	string	Unit type
- - unitSeq	string	Unit number
- - channelCount	string	Channel count
- - channels	array	Channel list
- - - channelSeq	string	Channel number
- - - channelName	string	Channel name
- subsystems	array	Subsystem Information
- - subsystemSeq	string	Subsystem seq
- - subsystemName	string	Subsystem name
- cardNo	string	LingGuan device card no.
- outRtspPort	string	External network mapping streaming port of onvif device

Parameter	Type	Description
- enableRtspTls	string	Onvif supports RTSP OVER TLS enable: 0=off, 1=on
- encryptType	string	Encryption method: 0=No encryption, 1=AES-CBC Nopadding
- accessKey	string	Access key
- secretKey	string	Secret key, AES encryption
- sourceType	string	Source type: 1=Platform, 3=Third-party
- sourceId	string	Source ID
- associateId	string	Third party device ID
- streamProtocol	string	Media transmission protocol: 1=TCP, 2=UDP, 3=HTTP, valid for ONVIF devices
- linkMode	string	Device connection mode: 0=long connection, 1=short connection
- maxBatchIssuePersons	string	The access control device supports the maximum number of personnel issued in batches, with a default of 1
- netWork	string	Device usage working network: 1=public network, 2=network card 1, 3=network card 2
- unlockModes	string	The combination of unlocking modes supported by the device, multiple separated by commas, such as 0,1,2,3
- exceedChannelCapability	string	Channel capacity set beyond platform limit, decimal value

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "deviceCode": "1000026",
    "deviceName": "dev1",
    "deviceIp": "192.168.1.1",
    "orgCode": "001",
    "deviceCategory": "1",
    "deviceType": "2",
    "devicePort": "37777",
    "outRtspPort": "0",
    "loginName": "admin",
```

```
"loginPass": null,
"protocolType": "1",
"deviceSeries": "0",
"deviceModel": "N5",
"mac": "4c:37:77:74:96:58",
"deviceSn": "sim-3777749658",
"capability": "0",
"registerId": "",
"manufacturer": "1",
"manufacturerName": "",
"serverCode": "video#local",
"sipId": null,
"sipPwd": null,
"enableBuilding": null,
"enableUnit": null,
"enablePhase": null,
"softwareVersion": "12345",
"faceAlgorithmVersion": "",
"hardwareVersion": null,
"domainId": null,
"isVideoNexus": null,
"proxyIp": null,
"proxyPort": "",
"addType": "1",
"snmpReadCredential": "",
"snmpWriteCredential": "",
"snmpAuthVersion": "",
"snmpAuthType": "",
"snmpUser": "",
"snmpPassword": "",
"snmpEncryptType": "",
"snmpEncryptSecretKey": "",
"devicePosition": "",
"palmVeinAlgorithmVersion": "v1.0",
"deviceContacts": [
  {
    "contactName": "Jack",
    "contactPhone": "12345678",
    "contactEmail": "test@123.com",
    "contactNote": ""
  }
],
"timezoneId": "100",
"timezoneName": "(UTC+08:00) Beijing, Chongqing, Hong Kong, Urumqi",
"timezoneOffset": "+08:00",
"enableDst": "0",
"dstType": "2",
"dstBeginTime": {
```

```
        "month": "1",
        "week": "1",
        "day": "1",
        "time": "00:00"
    },
    "dstEndTime": {
        "month": "1",
        "week": "1",
        "day": "2",
        "time": "00:00"
    },
    "timeSyncStatus": "1",
    "controllers": [],
    "encoderUnits": [
        {
            "unitType": "1",
            "unitSeq": "0",
            "channelCount": "1",
            "channels": [
                {
                    "channelSeq": "0",
                    "channelName": "Channel0",
                    "channelSn": null,
                    "channelType": "1",
                    "channelIp": null,
                    "channelMac": null,
                    "channelSubtype": null,
                    "remoteChannelType": null,
                    "cameraType": "1",
                    "keyCode": null,
                    "capability": "21",
                    "channelExt": {
                        "roadway": null,
                        "cartMaxSpeed": null,
                        "cartMinSpeed": null,
                        "dollyMaxSpeed": null,
                        "dollyMinSpeed": null,
                        "direction": null
                    },
                    "streamType": "3",
                    "rtspUrls": null,
                    "enableWeb": "0",
                    "encoderChannelType": "1",
                    "gpsX": "120.45",
                    "gpsY": "30.12",
                    "mapId": "0",
                }
            ]
        }
    ],
```

```
        "streamType": "2",
        "zeroChannelEncode": "0",
        "transcoder": "0"
    }
],
"decoderUnits": null,
"alarmInputUnits": [
    {
        "unitType": "3",
        "unitSeq": "0",
        "channelCount": "1",
        "channels": [
            {
                "channelSeq": "0",
                "channelName": "dev1_1",
                "alarmLevel": "1",
                "alarmType": "2",
                "parentTargetSeq": null,
                "parentTargetName": "dev1",
                "associateId": null,
                "subsystemSeq": null,
                "supportPartial": "0",
                "deviceIdentification": "",
                "extendZone": "0",
                "supportSnapshot": "1",
                "accessorySn": ""
            }
        ]
    }
],
"alarmOutputUnits": [
    {
        "unitType": "4",
        "unitSeq": "0",
        "channelCount": "1",
        "channels": [
            {
                "channelSeq": "0",
                "channelName": "dev1_1",
                "parentTargetSeq": null,
                "parentTargetName": null,
                "activeStatus": "1"
            }
        ]
    }
],
"accessUnits": null,
"virtualUnits": null,
```

```
"posUnits": null,
"ledUnits": null,
"cviAlarmUnits": null,
"boxAlarmUnits": null,
"boxAlarmOutUnits": null,
"audioLightAlarmUnits": null,
"enableLift": "0",
"liftUnits": null,
"parkingSpaceUnits": null,
"ledPartitionUnits": null,
"easUnits": null,
"uvssUnits": null,
"alarmSignalUnits": null,
"audioUnits": null,
"gateUnits": null,
"cardNo": "",
"enableRtspTls": "0",
"protocolVersion": "",
"encryptType": "0",
"accessKey": "",
"secretKey": "",
"sourceType": "1",
"sourceld": null,
"associateld": "",
"unlockModes": "",
"streamProtocol": "1",
"linkMode": "0",
"maxBatchIssuePersons": "1",
"subsystems": [],
"netWork": null,
"exceedChannelCapability": "0"
}
}
```

3.3.1.2 Get the List of Devices in Pages

Brief Description

- Get list of devices in pages. Calling interfaces requires administrator privileges.

Request Method

- GET

Request URL

- /brms/api/v1.1/device/page

Request Parameters

- Body Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orderType	No	string	Sorting type: 1 = device IP, 2 = device model, 3 = device code. If it is null, default sorting by online status
orderDirection	No	string	Sorting direction: 0 = Ascending sequence; 1 = Descending sequence, default 1
keyword	No	string	Keywords, fuzzy search supported, to match device name, IP, and device code
orgCode	No	string	Organization code
deviceCategory	No	string	Device category
deviceType	No	string	Device sub-category
status	No	string	Device status: 0 = Offline; 1 = Online; null means all
permitted	No	string	Permission filtering: 0 = Not filtered; 1 = Filtered; null means all
domain	No	string	Cascade device included or not: 0 = No; 1 = Yes; null means all
sourceType	No	string	Device channel type: 1=Platform, 3= Third-party, blank indicates querying all
containChild	No	string	Include subsets: 0=No, 1=Yes, default to 1
protocolTypes	No	string	Protocol type: 1=Dahua Protocol, 2=Hikvision Access, 99=Onvif Access, 185=Jiuzhou Protocol, multiple separated by commas, empty indicates all
manufacturers	No	string	Manufacturer: 1=Dahua, 2=Hikvision, 4=AXIS, 5=Uniview, 6=BOSCH, 185=JIUZHOU, 246=Other, multiple separated by commas, empty means query all

Request Example

```
GET /brms/api/v1.1/device/page?page=1&pageSize=20&orderType=&orderDirection=
&orgCode=001&deviceCategory=&deviceType=&status=&keyword=&permitted=&do
main=&sourceType=&containChild=&protocolTypes=&manufacturers= HTTP/1.1
Host: 192.168.1.1
```


Connection: close
 Accept-Language: en
 Content-Type: application/json;charset=UTF-8
 X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Total number
- pageData	object	None
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - deviceIp	string	Device IP
- - registerIp	string	Register IP
- - status	Integer	Device status: 0 = Offline; 1 = Online
- - deviceModel	string	Device model
- - orgCode	string	Organization number
- - orgName	string	Organization name
- - serverCode	string	Server code
- - serverName	string	Server name
- - deviceCategory	Integer	Device category
- - deviceCategoryName	string	Device category name
- - deviceType	string	Device sub-category
- - deviceTypeName	string	Device sub-category name
- - domainId	string	Cascade ID
- - offlineReasonCode	string	Offline reason code
- - offlineReasonDesc	string	Offline reason description
- - enableEdit	string	Enable edit: 0 = No; 1 = Yes
- - enableDelete	string	Enable delete: 0 = No; 1 = Yes
- - enableWeb	string	Whether to access device web: 0 = No; 1 = Yes
- - manufacturer	string	Manufacturer
- - manufacturerName	string	Manufacturer name
- - protocolType	string	Protocol
- - protocolTypeName	string	Protocol name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "deviceCode": "1000000",
        "deviceName": "test",
        "deviceIp": "192.168.1.1",
        "registerIp": "",
        "status": "1",
        "deviceModel": "ASI7213Y-V3",
        "orgCode": "001",
        "orgName": "root",
        "serverCode": "video#local",
        "serverName": "center server",
        "deviceCategory": "8",
        "deviceCategoryName": "Access Control",
        "deviceType": "5",
        "deviceTypeName": "Access Control 2nd",
        "domainId": null,
        "offlineReasonCode": "9",
        "offlineReasonDesc": "",
        "enableEdit": "1",
        "enableDelete": "1",
        "enableWeb": "1",
        "manufacturer": "1",
        "manufacturerName": "VMS",
        "protocolType": "1",
        "protocolTypeName": "VMS"
      }
    ]
  }
}
```

3.3.1.3 Get the List of Device Types

Brief Description

- Get the list of device types

Request Method

- GET

Request URL

- /brms/api/v1.1/definition/device-type/list

Request Parameter

None

Request Example

```
GET /brms/api/v1.1/definition/device-type/list HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- results	array	Capability set
- - deviceCategory	string	Device category ID
- - deviceCategoryName	string	Device category name
- - deviceTypes	array	Device sub-category
- - - deviceType	string	Device sub-category ID
- - - deviceTypeName	string	Device sub-category name
- - - source	string	Device subcategory source: 1=Platform, 3=Third-party, default platform
- - manufacturers	array	Manufacturer
- - - manufacturer	string	Manufacturer: 1 = VMS, 2 = HK, 4 = Axis, 5 = Yushi, 6 = Boshi, 185 = Jiuzhou, 246 = Others
- - - manufacturerName	string	Manufacturer name
- - protocolTypes	array	Protocol type
- - - protocolType	string	Protocol type: 1 = VMS, 2 = HK, 68 = RTSP, 98 = SNMP, 99 = Onvif, 100 = SIA (ADM-CID), 110 = SIA (DCS), 120 = DDIP, 185 = Jiuzhou
- - - protocolTypeName	string	Protocol type name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "deviceCategory": "1",
        "deviceCategoryName": "Encoder",
        "deviceTypes": [
          {
            "deviceType": "1",
            "deviceTypeName": "DVR",
            "source": "1"
          }
        ],
        "manufacturers": [
          {
            "manufacturer": "1",
            "manufacturerName": ""
          }
        ],
        "protocolTypes": [
          {
            "protocolType": "1",
            "protocolTypeName": ""
          }
        ]
      }
    ]
  }
}
```

3.3.1.4 Search Offline Reasons of Devices

Brief Description

- Search device offline reason

Request Method

- GET

Request URL

- /brms/api/v1.0/devicesManager/devices/offlineReasons?deviceCodes={deviceCodes}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
deviceCodes	Yes	string	Device code, separated by commas for multiple codes

Request Example

```
GET /brms/api/v1.0/devicesManager/devices/offlineReasons?deviceCodes=1000004 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: cbfd3c31ad7345bc807230b51b8eb7bf
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- offlineReasons	object	Device offline information
- - deviceCode	string	Device code
- - offlineReason	string	Offline error code, dictionary type. See [6.1.8 Offline Reasons of Devices].

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "offlineReasons": [
      {
        "deviceCode": "1000004",
        "offlineReason": "9"
      }
    ]
  }
}
```

3.3.1.5 Obtaining the Status of Devices and Channels in Batches

Brief Description

- Obtain the status of devices and channels in batches based on device codes.

Request Method

- POST

Request URL

- /brms/api/v1.1/device/status/fetch/batch/list

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
deviceCodes	No	array	The device code.

Request Example

```
POST /brms/api/v1.1/device/status/fetch/batch/list HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "deviceCodes": [
    "1000001",
    "1000002"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- results	object	Result
- - deviceCode	string	The device code.
- - status	int	The device status: 0 = Offline, 1 = Online.
- - sleepStatus	int	The sleep status: 0 = Not sleeping, 1 = Sleeping.

Parameter	Type	Description
-- channels	object	The status of device channels.
--- channelId	string	The channel code.
--- status	int	The channel status: 0 = Offline, 1 = Online.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "deviceCode": "1000001",
        "status": 1,
        "sleepStatus": 0,
        "channels": [
          {
            "channelId": "1000001$1$0$0",
            "status": 1
          },
          {
            "channelId": "1000001$1$0$1",
            "status": 0
          }
        ]
      }
    ]
  }
}
```

3.3.1.6 Capture the Picture of Real-time Video of the Specified Channel

Brief Description

- Capture real-time video pictures for video channels. It does not support business capture, such as re-capturing the license plate by ANPR. This interface does not support high frequency calls, please control the frequency of use. Note that the picture disk needs to be configured on the platform first.

Request Method

- POST

Request URL

- /brms/api/v1.1/device/channel/{channelId}/video/capture

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
channelId	Yes	string	Channel code

Request Example

```
POST /brms/api/v1.1/device/channel/1000001$1$0$0/video/capture HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: df0b13ee8eef4ea0a5a3ff412be05b5b

{
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data
- fileUrl	string	Picture url

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "fileUrl": "https://192.168.1.1/abc.jpg"
  }
}
```

3.3.2 Device Tree

3.3.2.1 Getting Device Tree (All)

Brief Description

- Getting device tree list

Request Method

- POST

Request URL

- /brms/api/v1.0/tree/devices

Request Parameters

- Body Parameters

Parameter	Required	Type	Description
deviceCodes	No	array	Device code, if it is null, all devices are obtained
categories	No	array	Device category, dictionary type. See [6.1.2 Device Type]
containVirtualDevice	No	string	Include virtual devices: 0=No, 1=Yes, default does not include virtual devices
sourceTypes	No	array	Source Type: 1=Platform, 3=Tripartite. If it is null, all devices are obtained
resourceTypes	No	array	Resource type: 1 = video resource, 2 = access control resource, 3 = ladder control resource, 4 = audio resource, 5 = EAS alarm channel, 6 = alarm input resource, 7 = alarm output resource, 8 = POS channel resource, 9 = radar resource, 10 = underbody scanning resource, 14 = decoding channel, 15 = sound and light warning channel, 16 = equipment; If it is empty, it means no resource type filtering will be performed.
filterAreaAuthority	No	string	Filter area permission: 0 = No, 1 = Yes, default 0

Request Example

```
POST /brms/api/v1.0/tree/devices HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 433b9cac285d434683643f79fb86be6b

{
  "deviceCodes": [],
  "categories": [],
```

```

    "containVirtualDevice": "1",
    "sourceTypes": [],
    "resourceTypes": [],
    "filterAreaAuthority": ""
  }

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- devices	array	Device Information
- - code	int	Device Code
- - name	string	Device name
- - category	string	Device category
- - type	string	Device type
- - model	string	Model
- - deviceModelStr	string	Model string
- - status	string	Device status: 0 = Offline; 1 = Online
- - offlineReason	string	Offline reason
- - protocol	string	Protocol
- - manufacturer	string	Manufacturer
- - deviceIp	string	Device IP
- - devicePort	string	Device port
- - registId	string	Register ID
- - proxyIp	string	Proxy IP, which is the ARS service IP when actively registering, and the PCPS service IP when using devices such as Hikvision, ONVIF, and GB28181
- - proxyPort	string	Proxy port
- - userName	string	User name
- - password	string	Password, using AES encryption
- - callNumber	string	SIP number
- - loginType	string	Login type
- - sn	string	SN number
- - alarmRecordPosition	string	Video storage location
- - capability	string	Device capability set, expressed in Decimal

Parameter	Type	Description
- - orgCode	string	Organization code
- - sipId	string	SIP ID
- - sipPwd	string	SIP password
- - bindingConfirmVtoDeviceCodes	array	Code list of secondary confirmation machine bound with indoor unit
- - unitEnable	string	Intercom unit enable
- - buildingEnable	string	Intercom building enable
- - liftEnable	string	Ladder control enable
- - softwareVersion	string	Software version
- - hardwareVersion	string	Hardware version
- - domainId	string	Domain ID
- - sourceType	string	Source type: 1=platform, 3=third-party
- - sourceId	string	Source ID
- - associateId	string	Third-party device ID
- - authority	string	Current user have permission for this device? 0=No, 1=Yes
- - modifyTime	string	Change date
- - isVideoNexus	string	Is video nexus: 1-Yes, 0-No
- - timeZoneId	string	Time zone ID
- - timeZoneName	string	Time zone name
- - timeZoneOffset	string	Time zone offset, such as +08:00
- - timeZoneDesc	string	Time zone description
- - enableDst	string	Enable DST: 0 - Disabled; 1 - Enabled
- - dstType	string	DST type: 1 - Date mode; 2 - Week mode
- - dstBeginTime	object	Start time of DST
- - - month	string	Month
- - - week	string	Week (the field value is 0 for the date mode); -1 means the final week
- - - day	string	Date (meaning a weekday in the week mode); 0 means Sunday
- - - time	string	Time, such as 09:30
- - dstEndTime	object	End time of DST
- - - month	string	Month
- - - week	string	Week (the field value is 0 for the date mode); -1 means the final week

Parameter	Type	Description
- - - day	string	Date (meaning a weekday in the week mode); 0 means Sunday
- - - time	string	Time, such as 09:30
- - serverCode	string	Server code
- - units	array	
- - - unitType	string	Unit type
- - - unitSeq	string	Unit sequence
- - - assistStream	string	Sub stream
- - - zeroChnEncode	string	Zero channel encode
- - - ssServiceId	string	SS service ID
- - - ptsServiceId	string	PTS service ID
- - - streamType	string	Code-stream type
- - - decodeMode	string	Decoding mode
- - - streamMode	string	Stream processing mode
- - - combineStatus	string	Video wall combining status, 0-combining is not supported, 1-combining is supported
- - - thirdPartyControl	string	Third party control
- - - voiceServerIp	string	Voice service IP
- - - voiceServerPort	string	Voice service port
- - - voiceStatusPort	string	Voice status port
- - - voiceClientIp	string	Voice client IP
- - - dynCode	string	Power environment resource code
- - - dynName	string	Power environment resource name
- - - dynType	string	Power environment resource type
- - - fingerPrintAuth	string	Whether fingerprint authentication is supported, 0-unknown, compatible with previous one (default), 1-not supported, 2-supported
- - - cardAuth	string	Whether card authentication is supported, 1-Yes, 0-No
- - - faceAuth	string	Whether face recognition authentication is supported, 1-Yes, 0-No
- - - userIsolate	string	Whether the user and card are separated, 1-Yes, 0-No
- - - unlockModes	string	The combination of unlocking modes supported by the device, multiple separated by commas, such as 0,1,2,3

Parameter	Type	Description
--- parkingStatisticsMode	string	Parking space statistics mode: 1=remaining space mode, 2=license plate mode
--- channels	array	Channel information
---- channelCode	string	Channel code
---- channelName	string	Channel name
---- channelSeq	string	Channel sequence, starting from 0
---- status	string	Channel status
---- sn	string	SN number
---- gpsX	string	GPS(X)
---- gpsY	string	GPS(X)
---- mapId	string	Raster map ID
---- cameraType	string	Camera type
---- channelType	string	Channel type
---- remoteType	string	Remote type
---- channelSubtype	string	Channel sub type, video channel [0=ordinary video 1-visible light 2-thermal imaging]
---- streamType	string	Channel code stream type: 1=main stream, 2=sub stream 1, 3=sub stream 2, default 1
---- cameraFunctions	string	Channel function set
---- multicastIp	string	Multicast IP
---- multicastPort	string	Multicast port
---- ipClp	string	IPC(IP)
---- recordType	string	Record type
---- forPeopleCount	string	Whether it is used for people count
---- maxSplitNum	string	Maximum number of video wall split
---- interfaceType	string	Interface type
---- alarmType	string	Alarm type
---- alarmLevel	string	Alarm level
---- signalType	string	Signal type
---- accessType	string	Access control type
---- posType	string	Pos type
---- posModel	string	Pos model
---- dynCode	string	Power environment resource
---- faceFunctions	string	Face image function
---- keyCode	string	Keyboard code

Parameter	Type	Description
- - - - db33Code	string	Landmark code
- - - - intelliState	string	Intelligent status
- - - - videoSource	string	Video source for virtual channel
- - - - targetDetection	string	Target detection function
- - - - capability	string	Capability set
- - - - domainId	string	Domain ID
- - - - floorNum	string	Number of floors
- - - - floorInfos	array	Floor information
- - - - - floorSerialNo	string	Floor number
- - - - - floorName	string	Floor name
- - - - parkingSpaceMode	string	Vehicle detector mode: 0 = parking space area mode or unknown, 1 = license plate mode, 2 = parking space statistics mode
- - - - parentTargetSeq	string	Target serial number
- - - - orgCode	string	Organizational code

Return example

```
{
  "code": 1000,
  "data": {
    "devices": [
      {
        "alarmRecordPosition": "3",
        "buildingEnable": "0",
        "capability": "2",
        "category": "1",
        "code": "1000000",
        "deviceIp": "10.55.51.82",
        "deviceModelStr": "IPC-HF7442F-FR",
        "devicePort": "37777",
        "domainId": "",
        "sourceType": "1",
        "sourceId": "",
        "associateId": "",
        "dstBeginTime": {
          "day": "1",
          "month": "1",
          "time": "00:00",
          "week": "1"
        },
        "dstEndTime": {
```

```
"day": "2",
"month": "1",
"time": "00:00",
"week": "1"
},
"dstType": "2",
"enableDst": "0",
"isVideoNexus": "0",
"liftEnable": "0",
"loginType": "0",
"manufacturer": "1",
"model": "0",
"modifyTime": "1686365433",
"name": "10.55.51.82",
"offlineReason": "0",
"orgCode": "001",
"password": "5f233d5a4ec184cce950a0d02d93a8af",
"protocol": "1",
"proxyPort": "",
"authority": "0",
"registId": "",
"serverCode": "video#local",
"sn": "5H000F4AAG6FAE9",
"softwareVersion": "2.800.0000000.14.R",
"status": "1",
"timezoneDesc": "Beijing",
"timezoneld": "100",
"timezoneName": "timezone.text.Asia.Shanghai",
"timezoneOffset": "+08:00",
"type": "2",
"unitEnable": "0",
"units": [
  {
    "accessControl": false,
    "alarmInput": false,
    "alarmOutput": false,
    "assistStream": "3",
    "box": false,
    "channels": [
      {
        "cameraFunctions": "0",
        "cameraType": "1",
        "capability": "197473",
        "channelCode": "1000000$1$0$0",
        "channelName": "... ..",
        "channelSeq": 0,
        "channelType": "1",
        "db33Code": "",
        "domainId": "",
```

```
        "faceFunctions": "2",
        "floorInfos": [

        ],
        "orgCode": "qawfq2",
        "floorNum": "",
        "forPeopleCount": "0",
        "intelliState": "",
        "status": "1",
        "streamType": "3",
        "targetDetection": "0"
    }
},
{
    "cvi": false,
    "decoder": false,
    "encoder": true,
    "liftController": false,
    "pos": false,
    "screenInput": false,
    "screenOutput": false,
    "streamType": "801",
    "transcoder": false,
    "unitSeq": "0",
    "unitType": "1",
    "virtual": false,
    "voice": false,
    "zeroChnEncode": "0"
},
{
    "accessControl": false,
    "alarmInput": true,
    "alarmOutput": false,
    "box": false,
    "channels": [
        {
            "alarmLevel": "1",
            "alarmType": "2",
            "capability": "0",
            "channelCode": "1000000$3$0$0",
            "channelName": "192.168.1.1_1",
            "channelSeq": 0,
            "db33Code": "",
            "domainId": "",
            "floorInfos": [

            ],
            "floorNum": "",
            "intelliState": "",
            "status": "1"
```



```
    },
    {
      "alarmLevel": "1",
      "alarmType": "2",
      "capability": "0",
      "channelCode": "1000000$3$0$1",
      "channelName": "192.168.1.1_2",
      "channelSeq": 1,
      "db33Code": "",
      "domainId": "",
      "floorInfos": [

      ],
      "floorNum": "",
      "intelliState": "",
      "status": "1"
    }
  ],
  "cvi": false,
  "decoder": false,
  "encoder": false,
  "liftController": false,
  "pos": false,
  "screenInput": false,
  "screenOutput": false,
  "transcoder": false,
  "unitSeq": "0",
  "unitType": "3",
  "virtual": false,
  "voice": false
},
{
  "accessControl": false,
  "alarmInput": false,
  "alarmOutput": true,
  "box": false,
  "channels": [
    {
      "capability": "0",
      "channelCode": "1000000$4$0$0",
      "channelName": "192.168.1.1_1",
      "channelSeq": 0,
      "db33Code": "",
      "domainId": "",
      "floorInfos": [

      ],
      "floorNum": "",
      "intelliState": "",
```

```

        "status": "1"
      },
      {
        "capability": "0",
        "channelCode": "1000000$4$0$1",
        "channelName": "192.168.1.1_2",
        "channelSeq": 1,
        "db33Code": "",
        "domainId": "",
        "floorInfos": [

        ],
        "floorNum": "",
        "intelliState": "",
        "status": "1"
      }
    ],
    "cvi": false,
    "decoder": false,
    "encoder": false,
    "liftController": false,
    "pos": false,
    "screenInput": false,
    "screenOutput": false,
    "transcoder": false,
    "unitSeq": "0",
    "unitType": "4",
    "virtual": false,
    "voice": false
  }
},
"user_name": "admin"
}
],
},
"desc": "Success"
}

```

3.3.2.2 Grading Device Organization Tree

Brief Description

- Grading device organization tree

Request Method

- POST

Request URL

- /brms/api/v1.1/tree/grade-device-tree

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
orgCode	Yes	string	Organization code
treeType	Yes	string	Tree Type: 1=device tree, 2=device channel tree, 3=organization tree
domain	No	string	Contain cascade devices: 0=no, 1=yes, empty for all
mapId	No	string	Map ID
filterEmptyDevice	No	string	When the device channel tree is selected, whether to filter empty devices: 0=no, 1= yes, default 1
assistFindSize	No	string	Assist in finding device channel number, range: 0-3000, default 0, 0 means no assistance for regular classification
conditions	No	array	
- deviceCategories	No	array	Device category. Dictionary type, see [6.1.2 Device Type]
- deviceTypes	No	array	Device type. Dictionary type, see [6.1.2 Device Type]
- deviceCapabilities	No	array	Device capability: 1=device image storage capability, 2=intelligence capability after face issuing, 3=new video recording plan configuration capability, 8=whether the device supports UTC time calibration capability, 9=whether it supports 4G capability, 10=whether it supports sleep capability ...
- unitTypes	No	array	Unit type. Dictionary type, see [6.1.3 Unit Type]. Applicable only when treeType=2
- channelCapabilities	No	array	Channel capability. Dictionary type, see [6.1.5 Encoding Channel Capability Set]. Applicable only when treeType=2
- channelTypes	No	array	Channel type: 1=video channel, 2=picture channel, 3=video-picture mixed channel. Applicable only when treeType=2
- channelSubtypes	No	array	Channel sub type: 0=normal video, 1=visible light, 2=thermal imaging. Applicable only when treeType=2

Parameter	Required	Type	Description
- channelRemoteTypes	No	array	Remote channel type: 0=normal video channel (default), 1=acquisition channel, 2=remote channel, 4=encoding channel. Applicable only when treeType=2
- cameraTypes	No	array	Camera type. Dictionary type, see [6.1.4 Camera Type]. Applicable only when treeType=2
- inputAlarmTypes	No	array	Alarm input channel alarm type: 2=external alarm, 19=infrared detection alarm, 28=remote external alarm. Applicable only when treeType=2
- excludeDeviceCategories	No	array	Exclude device category. Dictionary type, see [6.1.2 Device Type]
- excludeDeviceTypes	No	array	Exclude device type. Dictionary type, see [6.1.2 Device Type]. Applicable only when excludeDeviceCategories has value
- sourceTypes	No	array	Source Type: 1=platform, 3=tripartite
- protocolTypes	No	array	Protocol type: 1=VMS, 2=Hikvision, 98=SNMP, 99=Onvif, 100=SIA, 185=JIUZHOU
- excludeChannelCapabilities	No	array	Exclude channel capability, a single element of the array is separated by a comma to indicate the relationship. Dictionary type, see [6.1.5 Encoding Channel Capability Set]. Applicable only when treeType=2
- addTypes	No	array	Add type: 1=IP Address, 2=IP Segment, 3=Domain name, 4=Auto Registration, 5=P2P, 6=RTSP Address
- serverCodes	No	array	Server to which the device belongs
- resourceTypes	No	array	Resource types

Request Example

```
POST /brms/api/v1.1/tree/grade-device-tree HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ebd349c3fe4a4de591a880d6ed562cbd

{
  "orgCode": "001",
  "treeType": "1",
```

```
"domain": "1",
"mapId": "",
"filterEmptyDevice": "",
"assistFindSize": "",
"conditions": [
  {
    "deviceCategories": [
      "1",
      "5"
    ],
    "deviceTypes": [
      "1",
      "2"
    ],
    "deviceCapabilities": [
      "2"
    ],
    "unitTypes": [
      "1",
      "2"
    ],
    "channelTypes": [
      "1",
      "3"
    ],
    "channelSubtypes": [
    ],
    "channelCapabilities": [
      "1",
      "2"
    ],
    "channelRemoteTypes": [
      "0",
      "1"
    ],
    "resourceTypes": [
      "0",
      "1"
    ],
    "cameraTypes": [
      "1",
      "3"
    ],
    "inputAlarmTypes": [
      "2",
      "3"
    ],
    "excludeDeviceCategories": [
```

```

        "1"
      ],
      "excludeDeviceTypes": [
        "5"
      ],
      "sourceTypes": [
        "1"
      ],
      "protocolTypes": [
        "1"
      ],
      "excludeChannelCapabilities": [
      ]
    }
  ]
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- orgCode	string	Organization code
- orgName	string	Organization name
- domainId	string	Domain ID
- sourceType	string	Source Type: 1=platform, 3=tripartite
- sourceId	string	Source ID
- associateId	string	Tripartite organization ID
- authority	string	Current user have permission for this organization: 0=No, 1=Yes
- organizations	array	Organizations
- - orgCode	string	Organization code
- - orgName	string	Organization name
- - pld	string	Parent node
- - hasChild	string	Whether there are child nodes: 0=no, 1=yes
- - domainId	string	Domain ID
- - sourceType	string	Source Type: 1=platform, 3=tripartite
- - sourceId	string	Source ID
- - associateId	string	Tripartite organization ID

Parameter	Type	Description
- - checked	string	Selected Status: 0=no, 1=all selected, 2=half selected
- - autoAuthorize	string	Whether descendant nodes are automatically authorized: 0=no, 1=yes
- - authority	string	Current user have permission for this organization: 0=No, 1=Yes
- devices	array	Devices
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - deviceIp	string	Device IP
- - pld	string	Parent node
- - orgCode	string	Organization code
- - deviceCategory	string	Device category
- - deviceType	string	Device type
- - deviceModel	string	Device model
- - softwareVersion	string	Software version
- - domainId	string	Domain ID
- - status	string	Online Status: 0=offline, 1=online
- - addType	string	Add type: 1=IP Address, 2=IP Segment, 3=Domain name, 4=Auto Registration, 5=P2P, 6=RTSP Address
- - capability	string	Device capability set, decimal encoding
- - checked	string	Selected Status: 0=no, 1=all selected, 2=half selected
- - sourceType	string	Source Type: 1=platform, 3=tripartite
- - sourceId	string	Source ID
- - associateId	string	Tripartite device ID
- - protocolType	string	Protocol type: 1=VMS, 2=Hikvision, 98=SNMP, 99=Onvif, 100=SIA, 185=JIUZHOU
- - unlockModes	string	Unlock mode combinations supported by the device, multiple combinations are separated by commas, such as: 0,1,2,3
- - authority	string	Current user have permission for the device: 0=No, 1=Yes
- - channels	array	Channels
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - - pld	string	Parent node
- - - orgCode	string	Organization code

Parameter	Type	Description
- - - unitType	string	Unit type
- - - deviceCapability	string	Device capability set, decimal encoding
- - - capability	string	Channel capability set, decimal encoding
- - - channelType	string	Channel type: 1=video channel, 2=picture channel, 3=video-picture mixed channel
- - - cameraType	string	Camera type. Dictionary type, see [API Reference > Dictionary > Camera Type]
- - - channelSubtype	string	Channel sub type: 0=normal video, 1=visible light, 2=thermal imaging
- - - channelRemoteType	string	Remote channel type: 0=normal video channel (default), 1=acquisition channel, 2=remote channel, 4=encoding channel
- - - deviceCategory	string	Device category
- - - deviceType	string	Device type
- - - domainId	string	Domain ID
- - - status	string	Online Status: 0=offline, 1=online
- - - inputAlarmType	string	Alarm input channel alarm type: 2=external alarm, 19=infrared detection alarm, 28=remote external alarm
- - - streamType	string	Channel code stream type: 1=primary code stream, 2=secondary code stream 1, 3=secondary code stream 2, default primary code stream
- - - checked	string	Selected Status: 0=no, 1=all selected
- - - underDevice	string	Is belong to the device: 0=No, 1=Yes
- channels	array	Channels
- - channelId	string	Channel code
- - channelName	string	Channel name
- - pId	string	Parent node
- - orgCode	string	Organization code
- - unitType	string	Unit type
- - deviceCapability	string	Device capability set, decimal encoding
- - capability	string	Channel capability set, decimal encoding
- - channelType	string	Channel type: 1=video channel, 2=picture channel, 3=video-picture mixed channel
- - cameraType	string	Camera type. Dictionary type, see [API Reference > Dictionary > Camera Type]
- - channelSubtype	string	Channel sub type: 0=normal video, 1=visible light, 2=thermal imaging

Parameter	Type	Description
-- channelRemoteType	string	Remote channel type: 0=normal video channel (default), 1=acquisition channel, 2=remote channel, 4=encoding channel
-- deviceCategory	string	Device category
-- deviceType	string	Device type
-- domainId	string	Domain ID
-- status	string	Online Status: 0=offline, 1=online
-- inputAlarmType	string	Alarm input channel alarm type: 2=external alarm, 19=infrared detection alarm, 28=remote external alarm
-- streamType	string	Channel code stream type: 1=primary code stream, 2=secondary code stream 1, 3=secondary code stream 2, default primary code stream
-- checked	string	Selected Status: 0=no, 1=all selected
-- underDevice	string	Is belong to the device: 0=No, 1=Yes

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "orgCode": "001001",
    "orgName": "org11",
    "domainId": "0",
    "sourceType": "1",
    "sourceId": "",
    "associatedId": "",
    "authority": "0",
    "organizations": [
      {
        "orgCode": "001001",
        "orgName": "org11",
        "pId": "001",
        "hasChild": null,
        "domainId": "0",
        "sourceType": "1",
        "sourceId": "",
        "associatedId": "",
        "checked": null,
        "authority": "0",
        "autoAuthorize": null
      }
    ],
    "devices": [
```

```
{
  "deviceCode": "1000001",
  "pld": "001001001",
  "deviceName": "2",
  "deviceIp": "192.168.1.1",
  "checked": "1",
  "deviceCategory": "1",
  "deviceType": "6",
  "orgCode": "001",
  "deviceModel": "ITC237-PW6M-IRLZF",
  "softwareVersion": "2.623.0000000.0.R",
  "capability": "1031",
  "status": "1",
  "domainId": "0",
  "addType": "1",
  "sourceType": "1",
  "sourceId": "",
  "associateId": "",
  "protocolType": "1",
  "unlockModes": "",
  "authority": "0",
  "channels": [
    {
      "channelId": "1000001$1$0$0",
      "pld": "1000001",
      "channelName": "IPC",
      "unitType": "1",
      "channelType": "1",
      "channelSubtype": "0",
      "channelRemoteType": "",
      "deviceCapability": "1",
      "cameraType": "1",
      "checked": "0",
      "orgCode": "001",
      "deviceType": "6",
      "deviceCategory": "1",
      "inputAlarmType": "",
      "capability": "",
      "status": "1",
      "domainId": "0",
      "underDevice": "1",
      "streamType": "1"
    }
  ]
},
"channels": [
  {
    "channelId": "1000$1$0$0$0",
```

```

        "pld": "001",
        "channelName": "channel1",
        "unitType": "1",
        "channelType": "1",
        "channelSubtype": "1",
        "channelRemoteType": "1",
        "deviceCapability": "1",
        "cameraType": "1",
        "checked": "0",
        "orgCode": "001",
        "deviceType": "1",
        "deviceCategory": "1",
        "inputAlarmType": "",
        "capability": "1",
        "status": "1",
        "domainId": "0",
        "underDevice": "1",
        "streamType": "1"
    }
}
}

```

3.4 Live View

3.4.1 Start Real-time Video

Brief description

- Start real-time video. You also need to splice url and token. Please refer to the 7.1.2 section.

Request method

- POST

Request URL

- /brms/api/v1.0/MTS/Video/StartVideo

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None

Parameter	Required	Type	Description
- streamType	Yes	string	Stream type: 1=Main stream, 2=Sub stream. In multi-screen mode, the value range is 0-1024
- trackId	No	string	Track ID
- channelId	Yes	string	Video channel code
- keyCode	No	string	Keyboard code; if the parameter channel code is null, get it through the keyboard code
- dataType	Yes	string	Video type: 1=Video, 2=Audio, 3=Audio and video
- enableRtsp	No	string	Whether to enable stream encryption RTSPS: Null or 0=disabled, 1=enabled
- enableMulticast	No	string	Enable multicast mode: 0=disabled, 1=enabled
- fakeSdp	No	string	Forge SDP or not: 0 = No, 1 = Yes, Default 0

Request example

```
POST /brms/api/v1.0/MTS/Video/StartVideo HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "streamType": "1"
    "trackId": "",
    "channelId": "1000040$1$0$0",
    "keyCode": "",
    "dataType": "2",
    "enableRtsp": "0",
    "enableMulticast": "0",
    "fakeSdp": "0"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code

Parameter	Type	Description
desc	string	Result description
data	object	None
- minRate	string	Minimum stream, which should be provided by the customer
- protocol	string	Protocol: 1-CNM3,2-RTSP
- ip	string	IP address: The IP address of the forwarding server or device based on the type
- port	string	Port
- stunEnable	string	Whether to support the STUN protocol: 1: Yes, 0: No
- stunPort	string	STUN protocol port
- url	string	The RTSP address needs to be carried after the url when it is used "?token=xxx", see [5.3 Video Services] for details
- url2	string	Alternate RTSP address
- connectType	string	Type: 1: Forwarding mode, 2: Direct connection mode
- session	string	Forwarding session mark
- token	string	Token
- trackId	string	Track ID
- vkId	string	ID of secret key
- vkValue	string	Secret key
- multicastIp	string	Multicast IP
- multicastPort	string	Multicast port
- stopTime	string	Stop playback time, time stamp, unit: second, null means no need to stop

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "minRate": null,
    "protocol": null,
    "ip": null,
    "port": null,
    "stunEnable": null,
    "stunPort": null,
    "url": "rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000040%240%26substream=1",
    "url2": ""
  }
}
```

```

    "connectType": null,
    "session": "2204",
    "token": "2204",
    "trackId": "",
    "vkId": "ARSni0ctAQ+Y01sAz5BBCQ==",
    "vkValue": "mr9TWihPFfX3EfCPyS3NY10q9G8x6znmQn/OggHzwL4=",
    "multicastIp": "",
    "multicastPort": "",
    "stopTime": ""
  }
}

```

3.4.2 Start Talk

Brief Description

- Start talk. The usage method is similar to that of real-time video. You also need to splice url and token. Please refer to the 7.1.2 section.

Request Method

- POST

Request URL

- /brms/api/v1.0/MTS/Audio/StartTalk

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelSeq	No	string	Channel sequence (used for channel talk), the maximum value is 1024
- broadcastChannels	No	string	Broadcast channels (used for broadcast talk)
- deviceCode	Yes	string	Device code
- talkType	Yes	string	Talk type: 1-Device, 2-Channel, 3- National standard device broadcast
- audioType	No	string	Audio type: 0-DEFAULT, 1-PCM, 2-G711a, 3-AMR, 4-G711u, 5-G726...
- talkMode	No	string	Talk mode: 0-Talk (Default mode), 1-Broadcast (switching from broadcasting to intercom requires reconfiguration)

Parameter	Required	Type	Description
- audioBit	Yes	string	Represent with actual values. If it is 8 digits, the value is 8
- sampleRate	No	string	The sampling rate, such as 16k, has a value of 16000
- enableRtsp	No	string	Whether to enable RTSPS: 0 = disabled, 1 = enabled, default 0

Request Example

```
POST /brms/api/v1.0/MTS/Audio/StartTalk HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "channelSeq": "0",
    "broadcastChannels": "",
    "deviceCode": "1000040",
    "talkType": "1",
    "audioType": "1",
    "talkMode": "",
    "audioBit": "16",
    "sampleRate": "8000",
    "enableRtsp": "0"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- deviceCode	string	Device code
- url	string	RTSP address. It is mutually exclusive with VDTs IP/Port
- token	string	RTSP token
- session	string	Talk session
- audioType	string	If the audioType supported by the client request is available, it will be used as is. Otherwise, the audio will be sent in the following order of priority: G711a, G711u, PCM, G726, AMR, DEFAULT. If

Parameter	Type	Description
		none of these audio types are supported by the device, the first available audioType provided by the device will be selected
- audioBit	string	Represent with actual values. If it is 8 digits, the value is 8
- sampleRate	string	The sampling rate, such as 16k, has a value of 16000
- protocolType	string	Protocol type version

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "deviceCode": "1000000",
    "url": "rtsp://192.168.1.1:9100/talk/pu/26",
    "token": "0",
    "session": "26",
    "audioType": "1",
    "audioBit": "16",
    "sampleRate": "8000",
    "protocolType": "2"
  }
}
```

3.4.3 Stop Talk

Brief Description

- Stop talk

Request Method

- POST

Request URL

- /brms/api/v1.0/MTS/Audio/StopTalk

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None

Parameter	Required	Type	Description
- talkType	Yes	string	National Standard Code, client number, usually consists of 18 digits
- deviceCode	Yes	string	Device code
- session	Yes	string	Forward session ID
- channelSeq	No	string	Channel sequence (used for channel talk) , the maximum value is 1024

Request Example

```
POST /brms/api/v1.0/MTS/Audio/StopTalk HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "data": {
    "talkType": "1",
    "deviceCode": "1000040",
    "session": "2380",
    "channelSeq": "1"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.4.4 Acquiring Streaming Media URL for Live View(FLV)

Brief Description

- Acquiring FLV streaming media address for live view. You need to install the StreamingPlugin plug-in before you can use it.

Request Method

- GET

Request URL

- /brms/api/v1.1/video/live/channel/{channelId}/flv?protocol={protocol}&streamType={streamType}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
channelId	Yes	string	Channel ID
protocol	Yes	string	Protocol type: http, https
streamType	Yes	string	Stream type: 1= Main stream, 2 = Sub stream 1, 3 = Sub stream 2

Request Example

```
GET /brms/api/v1.1/video/live/channel/1000001$1$0$0/flv?streamType=1&protocol=ht
tps HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0e6d90f8a5194e61b86eb586d754e1b8
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- streamUrl	string	Streaming media URL

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
```

```

    "streamUrl": "http://192.168.1.1:7886/live/cameraid/1000001%240/substream/
    1.flv?token=ff435202efab4a54af01c5c0a4e65abe"
  }
}

```

3.4.5 Acquiring Streaming Media URL for Live View(HLS)

Brief Description

- Acquiring HLS streaming media address for live view. You need to install the StreamingPlugin plug-in before you can use it.

Request Method

- GET

Request URL

- /brms/api/v1.1/video/live/channel/{channelId}/hls?protocol={protocol}&streamType={streamType}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
channelId	Yes	string	Channel ID
protocol	Yes	string	Protocol type: http, https
streamType	Yes	string	Stream type: 1= Main stream, 2 = Sub stream 1, 3 = Sub stream 2

Request Example

```

GET /brms/api/v1.1/video/live/channel/1000001$1$0$0/hls?streamType=1&protocol=h
ttps HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0e6d90f8a5194e61b86eb586d754e1b8

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description

Parameter	Type	Description
data	object	None
- streamUrl	string	Streaming media URL

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "streamUrl": "http://192.168.1.1:7096/live/cameraid/1000001%240/substream/1.m3u8?token=ff435202efab4a54af01c5c0a4e65abe"
  }
}
```

3.5 Record Playback

3.5.1 Search Recording

Brief Description

- Search recording

Request Method

- POST

Request URL

- /brms/api/v1.0/SS/Record/QueryRecords

Request Parameter

- Body Parameters

Parameter	Required Type		Description
data	Yes	object	None
- endTime	Yes	string	End time (seconds)
- streamType	Yes	string	Stream type: 1: Main stream, 2: Sub stream
- recordType	Yes	string	Recording type: 0: All, 1: Manual recording, 2: Alarm recording, 3: Dynamic monitoring, 4: Video loss, 5: Video tampering, 6: Scheduled recording, 7: All-weather recording, 8: File recording conversion

Parameter	Required	Type	Description
- recordSource	Yes	string	Recording source: 2: Device, 3: Center
- startTime	Yes	string	Start time (seconds)
- channelId	Yes	string	Channel code
- page	No	string	Page number, starting from 1
- session	No	string	Session ID, required when page is not empty

Request Example

```
POST /brms/api/v1.0/SS/Record/QueryRecords HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "endTime": "1540889999",
    "streamType": "1",
    "recordType": "0",
    "recordSource": "3",
    "startTime": "1540828800",
    "channelId": "1000067$1$0$1",
    "page": "1",
    "session": "5fad3e259cbc497fa3e2ece67676469c"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data
- session	string	Session ID
- pageCount	string	Page count
- records	array	Record list
- - channelId	string	Channel code
- - recordSource	string	Recording source: 2: Device, 3: Center
- - recordType	string	Recording type: 0: All, 1: Manual recording, 2: Alarm recording, 3: Dynamic monitoring, 4: Video loss, 5: Video tampering, 6:

Parameter	Type	Description
		Scheduled recording, 7: All-weather recording, 8: File recording conversion
- - startTime	string	Start time (seconds)
- - endTime	string	End time (seconds)
- - recordName	string	Recording name (different manufacturers have different file identifications)
- - fileLength	string	File length
- - planId	string	Recording plan ID
- - sslId	string	Storage service ID
- - diskId	string	Disk ID
- - streamId	string	Stream processing (StreamId)
- - forgotten	string	Forgot or not
- - streamType	string	Stream type: 1: Main stream, 2: Sub stream
- - recordTypeList	string	Video type (dynamic detection, alarm, etc.)

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "session": "5fad3e259cbc497fa3e2ece67676469c",
    "pageCount": "1",
    "records": [
      {
        "channelId": "1000067$1$0$1",
        "recordSource": "3",
        "recordType": "6",
        "startTime": "1540886244",
        "endTime": "1540886303",
        "recordName": "{91254f10-b054-4513-96ac-61a008cb1cb2}",
        "fileLength": "16384",
        "planId": "0",
        "sslId": "1001",
        "diskId": "1540886244-1540886303",
        "streamId": "2726",
        "forgotten": "0",
        "streamType": "1",
        "recordTypeList": []
      },
      {
        "channelId": "1000067$1$0$1",
```

```

        "recordSource": "3",
        "recordType": "6",
        "startTime": "1540888200",
        "endTime": "1540888387",
        "recordName": "{2b457c5f-7aed-4ac0-bb08-359a1b953904}",
        "fileLength": "32768",
        "planId": "0",
        "ssId": "1001",
        "diskId": "1540888200-1540888387",
        "streamId": "2726",
        "forgotten": "0",
        "streamType": "1",
        "recordTypeList": []
      }
    ]
  }
}

```

3.5.2 Video Playback by Time

Brief Description

- Play back the recording in the form of time, suitable for recording storage on the device

Request Method

- POST

Request URL

- /brms/api/v1.0/SS/Playback/StartPlaybackByTime

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- ssId	No	string	SS service ID
- recordType	Yes	string	Recording type: 1: General recording, 2: Alarm recording
- streamType	Yes	string	Stream type: 1: Main stream, 2: Sub stream
- recordSource	Yes	string	Recording source: 2: Device, 3: Center

Parameter	Required	Type	Description
- channelId	Yes	string	Channel code
- endTime	Yes	string	End time (seconds)
- startTime	Yes	string	Start time (seconds)
- streamId	Yes	string	Stream ID
- enableRtsp	No	string	Is RTSPS enabled? 0=disabled, 1=enabled
- fakeSdp	No	string	Forge SDP: 0=No, 1=Yes, default 0

Request Example

```
POST /brms/api/v1.0/SS/Playback/StartPlaybackByTime HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "data": {
    "ssId": "1001",
    "recordType": "1",
    "streamType": "1",
    "recordSource": "2",
    "channelId": "1000067$1$0$0",
    "endTime": "1540915199",
    "startTime": "1540828800",
    "streamId": "112",
    "enableRtsp": "0",
    "fakeSdp": "0"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- url	string	The RTSP address needs to be carried after the url when it is used "?token=xxx", see [5.3 Video Services] for details
- url2	string	Alternate RTSP address
- session	string	Recording playback session ID
- protocol	string	Protocol type, 0 = RTSP

Parameter	Type	Description
- token	string	Token
- fromDevice	string	Whether to acquire stream from the device (acquire stream from MTS by default): 0: Acquire stream from the MTS platform, 1: Directly acquire stream from the device (such as EVS)
- vks	array	Stream VK information
- - vkId	string	ID of secret key
- - vkValue	string	Secret key

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "url": "rtsp://192.168.1.1:9320/playback/pu/94",
    "session": "94",
    "protocol": "0",
    "token": "94",
    "url2": "",
    "fromDevice": "",
    "vks": [
      {
        "vkId": "AfSxwIDPsAjICAFAZAU4EQ==",
        "vkValue": "m8ZR3B4OHFCfDvRTia54hXKic0t8L5ttySCU50apq8w="
      }
    ]
  }
}
```

3.5.3 Video Playback by File

Brief Description

- Play back recordings by file (applicable to store recordings in the center)

Request Method

- POST

Request URL

- /brms/api/v1.0/SS/Playback/StartPlaybackByFile

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	Data
- sslId	Yes	string	SS service ID
- endTime	No	string	End time (second)
- fileName	Yes	string	Recording file information
- diskId	Yes	string	Disk ID
- startTime	No	string	Start time (seconds)
- recordSource	No	string	Recording source: 2: Device, 3: Center
- channelId	Yes	string	Channel code
- streamId	No	string	Stream processing (StreamId)
- enableRtsp	No	string	Is RTSPS enabled? 0=disabled, 1=enabled
- playbackType	No	string	Playback type: 0 = file playback, 1 = time playback, 2 = RTSP playback, 3 = device secondary compression playback by time, 4 = local file playback (DAV and MP4 format), default 0
- fakeSdp	No	string	Forge SDP: 0=No, 1=Yes, default 0

Request Example

```
POST /brms/api/v1.0/SS/Playback/StartPlaybackByFile HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "data": {
    "sslId": "1001",
    "endTime": "1540853395",
    "fileName": "{fe69f729-9d4b-42d4-b6a0-56189aaa4e1e}",
    "diskId": "1540852944-1540853395",
    "startTime": "1540852944",
    "recordSource": "3",
    "channelId": "1000040$1$0$0",
    "streamId": "5875",
    "enableRtsp": "0",
    "playbackType": "0",
    "fakeSdp": "0"
  }
}
```

```
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- url	string	The RTSP address needs to be carried after the url when it is used "?token=xxx", see [5.3 Video Services] for details
- url2	string	RTSP URLs that can be directly accessed
- fromDevice	string	Whether to pull stream from device: 0- pull stream from MTS platform, 1- directly pull stream from device (such as EVS)
- session	string	Recording playback session ID
- protocol	string	Protocol type: 0 = RTSP
- token	string	Token
- vks	object	Stream VK information
- - vkId	string	ID of secret key
- - vkValue	string	Secret key

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "url": "rtsp://192.168.1.1:9320/playback/center/5",
    "url2": "",
    "fromDevice": "",
    "session": "5",
    "protocol": "0",
    "token": "5",
    "vks": [
      {
        "vkId": "AfSxwIDPsAjICWUAZzs4EQ=",
        "vkValue": "m8ZR3B4OHFCfDvRTiaOGhXKic0t8L5ghySCU50apq8w="
      }
    ]
  }
}
```

3.5.4 Acquiring Stream Media URL for Video Playback(HLS)

Brief Description

- Acquiring HLS stream media address for video playback. The parameter comes from the data returned by the [3.5.1 Search Recording] interface. You need to install the StreamingPlugin plug-in before you can use it.

Request Method

- GET

Request URL

- /brms/api/v1.1/video/playback/channel/{channelId}/hls?recordSource={recordSource}&streamId={streamId}&recordType={recordType}&streamType={streamType}&startTime={startTime}&endTime={endTime}&protocol={protocol}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
channelId	Yes	string	Channel ID
protocol	Yes	string	Protocol Type: Http, https
recordSource	Yes	string	Video source: 2 = Device, 3 = Central server
streamId	No	string	Video index (required when the video is stored in central server)
recordType	No	string	Video type: 0 = All, 1 = Manual recording, 2 = Alarm video, 3 = Dynamic monitoring, 4 = Video loss, 5 = Video block, 6 = Scheduled recording, 7 = All-day recording, 8 = File recording switch (required when the video is stored in the device)
streamType	Yes	string	Stream type: 1 = Main stream, 2 = Sub stream 1, 3 = Sub stream 2
startTime	Yes	string	Start time, time stamp (in seconds)
endTime	Yes	string	End time, time stamp (in seconds)
ssId	No	string	Storage service ID. For example: 100101@video#local needs to be transcoded to 100101%40video%23local, or parameters following # will be cut off.

Request Example

```
GET /brms/api/v1.1/video/playback/channel/1000014$1$0$0/hls?recordSource=3&streamId=39977&recordType=3&streamType=1&startTime=1621302659&endTime=1621302917&protocol=http HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0e6d90f8a5194e61b86eb586d754e1b8
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- streamUrl	string	Streaming media URL

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "streamUrl": "http://192.168.1.1:7086/vod/center/cameraid/1000014%240/substream/1/recordtype/2/ssid/100101/totallength/258/begintime/1621302659/endtime/1621302917.m3u8?token=ff435202efab4a54af01c5c0a4e65abe"
  }
}
```

3.6 Alarm(Event Center)

3.6.1 Subscribe to Alarms

Brief Description

- Subscribe alarm, push alarm information through callback, the requester needs to define the callback interface according to the specified protocol. The same user only supports subscribing once, and repeating the subscription will overwrite the results of the previous subscription.
- An alarm message might require multiple callbacks to complete, usually no more than 2. The structure of multiple callbacks is the same, and the message type is distinguished by the callbackType field. When callbackType=1, it is the main content of the alarm message, and all fields need to be paid attention to, including alarmPictures (the image address carried by the alarm itself); when callbackType=2, it is the linked image information, usually the asynchronous obtained alarm linkage image, at this time, only alarmCode and alarmPictures need to be updated, other fields should not be updated, otherwise the original

alarm information will be cleared. It should be noted that when updating alarmPictures, the increment method should be used, otherwise it will overwrite the image carried by the original alarm itself.

Callback format:

```
POST callbackUrl HTTP/1.1
Content-Type: application/json;charset=UTF-8)
{
  "callbackType": "Callback message type: 1 = alarm message body, 2 = linked image (usually asynchronously obtained alarm linkage image, in this case only the alarmPictures field needs to be incremented), default 1, required",
  "alarmCode": "Alarm code, required",
  "sourceCode": "Alarm source code, required",
  "sourceName": "Alarm source name, optional",
  "alarmType": "Alarm type code, required",
  "alarmTypeName": "Alarm type name, optional",
  "alarmGrade": "Alarm level: 1=High, 2=Medium, 3=Low, required",
  "alarmStatus": "Alarm status: 1=Alarm generated, 2=Alarm cleared, required",
  "alarmTime": "Alarm time, timestamp, required",
  "alarmPictures": [
    "Alarm image, Base64 encoding, optional"
  ],
  "remark": "Alarm scheme remarks, optional",
  "eventRemark": "Alarm event remarks, optional",
  "extData": "Extended data, in JavaScript Object Notation (JSON) format, may contain some business-related information. The data structure may vary depending on the business, optional.",
  "signature": "Signature data, optional"
}
```

Request Method

- POST

Request URL

- /brms/api/v1.1/push-data/alarm/subscribe

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
callbackUrl	Yes	string	Callback address, supports http and https
action	Yes	string	Action: 1 = subscribe, 0 = unsubscribe
signature	No	string	Signature data, carried during callback, can be used for authentication, empty means callback interface does

Parameter	Required	Type	Description
			not require authentication. The format is determined by the requester, generally using a random string is sufficient.

Request Example

```
POST /brms/api/v1.1/push-data/alarm/subscribe HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 26aa1095c9364d6b94be425d5b8798b7

{
  "callbackUrl": "https://192.168.1.1:443/brms/api/v1.1/push-data/alarm/push",
  "action": "1",
  "signature": "c743b433407be68ee62c46ede8fb5a0c"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.6.2 Get Alarm Messages in Pages

Brief Description

- Get alarm information in pages

Request Method

- POST

Request URL

- /eams/api/v1.1/alarm/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
alarmCode	No	string	Alarm ID
deviceCodes	No	array	Device code
channelIds	No	array	Channel code
alarmStatus	No	array	Alarm status: 1 = Alarm generation; 2 = Alarm disappearance
alarmTypes	No	array	Alarm type
startAlarmTime	Yes	string	Timestamp of alarm starting (unit: Second)
endAlarmTime	Yes	string	Timestamp of alarm ending (unit: Second)
alarmGrade	No	array	Alarm grade
handleUser	No	string	Alarm handler (username)
handleStatus	No	array	Handling status: 1 = Handling; 2 = Solved; 3 = False alarm; 4 = Ignored; 5 = Not solved
splitTime	No	string	Timestamp of page splitting alarms in pages (unit: Second), required during page turning
splitId	No	string	ID of page splitting alarms, required during page turning
currentPage	Yes	string	Current page number, required during page turning
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orderType	Yes	string	Sort type: 1 = alarm time, other values are processed as 1 by default
orderDirection	Yes	string	Sorting direction: 0 = Ascending sequence; 1 = Descending sequence
handleMessage	No	string	Alarm handling suggestions

Request Example

```
POST /eams/api/v1.1/alarm/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
```


Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

```
{
  "alarmCode": "",
  "deviceCodes": [],
  "channelIds": [],
  "alarmStatus": [],
  "alarmTypes": [],
  "startAlarmTime": "1610991072",
  "endAlarmTime": "1611423072",
  "alarmGrade": [],
  "handleUser": "",
  "handleStatus": [],
  "splitTime": "",
  "splitId": "",
  "currentPage": "1",
  "pageSize": "20",
  "page": "1",
  "orderType": "1",
  "orderDirection": "1",
  "handleMessage": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Results
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	object	Results
- - alarmId	string	Alarm ID
- - alarmCode	string	Alarm code
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - channelId	string	Channel code
- - channelName	string	Channel name
- - alarmStatus	string	Alarm status: 1 = Alarm generation; 2 = Alarm disappearance

Parameter	Type	Description
- - alarmType	string	Alarm type ID, dictionary type. See [6.1.7 Alarm Type]
- - alarmDate	string	Alarm timestamp (unit: Second)
- - alarmGrade	string	Alarm grade
- - handleStatus	string	Handling statue: 1 = Handling; 2 = Solved; 3 = False alarm; 4 = Ignored; 0 = Not solved
- - handleUser	string	Alarm handler (username)
- - handleDate	string	Handling timestamp (unit: Second)
- - handleMessage	string	Handling opinion
- - picture	string	Alarm snapshot
- - pictureSize	string	Size of alarm snapshot
- - mailReceivers	string	Email recipient (separated by commas for multiple recipients)
- - memo	string	Alarm plan remarks
- - eventRemark	string	Alarm event remarks
- - extData	string	Extended data: JSON format
- - linkRecordChannels	array	Linkage video channel list
- - - channelId	string	Channel code
- - - alarmRecordPosition	string	Alarm video storage location: 1 = all, 2 = equipment, 3 = center
- - - streamType	string	Stream type
- - - recordTime	string	Recording time, in seconds
- - - preRecordTime	string	Prerecorded time, in seconds

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-120",
    "pageData": [
      {
        "deviceCode": "1000002",
        "deviceName": "tm",
        "channelName": "tm_1",
        "channelId": "1000002$1$0$0",
        "alarmStat": "1",
        "alarmType": "3",
        "alarmDate": "1611048504",
```

```
    "alarmGrade": "1",
    "handleDate": null,
    "handleMessage": null,
    "alarmCode": "{80C37A23-5C5C-4D68-8389-ED705C1F9430}",
    "mailReceivers": null,
    "memo": "Plan remark.",
    "eventRemark": "Event remark.",
    "alarmId": "106443485379960859",
    "handleStatus": "0",
    "handleUser": null,
    "picture": "",
    "pictureSize": null,
    "extData": null,
    "linkRecordChannels": [
      {
        "channelId": "0",
        "alarmRecordPosition": "1",
        "streamType": "1",
        "recordTime": "10",
        "preRecordTime": "5"
      }
    ]
  }
}
```

3.6.3 Handle Alarms

Brief Description

- Process alarms. The processing results are returned through MQ, see [4.2.2 Notification of Handling Alarms]

Request Method

- POST

Request URL

- /eams/api/v1.0/BRM/Alarm/HandleAlarm

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	Data
- handleUser	No	string	The handler (username), which when blank indicates the current login user
- comment	No	string	Remarks, which are customized on the client and can be used to expand the alarm linkage video and other information
- handleStatus	Yes	string	Handling status: 0 = Send Message, 1 = Forward User, Claim, 2 = Resolved, 3 = Ignored, 4 = False Positive, 5 = Unprocessed
- handleMessage	No	string	Handling comments, maximum length 1024. If the processing action is to send an email, the email address needs to be sent here
- alarmCode	Yes	string	Alarm code
- alarmDate	Yes	string	Alarm date, in seconds
- deviceCode	No	string	Device code

Request Example

```
POST /eams/api/v1.0/BRM/Alarm/HandleAlarm HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84
```

```
{
  "data": {
    "handleUser": "system",
    "comment": "",
    "handleStatus": "1",
    "handleMessage": "",
    "alarmCode": "{7962659F-9AD1-4E48-899D-82E548B5D721}",
    "alarmDate": "1676108166",
    "deviceCode": ""
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.6.4 Get Alarm Related Face Snapshot Information

Brief Description

- Get alarm related face snapshot information

Request Method

- POST

Request URL

- /eams/api/v1.0/BRM/Alarm/GetAlarmFaceDetectionInfo

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	Data
- alarmCode	Yes	string	Alarm code
- alarmDate	Yes	string	Alarm date, in seconds
- deviceCode	No	string	Device channel code

Request Example

```
POST /eams/api/v1.0/BRM/Alarm/GetAlarmFaceDetectionInfo HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84

{
  "data": {
    "alarmCode": "{1f4a86fd-d1c4-4777-ad8f-1659aa3f036b}",
    "alarmDate": "1666668815",
    "deviceCode": "1000000$1$0$0"
```

```
}
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- id	string	Primary key
- code	string	ID of the person snapshot information
- deviceCode	string	Device code
- alarmCode	string	Alarm code
- channelSeq	string	Channel number
- channelId	string	Channel code
- facelImageUrl	string	The URL of the face snapshot
- pictureUrl	string	The URL of the scene image
- appearTime	string	Appearance times
- age	string	Age
- gender	string	Gender: 0: Unknown, 1: Male, 2: Female
- fringe	string	With fringe or not: 0=No, 1=Yes
- eye	string	Eyes: 1=Unidentified, 2=Eyes closed, 3=Eyes open
- mouth	string	Mouth: 1=Unidentified, 2=Mouth closed, 3=Mouth open
- mask	string	Wearing a mask or not: 1=Unidentified, 2=No, 3=Yes
- beard	string	With beard or not: 1=Unidentified, 2=No, 3=Yes
- glasses	string	Wearing glasses or not: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
- emotion	string	Expressions: 0=Smile, 1=Angry, 2=Sad, 3=Disgusted, 4=Scared, 5=Surprised, 6=Normal, 7=Laughing, 8=Happy, 9=Confused, 10=Screaming
- alarmTime	string	Alarm time (second)

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "12",
```

```

    "code": "26",
    "deviceCode": "1000000",
    "alarmCode": "{1f4a86fd-d1c4-4777-ad8f-1659aa3f036b}",
    "channelSeq": "0",
    "channelId": "1000000$1$0$0",
    "facelmaUrl": "http://192.168.1.1:8081/d/1000003$1$0$0/20181121/11/1528-4800-0.jpg?streamID:37@capTime:1542770166",
    "pictureUrl": "http://192.168.1.1:8081/d/1000003$1$0$0/20181121/11/1528-4800-0.jpg?streamID:37@capTime:1542770166",
    "appearTime": "2",
    "age": "23",
    "gender": "0",
    "fringe": "0",
    "eye": "3",
    "mouth": "2",
    "mask": "3",
    "beard": "1",
    "glasses": "2",
    "emotion": "6",
    "alarmTime": "1564208732"
  }
}

```

3.6.5 Get Alarm Related Face Recognition Information

Brief Description

- Get alarm-associated face recognition information

Request Method

- POST

Request URL

- /eams/api/v1.0/BRM/Alarm/GetAlarmFaceRecognitionInfo

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	Data
- alarmCode	Yes	string	Alarm code
- alarmDate	Yes	string	Alarm date, in seconds

Parameter	Required	Type	Description
- deviceCode	No	string	Device channel code

Request Example

```
POST /eams/api/v1.0/BRM/Alarm/GetAlarmFaceRecognitionInfo HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84

{
  "data": {
    "alarmCode": "{1f4a86fd-d1c4-4777-ad8f-1659aa3f036b}",
    "alarmDate": "1666668815",
    "deviceCode": "1000000$1$0$0"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- alarmCode	string	Alarm code
- alarmTime	string	Alarm time (second)
- name	string	Name
- repositoryName	string	Face database name
- personId	string	Person ID
- personTypeName	string	Person attribute name
- gender	string	Gender, 1: Male; 2: Female
- birthday	string	Date of birth, yyyy-MM-dd
- nationality	string	Nationality
- detectionImageUrl	string	Snapshot URL
- repositoryImageUrl	string	Face image URL
- similarity	string	Similarity
- tel	string	Phone number

Return Example


```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "alarmCode": "{1f4a86fd-d1c4-4777-ad8f-1659aa3f036b}",
    "alarmTime": "1564208732",
    "name": "zhangsan",
    "repositoryName": "lib01",
    "personId": "12",
    "personTypeName": "test type",
    "gender": "1",
    "birthday": "2000-05-12",
    "nationality": "China",
    "detectionImageUrl": "http://192.168.1.1:8081/d/1000003$1$0$0/20181121/11/1528-4800-0.jpg?streamID:37@capTime:1542770166",
    "repositoryImageUrl": "http://192.168.1.1:8081/d/1000003$1$0$0/20181121/11/1528-4800-0.jpg?streamID:37@capTime:1542770166",
    "similarity": "67",
    "tel": "1234567"
  }
}
```

3.6.6 Get Alarm Related Access Control Information

Brief Description

- Get alarm related access control information

Request Method

- POST

Request URL

- /eams/api/v1.0/BRM/Alarm/GetAlarmAccessRecordInfo

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	Data
- alarmCode	Yes	string	Alarm code
- alarmDate	Yes	string	Alarm time, unit: second
- deviceCode	Yes	string	Device code

Request Example

```
POST /eams/api/v1.0/BRM/Alarm/GetAlarmAccessRecordInfo HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 841e473bacef4b31abc5f53272afeb84
```

```
{
  "data": {
    "alarmCode": "{E381EFF0-5A82-4502-8CCA-89E14AE34BBB}",
    "alarmDate": "1691562669",
    "deviceCode": "1000000"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- alarmCode	string	Alarm code
- alarmTime	string	Alarm time (seconds)
- deviceCode	string	Device code
- channelId	string	Channel code
- deviceName	string	Device name
- channelName	string	Channel name
- facelImageUrl	string	Face image
- captureImageUrl	string	Face snapshot
- personId	string	Person ID
- firstName	string	The first name of the person
- lastName	string	The last name of the person
- stageId	string	Stage number
- buildingId	string	Building number
- unitId	string	Unit number
- roomId	string	Room number
- gender	string	Gender: 0: Female, 1: Male, 2: Unknown
- idNum	string	ID number

Parameter	Type	Description
- departmentName	string	Department name
- orgInfos	array	Personnel group information
- - orgCode	string	Personnel group code
- - orgName	string	Personnel group name
- tel	string	Phone number
- cardNum	string	Card number
- linkChannels	array	Channel for alarm linkage
- bindChannels	array	Channel for resource binding
- antiPassType	string	Alarm handling suggestions, optional anti submarine type: 0=local or non anti submarine, 1=global
- operateType	string	Operation type: 0=other, 1=user, 2=room, 3=management machine
- operateInfo	string	Operation info

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "alarmCode": "{E381EFF0-5A82-4502-8CCA-89E14AE34BBB}",
    "alarmTime": "1582369632",
    "deviceCode": "1000000",
    "channelId": "1000000$7$1$0",
    "deviceName": "0",
    "channelName": "1",
    "facelImageUrl": "",
    "captureImageUrl": "",
    "personId": "1",
    "firstName": "1",
    "lastName": "0",
    "stageId": "1111",
    "buildingId": "001",
    "unitId": "24",
    "roomId": "25",
    "gender": "0",
    "idNum": "11011",
    "departmentName": "001",
    "orgInfos": [
      {
        "orgCode": "001001",
        "orgName": "Root"
      }
    ]
  }
}
```

```

    }
  ],
  "tel": "111111124",
  "cardNum": "25",
  "linkChannels": [],
  "bindChannels": [],
  "antiPassType": "0",
  "operateType": "2",
  "operateInfo": "1#24#25"
}
}

```

3.6.7 Obtain the Alarm Details of the Entrance and Exit

Brief Description

- Obtain the alarm details of the entrance and exit (blocklist, no entry and exit of vehicles, parking overtime, parking overtime, exclusive parking space occupation, parking crossing line, parking full, unfamiliar vehicles)

Request Method

- GET

Request URL

- /eams/api/v1.1/alarm/record/entrance/detail?alarmCode={alarmCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
alarmCode	Yes	string	Alarm code

Request Example

```

GET /brms/api/v1.1/alarm/record/entrance/detail?alarmCode={1f4a86fd-d1c4-4777-ad8f-1659aa3f036b} HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ca19ae28474040f192b9fc8cc7b9c69a

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- alarmCode	string	Alarm code
- alarmType	string	Alarm type
- recordType	string	Record type: 1=entrance and exit capture event, 2=license plate mode parking space capture event, 3=rest statistics mode event
- plateNo	string	License plate number
- carBrand	string	Vehicle brand
- carColor	string	Vehicle color
- personId	string	Person ID
- personName	string	Name of vehicle owner
- parkingLotId	string	Parking lot ID
- parkingLotName	string	Parking lot name
- parkingLotOrgCode	string	Parking lot organization code
- parkingLotOrgName	string	Parking lot organization name
- parentParkingLotId	string	Parent parking lot ID
- parentParkingLotName	string	Parent parking lot name
- positionId	string	Point ID
- positionName	string	Point name
- pointId	string	Gate ID
- pointName	string	Gate name
- vehiclePicture	string	Vehicle picture Url
- plateNoPicture	string	License plate picture Url
- channelId	string	Entry snapshot channel code
- channelName	string	Entry snapshot channel name
- captureTime	string	Capture time(seconds)
- parkingTime	string	Parking time(seconds)
- ParkingOverStandardTime	string	Parking timeout base time length (seconds)
- unrecordedTime	string	Duration without entry and exit records (days)
- unrecordedStandardTime	string	Time length without entry and exit record reference (days)
- parkingSpaceId	string	Parking space ID

Parameter	Type	Description
- parkingSpaceName	string	Parking space name
- layerId	string	Layer ID
- layerName	string	Layer name
- layerPicture	string	Layer picture Url
- height	string	Alarm point height
- width	string	Alarm point width
- locationX	string	X-axis coordinate of alarm point
- locationY	string	Y-axis coordinate of alarm point
- angle	string	Alarm point angle
- vehicleType	string	Vehicle type: 1=visitor, 2=allowlist, 4=blocklist, 5=temporary car
- entranceGroups	array	Entrance and exit groups
- - entranceGroupId	string	Entrance and exit group ID
- - entranceGroupName	string	Entrance and exit group name

Return Example

```
{
  "code":1000,
  "desc":"Success",
  "data":{
    "plateNo":"A12A26",
    "personName":"Jack",
    "parkingLotName":"parking lot 1",
    "parkingLotOrgCode": "001001",
    "parkingLotOrgName": "parking lot org 1",
    "pointName":"point1",
    "positionName":"position1",
    "plateNoPicture":"http://192.168.1.1:80/v1/s/192.168.1.1/9901/5fc43bb3-93a5-11ec-804a-204747922ee9/20220402/3/dsf_2788298d-b233-11ec-829a-204747922ee9_28893054_28897499.jpg",
    "parkingTime":"519169",
    "standardTime":"30",
    "channelId":"1000580$1$0$0",
    "enterTime":"1648870454",
    "alarmFrequency":"2",
    "layerId":"6",
    "layerName":"layer1",
    "layerPicture":"http://192.168.1.1:80/upload/ipms/layer/1649389568828/14.9MB_8100_8100.jpg",
    "height":"32",
    "width":"32",
```

```
{
  "locationX": "459",
  "locationY": "215",
  "angle": "0",
  "entranceGroups": [
    {
      "entranceGroupId": "5",
      "entranceGroupName": "group1"
    }
  ],
  "vehicleType": "2"
}
```

3.6.8 Obtain a List of Alarm Types

Brief Description

- Obtain a list of alarm types

Request Method

- GET

Request URL

- /eams/api/v1.1/alarm/plan/alarm-type/list

Request Parameters

None

Request Example

```
GET /eams/api/v1.1/alarm/plan/alarm-type/list HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 706c5572ab5a4ad9ac2471a766b28c29
```

Return Parameters

Parameter	Type	Description
code	int	The error code
desc	string	The result description
data	object	Data

Parameter	Type	Description
- results	array	Result
- - sourceTypeId	string	Alarm source type ID: 1=device, 2=encoding channel, 3=alarm input channel, 4=access control channel, 6=radar, 7= parking lot of entrance and exit, 9=lift control channel, 10=software trigger, 11=POS, 12=EAS channel, 13=combined event, 14=custom transparent alarm, 15=inspection point
- - sourceTypeName	string	Alarm source type name
- - groups	array	Group information
- - - groupId	string	Alarm group ID: 1=device alarm, 2=encoding channel alarm, 3=alarm input channel alarm, 4=intelligent channel alarm, 5=access control channel alarm, 6=system alarm, 7=dynamic environment alarm, 8=alarm host alarm, 9=on board alarm, 10=video intercom alarm, 11=thermal imaging alarm
- - - groupName	string	Alarm group name
- - - alarmTypes	array	Alarm type information
- - - - alarmTypeId	string	Alarm type ID, see Alarm Type Dictionary for details
- - - - alarmTypeName	string	Alarm type name
- - - - icon	string	Alarm type icon
- - - - allowDelete	string	Allow deletion: 0=No, 1=Yes

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "sourceTypeId": "1",
        "sourceTypeName": "Device",
        "groups": [
          {
            "groupId": "101",
            "groupName": "Common Device Alarm",
            "alarmTypes": [
              {
                "alarmTypeId": "16",
                "alarmTypeName": "Device Offline",

```



```
        "icon": "",
        "allowDelete": "1"
    }
}
]
```

3.7 Access Control

3.7.1 Unlock and Lock a Single Door

Brief Description

- Single door opening and closing

Request Method

- POST

Request URL

- /obms/api/v1.0/accessControl/door/control

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
status	Yes	string	Single door status: 0: Closed; 1: Opened
channelId	Yes	string	Access control channel code

Request Example

```
POST /obms/api/v1.0/accessControl/door/control HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 1b90ee2b0a5f4f7cb44149b55b5f973f

{
  "status": "1",
  "channelId": "1000006$7$0$0"
```

```
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- session	string	Session identifier, only used for asynchronous message push of third-party device door control failure

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "session": "1691805486588"
  }
}
```

3.7.2 Configuring the Global Status of Access Control and Elevator Control for Access Control Zones

Brief Description

- Configuring the global status of access control and elevator control for access control zones.

Request Method

- PUT

Request URL

- /obms/api/v1.1/acp/area/{areaCode}/global/status

Request Parameters

- Path Parameters

Parameter	Required	Type	Description
areaCode	Yes	string	Area coding, root node using 001

- Body Parameters

Parameter	Required	Type	Description
globalStatus	Yes	string	Global status: 1 = Restore, 2 = Normally Open, 3 = Normally Closed, 4 = Authentication Mode, 5 = No Authentication Mode, 6 = Lock Mode

Request Example

```
PUT /obms/api/v1.1/acp/area/001/global/status HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: d4c4d0989705492eb99202546888f7b8

{
  "globalStatus": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.7.3 Obtaining a List of Access Rules that Were Not Configured for Resources

Brief Description

- Obtaining a list of access rules that were not configured for resources.

Request Method

- GET

Request URL

- /obms/api/v1.1/acp/passage/rule/exclude/page?page={page}&pageSize={pageSize}&resourceType={resourceType}&resourceCode={resourceCode}&keyword={keyword}

Request Parameters

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
resourceType	Yes	string	Resource type: 1 = Zone, 2= Access Point, 3 = Person Group, 4 = Person
resourceCode	Yes	string	According to the resourceType, the resourceCode are areaCode, pointId, orgCode and personId respectively. When the resourceType is 4, it is not required.
keyword	No	string	Fuzzy search based on the access rule name

Request Example

```
GET /obms/api/v1.1/acp/passage/rule/exclude/page?page=1&pageSize=20&keyword=&resourceType=4&resourceCode=67312393 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept: application/json, text/plain
Accept-Language: en
X-Subject-Token: 783f732852ce451c871ac7ba2d78760f
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- nextPage	string	Next page, -1 means no next page
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.

Parameter	Type	Description
- pageData	array	Page data
- - id	string	Access rule ID
- - ruleName	string	Access rule name
- - ruleType	string	Rule type: 1 = General Authentication, 2 = Normally Open, 3 = Normally Closed, 4 = First Card, 5 = Multiple Cards, 6 = Anti-passback, 7 = Multi-door Interlock, 8 = Remote Verification
- - pointType	string	Access point type: 1 = Door, 2 = Elevator
- - timeTemplateId	string	Time template ID
- - timeTemplateName	string	Time template name
- - holidayPlanId	string	Holiday plan ID
- - holidayPlanName	string	Holiday plan name
- - pointCount	string	Number of access points
- - personCount	string	Number of persons

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "nextPage": "-1",
    "totalCount": "2",
    "pageData": [
      {
        "id": "8",
        "ruleName": "rule1",
        "ruleType": "1",
        "pointType": "1",
        "timeTemplateId": "1",
        "timeTemplateName": "time.template.name.all",
        "holidayPlanId": null,
        "holidayPlanName": null,
        "personCount": 1,
        "pointCount": 0
      }
    ]
  }
}
```

3.7.4 Get Access Control Area List

Brief Description

- Get access control area list

Request Method

- GET

Request URL

- /obms/api/v1.1/acp/area/list

Request Parameter

None

Request Example

```
GET /obms/api/v1.1/acp/area/list HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
X-Subject-Token: c4aabfe1f9814aa8bba92992b921271a
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- results	array	Result
- - areaCode	string	Area code
- - parentAreaCode	string	Parent area code
- - areaName	string	Area name
- - icon	string	Icon
- - remark	string	Remark
- - globalStatus	string	Global status: 1 = Restore, 2 = Normally Open, 3 = Normally Closed, 4 = Authentication Mode, 5 = No Authentication Mode, 6 = Lock Mode. Multiple states separated by commas, for example: 1,2,3
- - mapId	string	Map ID
- - parentMapId	string	Parent map ID
- - childNum	string	Number of sub areas
- - authority	string	Has authority, 0=No; 1=Yes

Parameter	Type	Description
unassignedPointCount	string	Number of unassigned access points

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "areaCode": "001",
        "parentAreaCode": null,
        "areaName": "Site",
        "icon": "JelconAreaRoot",
        "remark": "",
        "mapId": "",
        "parentMapId": "",
        "globalStatus": null,
        "childNum": "1",
        "authority": "1",
        "unassignedPointCount": "2"
      }
    ]
  }
}
```

3.7.5 Get the Access Point List of the Specified Area of Access Control by Paging

Brief Description

- Get the access point list of the specified area of access control by paging

Request Method

- GET

Request URL

- /obms/api/v1.1/acp/area/{areaCode}/point/page?page={page}&pageSize={pageSize}&containChild={containChild}&boundary={boundary}&pointTypes={pointTypes}&pointName={pointName}&excludeRuleType={excludeRuleType}&supportGlobalAnti={supportGlobalAnti}&ruleId={ruleId}&sourceTypes={sourceTypes}&supportHoliday={supportHoliday}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
areaCode	Yes	string	Area code
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
containChild	No	string	Whether to include the sub-area's passpoints: 0=no, 1=yes, default 0
boundary	No	string	Is it an edge point: 0=no, 1=yes, blank means all
pointTypes	No	string	Access point type: 1=access control, 2=lift, multiple separated by commas, blank means all
pointName	No	string	Passpoint name, fuzzy matching
excludeRuleType	No	string	Exclude rule type: 1=normal authorization, 2=keep open, 3=keep closed, 4=first card, 5=multiple cards, 6=anti-passback, 7=multi-door interlock, 8=remote verification
includeRuleType	No	string	Include rule type: 1=normal authorization, 2=keep open, 3=keep closed, 4=first card, 5=multiple cards, 6=anti-passback, 7=multi-door interlock, 8=remote verification
ruleId	No	string	Pass rule ID
sourceTypes	No	string	Source type: 1=platform, 2=domain, 3=third-party, separate multiple with commas, empty for all
supportHoliday	No	string	Whether to support holiday schedule: 0=no, 1=yes

Request Example

```
GET /obms/api/v1.1/acp/area/001/point/page?page=1&pageSize=100&containChild=
&boundary=&pointTypes=&pointName=&excludeRuleType=&includeRuleType=&ruleI
d=&sourceTypes=&supportHoliday=1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 172CA29640AE4958bbb79f0c500b4a44
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- totalCount	string	Record total count
- pageData	array	Paging data
- - id	string	Passpoint ID
- - areaCode	string	Area code
- - areaName	string	Area name
- - areaPath	string	Area path, for example: Area1399-BuildingA-Floor4
- - pointName	string	Passpoint name
- - pointType	string	Access point type: 1=access control, 2=lift
- - boundary	string	Is it an edge point: 0=no, 1=yes
- - remark	string	Remark
- - mapId	string	Map ID
- - gpsX	string	X coordinate
- - gpsY	string	Y coordinate
- - globalStatus	string	Global status: 1=normal, 2=keep Open, 3=keep closed, 4=authentication mode, 5=non-Authentication mode, 6=lock mode. Multiple statuses are separated by commas, for example: 1,2,3
- - sourceType	string	Data source type: 1=platform, 2=domain, 3=third party
- - sourceId	string	Source ID
- - associateId	string	Associated external ID, unique ID within the third-party system
- - relateObjects	array	Related objects
- - - objectType	string	Resource type: 1=access control, 2=lift, 3=video, 4=video intercom
- - - objectCode	string	Resource code
- - - objectName	string	Resource name
- - - readers	array	Readers
- - - - readerId	string	Reader ID

Parameter	Type	Description
direction	string	Reader direction: 0=in, 1=out

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "id": "19",
        "areaCode": "001",
        "areaName": "Site",
        "areaPath": "Site",
        "pointName": "VTO82_Door1",
        "pointType": "1",
        "remark": null,
        "mapId": "1680079222940",
        "gpsX": "-187.50000000000001",
        "gpsY": "-59.765624999999986",
        "boundary": "0",
        "globalStatus": "",
        "sourceType": "1",
        "sourceld": "",
        "associateld": "",
        "relateObjects": [
          {
            "objectType": "1",
            "objectCode": "1000006$7$0$0",
            "objectName": "Door1",
            "readers": [
              {
                "readerId": "1",
                "direction": "0"
              }
            ]
          }
        ]
      },
      {
        "objectType": "4",
        "objectCode": "1000006",
        "objectName": "VTO82",
        "readers": [
          {
            "readerId": "1",
            "direction": "0"
          }
        ]
      }
    ]
  }
}
```

```
}
{
  }
  ]
  }
  ]
  }
  ]
  }
}
```

3.7.6 Get the Access Control Point Access Status

Brief Description

- Get the access control point access status(synchronous)

Request Method

- POST

Request URL

- /obms/api/v1.1/acp/point/door/status

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
pointIds	Yes	array	Pass point ID, up to 100

Request Example

```
POST /obms/api/v1.0/accessControl/door/control HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 1b90ee2b0a5f4f7cb44149bbbb5f973f
```

```
{
  "pointIds": [
    "2"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- results	array	Results
- - pointId	string	Pass point ID
- - status	string	Access status: 1=open, 2=closed

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "pointId": "2",
        "status": "2"
      }
    ]
  }
}
```

3.8 Person

3.8.1 Person Group

3.8.1.1 Add a Person Group

Brief Description

- Add person group

Request Method

- POST

Request URL

- /obms/api/v1.1/acs/person-group

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
parentOrgCode	Yes	string	The parent group contains a maximum of nine layers and 27 digits. Each layer consists of three digits, for example, 001001
orgName	Yes	string	Person group name , is 1 to 50 characters long ,/
remark	No	string	Remarks, Maximum length 255
roleIds	Yes	array	Role ID
- sourceType	No	string	Data source: 1=platform, 2=domain, 3=third-party
- sourceId	No	string	Data source ID, Maximum length 20,mandatory for third-party platforms
- associateId	No	string	Associated external ID, Maximum length 100, unique ID within the third-party system

Request Example

```
POST /obms/api/v1.1/acs/person-group HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 4930914703254910bb0ff273471c744c
```

```
{
  "parentOrgCode": "001",
  "orgName": "test",
  "remark": "remark",
  "roleIds": [
    "1",
    "2"
  ],
  "sourceType": "1",
  "sourceId": "",
  "associateId": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- orgCode	string	Person group ID
- overLimitRoleName	string	Normal role names exceeding the limit of 500 personnel groups

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "orgCode": "001002",
    "overLimitRoleName": ""
  }
}
```

3.8.1.2 Update a Person Group

Brief Description

- Update the person group

Request Method

- PUT

Request URL

- /obms/api/v1.1/acs/person-group/{orgCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
orgCode	Yes	string	Person group ID, required ,contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001

- Body Parameters

Parameter	Required	Type	Description
orgName	Yes	string	Person group name, is 1 to 50 characters long
remark	No	string	Remarks, maximum length 255
roleIds	Yes	array	Role ID
- sourceType	No	string	Data source: 1=platform, 2=domain, 3=third-party
- sourceId	No	string	Data source ID, Maximum length 20, mandatory for third-party platforms
- associateId	No	string	Associated external ID, Maximum length 100, unique ID within the third-party system

Request Example

```
PUT /obms/api/v1.1/acs/person-group/001002 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 4930914703254910bb0ff273471c744c
```

```
{
  "orgName": "test",
  "remark": "remark",
  "roleIds": [
    "1",
    "2"
  ],
  "sourceType": "1",
  "sourceId": "",
  "associateId": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.8.1.3 Delete a Person Group

Brief Description

- Delete person group

Request Method

- DELETE

Request URL

- /obms/api/v1.1/acs/person-group/{orgCode}?mode={mode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
orgCode	Yes	string	Person group ID, required ,contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001
mode	No	string	Delete Mode: 1=Normal Mode, 2=Strong Delete Mode, default to Normal Delete

Request Example

```
DELETE /obms/api/v1.1/acs/person-group/001002 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 4930914703254910bb0ff273471c744c
```

Return Parameter

Parameter	Type	Description
code	int	Error code

Parameter	Type	Description
desc	string	Result description
data	object	Return information
session	string	Session mark

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "session": "1634129385561"
  }
}
```

3.8.1.4 Get the List of Person Groups

Brief Description

- Get the list of the person groups

Request Method

- GET

Request URL

- /obms/api/v1.1/acs/person-group/list

Request Parameter

None

Request Example

```
GET /obms/api/v1.1/acs/person-group/list HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 12c46ebc2cd440ce84c77fb2ce69f114
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- results	object	Returned result
- - parentOrgCode	string	Code of parent organization
- - orgCode	string	Code of person group
- - orgName	string	Name of person group
- - remark	string	Remarks
- - childNum	string	Number of child nodes
- - authority	string	Whether the current user is authorized: 0=No, 1=Yes
- - sourceType	string	Data source: 1=platform, 2=domain, 3=third-party
- - sourceId	string	Data source ID
- - associateId	string	Associated external ID, unique ID within the third-party system

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "parentOrgCode": "001",
        "orgCode": "001001",
        "orgName": "test",
        "remark": "",
        "childNum": "0",
        "authority": "1",
        "sourceType": "1",
        "sourceId": "",
        "associateId": ""
      }
    ]
  }
}
```

3.8.2 Person Information

3.8.2.1 Add a Person

Brief Description

- Add person

Request Method

- POST

Request URL

- /obms/api/v1.1/acs/person

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
baseInfo	Yes	object	Basic person information
- personId	Yes	string	Person ID, contain a maximum of 30 letters and numbers
- lastName	No	string	Last name, contains a maximum of 127 characters
- firstName	Yes	string	First name, contains a maximum of 127 characters
- gender	Yes	string	Gender: 0 = Unknown, 1 = Male, 2 = Female; 0 by default
- orgCode	Yes	string	Person group ID, contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001
- orgCodes	No	array	Personnel multi organization code
- email	No	string	Email
- tel	No	string	Mobile phone number, Maximum length 20
- remark	No	string	Remarks, Maximum length 255
- facePictures	No	array	Face image data Base64
- source	Yes	string	Person source, 0 = People management list, 1 = Access control group, 2 = Face comparison group, 3 = Entrance and exit vehicle group; 0 by default

Parameter	Required	Type	Description
- sourceType	No	string	Data source: 1=platform, 2=domain, 3=third-party, default 1
- sourceId	No	string	Data source ID, Maximum length 20, mandatory for third-party platforms
- associateId	No	string	Associated external ID, Maximum length 100, unique ID within the third-party system
extensionInfo	Yes	object	Extension information
- nickName	No	string	Nickname, Maximum length 255
- address	No	string	Address, Maximum length 255
- idType	Yes	string	ID type, 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others; 6 by default
- idNo	No	string	ID number, Maximum length 50
- nationalityId	Yes	string	Country/region ID, Unknown=9999
- birthday	No	string	Birthday, year-month-day string, length 10
- companyName	No	string	Company name, Maximum length 255
- department	No	string	Department name, deprecated, just leave it blank
- position	No	string	Position, Maximum length 255
userDefineFields	No	array	User define fields
- fieldId	No	string	Field ID, be composed of numbers
- fieldName	No	string	Field name
- fieldValue	No	string	Field name
residentInfo	No	object	Resident information
- houseHolder	No	string	Video intercom householder or not, 0 = No, 1 = Yes; 0 by default
- sipId	No	string	Building#unit#room number, 1#1#1001

Parameter	Required	Type	Description
- vdpUser	No	string	Register VDP APP user, 0=No, 1=Yes, default 0
- inheritHouseHolder	No	string	Inherit owner permission: 0 = No, 1 = Yes, default 0
authenticationInfo	Yes	object	Authorization information
- startTime	Yes	string	Start time of the validity period, timestamp, in seconds
- endTime	Yes	string	End time of the validity period, timestamp, in seconds
- combinationPassword	No	string	Combination unlocking password, AES encryption
- cards	No	array	Access card list
- - cardNo	No	string	Card No.,consists of 0-9A-F characters and is 8-16 in length
- - duressFlag	No	string	Duress card or not, 0 = No, 1 = Yes
- fingerprints	No	arrays	Fingerprint list
- - fingerprint	No	string	Fingerprint data Base64
- - name	No	string	Fingerprint name, Maximum length 255
- - duressFlag	No	string	Duress fingerprint or not, 0 = No, 1 = Yes
- palmVeins	No	array	
- - palmType	No	string	Palm type: 1=left, 2=right
- - style	No	string	Palm vein style: 1=red light, 2=white light
- - feature	No	string	Palm vein features Base64
- - version	No	string	Palm vein feature version
- - picture	No	string	Palm vein image Base64
accessInfo	Yes	object	Access control information
- accessType	Yes	string	Access control person type, 0 = Normal, 1 = Blocklist, 3= Patrol, 4 = VIP, 5 = Others; 0 by default
- guestUseTimes	No	string	Number of times visitors can unlock; max 200
- passageRuleIds	No	array	Access rule ID
faceComparisonInfo	Yes	object	Face comparison group information

Parameter	Required	Type	Description
- enableFaceComparisonGroup	Yes	string	Enable face comparison group, 0 = No, 1 = Yes
- faceComparisonGroupId	No	string	Face comparison group ID, the reserved fields are forward-compatible, can continue to be assigned, and the automatic deduplication will be converted into the faceComparisonGroupIds
- faceComparisonGroupIds	No	array	Face comparison group ID
entranceInfo	Yes	object	Entrance and exit information
- parkingSpaceQuotas	No	array	
- - parkingLotId	No	string	Parking lot ID
- - parkingSpaceQuota	No	string	Number of parking spaces available
- vehicles	No	arrays	Vehicle information list
- - id	No	string	Vehicle ID
- - plateNo	No	string	License plate, Length: 1 to 15
- - vehicleColor	No	string	Vehicle color, dictionary type. See [6.1.10 Color of motor vehicles]
- - vehicleBrand	No	string	Vehicle brand, dictionary type. See [6.1.12 Brand of motor vehicles]
- - companyTelephone	No	string	Company telephone
- - companyOwner	No	string	Company owner
- - affiliatedCompany	No	string	Affiliated company
- - bindingDeviceCode	No	string	Vehicle mounted device code
- - remark	No	string	Remarks, Maximum length 255
- - entranceGroups	No	array	Vehicle entrance and exit groups
- - - parkingLotId	No	string	Parking lot ID
- - - entranceGroupIds	No	array	Vehicle entrance and exit group ID
- - - entranceLongTerm	No	string	Long-term vehicle or not: 0 = No, 1 = Yes
- - - entranceStartTime	No	string	Start time of the validity period, timestamp, in seconds; required for non-long-term vehicles

Parameter	Required	Type	Description
- - - entranceEndTime	No	string	End time of the validity period, timestamp, in seconds; required for non-long-term vehicles

Request Example

```
POST /obms/api/v1.1/acs/person HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 522eb5ed8e014c809978b7f64ae9cdd1
```

```
{
  "baseInfo": {
    "personId": "16448790",
    "lastName": "",
    "firstName": "test",
    "gender": "0",
    "orgCode": "001",
    "orgCodes": [
      "001"
    ],
    "email": "aa@test.com",
    "tel": "18888888888",
    "remark": "asd",
    "source": "0",
    "facePictures": [

    ]
  },
  "extensionInfo": {
    "nickName": "nnnn",
    "address": "qqq",
    "idType": "0",
    "idNo": "123123",
    "nationalityId": "9999",
    "birthday": "1993-10-10",
    "companyName": "cccc",
    "department": "bbb",
    "position": "zzzz"
  },
  "userDefineFields": [
    {
      "fieldId": "1",
      "fieldName": "wse",
      "fieldValue": "wse"
    }
  ]
}
```

```
    }
  ],
  "residentInfo": {
    "houseHolder": "1",
    "sipId": "1#1#1",
    "vdpUser": "1",
    "inheritHouseHolder": "0"
  },
  "authenticationInfo": {
    "combinationPassword": "59a4ac79560526b7495585c572b87781",
    "cards": [
      {
        "cardNo": "AAAA1111",
        "duressFlag": "0"
      }
    ],
    "palmVeins": [
      {
        "palmType": "1",
        "style": "1",
        "feature": "sfdwffq",
        "version": "1.1",
        "picture": "sdfberthert"
      }
    ],
    "fingerprints": [

  ],
  "startTime": "1711296000",
  "endTime": "2026915199"
},
"accessInfo": {
  "accessType": "0",
  "guestUseTimes": "200",
  "passageRuleIds": [
    "1"
  ]
},
"faceComparisonInfo": {
  "enableFaceComparisonGroup": "1",
  "faceComparisonGroupId": "1"
},
"entranceInfo": {
  "parkingSpaceQuotas": [
    {
      "parkingLotId": "2",
      "parkingSpaceQuota": 11
    }
  ],

```



```

"vehicles": [
  {
    "id": "1",
    "plateNo": "ASD1212",
    "vehicleColor": "100",
    "vehicleBrand": "-1",
    "remark": "",
    "bindingDeviceCode": "1000000",
    "companyTelephone": "2343545423",
    "companyOwner": "abc",
    "affiliatedCompany": "BBC",
    "entranceGroups": [
      {
        "parkingLotId": "1",
        "entranceGroupIds": [
          ],
        "entranceLongTerm": "1",
        "entranceStartTime": "-1",
        "entranceEndTime": "-1"
      }
    ]
  }
]
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- existPlateNos	array	Existing license plates
- existCardNos	array	Existing card No.
- overstepPlateNos	array	License plates exceeding the quantity
- limitPersonCount	string	Maximum number of people

Return Example

```

{
  "code": 1000,
  "desc": "Success",
  "data": {
    "existPlateNos": [

```

```

        "JA12A26"
    ],
    "existCardNos": [
        "ABCDEF12"
    ],
    "overstepPlateNos": [
        "JA12A27"
    ],
    "limitPersonCount": "999"
}
}

```

3.8.2.2 Update a Person

Brief Description

- Update person information

Request Method

- PUT

Request URL

- /obms/api/v1.1/acs/person/{personId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
personId	Yes	String	Person ID, contain a maximum of 30 letters and numbers

- Body Parameters

Parameter	Required	Type	Description
baseInfo	Yes	object	Basic person information
- personId	Yes	string	Person ID, contain a maximum of 30 letters and numbers
- lastName	No	string	Last name, contains a maximum of 127 characters
- firstName	Yes	string	First name, contains a maximum of 127 characters
- gender	Yes	string	Gender, 0 = Female; 1 = Male; 2 = Unknown; 2 by default

Parameter	Required	Type	Description
- orgCode	Yes	string	Person group ID, required ,contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001
- orgCodes	No	array	Personnel multi organization code
- email	No	string	Email
- tel	No	string	Mobile phone number, Maximum length 20
- updateFacePictures	No	string	Update facial image: 0=No, 1=Yes, default 1
- remark	No	string	Remarks, Maximum length 255
- facePictures	No	array	Face image data Base64
- source	Yes	string	Person source, write 0 directly
extensionInfo	No	object	Extension information
- nickName	No	string	Nickname, Maximum length 255
- address	No	string	Address, Maximum length 255
- idType	Yes	string	ID type, 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others; 6 by default
- idNo	No	string	ID number, Maximum length 50
- nationalityId	Yes	string	Country/region ID, Unknown=9999
- birthday	No	string	Birthday, year-month-day string,length 10
- companyName	No	string	Company name, Maximum length 255
- department	No	string	Department name, deprecated, just leave it blank
- position	No	string	Position, Maximum length 255
userDefineFields	No	array	User define fields
- fieldId	No	string	Field ID, be composed of numbers
- fieldName	No	string	Field name

Parameter	Required	Type	Description
- fieldValue	No	string	Field name
residentInfo	No	object	Resident information
- houseHolder	No	string	Video intercom householder or not, 0 = No, 1 = Yes; 0 by default
- sipId	No	string	Building#unit#room number, 1#1#1001
- vdpUser	No	string	Register VDP APP user, 0=No, 1=Yes, default 0
authenticationInfo	Yes	object	Authorization information
- startTime	Yes	string	Start time of the validity period, timestamp, in seconds
- endTime	Yes	string	End time of the validity period, timestamp, in seconds
- combinationPasswordChanged	No	string	Indicate whether the combination unlocking password is changed. 0=No, 1=Yes. The default value is 0.
- combinationPassword	No	string	Combination unlocking password, AES encryption
- cards	No	array	Access card list
- - cardNo	No	string	Card No., consists of 0-9A-F characters and is 8-16 in length
- - duressFlag	No	string	Duress card or not, 0 = No, 1 = Yes
- fingerprints	No	array	Fingerprint list
- - duressFlag	No	string	Duress fingerprint or not, 0 = No, 1 = Yes
- - name	No	string	Fingerprint name, Maximum length 255
- - fingerprint	No	string	Fingerprint data Base64
- updatePalmVeins	No	string	Update palmar vein: 0=No, 1=Yes, default 1
- palmVeins	No	array	
- - palmType	No	string	Palm type: 1=left palm, 2=right palm
- - style	No	string	Palm vein style: 1=red light, 2=white light

Parameter	Required	Type	Description
- - feature	No	string	Palm vein features Base64
- - version	No	string	Palm vein feature version
- - picture	No	string	Palm vein image Base64
accessInfo	Yes	object	Access control information
- accessType	Yes	string	Access control person type, 0 = Normal, 1 = Blocklist, 3 = Patrol, 4 = VIP, 5 = Others; 0 by default
- guestUseTimes	No	string	Number of times visitors can unlock; max 200
- passageRuleIds	No	array	Access rule ID
faceComparisonInfo	Yes	object	Face comparison group information
- enableFaceComparisonGroup	Yes	string	Enable face comparison group, 0 = No, 1 = Yes
- faceComparisonGroupId	No	string	Face comparison group ID
entranceInfo	Yes	object	Entrance and exit information
- parkingSpaceQuotas	No	array	None
- - parkingLotId	No	string	Parking lot ID
- - parkingSpaceQuota	No	string	Number of parking spaces available
- vehicles	No	array	Vehicle information list
- - id	No	string	Vehicle ID
- - plateNo	No	string	License plate, Length: 1 to 15
- - vehicleColor	No	string	Vehicle color, dictionary type. See [6.1.10 Color of motor vehicles]
- - vehicleBrand	No	string	Vehicle brand, dictionary type. See [6.1.12 Brand of motor vehicles]
- - remark	No	string	Remarks, Maximum length 255
- - bindingDeviceCode	No	string	Vehicle mounted device code
- - companyTelephone	No	string	Company telephone
- - companyOwner	No	string	Company owner
- - affiliatedCompany	No	string	Affiliated company
- - entranceGroups	No	array	Vehicle entrance and exit groups
- - - parkingLotId	No	string	Parking lot ID

Parameter	Required	Type	Description
- - - entranceGroupIds	No	array	Vehicle entrance and exit group ID
- - - entranceLongTerm	No	string	Long-term vehicle or not: 0 = No, 1 = Yes
- - - entranceStartTime	No	string	Start time of the validity period, timestamp, in seconds; required for non-long-term vehicles
- - - entranceEndTime	No	string	End time of the validity period, timestamp, in seconds; required for non-long-term vehicles

Request Example

```
PUT /obms/api/v1.1/acs/person/19476543 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: eb4ba8608be84c928630860d86261d34
```

```
{
  "baseInfo": {
    "personId": "19476543",
    "lastName": "",
    "firstName": "Lily",
    "gender": "0",
    "orgCode": "001",
    "email": "1@2.com",
    "tel": "10086",
    "updateFacePictures": "0",
    "remark": "",
    "facePictures": [
      "E6D8F09...F97FL/D0A91B"
    ],
    "source": "0"
  },
  "extensionInfo": {
    "nickName": "sunny",
    "address": "",
    "idType": "6",
    "idNo": "1234567890",
    "nationalityId": "9999",
    "birthday": "2001-01-01",
```

```
"companyName": "",
"department": "",
"position": "boss"
},
"userDefineFields": [
  {
    "fieldId": "1",
    "fieldName": "userDefined1",
    "fieldValue": "value1"
  }
],
"residentInfo": {
  "houseHolder": "0",
  "sipId": "10#1#1001",
  "vdpUser": "1"
},
"authenticationInfo": {
  "startTime": "1615824000",
  "endTime": "1931443199",
  "combinationPasswordChanged": "0",
  "combinationPassword": "AES ciphertext",
  "cards": [
    {
      "cardNo": "ABCDEF12",
      "duressFlag": "0"
    }
  ],
  "updatePalmVeins": "0",
  "palmVeins": [
    {
      "palmType": "1",
      "style": "1",
      "feature": "ssdvr",
      "version": "1.1",
      "picture": "sgsw454"
    }
  ],
  "fingerprints": [
    {
      "fingerprint": "E6D8F09...F97FL/D0A91B",
      "name": "fg1",
      "duressFlag": "whether or not duress fingerprints, 0=NO, 1=YES"
    }
  ]
},
"accessInfo": {
  "accessType": "0",
  "guestUseTimes": "0",
  "passageRuleIds": [
```

```

    "2"
  ]
},
"faceComparisonInfo": {
  "enableFaceComparisonGroup": "1",
  "faceComparisonGroupId": "1"
},
"entranceInfo": {
  "parkingSpaceQuotas": [
    {
      "parkingLotId": "2",
      "parkingSpaceQuota": 11
    }
  ],
  "vehicles": [
    {
      "id": "1",
      "plateNo": "A12A26",
      "vehicleColor": "100",
      "vehicleBrand": "-1",
      "bindingDeviceCode": "1000000",
      "companyTelephone": "234453",
      "companyOwner": "aab",
      "affiliatedCompany": "bbc",
      "remark": "",
      "entranceGroups": [
        {
          "parkingLotId": "1",
          "entranceGroupIds": [

          ],
          "entranceLongTerm": "1",
          "entranceStartTime": "-1",
          "entranceEndTime": "-1"
        }
      ]
    }
  ]
}
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information

Parameter	Type	Description
- existPlateNos	array	Existing license plate
- existCardNos	array	Existing card number
- overstepPlateNos	array	License plates exceeding the quantity
- limitPersonCount	string	Maximum number of people

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "existPlateNos": [
      "JA12A26"
    ],
    "existCardNos": [
      "ABCDEF12"
    ],
    "overstepPlateNos": [
      "JA12A27"
    ],
    "limitPersonCount": "999"
  }
}
```

3.8.2.3 Batch Delete Persons

Brief Description

- Batch delete persons

Request Method

- POST

Request URL

- /obms/api/v1.1/acs/person-group/person/delete/batch

Request Parameter

Parameter	Required	Type	Description
personIds	Yes	array	Person ID, contain a maximum of 30 letters and numbers

Parameter	Required	Type	Description
mode	No	string	Delete Mode: 1=Normal Mode, 2=Strong Delete Mode, default to Normal Delete
orgCode	No	string	Person group ID, required ,contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001
containChild	No	string	Include child organizations: 0 = No, 1 = Yes, default 0

Request Example

```
POST /obms/api/v1.1/acs/person-group/person/delete/batch HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 948ad07052e8495dba0c422992907e0e
```

```
{
  "personIds": [
    "29282770"
  ],
  "mode": "1",
  "orgCode": "001001",
  "containChild": "0"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.8.2.4 Get the Details of a Person

Brief Description

- Get details of persons

Request Method

- GET

Request URL

- /obms/api/v1.1/acs/person/{personId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
personId	Yes	string	Person ID, contain a maximum of 30 letters and numbers

Request Example

```
GET /obms/api/v1.1/acs/person/19476543 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 948ad07052e8495dba0c422992907e0e
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- baseInfo	object	Basic person information
- - personId	string	Person ID
- - lastName	string	Last name
- - firstName	string	First name
- - gender	string	Gender, 0 = Female; 1 = Male; 2 = Unknown; 2 by default

Parameter	Type	Description
-- orgCode	string	Person group ID, 001 by default
-- orgName	string	Person group name
-- orgInfos	array	Personnel multi organization information
-- -- orgCode	string	Person group code
-- -- orgName	string	Person group name
-- -- hasAuthority	string	Has authority, 0=No; 1=Yes
-- email	string	Email
-- tel	string	Mobile phone number
-- remark	string	Remarks
-- facePictures	array	Face picture URL, such as https://192.168.1.1:443/upload/person/xxx.jpg
-- sourceType	string	Data source: 1=platform, 2=domain, 3=third-party
-- sourceId	string	Data source ID, mandatory for third-party platforms
-- associateId	string	Associated external ID, unique ID within the third-party system
- extensionInfo	object	Extension information
-- nickName	string	Nickname
-- address	string	Address
-- idType	string	ID type, 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others; 6 by default
-- idNo	string	ID number
-- nationalityId	string	Country/region ID, Unknown=9999
-- nationalityName	string	Country/region name
-- birthday	string	Birthday, year-month-day string
-- companyName	string	Company name
-- position	string	Position
-- department	string	Department
- userDefineFields	array	User define fields
-- fieldId	string	Field ID
-- fieldName	string	Field name
-- fieldValue	string	Field name

Parameter	Type	Description
- residentInfo	object	Resident information
- - houseHolder	string	Video intercom householder or not, 0 = No, 1 = Yes; 0 by default
- - sipId	string	Building#unit#room number, 1#1#1001
- - vdpUser	string	Register VDP APP user, 0=No, 1=Yes, default 0
- - vdpUserName	string	VDP user name
- authenticationInfo	object	Authorization information
- - startTime	string	Start time of the validity period, timestamp, in seconds
- - endTime	string	End time of the validity period, timestamp, in seconds
- - expired	string	Expired or not: 1: Yes; 0: No
- - haveCombinationPassword	string	Combination password or not: 0 = No; 1 = Yes; 0 by default
- - combinationPassword	string	Combination unlocking password, AES encryption
- - cards	array	Access card list
- - - cardNo	string	Card No.
- - - duressFlag	string	Duress card or not, 0 = No, 1 = Yes
- - fingerprints	array	Fingerprint list
- - - fingerprint	string	Fingerprint data Base64
- - - name	string	Fingerprint name
- - - duressFlag	string	Duress fingerprint or not, 0 = No, 1 = Yes
- - palmVeins	array	
- - - palmType	string	Palm type: 1=left palm, 2=right palm
- - - style	string	Palm vein style: 1=red light, 2=white light
- - - feature	string	Palm vein features Base64
- - - version	string	Palm vein feature version
- - - picture	string	Palm vein image URL
- accessInfo	object	Access control information
- - accessType	string	Access control person type, 0 = Normal, 1 = Blocklist, 2 = Visitor, 3 = Patrol, 4 = VIP, 5 = Others; 0 by default

Parameter	Type	Description
-- guestUseTimes	string	Number of times visitors can unlock; max 200
-- passageRules	array	Access rule
-- -- id	string	Rule ID
-- -- ruleName	string	Rule name
-- -- ruleType	string	Rule types: 1=Normal Authorization, 2=Normally Open, 3=Normally Closed, 4=First Card, 5=Multi Card, 6=Anti Submarine, 7=Multi Door Interlock, 8=Remote Verification
-- -- pointCount	string	Access point count
-- -- personCount	string	Person count
-- -- timeTemplateId	string	Time template ID
-- -- timeTemplateName	string	Time template name
-- -- holidayPlanId	string	Holiday plan ID
-- -- holidayPlanName	string	Holiday plan
-- -- inherited	string	Inherited: 0=No, 1=Yes
- faceComparisonInfo	object	Face comparison group information
-- enableFaceComparisonGroup	string	Enable face comparison group, 0 = No, 1 = Yes
-- faceComparisonGroupId	string	Face comparison group ID. Forward compatibility of reserved fields
-- faceComparisonGroupName	string	Face comparison group name. Forward compatibility of reserved fields
-- faceComparisonGroups	array	Face comparison group
-- -- faceComparisonGroupId	string	Face comparison group ID
-- -- faceComparisonGroupName	string	Face comparison group name
- entranceInfo	object	Entrance and exit information
-- enableEntranceGroup	string	Enable vehicle entrance and exit group, 0 = No, 1 = Yes
-- parkingSpaceQuotas	array	Number of parking spaces
-- -- parkingLotId	string	Parking lot ID
-- -- parkingLotName	string	Parking lot name
-- -- parentParkingLotId	string	Parent parking lot ID
-- -- parentParkingLotName	string	Parent parking lot name
-- -- parkingSpaceQuota	string	Number of parking spaces
-- vehicles	array	Vehicle information list

Parameter	Type	Description
- - - id	string	Vehicle ID
- - - plateNo	string	License plate
- - - vehicleColor	string	Vehicle color, dictionary type. See [6.1.10 Color of motor vehicles]
- - - bindingDeviceCode	string	Binding device code
- - - companyTelephone	string	Company telephone
- - - companyOwner	string	Company owner
- - - affiliatedCompany	string	Affiliated company
- - - vehicleBrand	string	Vehicle brand, dictionary type. See [6.1.12 Brand of motor vehicles]
- - - remark	string	Remarks
- - - entranceGroups	array	Vehicle entrance and exit group
- - - - parkingLotId	string	Parking lot ID
- - - - parkingLotName	string	Parking lot name
- - - - groupId	string	Vehicle entrance and exit group ID
- - - - groupName	string	Vehicle entrance and exit group name
- - - - groupColor	string	Vehicle entrance and exit group background color
- - - - remark	string	Remark
- - - - level	string	Level
- - - entranceLongTerm	string	Long-term vehicle or not: 0 = No, 1 = Yes
- - - entranceStartTime	string	Start time of the validity period, timestamp, in seconds; required for non-long-term vehicles
- - - entranceEndTime	string	End time of the validity period, timestamp, in seconds; required for non-long-term vehicles

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "baseInfo": {
      "personId": "19476543",
      "lastName": "",
      "firstName": "Lily",
      "gender": "0",

```

```
"orgCode": "001",
"orgName": "All Persons",
"orgInfos": [
  {
    "orgCode": "001",
    "orgName": "All Persons and Vehicles",
    "hasAuthority": "1"
  }
],
"email": "1@2.com",
"tel": "10086",
"remark": "",
"facePictures": [
  "https://192.168.1.1:8443/upload/obms/headPic/19476543@1@1615886781645.jpg"
],
"sourceType": "1",
"sourceld": "",
"associateld": ""
},
"extensionInfo": {
  "nickName": "",
  "address": "",
  "idType": "6",
  "idNo": "1234567890",
  "nationalityId": "9999",
  "nationalityName": "Other",
  "birthday": "2001-01-01",
  "companyName": "",
  "position": "",
  "department": ""
},
"residentInfo": {
  "houseHolder": "0",
  "sipld": "10#1#1001",
  "vdpUser": "1",
  "vdpUserName": "Jack"
},
"authenticationInfo": {
  "startTime": "1615824000",
  "endTime": "1931443199",
  "expired": "0",
  "haveCombinationPassword": "1",
  "combinationPassword": null,
  "cards": [
    {
      "cardNo": "ABCDEF12",
      "duressFlag": "0"
    }
  ]
}
```



```
    ],
    "fingerprints": [
      {
        "fingerprint": "E6D8F09...F97FL/D0A91B",
        "name": "fg1",
        "duressFlag": "0"
      }
    ],
    "palmVeins": [
      {
        "palmType": "1",
        "style": "1",
        "feature": "asefqw2",
        "version": "1",
        "picture": "http://ms/palm/00000786@1@1641891236167.jpg"
      }
    ]
  },
  "accessInfo": {
    "accessType": "0",
    "guestUseTimes": "0",
    "passageRules": [
      {
        "id": "1",
        "ruleName": "test",
        "ruleType": "1",
        "pointCount": "2",
        "personCount": "50",
        "timeTemplateId": "1",
        "timeTemplateName": "workTimeTemplate",
        "holidayPlanId": "2",
        "holidayPlanName": "autumn",
        "inherited": "1"
      }
    ]
  },
  "faceComparisonInfo": {
    "enableFaceComparisonGroup": "1",
    "faceComparisonGroupId": "1",
    "faceComparisonGroupName": "1",
    "faceComparisonGroups": [
      {
        "faceComparisonGroupId": "1",
        "faceComparisonGroupName": "bc"
      }
    ]
  },
  "entranceInfo": {
    "enableEntranceGroup": "1",
```

```
    "parkingSpaceQuotas": [
      {
        "parkingLotId": "2",
        "parkingLotName": "parkinglot2",
        "parentParkingLotId": "1",
        "parentParkingLotName": "parkinglot1",
        "parkingSpaceQuota": "11"
      }
    ],
    "vehicles": [
      {
        "id": 1,
        "plateNo": "A12A26",
        "vehicleColor": "100",
        "vehicleBrand": "-1",
        "remark": "",
        "bindingDeviceCode": "23232w",
        "companyTelephone": "23233",
        "companyOwner": "aac",
        "affiliatedCompany": "bbc",
        "entranceGroups": [
          {
            "entranceGroupId": "5",
            "entranceGroupName": "123",
            "parkingLotId": "2",
            "parkingLotName": "parkinglot2",
            "entranceStartTime": "1711445480402",
            "entranceEndTime": "2026978280402",
            "entranceLongTerm": "1"
          }
        ],
        "entranceLongTerm": "1",
        "entranceStartTime": "0",
        "entranceEndTime": "0"
      }
    ]
  }
}
```

3.8.2.5 Get the List of Persons in Pages

Brief Description

- Get the list of persons in pages

Request Method

- GET

Request URL

- /obms/api/v1.1/acs/person/page?page={page}&pageSize={pageSize}&orgCode={orgCode}&containChild={containChild}&personId={personId}&cardNo={cardNo}&personName={personName}&sipld={sipld}&associated={associated}&hasVehicle={hasVehicle}&isHouseHolder={isHouseHolder}&isVdpUser={isVdpUser}&accessType={accessType}&expired={expired}&inFaceRepository={inFaceRepository}&hasParkingSpace={hasParkingSpace}&hasFace={hasFace}&hasCard={hasCard}&hasFingerprint={hasFingerprint}&plateNo={plateNo}&plateNoMatchMode={plateNoMatchMode}&filterAuthorityOrg={filterAuthorityOrg}&company={company}&department={department}&idNo={idNo}&defineFieldId={defineFieldId}&defineFieldValue={defineFieldValue}&associateId={associateId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orgCode	Yes	string	Person group ID, required, contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001
containChild	Yes	string	Whether to associate with subordinate organizations, 0=No, 1=Yes
personId	No	string	Personnel ID, precise matching
cardNo	No	string	Card number, precise matching
personName	No	string	Name keywords, fuzzy matching
sipld	No	string	Room number, fuzzy matching
associated	No	string	Is there a tripartite association ID: 0=No, 1=Yes
hasVehicle	No	string	Is there a vehicle: 0=No, 1=Yes
isHouseHolder	No	string	Is it the head of household: 0=No, 1=Yes
isVdpUser	No	string	Is it associated with an APP account: 0=No, 1=Yes
accessType	No	string	Access types: 0=Normal, 1=Blocklist, 2=Visitor, 3=Patrol, 4=VIP, 5=Delayed opening of doors
expired	No	string	Is the access permission valid: 0=expired, 1=in effect, 2=not in effect

Parameter	Required	Type	Description
inFaceRepository	No	string	Is it in the facial control group: 0=No, 1=Yes
hasParkingSpace	No	string	Is there a parking space:0=No,1=Yes
hasFace	No	string	Is there a face: 0=No, 1=Yes
hasCard	No	string	Is there a card: 0=No, 1=Yes
hasFingerprint	No	string	Is there a fingerprint: 0=No, 1=Yes
hasPalmVein	No	string	Has palmar vein: 0=No, 1=Yes
plateNo	No	string	License plate number, precise matching
plateNoMatchMode	No	string	License plate number matching mode, 0=fuzzy match, 1=precise match
filterAuthorityOrg	No	string	Filter organizational permissions: 0=No, 1=Yes, default to 1
company	No	string	Company, fuzzy matching
department	No	string	Department, fuzzy matching
idNo	No	string	Certificate number, fuzzy matching
defineFieldId	No	string	Custom field serial number, 1 to 10, associated with the custom field column of the personnel table, precise matching
defineFieldValue	No	string	Custom field content, fuzzy matching
associateId	No	string	Associated external ID, unique ID within the third-party system

Request Example

```
GET /obms/api/v1.1/acs/person/page?cardNo=&containChild=0&page=1&personId=
&personName=&orgCode=001&pageSize=20&hasPalmVein= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 948ad07052e8495dba0c422992907e0e
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are

Parameter	Type	Description
		used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	Person page data
- - baseInfo	object	Basic person information
- - - personId	string	Person ID
- - - lastName	string	Last name
- - - firstName	string	First name
- - - orgCode	string	Person group ID, 001 by default
- - - orgName	string	Person group name
- - - sourceType	string	Data source: 1=platform, 2=domain, 3=third-party
- - - sourceId	string	Data source ID, mandatory for third-party platforms
- - - associateId	string	Associated external ID, unique ID within the third-party system
- - - orgInfos	array	Personnel multi organization information
- - - - orgCode	string	Person group code
- - - - orgName	string	Person group name
- - - - hasAuthority	string	Has authority, 0=No; 1=Yes
- - - facePicture	string	The URL of the face image. For example, https://192.168.1.1:443/upload/person/xxx.jpg
- - - tel	string	Mobile phone number
- - - email	string	Email
- - - companyName	string	Company name
- - - department	string	Department
- - - idNo	string	ID number
- - authenticationInfo	object	Authorization information
- - - expired	string	Expired or not: 1: Yes; 0: No
- - - cardNum	string	Number of cards
- - - fingerprintNum	string	The number of fingerprints
- - - passwordNum	string	The number of passwords
- - - cardNo	string	Card number (main card/the first card). A QR code can be generated.
- - - faceNum	string	The number of faces
- - - palmVeinNum	string	Number of palmar veins
- - - cards	array	Card information

Parameter	Type	Description
- - - mainFlag	string	Whether it is the main card (first generation card), 0=no, 1=yes
- - - duressFlag	string	Duress fingerprint or not, 0 = No, 1 = Yes
- - residentInfo	object	Resident information
- - - houseHolder	string	Indicate whether this is a householder of video call: 0=No, 1=Yes. The default value is 0.
- - - sipId	string	Building#unit#room number, 1#1#1001
- - - isVdpUser	string	Is it associated with an APP account: 0=No, 1=Yes
- - accessInfo	object	Access control information
- - - accessType	string	Access control person type, 0 = Normal, 1 = Blocklist, 2 = Visitor, 3= Patrol, 4 = VIP, 5 = Others; 0 by default
- - - guestUseTimes	string	Number of times visitors can unlock; max 200
- - faceComparisonInfo	object	Face comparison group information
- - - enableFaceComparisonGroup	string	Enable face comparison group, 0 = No, 1 = Yes
- - - faceComparisonGroupId	string	Face comparison group ID
- - - faceComparisonGroupName	string	Face comparison group name
- - - faceComparisonGroups	array	Face comparison group
- - - - faceComparisonGroupId	string	Face comparison group ID
- - - - faceComparisonGroupName	string	Face comparison group name
- - entranceInfo	object	Entrance and exit information
- - - parkingSpaceQuotas	array	Number of parking spaces
- - - - parkingLotId	string	Parking lot ID
- - - - parkingLotName	string	Parking lot name
- - - - parentParkingLotId	string	Parent parking lot ID
- - - - parentParkingLotName	string	Parent parking lot name
- - - - parkingSpaceQuota	string	Number of parking spaces
- - - vehicles	array	Vehicle information list

Parameter	Type	Description
plateNo	string	License plate
entranceGroups	array	Vehicle entrance and exit group
parkingLotId	string	Parking lot ID
parkingLotName	string	Parking lot name
entranceGroupId	string	Vehicle entrance and exit group ID
entranceGroupName	string	Vehicle entrance and exit group name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "baseInfo": {
          "personId": "19476543",
          "lastName": "",
          "firstName": "Lily",
          "orgCode": "001",
          "orgName": "All Persons",
          "sourceType": "1",
          "sourceId": "0",
          "associateId": null,
          "orgInfos": [
            {
              "orgCode": "001",
              "orgName": "All Persons and Vehicles",
              "hasAuthority": "1"
            }
          ],
          "facePicture": "https://192.168.1.1:8443/upload/obms/headPic/19476543@1@1615886781645.jpg",
          "tel": null,
          "email": null,
          "companyName": "",
          "department": "",
          "idNo": null
        },
        "authenticationInfo": {
          "expired": "0",
          "cardNum": "1",
          "fingerprintNum": "0",

```

```
"passwordNum": "2",
"cardNo": "ABCDEF12",
"faceNum": "0",
"palmVeinNum": "12",
"cards": null
},
"userDefineFields": [
  {
    "fieldId": "1",
    "fieldName": "userDefined1",
    "fieldValue": "value1"
  }
],
"residentInfo": {
  "houseHolder": "0",
  "sipId": "10#1#1001",
  "isVdpUser": "0"
},
"accessInfo": {
  "accessType": "0",
  "guestUseTimes": "0"
},
"faceComparisonInfo": {
  "enableFaceComparisonGroup": "1",
  "faceComparisonGroupId": "1",
  "faceComparisonGroupName": "1",
  "faceComparisonGroups": [
    {
      "faceComparisonGroupId": "1",
      "faceComparisonGroupName": "bc"
    }
  ]
},
"entranceInfo": {
  "parkingSpaceQuotas": [
    {
      "parkingLotId": "2",
      "parkingLotName": "parkinglot2",
      "parentParkingLotId": "1",
      "parentParkingLotName": "parkinglot1",
      "parkingSpaceQuota": "11"
    }
  ]
},
"vehicles": [
  {
    "plateNo": "A12A26",
    "entranceGroups": [
      {
        "entranceGroupId": "5",
```



```

        "entranceGroupName": "123",
        "parkingLotId": "2",
        "parkingLotName": "parkinglot2"
      }
    ]
  }
}

```

3.8.2.6 Get the List of Persons in Pages (Brief Information)

Brief Description

- Get the person list in pages (brief information). For scenarios where only brief information such as person ID and person name needs to be obtained.

Request Method

- GET

Request URL

- /obms/api/v1.1/acs/person/summary/page?page={page}&pageSize={pageSize}&orgCode={orgCode}&keyword={keyword}&faceComparisonGroupId={faceComparisonGroupId}&faceIssueResult={faceIssueResult}&deviceCode={deviceCode}&channelId={channelId}&accessTypes={accessTypes}&inMultiCardGroups={inMultiCardGroups}&inMultiCardGroupId={inMultiCardGroupId}&containChild={containChild}&withMainOrg={withMainOrg}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orgCode	Yes	string	Person group ID, required ,contains a maximum of ten layers and 30 digits. Each layer consists of three digits, for example, 001001

Parameter	Required	Type	Description
keyword	No	string	Keyword, precise search of person ID, fuzzy search of name
faceComparisonGroupId	No	string	Face comparison group ID; null means all
faceIssueResult	No	string	Face database arming sending result, 0 = Failure, 1 = Success, empty means all
deviceCode	No	string	Device code associated with the person, which is used when the anti-passback module is associated with a device
channelId	No	string	Channel code associated with the person, which is used when the advanced function of access control filters successfully issued persons
accessTypes	No	string	Person type: -1 = All, 0 = Normal, 1 = Blocklist, 2 = Visitor, 3 = Patrol, 4 = VIP; when it is not all, multiple types are separated with commas
inMultiCardGroups	No	string	Whether multi card door opening has been configured: 0=No, 1=Yes
inMultiCardGroupId	No	string	ID of multi card door opening personnel group
containChild	No	string	Include sub personnel groups: 0=No, 1=Yes, default 1
withMainOrg	No	string	Whether to query by personnel primary group: 0=No, 1=Yes, default 0

Request Example

```
GET /obms/api/v1.1/acs/person/summary/page?faceIssueResult=&deviceCode=&page=1&accessTypes=&keyword=&orgCode=001&channelId=&pageSize=20&faceComparisonGroupId=&inMultiCardGroups=&inMultiCardGroupId=&containChild=&withMainOrg= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 4b6b124e0a08418c98ed303e87c3808a
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	Person page data
- - personId	string	Person ID
- - lastName	string	Last name
- - firstName	string	First name
- - orgCode	string	Person group ID, 001 by default
- - orgName	string	Person group name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - - hasAuthority	string	Has authority, 0=No; 1=Yes
- - facePicture	string	Face image URL, such as: https://192.168.1.1:443/upload/person/xxx.jpg
- - companyName	string	Company name
- - department	string	Department
- - email	string	Email
- - tel	string	Mobile phone number
- - enableParkingSpaceQuota	string	Parking space enable, 0=no, 1=yes
- - parkingSpaceQuota	string	Parking space count
- - faceIssueResult	string	Face database arming sending result, 0 = Failure, 1 = Success
- - sipId	string	Building#unit#room number, 1#1#1001
- - accessType	array	Access control person type, 0 = Normal, 1 = Blocklist, 2 = Visitor, 3 = Patrol, 4 = VIP, 5 = Others; 0 by default
- - remark	string	Remarks

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "personId": "19476543",
        "lastName": "",
        "firstName": "Lily",
        "orgCode": "001",
        "orgName": "All Persons",
        "orgInfos": [
          {
            "orgCode": "001",
            "orgName": "All Persons",
            "hasAuthority": "1"
          }
        ],
        "facePicture": "https://192.168.1.1:8443/upload/obms/headPic/19476543@1@1615886781645.jpg",
        "companyName": "",
        "department": "",
        "email": "1@2.com",
        "tel": "10086",
        "enableParkingSpaceQuota": "0",
        "parkingSpaceQuota": "0",
        "faceIssueResult": "1",
        "simpld": "10#1#1001",
        "accessType": "0",
        "remark": ""
      }
    ]
  }
}
```

3.8.2.7 Get the List of Exceptions in Sending Data of Persons

Brief Description

- Get the list of exceptions in sending data of persons

Request Method

- GET

Request URL

- /obms/api/v1.1/acs/person/authorize/failure/list?personIds={personIds}&dataType={dataType}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
personIds	Yes	string	Required. Person ID, contain a maximum of 30 letters and numbers, separated by commas if many. Support up to 100 person IDs.
dataType	No	string	Exception type: 0=All (access control group + face comparison group + entrance and exit group), 1=Access control group, 2=Face comparison group, 3=Entrance and exit vehicle group, 5=Visitor. The default value is 0.

Request Example

```
GET /obms/api/v1.1/acs/person/authorize/failure/list?personIds=21600438&dataType=0 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 41cce2260b9c4d17917e660fb924d223
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- results	array	The returned result
- - personId	string	Person ID
- - lastName	string	The last name
- - firstName	string	The first name
- - accessInfos	array	Access control information
- - - deviceCode	string	Device code
- - - deviceName	string	Device name

Parameter	Type	Description
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - - action	string	Operation type: 0=Cancel, 1=Send, 2=Update
- - - userErrorCode	string	Error code of person exceptions
- - - fingerprintErrorCode	string	Error code of fingerprint exceptions
- - - faceErrorCode	string	Error code of face exceptions
- - - palmVeinErrorCode	string	Error code of palmar vein exceptions
- - - combinationPasswordErrorCode	string	Error code of combination unlocking password exceptions
- - - unlockingPasswordErrorCode	string	Error code of unlocking password exceptions
- - - cardErrors	array	Card number exception
- - - - cardNo	string	Card number
- - - - errorCode	string	Error code of card number exceptions
- - faceComparisonInfos	array	Face comparison information
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - - faceComparisonGroupId	string	Face comparison group ID
- - - faceComparisonGroupName	string	Face comparison group name
- - - errorCode	string	Error code of face comparison exceptions
- - entranceInfos	array	Entrance and exit information
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - - vehicleErrors	array	Error code of license plate exceptions
- - - - plateNo	string	License plate number
- - - - errorCode	string	Error code of license plate number exceptions
- - passageInfos	array	Access point informations
- - - pointId	string	Access point ID

Parameter	Type	Description
- - - pointName	string	Access point name
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - - action	string	Operation type: 0=cancel, 1=distribute, 2=update
- - - userErrorCode	string	Personnel exception error code
- - - fingerprintErrorCode	string	Fingerprint exception error code
- - - faceErrorCode	string	Face exception error code
- - - combinationErrorCode	string	Combination door opening password exception error code
- - - tips	string	Exception handling prompt: 1=Please try again later, 2=Update personnel information and try again, 3=Clear some data and try again, 4=Try again after the device is online
- - - cardErrors	array	Card number exception
- - - - cardNo	string	Card
- - - - errorCode	string	Card number exception error code
- - liftInfos	array	Lift control info
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - - action	string	Operation type: 0=cancel, 1=distribute, 2=update
- - - userErrorCode	string	Personnel exception error code
- - - cardErrors	array	Card number exception
- - - - cardNo	string	Card
- - - - errorCode	string	Card number exception error code

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
```

```
"results": [  
  {  
    "personId": "21600438",  
    "lastName": "",  
    "firstName": "Jack",  
    "passageInfos": [  
      {  
        "pointId": "1",  
        "pointName": "test",  
        "deviceCode": "100001",  
        "deviceName": "testDevice",  
        "action": "2",  
        "userErrorCode": "141002",  
        "fingerprintErrorCode": "141002",  
        "faceErrorCode": "141002",  
        "combinationErrorCode": "141002",  
        "tips": "1",  
        "cardErrors": [  
          {  
            "cardNo": "123456AB",  
            "errorCode": "141002"  
          }  
        ]  
      }  
    ],  
    "faceComparisonInfos": [  
      {  
        "deviceCode": "1000002",  
        "deviceName": "Device2",  
        "channelId": "1000002$1$0$4",  
        "channelName": "IPC1",  
        "faceComparisonGroupId": "1",  
        "faceComparisonGroupName": "face-test",  
        "errorCode": "141002"  
      }  
    ],  
    "entranceInfos": [  
      {  
        "deviceCode": "1000003",  
        "deviceName": "Device3",  
        "vehicleErrors": [  
          {  
            "plateNo": "A12A25",  
            "errorCode": "141002"  
          }  
        ]  
      }  
    ],  
    "accessInfos": [  
      {  
        "deviceCode": "1000004",  
        "deviceName": "Device4",  
        "channelId": "1000004$1$0$4",  
        "channelName": "IPC1",  
        "faceComparisonGroupId": "1",  
        "faceComparisonGroupName": "face-test",  
        "errorCode": "141002"  
      }  
    ]  
  }  
]
```



```

{
  {
    "deviceCode": "100001",
    "deviceName": "testDevice",
    "channelId": "1000001$7$0$0",
    "channelName": "Channel1",
    "action": "1",
    "userErrorCode": "141002",
    "fingerprintErrorCode": "141002",
    "faceErrorCode": "141002",
    "palmVeinErrorCode": "141002",
    "combinationPasswordErrorCode": "141002",
    "unlockingPasswordErrorCode": "141002",
    "cardErrors": [
      {
        "cardNo": "123456AB",
        "errorCode": "141002"
      }
    ]
  },
  "liftInfos": [
    {
      "deviceCode": "100002",
      "deviceName": "testDevice2",
      "channelId": "1000002$7$0$0",
      "channelName": "Channel1",
      "action": "1",
      "userErrorCode": "141002",
      "cardErrors": [
        {
          "cardNo": "123456AB",
          "errorCode": "141002"
        }
      ]
    }
  ]
}

```

3.8.2.8 Resend the Persons with Exceptions in Sending Data

Brief Description

- Resend the persons with exceptions in sending data

Request Method

- POST

Request URL

- /obms/api/v1.1/acs/person/authorize/failure/reissue

Request Parameter

Parameter	Required	Type	Description
- personIds	No	array	The person ID, contain a maximum of 30 letters and numbers. If it is null, it indicates all failed persons.
- ruleId	No	string	Pass rule ID
- dataType	Yes	string	Exception type: 0=All (access control group + face comparison group + entrance and exit group), 1=Access control group, 2=Face comparison group, 3=Entrance and exit vehicle group. The default value is 0.

Request Example

```
POST /obms/api/v1.1/acs/person/authorize/failure/reissue HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b192baa9e3ee47d38bce9cfdfa3dd328
```

```
{
  "personIds": [
    "21600438"
  ],
  "ruleId": "",
  "dataType": "0"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.8.2.9 Locking and Unlocking Access to Personnel

Brief Description

- Locking and unlocking access to personnel

Request Method

- POST

Request URL

- /obms/api/v1.1/acs/person/access/authority/freeze

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
personId	Yes	string	Person ID
action	Yes	string	Action: 0 = unlock, 1 = lock

Request Example

```
POST /obms/api/v1.1/acs/person/access/authority/freeze HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: df543ad16c2049c885106d4f211e091f
```

```
{
  "personId": "00029949",
  "action": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code

Parameter	Type	Description
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.8.3 Access Control Record

3.8.3.1 Obtaining Access Control Records by Page

Brief Description

- Obtain access control records by page. The obtained records are used for retrieving personnel archives.

Request Method

- POST

Request URL

- /obms/api/v1.1/acs/access/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
channelIds	No	array	The channel code
alarmTypeIds	No	array	Alarm type id. Dictionary type, see [6.1.7 Alarm Type]
personId	No	string	The personnel ID. Exact search is supported.
startTime	Yes	string	The start time (in seconds)

Parameter	Required	Type	Description
endTime	Yes	string	The end time (in seconds)
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster

Request Example

```
POST /obms/api/v1.1/acs/access/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0e9720a36c304014a2efd5b7cf3bec
Accept-Language: en
```

```
{
  "page": "1",
  "pageSize": "20",
  "channelIds": [
    "1000002$7$0$0"
  ],
  "alarmTypeIds": [],
  "personId": "12345",
  "startTime": "1610932954",
  "endTime": "1611019354",
  "splitTime": "",
  "splitId": "",
  "currentPage": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.

Parameter	Type	Description
data	object	N/A
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	Access record page data
- - id	string	The record ID.
- - alarmTime	string	The start time, in seconds.
- - deviceCode	string	The device code.
- - deviceName	string	The device name.
- - channelId	string	The channel code.
- - channelName	string	The channel name.
- - pointName	string	Passing point name.
- - alarmTypeId	string	The ID of the event type.
- - alarmTypeName	string	The name of the event type.
- - personId	string	The personnel ID.
- - firstName	string	The first name of the personnel.
- - lastName	string	The last name of the personnel.
- - direction	string	In and out direction: 0=in, 1=out
- - captureImageUrl	string	The face snapshot.
- - temperature	string	Human body temperature
- - temperatureUnit	string	Temperature units: 0=Celsius, 1=Fahrenheit, 2=Kelvin
- - gpsX	string	Longitude
- - gpsY	string	Latitude
- - mapId	string	Map id

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "id": "123",
```

```

        "alarmTime": "1610715844",
        "deviceCode": "1000002",
        "deviceName": "167ASC",
        "channelId": "1000002$7$0$0",
        "channelName": "channel_name",
        "pointName": "point_name",
        "alarmTypeId": "13100",
        "alarmTypeName": "",
        "personId": "12345",
        "firstName": "firstName",
        "lastName": "lastName",
        "direction": "0",
        "captureImageUrl": "https://192.168.1.1:9901/604cae90-56e4-11eb-
a76e-02004c4f4f50/20210114/2/dsf_1fb9c7e9-5700-11eb-9a39-
02004c4f4f50_842760_921850.jpg",
        "temperature": "37",
        "temperatureUnit": "0",
        "gpsX": "123.2323",
        "gpsY": "30.2323",
        "mapId": "34343"
    }
}
}
}

```

3.8.3.2 Obtaining the Details of Access Control Records

Brief Description

- Obtain the details of access control records. The obtained records are used for retrieving personnel archives.

Request Method

- GET

Request URL

- /obms/api/v1.1/acs/access/record/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	The record ID, be composed of numbers

Request Example

```
GET /obms/api/v1.1/acs/access/record/123 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0e9720a36c304014a2efd5b7cfeb3bec
Accept-Language: en
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	Response data
- id	string	The record ID.
- alarmTime	string	The start time, in seconds.
- deviceCode	string	The device code.
- deviceName	string	The device name.
- channelId	string	The channel code.
- channelName	string	The channel name.
- alarmTypeeld	string	The ID of the event type. Dictionary type, see [6.1.7 Alarm Type]
- alarmTypeName	string	Event type name
- personId	string	The personnel ID.
- firstName	string	The first name of the personnel.
- lastName	string	The last name of the personnel.
- orgCode	string	The code of the personnel group.
- orgName	string	The name of the personnel group.
- orgInfos	array	Personnel multi organization infomation
- - orgCode	string	Person group code
- - orgName	string	Person group name
- sipId	string	The SIP-ID of the visual intercom.
- gender	string	The gender: 0 = Female, 1 = Male, 2 = Unknown.
- idType	string	The certificate type: 0 = ID card, 1 = Certificate of officer, 2 = Student card, 3 = Driving license, 4 = Passport, 5 = Taxpayer identification number, 6 = Others. The default value is 6.

Parameter	Type	Description
- idNo	string	The ID number.
- tel	string	The phone number.
- cardNo	string	The card number.
- facePicture	string	The face picture.
- captureImageUrl	string	The face snapshot.
- operateType	string	Operation type: 0=other, 1=user, 2=room, 3=management machine
- operateInfo	string	Operational information

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "123",
    "alarmTime": "1610953106",
    "deviceCode": "1000000",
    "deviceName": "86",
    "channelId": "1000000$7$0$0",
    "channelName": "channel1",
    "alarmTypeId": "51",
    "alarmTypeName": "",
    "personId": "13646494",
    "firstName": "1610785535",
    "lastName": "1610785535",
    "orgCode": "001002",
    "orgName": "222",
    "orgInfos": [
      {
        "orgCode": "001",
        "orgName": "All Persons",
        "hasAuthority": "1"
      }
    ],
    "sipld": "1#1#1001",
    "gender": "2",
    "idType": "1",
    "idNo": "1610785535",
    "tel": "1610785535",
    "cardNo": "13646494",
    "facePicture": "https://192.168.1.1:443/upload/person/123.jpg",
    "captureImageUrl": "https://192.168.1.1:9901/604cae90-56e4-11eb-a76e-02004c4f4f50/20210114/2/dsf_1fb9c7e9-5700-11eb-9a39-02004c4f4f50_842760_921850.jpg"
  },
  "captureImageUrl": "https://192.168.1.1:9901/604cae90-56e4-11eb-a76e-02004c4f4f50/20210114/2/dsf_1fb9c7e9-5700-11eb-9a39-02004c4f4f50_842760_921850.jpg"
}
```

```

    "operateType": "1",
    "operateInfo": ""
  }
}

```

3.9 Visitor

3.9.1 Search the List of Visitors in Pages

Brief Description

- Search list of visitors in pages (appointment, in-visit status)

Request Method

- GET

Request URL

- /obms/api/v1.0/visitors/visitor/page?direction={direction}&orderType={orderType}&endTime={endTime}&timeType={timeType}&page={page}&key={key}&status={status}&startTime={startTime}&pagesize={pagesize}&visitedPersonOrgCode={visitedPersonOrgCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pagesize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
startTime	No	string	Start time, timestamp, unit: Second. Required when timeType! = -1
endTime	No	string	End time, timestamp, unit: Second. Required when timeType! = -1
timeType	Yes	string	Time type, 0 = Appointed time of visit; 1 = Time of visit; 2 = Appointment leaving time; -1 = No restriction
status	Yes	string	Visitor status, 0 = Appointed; 1 = In visit; 4 = Appointment to be

Parameter	Required	Type	Description
			approved; 6 = Overstayed; -1 = No restriction
key	No	string	Keyword, fuzzy search by visitor name, visited person name, organization of visited person, precise search by contact phone number, email address, ID number, card number of visitor
orderType	Yes	string	Sorting field, 0 = Visitor ID; 1 = Name of visited person; 2 = Organization of visited person; 3 = Appointed time of visit; 4 = Time of visit; 5 = Status, 0 by default
direction	Yes	string	Sorting direction, 0 = Ascending sequence; 1 = Descending sequence, 1 by default
visitedPersonOrgCode	No	string	Host group code

Request Example

```
GET /obms/api/v1.0/visitors/visitor/page?direction=1&orderType=0&endTime=1615824000&timeType=-1&page=1&key=&status=-1&startTime=1615651200&pagesize=20&visitedPersonOrgCode= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b1179ba949cf4549af7eec7d734f5dbb
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- nextPage	string	Next page, -1: No next page
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying

Parameter	Type	Description
		paginated controls. Cannot be used for data statistics.
- pageData	array	Visitor information
- - visitorId	string	Visitor ID
- - status	string	Visitor status: 0: Appointed; 1: In visit
- - source	string	Source of visitor, 1 = Client; 2 = VDPAPP
- - registerType	string	Appointment registration method: 0 = visitor web self-service reservation, 1 = interviewee web invitation, 2 = VDP reservation, 3 = client reservation, 4 = client registration, 5 = Agile APP reservation, 6 = Agile APP registration
- - registerDetail	string	Appointment registration details
- - approvable	string	Approval Authority: 0 = no Approval Authority, 1 = approval Authority
- - createTime	string	Creation time, timestamp, unit: second
- - visitorName	string	Visitor name
- - visitorOrgName	string	Organization of visitor
- - visitedName	string	Name of visited person
- - visitedOrgName	string	Organization of visited person
- - visitedEmail	string	Email of visited person
- - visitedPersonId	string	ID of visited person
- - visitedPersonOrgCode	string	Host group code
- - visitedPersonOrgNames	string	Perosn group of visited person
- - useTimes	string	Number of uses, default 200
- - idType	string	Certificate type: 0: ID card; 1: Officer ID card; 2: Student ID card; 3: Driver license; 4: Passport; 5: Tax number; 6: Other
- - idNum	string	ID number
- - tel	string	Contact number
- - email	string	Email address
- - expectArrivalTime	string	Appointed time of visit (second)
- - arrivalTime	string	Actual time of visit, timestamp, unit: Second
- - expectLeaveTime	string	Appointed visitor leaving time, timestamp, unit: Second

Parameter	Type	Description
-- leaveTime	string	Actual visitor leaving time, timestamp, unit: Second
-- timeTemplateId	string	Time template ID
-- plateNo	string	Plate number
-- reason	string	Reason for visit
-- remark	string	Remarks
-- personId	string	ID of associated person
-- authInfo	object	Certification information
--- sipId	string	Room number
--- cardNo	string	Card number
--- qrCode	string	QR code data
--- passportCardNo	string	Card number in pass certificate
--- facePictures	array	Face picture (URL)
--- idPicture	string	ID photo
-- rightInfo	object	Authorization information
--- inheritVisitedAuthority	string	Whether to inherit the respondent's permissions, 0 = No, 1 = Yes
--- authorityGroups	array	
---- groupId	string	Authority group ID
---- groupName	string	Authority group name
--- acsChannels	array	Channel of access control devices
---- deviceCode	string	Device code
---- deviceName	string	Device name
---- channelId	string	Channel code
---- channelName	string	Channel name
--- vtoChannels	array	Access control channel of VTO devices
---- deviceCode	string	Device code
---- deviceName	string	Device name
---- channelId	string	Channel code
---- channelName	string	Channel name
--- positions	array	Entrance and exit
---- positionId	string	Entrance and exit ID
---- positionName	string	Entrance and exit name
--- liftChannels	array	Lift control channel
---- deviceCode	string	Device code

Parameter	Type	Description
deviceName	string	Device name
channels	array	Channel
channelId	string	Channel code
channelName	string	Channel name
floors	array	Floor
floorId	string	Floor ID
floorName	string	Floor name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "nextPage": "-1",
    "totalCount": "1",
    "pageData": [
      {
        "visitorId": "57",
        "status": "1",
        "source": "1",
        "registerType": "0",
        "registerDetail": "appointment",
        "approvable": "1",
        "createTime": "1615788430",
        "visitorName": "xxx",
        "visitorOrgName": "",
        "visitedName": "xxx",
        "visitedOrgName": "",
        "visitedEmail": "test@test.com",
        "visitedPersonId": "",
        "visitedPersonOrgCode": "",
        "visitedPersonOrgNames": "",
        "useTimes": "200",
        "idType": "0",
        "idNum": "",
        "tel": "",
        "email": "",
        "expectArrivalTime": "0",
        "arrivalTime": "1615787229",
        "expectLeaveTime": "1615794429",
        "leaveTime": "0",
        "timeTemplateId": "1",
        "plateNo": "",
        "reason": ""
      }
    ]
  }
}
```

```
"remark": "",
"personId": "99999999",
"authInfo": {
  "sipId": "1#1#1001",
  "cardNo": "",
  "qrcode": "",
  "passportCardNo": "",
  "facePictures": [],
  "idPicture": ""
},
"rightInfo": {
  "inheritVisitedAuthority": "0",
  "authorityGroups": [
    {
      "groupId": "1",
      "groupName": "default group"
    }
  ],
  "acsChannels": [
    {
      "deviceCode": "1000000",
      "deviceName": "asc1",
      "channelId": "1000000$7$0$0",
      "channelName": "asc1_1"
    }
  ],
  "vtoChannels": [
    {
      "deviceCode": "1000002",
      "deviceName": "vto1",
      "channelId": "1000002$7$0$0",
      "channelName": "vto1_1"
    }
  ],
  "positions": [
    {
      "positionId": "1",
      "positionName": "pos1"
    }
  ],
  "liftChannels": [
    {
      "deviceCode": "1000001",
      "deviceName": "tk1",
      "channels": [
        {
          "channelId": "1000001$50$0$0",
          "channelName": "tk1_1"
        }
      ]
    }
  ]
}
```

```

    ],
    "floors": [
      {
        "floorId": "1",
        "floorName": "Floor 1"
      }
    ]
  }
}

```

3.9.2 Search for a Specified Visitor

Brief Description

- Search a visitor

Request Method

- GET

Request URL

- /obms/api/v1.0/visitors/visitor/{visitorId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
visitorId	Yes	string	Visitor ID

Request Example

```

GET /obms/api/v1.0/visitors/visitor/1 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b231ee5551ee40609bbf875680b1cb81

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- visitorId	string	Visitor ID
- status	string	Visitor status: 0 - Appointed; 1 - In visit
- source	string	Source of visitor: 1 - Client; 2 - VDPAPP
- registerType	string	Appointment registration method: 0 = visitor web self-service reservation, 1 = interviewee web invitation, 2 = VDP reservation, 3 = client reservation, 4 = client registration, 5 = Agile APP reservation, 6 = Agile APP registration
- registerDetail	string	Appointment registration details
- approvable	string	Approval Authority: 0 = no Approval Authority, 1 = approval Authority
- createTime	string	Creation time, timestamp, unit: second
- visitorName	string	Visitor name
- visitorOrgName	string	Organization of visitor
- visitedName	string	Name of visited person
- visitedOrgName	string	Organization of visited person
- visitedEmail	string	Email of visited person
- visitedPersonId	string	ID of visited person
- visitedPersonOrgCode	string	Host group code
- visitedPersonOrgNames	string	Perosn group of visited person
- useTimes	string	Number of uses, default 200
- idType	string	Certificate type: 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others
- idNum	string	ID number
- tel	string	Contact number
-email	string	Email address
- expectArrivalTime	string	Appointed time of visit, timestamp (second)
- arrivalTime	string	Actual time of visit, timestamp (second)
- expectLeaveTime	string	Appointed visitor leaving time, timestamp (second)

Parameter	Type	Description
- leaveTime	string	Actual visitor leaving time, timestamp (second)
- timeTemplateId	string	Time template ID
- plateNo	string	Plate number
- reason	string	Reason for visit
- remark	string	Remarks
- personId	string	ID of associated person
- authInfo	object	Certification information
- - sipId	string	Room number
- - cardNo	string	Card number
- - qrcode	string	QR code data
- - passportCardNo	string	Card number in pass certificate
- - facePictures	array	Face picture (URL)
- - idPicture	string	ID photo
- rightInfo	object	Authorization information
- - inheritVisitedAuthority	string	Whether to inherit the respondent's permissions, 0 = No, 1 = Yes
- - authorityGroups	array	
- - - groupId	string	Authority group ID
- - - groupName	string	Authority group name
- - acsChannels	array	
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - vtoChannels	array	
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - positions	array	
- - - positionId	string	Position ID
- - - positionName	string	Position name
- - liftChannels	array	
- - - deviceCode	string	Device code

Parameter	Type	Description
- - - deviceName	string	Device name
- - - floors	array	
- - - - floorId	string	Floor ID
- - - - floorName	string	Floor name
- - - channels	array	
- - - - channelId	string	Channel code
- - - - channelName	string	Channel name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "visitorId": "1",
    "status": "1",
    "source": "1",
    "registerType": "0",
    "registerDetail": "appointment",
    "approvable": "1",
    "createTime": "1615947987",
    "visitorName": "Lily",
    "visitorOrgName": "",
    "visitedName": "Amy",
    "visitedOrgName": "",
    "visitedEmail": "test@test.com",
    "visitedPersonId": "",
    "visitedPersonOrgCode": "",
    "visitedPersonOrgNames": "",
    "useTimes": "200",
    "idType": "0",
    "idNum": "340521199504214011",
    "tel": "13170055715",
    "email": "test@qq.com",
    "expectArrivalTime": "0",
    "arrivalTime": "1615947987",
    "expectLeaveTime": "1615955187",
    "leaveTime": "0",
    "timeTemplateId": "1",
    "plateNo": "A987123",
    "reason": "for interview",
    "remark": "",
    "personId": "999999999",
    "authInfo": {
      "sipId": "1#1#102",
```

```
"cardNo": "0C987123",
"qrCode": "a+PcZsF7BWo=",
"passportCardNo": "AC54FFDA",
"facePictures": [],
"idPicture": ""
},
"rightInfo": {
  "inheritVisitedAuthority": "0",
  "authorityGroups": [
    {
      "groupId": "1",
      "groupName": "default group"
    }
  ],
  "acsChannels": [
    {
      "deviceCode": "1000000",
      "deviceName": "asc1",
      "channelId": "1000000$7$0$0",
      "channelName": "asc1_1"
    }
  ],
  "vtoChannels": [
    {
      "deviceCode": "1000002",
      "deviceName": "vto1",
      "channelId": "1000002$7$0$0",
      "channelName": "vto1_1"
    }
  ],
  "positions": [
    {
      "positionId": "1",
      "positionName": "pos1"
    }
  ],
  "liftChannels": [
    {
      "deviceCode": "1000001",
      "deviceName": "tk1",
      "channels": [
        {
          "channelId": "1000001$50$0$0",
          "channelName": "tk1_1"
        }
      ]
    }
  ],
  "floors": [
    {
      "floorId": "1",
```


Parameter	Required	Type	Description
visitedPersonOrgNames	No	string	Person group of visited person, length not exceeding 255
useTimes	No	string	Number of uses, min 1, max 500, default 200
idType	Yes	string	ID type: 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others
idNum	No	string	ID number, length not exceeding 50
tel	No	string	Contact number, length not exceeding 20
email	No	string	Email address, length not exceeding 50
expectArrivalTime	No	string	Visit time by appointment, timestamp (seconds)
arrivalTime	No	string	Actual visit time, timestamp (seconds)
expectLeaveTime	No	string	Leave time by appointment, timestamp, (seconds)
leaveTime	No	string	Actual leave time, timestamp (seconds)
timeTemplateId	No	string	Time template ID
plateNo	No	string	Plate number, length not exceeding 15
reason	No	string	Reason for visit, length not exceeding 50
remark	No	string	Remarks, length not exceeding 255
authInfo	No	object	Authentication information
- sipId	No	string	Room number, length not exceeding 13
- cardNo	No	string	Card No. length not exceeding 16, only contains characters 0-9 A-F
- qrCode	No	string	QR code data in pass permit. See [3.9.5 Generate the QR Code Data and Card Number in the Visitor Pass Permit]. Length not exceeding 128.
- passportCardNo	No	string	Card No. in pass permit, see [3.9.5 Generate the QR Code Data and

Parameter	Required	Type	Description
			Card Number in the Visitor Pass Permit]. Length not exceeding 16, only contains characters 0-9 A-F
- facePictures	No	array	Face image (BASE64)
- idPicture	No	string	Certificate photo, can be directly sent empty
rightInfo	No	object	Authorization information
- inheritVisitedAuthority	No	string	Whether to inherit the respondent's permissions, 0 = No, 1 = Yes
- authorityGroupIds	No	string	Authority group ID
- acsChannelIds	No	array	Access control channel of access control device
- vtoChannelIds	No	array	Access control channel of VTO device
- positionIds	No	array	Entrance and exit position ID
- liftChannels	No	array	Lift control channel
- - channelId	No	string	Lift control channel code, such as 1000000\$50\$0\$0
- - floors	No	string	Lift control floor, such as 1,2,3

Request Example

```
POST /obms/api/v1.0/visitors/visitor HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 78919fbe0862459ab4c4d67dbf273f03
```

```
{
  "status": "1",
  "visitorName": "Lily",
  "visitorOrgName": "",
  "visitedName": "Amy",
  "visitedOrgName": "",
  "visitedEmail": "test@test.com",
  "visitedPersonId": "",
  "visitedPersonOrgCode": "",
  "visitedPersonOrgNames": "",
  "useTimes": "200",
  "idType": "0",
  "idNum": "340521199504214011",
```

```

"tel": "13170055715",
"email": "test@qq.com",
"expectArrivalTime": "0",
"arrivalTime": "1615947987",
"expectLeaveTime": "1615955187",
"leaveTime": "0",
"timeTemplatId": "1",
"plateNo": "A987123",
"reason": "",
"remark": "",
"authInfo": {
  "sipId": "1#1#102",
  "cardNo": "0C987123",
  "qrcode": "a+PcZsF7BW0=",
  "passportCardNo": "AC54FFDA",
  "facePictures": [],
  "idPicture": ""
},
"rightInfo": {
  "inheritVisitedAuthority": "0",
  "authorityGroupIds": ["1"],
  "acsChannelIds": [
    "1000174$7$0$0"
  ],
  "vtoChannelIds": [
    "1000158$7$0$0",
    "1000158$7$0$1"
  ],
  "positionIds": [
    "1"
  ],
  "liftChannels": [
    {
      "channelId": "1000000$50$0$0",
      "floors": "1,2,3"
    }
  ]
}
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- personId	string	Related person ID

Parameter	Type	Description
- visitorId	string	Visitor ID

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "personId": "999999999",
    "visitorId": "1"
  }
}
```

3.9.4 Update a Visitor

Brief Description

- Update visitor (visit registration of appointed visitor, leaving registration of leaving visitor, appointment cancellation of appointed visitor, information update of in-visit/appointed visitor)

Request Method

- PUT

Request URL

- /obms/api/v1.0/visitors/visitor/{visitorId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
visitorId	Yes	string	Visitor ID

- Body Parameters

Parameter	Required	Type	Description
status	Yes	string	Visitor status: 0: Appointment, 1: In visit, 3: Appointment cancelled
visitorName	Yes	string	Visitor name, length not exceeding 127

Parameter	Required	Type	Description
visitorOrgName	No	string	Visitor unit, length not exceeding 50
visitedName	Yes	string	Receiver name, length not exceeding 127
visitedOrgName	No	string	Receiver organization, length not exceeding 512
visitedEmail	No	string	Email of visited person, length not exceeding 50
visitedPersonId	No	string	ID of visited person, length not exceeding 64
visitedPersonOrgCode	No	string	Host group code, length not exceeding 90
visitedPersonOrgNames	No	string	Person group of visited person, length not exceeding 255
useTimes	No	string	Number of uses, min 1, max 500, default 200
idType	Yes	string	ID type: 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others
idNum	No	string	ID number, length not exceeding 50
tel	No	string	Contact number, length not exceeding 20
email	No	string	Email address, length not exceeding 50
expectArrivalTime	No	string	Visit time by appointment, timestamp (seconds)
arrivalTime	No	string	Actual visit time, timestamp (seconds)
expectLeaveTime	No	string	Leave time by appointment, timestamp, (seconds)
leaveTime	No	string	Actual leave time, timestamp (seconds)
timeTemplateId	No	string	Time template ID
plateNo	No	string	Plate number, length not exceeding 15

Parameter	Required	Type	Description
reason	No	string	Reason for visit, length not exceeding 50
remark	No	string	Remarks, length not exceeding 255
authInfo	No	object	Authentication information
- sipId	No	string	Room number, length not exceeding 13
- cardNo	No	string	Card No. length not exceeding 16, only contains characters 0-9 A-F
- qrCode	No	string	QR code data in pass permit. See [3.9.5 Generate the QR Code Data and Card Number in the Visitor Pass Permit]. Length not exceeding 128.
- passportCardNo	No	string	Card No. in pass permit, see [3.9.5 Generate the QR Code Data and Card Number in the Visitor Pass Permit]. Length not exceeding 16, only contains characters 0-9 A-F
- facePictures	No	array	Face image (BASE64)
- idPicture	No	string	Certificate photo, can be directly sent empty
rightInfo	No	object	Authorization information
- inheritVisitedAuthority	No	string	Whether to inherit the respondent's permissions, 0 = No, 1 = Yes
- authorityGroupIds	No	string	Authority group ID
- acsChannelIds	No	array	Access control channel of access control device
- vtoChannelIds	No	array	Access control channel of VTO device
- positionIds	No	array	Entrance and exit position ID
- liftChannels	No	array	Lift control channel
- - channelId	No	string	Lift control channel code, such as 1000000\$50\$0\$0
- - floors	No	string	Lift control floor, such as 1,2,3

Request Example

PUT /obms/api/v1.0/visitors/visitor/1 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b231ee5551ee40609bbf875680b1cb81

```
{
  "visitorId": "1",
  "status": "1",
  "visitorName": "Lily",
  "visitorOrgName": "",
  "visitedName": "Amy",
  "visitedOrgName": "",
  "visitedEmail": "test@test.com",
  "visitedPersonId": "",
  "visitedPersonOrgCode": "",
  "visitedPersonOrgNames": "",
  "useTimes": "200",
  "idType": "0",
  "idNum": "340521199504214011",
  "tel": "13170055715",
  "email": "test@qq.com",
  "expectArrivalTime": "0",
  "arrivalTime": "1615947987",
  "expectLeaveTime": "1615955187",
  "leaveTime": "0",
  "timeTemplateId": "1",
  "plateNo": "A987123",
  "reason": "",
  "remark": "",
  "authInfo": {
    "sipId": "1#1#102",
    "cardNo": "0C987123",
    "qrcode": "a+PcZsF7BWo=",
    "passportCardNo": "AC54FFDA",
    "facePictures": [],
    "idPicture": ""
  },
  "rightInfo": {
    "inheritVisitedAuthority": "0",
    "authorityGroupIds": [
      "1"
    ],
    "acsChannelIds": [
      "1000174$7$0$0"
    ],
    "vtoChannelIds": [
      "1000158$7$0$0",
      "1000158$7$0$1"
    ]
  }
}
```

```

    ],
    "positionIds": [
        "1"
    ],
    "liftChannels": [
        {
            "channelId": "1000000$50$0$0",
            "floors": "1,2,3"
        }
    ]
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- personId	string	Related person ID
- visitorId	string	Visitor ID

Return Example

```

{
  "code": 1000,
  "desc": "Success",
  "data": {
    "personId": "999999999",
    "visitorId": "1"
  }
}

```

3.9.5 Generate the QR Code Data and Card Number in the Visitor Pass Permit

Brief Description

- Generate the QR code data and card number in the visitor pass permit

Request Method

- GET

Request URL

- /obms/api/v1.0/visitors/visitor/passport/generate

Request Parameter

None

Request Example

```
GET /obms/api/v1.0/visitors/visitor/passport/generate HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b231ee5551ee40609bbf875680b1cb81
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- qrcode	string	The QR code data in the pass permit
- passportCardNo	string	Card No. in pass permit

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "qrcode": "r0x6kgQsJ0Q=",
    "passportCardNo": "AC022177"
  }
}
```

3.9.6 Get the Number of Appointments and On-site Visitors

Brief Description

- Get the number of visitors who have made appointment or who are in visit

Request Method

- GET

Request URL

- /obms/api/v1.0/visitors/visitor/status/count

Request Parameter

None

Request Example

```
GET /obms/api/v1.0/visitors/visitor/status/count HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b231ee5551ee40609bbf875680b1cb81
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- reserved	string	Number of appointed visitors
- visiting	string	Number of in-visit visitors
- approving	string	Number of visitors to be approved
- retention	string	Number of stranded visitors

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "reserved": "1",
    "visiting": "1",
    "approving": "1",
    "retention": "1"
  }
}
```

3.9.7 Get the List of Visitor History Records in Pages

Brief Description

- Get the visitor history list in pages (including leave and visit|appointment cancellation status)

Request Method

- GET

Request URL

- /obms/api/v1.1/visitor/history/record/page?page={page}&pageSize={pageSize}&visitorName={visitorName}&tel={tel}&email={email}&cardNo={cardNo}&idNo={idNo}&plateNo={plateNo}&visitedName={visitedName}&visitedCompany={visitedCompany}&startTime={startTime}&endTime={endTime}&status={status}¤tPage={currentPage}&splitId={splitId}&visitedPersonOrgCode={visitedPersonOrgCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
visitorName	No	string	Visitor name, fuzzy search, length not exceeding 127
tel	No	string	Contact number, precise search, length not exceeding 20, only contains characters 0-9 - +
email	No	string	Email address, precise search, length not exceeding 50
cardNo	No	string	Card No., precise search, length not exceeding 16, only contains characters 0-9 A-F
idNo	No	string	ID number, precise search, length not exceeding 50
plateNo	No	string	License plate number, accurate search, length not exceeding 15
visitedName	No	string	Receiver name, fuzzy search, length not exceeding 127
visitedCompany	No	string	Receiver organization, fuzzy search, length not exceeding 50
startTime	Yes	string	Start time, including visit by appointment and actual visit, timestamp
endTime	Yes	string	End time, including visit by appointment and actual visit, timestamp

Parameter	Required Type		Description
status	Yes	string	Visitor status: 2 = Leave, 3 = Cancel appointment, 5 = Access denied, 7 = Appointment expiration due to no arrival, -1 = Unlimited
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
visitedPersonOrgCode	No	string	Host group code, length not exceeding 90

Request Example

```
GET /obms/api/v1.1/visitor/history/record/page?cardNo=&idNo=&tel=&visitorName=
&page=1&endTime=1615823999&visitedCompany=&startTime=1615737600&status=-
1&visitedName=&pageSize=20&email=&currentPage=&splitId=&visitedPersonOrgCod
e= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: cc5757aa66d44b30a93326a16585dff4
```

Return Parameter

Parameter	Type	Description
totalCount	string	Total number of records
pageData	object	None
- id	string	Record ID
- visitorId	string	Visitor ID
- status	string	Visitor status: 2 = Leave, 3 = Cancel appointment, 5 = Access denied, 7 = Appointment expiration due to no arrival
- visitorName	string	Visitor name
- visitedCompany	string	Visitor unit
- idType	string	ID type: 0 = ID card, 1 = Officer ID card, 2 = Student ID card, 3 = Driver's license, 4 = Passport, 5 = Tax ID, 6 = Others
- idNo	string	ID number
- plateNo	string	License plate number

Parameter	Type	Description
- tel	string	Contact number
- email	string	Email address
- expectArrivalTime	string	Visit time by appointment, timestamp, in seconds
- arrivalTime	string	Actual visit time, timestamp, in seconds
- expectLeaveTime	string	Leave time by appointment, timestamp, in seconds
- leaveTime	string	Actual leave time, timestamp, in seconds
- reason	string	Reason for visit
- remark	string	Remarks
- personId	string	Related person ID
- registerType	string	Appointment registration method: 0 = visitor web self-service reservation, 1 = interviewee web invitation, 2 = VDP reservation, 3 = client reservation, 4 = client registration, 5 = Agile APP reservation, 6 = Agile APP registration
- registerDetail	string	Appointment registration details
- visitedPersonOrgCode	string	Host group code
- visitedPersonOrgName	string	Host group name
- timeTemplateName	string	Time template name
- authInfo	string	None
- - cardNo	string	Card No.
- - facePicture	string	Face URL

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-1",
    "pageData": [
      {
        "id": "28",
        "visitorId": "28",
        "status": "2",
        "registerType": "0",
        "registerDetail": "appointment",
        "visitorName": "szt222",
        "visitorCompany": "",
        "visitedName": "sfdsd",

```

```

"visitedCompany": "",
"idType": "0",
"idNo": "",
"plateNo": "",
"tel": "13170055715",
"email": "",
"expectArrivalTime": "0",
"arrivalTime": "1615787523",
"expectLeaveTime": "1616658723",
"leaveTime": "1615528665",
"reason": "",
"remark": "",
"personId": "99999999",
"visitedPersonOrgCode": "001",
"visitedPersonOrgName": "group1",
"timeTemplateName": "",
"authInfo": {
  "cardNo": "364CECA4",
  "facePicture": "http://192.168.1.1:80/upload/obms/visitorFacePic/
1615528664605@1.jpg"
}
}
]
}
}

```

3.9.8 Sign off Expired Visitors

Brief Description

- Immediately log out expired visitors

Request Method

- POST

Request URL

- /obms/api/v1.0/visitors/visitor/overdue/clear

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
visitorIds	No	array	Visitor ID, which is blank. Sign-off all expired visitors

Request Example

```
POST /obms/api/v1.0/visitors/visitor/overdue/clear HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b231ee5551ee40609bbf875680b1cb81
```

```
{
  "visitorIds": [
    "1"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.10 Attendance

3.10.1 Get Card Swiping Records in Pages

Brief Description

- Get the card swiping records in pages

Request Method

- GET

Request URL

- /obms/api/v1.0/attendance/swiping-card-report/page?startTime={startTime}&endTime={endTime}&personId={personId}&personName={personName}&deptId={deptId}&eventType={eventType}&page={

page}&pageSize={pageSize}&displayServerTimezoneOffset={displayServerTimezoneOffset}¤tPage={currentPage}&splitId={splitId}&splitTime={splitTime}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
startTime	Yes	string	The start time in format of yyyy-MM-dd HH:mm:ss
endTime	Yes	string	The end time in format of yyyy-MM-dd HH:mm:ss
personId	No	string	Person ID keyword, exact matching
personName	No	string	Name keyword, fuzzy matching
deptId	Yes	string	Department ID, such as: 001001, the orgCode field of the corresponding personnel group
eventType	No	string	The attendance events: 0=All, 1=Check in, 2=Check out, 3=Outing, 4=Return from outing
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
displayServerTimezoneOffset	No	string	Whether to display the server time zone, 0 = No, 1 = Yes, default 0
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster

Request Example

```
GET /obms/api/v1.0/attendance/swiping-card-report/page?startTime=2023-07-12%2000%3A00%3A00&endTime=2023-07-12%2023%3A59%3A59&personName=&personId=&deptId=001&eventType=0&page=1&pageSize=20&displayServerTimezoneOffset=0&currentPage=&splitId=&splitTime= HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
Time-Zone: Asia/Shanghai
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	Record ID
- - swipeTime	string	Card swiping time, such as 2023-07-12 12:23:23
- - code	string	Person ID
- - name	string	Person name
- - deptName	string	Department name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - cardNumber	string	Card number
- - personPic	string	The person's avatar. Reserved field.
- - swipeLocation	string	The position of the device on which the card is swiped
- - eventName	string	The attendance events: 0=Empty, 1=Check in, 2=Check out, 3=Outing, 4=Return from outing

Return Example

```
{
```

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-121",
    "pageData": [
      {
        "id": "1",
        "swipeTime": "2023-07-12 12:23:23",
        "code": "123",
        "name": "test",
        "deptName": "Root",
        "orgInfos": [
          {
            "orgCode": "001",
            "orgName": "All Persons and Vehicles"
          }
        ],
        "cardNumber": "12345678",
        "personPic": "",
        "swipeLocation": "test",
        "eventName": "1"
      }
    ]
  }
}
```

3.10.2 Search Attendance Reports in Pages

Brief Description

- Search attendance reports in pages

Request Method

- GET

Request URL

- /obms/api/v1.0/attendance/record-info-report/page?startTime={startTime}&endTime={endTime}&personId={personId}&personName={personName}&deptId={deptId}&page={page}&pageSize={pageSize}&timeFormat={timeFormat}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
startTime	Yes	string	The start time in format of yyyy-MM-dd HH:mm:ss
endTime	Yes	string	The end time in format of yyyy-MM-dd HH:mm:ss
personId	No	string	Person ID keyword, exact matching
personName	No	string	Name keyword, fuzzy matching
deptId	Yes	string	Department ID, such as: 001001,the orgCode field of the corresponding personnel group
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
timeFormat	No	string	Time format of time length of absence, working hours and overtime, 0 = hour plus minute, 1 = minute, default 0

Request Example

```
GET /obms/api/v1.0/attendance/record-info-report/page?startTime=2023-07-12%2000%3A00%3A00&endTime=2023-07-12%2023%3A59%3A59&personName=&personId=&deptId=001&page=1&pageSize=20&timeFormat= HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	object	None
- - id	string	Attendance information ID

Parameter	Type	Description
-- attendanceDate	string	Date, such as 2018-09-20
-- code	string	Person ID
-- name	string	Person name
-- deptName	string	Department name
-- orgInfos	array	Personnel multi organization information
-- orgCode	string	Person group code
-- orgName	string	Person group name
-- signOutTime	string	Check-out time, such as 13:10
-- week	string	Week: 0: Sunday, 1: Monday,..., 6: Saturday
-- outTime	string	Outing duration, such as 1 hour and 10 minutes, 10 minutes, 1 hour or 0 minute
-- workTime	string	Working duration (minutes)
-- workOverTime	string	Overtime duration (minutes)
-- abnormalStatus	string	Exception type: 1 = late, 2 = leave early, 3 = absent, 4 = late and leave early, 5 = forget to punch in, 6 = late and forget to punch in, 7 = forget to punch in and leave early

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-121",
    "pageData": [
      {
        "id": "1",
        "attendanceDate": "2023-07-12",
        "code": "123",
        "name": "test",
        "deptName": "Root",
        "orgInfos": [
          {
            "orgCode": "001",
            "orgName": "All Persons"
          }
        ],
        "orderName": "",
        "signInTime": "12:10",

```

```

        "signOutTime": "13:10",
        "week": "0",
        "outTime": "1 hour and 10 minutes ",
        "workTime": "12",
        "workOverTime": "13",
        "abnormalStatus": "1"
      }
    ]
  }
}

```

3.10.3 Search Exception Records in Pages

Brief Description

- Search abnormal records in pages

Request Method

- GET

Request URL

- /obms/api/v1.0/attendance/abnormal-report/page?startTime={startTime}&endTime={endTime}&personId={personId}&personName={personName}&deptId={deptId}&eventType={eventType}&page={page}&pageSize={pageSize}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
startTime	Yes	string	The start time in format of yyyy-MM-dd HH:mm:ss
endTime	Yes	string	The end time in format of yyyy-MM-dd HH:mm:ss
personId	No	string	Person ID keyword, exact matching
personName	No	string	Name keyword, fuzzy matching
deptId	Yes	string	Department ID, such as: 001001,the orgCode field of the corresponding personnel group
eventType	No	string	Attendance event type: 0: All; 1: Late; 2: Leave early; 3: Unauthorized absence; 4: Late and leave early; 5: Forget to punch

Parameter	Required	Type	Description
			in/out; 6: Late and forget to punch in; 7: Forget to punch out and leave early
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000

Request Example

```
GET /obms/api/v1.0/attendance/abnormal-report/page?startTime=2023-07-12%2000%3A00%3A00&endTime=2023-07-12%2023%3A59%3A59&personName=&personId=&deptId=001&eventType=0&page=1&pageSize=20 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	object	None
- - id	string	Abnormal record ID
- - attendanceDate	string	Date, format: 2023-07-12
- - code	string	Person ID
- - name	string	Person name
- - deptName	string	Department name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - orderName	string	Shift name, reserved field
- - signInTime	string	Sign-in time, format:12:10

Parameter	Type	Description
- - signOutTime	string	Sign-out time, format: 13:10
- - week	string	Weekday: 0: Sunday; 1: Monday...; 6: Saturday
- - outTime	string	Outing duration, format: 1 hour 10 minutes, 10 minutes, 1 hour, 0 minute
- - workTime	string	Work duration (minute)
- - workOverTime	string	Overtime duration (minute)
- - abnormalStatus	string	Exception type: 1: Late; 2: Leave early; 3: Unauthorized absence; 4: Late and leave early; 5: Forget to punch in/out; 6: Late and forget to punch in; 7: Forget to punch out and leave early

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-121",
    "pageData": [
      {
        "id": "1",
        "attendanceDate": "2023-07-12",
        "code": "123",
        "name": "test",
        "deptName": "Root",
        "orgInfos": [
          {
            "orgCode": "001",
            "orgName": "All Persons and Vehicles"
          }
        ],
        "orderName": "",
        "signInTime": "12:10",
        "signOutTime": "13:10",
        "week": "0",
        "outTime": "1 hour and 10 minutes",
        "workTime": "12",
        "workOverTime": "13",
        "abnormalStatus": "1"
      }
    ]
  }
}
```

3.10.4 Search Overtime Records in Pages

Brief Description

- Search overtime records in pages

Request Method

- GET

Request URL

- /obms/api/v1.0/attendance/work-overtime-report/page?startTime={startTime}&endTime={endTime}&personId={personId}&personName={personName}&deptId={deptId}&workOverType={workOverType}&page={page}&pageSize={pageSize}&timeFormat={timeFormat}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
startTime	Yes	string	The start time in format of yyyy-MM-dd HH:mm:ss
endTime	Yes	string	The end time in format of yyyy-MM-dd HH:mm:ss
personId	No	string	Person ID keyword, exact matching
personName	No	string	Name keyword, fuzzy matching
deptId	Yes	object	Department ID, such as: 001001, the orgCode field of the corresponding personnel group
workOverType	No	string	Overtime type: 1: Workday; 2: Holiday; 3: Restday
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
timeFormat	No	string	Time format of time length of absence, working hours and overtime, 0 = hour plus minute, 1 = minute, default 0

Request Example

```
GET /obms/api/v1.0/attendance/work-overtime-report/page?startTime=2023-07-12%2000%3A00%3A00&endTime=2023-07-12%2023%3A59%3A59&personId=&personName=&deptId=001001&workOverType=0&page=1&pageSize=20&timeFormat=0
```

Host: 192.168.1.1
 Connection: close
 Content-Type: application/json;charset=UTF-8
 X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	object	None
- - id	string	Overtime record ID
- - date	string	Date 2018-09-20
- - code	string	Person ID
- - name	string	Person name
- - deptName	string	Department name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - workOverTime	string	Overtime duration (minute)
- - workOverType	string	Overtime type: 1: Workday; 2: Holiday; 3: Restday

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-121",
    "pageData": [
      {
        "id": "1",
        "date": "2023-07-12",
        "code": "56852",
        "name": "test",
```

```

    "deptName": "Root",
    "orgInfos": [
      {
        "orgCode": "001",
        "orgName": "All Persons and Vehicles"
      }
    ],
    "workOverTime": "65",
    "workOverType": "1"
  }
]
}

```

3.10.5 Get Outing Records in Pages

Brief Description

- Get outing records in pages

Request Method

- GET

Request URL

- /obms/api/v1.0/attendance/outing-report/page?startTime={startTime}&endTime={endTime}&personId={personId}&personName={personName}&deptId={deptId}&page={page}&pageSize={pageSize}&timeFormat={timeFormat}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
startTime	Yes	string	The start time in format of yyyy-MM-dd HH:mm:ss
endTime	Yes	string	The end time in format of yyyy-MM-dd HH:mm:ss
personId	No	string	Person ID keyword, exact matching
personName	No	string	Name keyword, fuzzy matching
deptId	Yes	string	Department ID, such as: 001001, the orgCode field of the corresponding personnel group

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
timeFormat	No	string	Time format of time length of absence, working hours and overtime, 0 = hour plus minute, 1 = minute, default 0

Request Example

```
GET /obms/api/v1.0/attendance/outing-report/page?startTime=2023-07-12%2000%3A00%3A00&endTime=2023-07-12%2023%3A59%3A59&personId=&personName=&deptId=001&page=1&pageSize=20&timeFormat=0 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	object	None
- - id	string	Outing record ID
- - date	string	Date, such as 2018-09-20
- - code	string	Person ID
- - name	string	Person name
- - deptName	string	Department name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name

Parameter	Type	Description
- - cardNum	string	Card number
- - leaveTime	string	Leaving time, such as 12:10
- - backTime	string	Return time, such as 13:10
- - outingTime	string	Outing period, such as 1 hour and 10 minutes, 10 minutes, 1 hour, 0 minutes

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-121",
    "pageData": [
      {
        "id": "1",
        "date": "2023-07-12",
        "code": "123",
        "name": "test",
        "deptName": "Root",
        "orgInfos": [
          {
            "orgCode": "001",
            "orgName": "All Persons and Vehicles"
          }
        ],
        "cardNum": "11122223",
        "leaveTime": "12:10",
        "backTime": "13:10",
        "outingTime": "1 hour and 10 minutes"
      }
    ]
  }
}
```

3.11 Vehicle(Entrance and Exit)

3.11.1 Vehicle Group

3.11.1.1 Add a Vehicle Entrance and Exit Group

Brief Description

- Add a Vehicle Entrance and Exit Group

Request Method

- POST

Request URL

- /ipms/api/v1.1/entrance-group

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
parkingLotIds	Yes	array	Primary parking ID
groupName	Yes	string	Group name, Cannot exceed 50 characters
groupColor	Yes	string	Group color: 1=gray, 2=red, 3=orange, 4=yellow, 5=green, 6=green, 7=blue, 8=purple
remark	No	string	Remarks, the maximum length is 1024 bytes

Request Example

```
POST /ipms/api/v1.1/entrance-group HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

```
{
  "parkingLotIds": [
    "1",
    "2"
  ],
  "groupName": "test",
  "groupColor": "1",
  "remark": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- id	string	The ID of a specified group

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "6"
  }
}
```

3.11.1.2 Update a Vehicle Entrance and Exit Group

Brief Description

- Update vehicle entrance and exit group

Request Method

- PUT

Request URL

- /ipms/api/v1.1/entrance-group/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	Vehicle entrance and exit group ID

- Body Parameters

Parameter	Required	Type	Description
groupName	Yes	string	Group name, Cannot exceed 50 characters
groupColor	Yes	string	Group color: 1=gray, 2=red, 3=orange, 4=yellow, 5=green, 6=green, 7=blue, 8=purple
remark	No	string	Remarks, The maximum length is 1024 bytes

Request Example

```
PUT /ipms/api/v1.1/entrance-group/7 HTTP/1.1
Host: 192.168.1.1
```

Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463

```
{  
  "groupName": "test2",  
  "groupColor": "3",  
  "remark": ""  
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{  
  "code": 1000,  
  "desc": "Success",  
  "data": null  
}
```

3.11.1.3 Delete a Vehicle Entrance and Exit Group

Brief Description

- Delete vehicle entrance and exit group

Request Method

- DELETE

Request URL

- /ipms/api/v1.1/entrance-group?ids={ids}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
ids	Yes	string	Vehicle entrance and exit group ID, separated by commas for multiple IDs

Request Example

```
DELETE /ipms/api/v1.1/entrance-group?ids=13 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.11.1.4 Get the List of Entrance and Exit Groups of Vehicles

Brief Description

- Get the list of entrance and exit Groups of vehicles

Request Method

- GET

Request URL

- /ipms/api/v1.1/entrance-group/list?parkingLotId={parkingLotId}&keyword={keyword}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
parkingLotId	No	string	Parking Lot ID

Parameter	Required	Type	Description
keyword	No	string	Keyword, supporting fuzzy search, matching vehicle group name
orgCode	No	string	Parking lot organization code

Request Example

```
GET /ipms/api/v1.1/entrance-group/list?parkingLotId=1&keyword= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: zh
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Description of results
data	object	Data
- results	array	Result
- - parkingLotId	string	Parking lot ID
- - parkingLotName	string	Parking lot name
- - groupId	string	Group ID
- - groupName	string	Group name
- - groupColor	string	Group color: 1=gray, 2=red, 3=orange, 4=yellow, 5=green, 6=green, 7=blue, 8=purple
- - groupType	string	Group type: 1 = visitor, 2 = vehicle group allowlist, 3 = control group, 4 = vehicle group blocklist, 5 = temporary vehicle, 6 = private parking space group
- - remark	string	Remark
- - defaultGroup	string	Whether it is a default group: 0 = no, 1 = yes
- - vehicleCount	string	Total number of vehicles in the current group
- - effectiveCount	string	Number of current group valid vehicles
- - invalidCount	string	Current group invalid number of vehicles

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "parkingLotId": "1",
        "parkingLotName": "test",
        "groupId": "4",
        "groupName": "Blocklist",
        "groupColor": "2",
        "groupType": "4",
        "remark": "",
        "defaultGroup": "1",
        "vehicleCount": "10",
        "effectiveCount": "6",
        "invalidCount": "4"
      },
      {
        "parkingLotId": "1",
        "parkingLotName": "test",
        "groupId": "3",
        "groupName": "VIP",
        "groupColor": "4",
        "groupType": "2",
        "remark": "",
        "defaultGroup": "1",
        "vehicleCount": "10",
        "effectiveCount": "6",
        "invalidCount": "4"
      },
      {
        "parkingLotId": "1",
        "parkingLotName": "test",
        "groupId": "2",
        "groupName": "General",
        "groupColor": "1",
        "groupType": "2",
        "remark": "",
        "defaultGroup": "1",
        "vehicleCount": "10",
        "effectiveCount": "6",
        "invalidCount": "4"
      }
    ]
  }
}
```

3.11.1.5 Obtain the List of Vehicle Survey Group List

Brief Description

- Obtain the list of vehicle survey group list

Request Method

- GET

Request URL

- /ipms/api/v1.1/survey-group/list

Request Parameter

None

Request Example

```
GET /ipms/api/v1.1/survey-group/list HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ca19ae28474040f192b9fc8cc7b9c69a
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- results	array	Result
- - groupId	string	Group ID
- - groupName	string	Group name
- - groupColor	string	Group color
- - remark	string	Remark
- - defaultGroup	string	Default survey group: 0=no, 1=yes

Return Example

```
{
  "code":1000,
  "desc":"Success",
  "data":{
    "results":[
```



```
{
  "groupId": "1",
  "groupName": "High Risk Vehicle",
  "groupColor": "2",
  "remark": "",
  "defaultGroup": "1"
}
```

3.11.2 Vehicle Information

3.11.2.1 Batch Save Vehicle Information

Brief Description

- Save vehicle information in batches

Request Method

- POST

Request URL

- /ipms/api/v1.1/vehicle/save/batch

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
person	No	object	Person information
- personId	No	string	Person ID
- remark	No	string	Remarks
- companyName	No	string	Company name
- email	No	string	Email address
- tel	No	string	Phone number
vehicles	Yes	array	Vehicle information
- id	No	string	Vehicle ID, it is required when editing a vehicle
- plateNo	Yes	string	Plate number, Cannot exceed 15 characters

Parameter	Required	Type	Description
- vehicleColor	No	string	Vehicle color, dictionary type. See [6.1.10 Color of motor vehicles]
- vehicleBrand	No	string	Vehicle trademark, dictionary type. See [6.1.12 Brand of motor vehicles]
- orgCode	Yes	string	Personnel group code
- orgCodes	No	array	Personnel multi organization code
- bindingDeviceCode	No	string	Bind vehicle-mounted equipment code
- companyTelephone	No	string	Company phone
- companyOwner	No	string	Responsible person of the company
- affiliatedCompany	No	string	Belonging company
- remark	No	string	Remarks. Cannot exceed 255 characters
- surveyStartTime	No	string	The start time of the validity period of the arming group (in seconds). Enter 1 when it is empty.
- surveyEndTime	No	string	The end time of the validity period of the arming group (in seconds). Enter 1 when it is empty.
- surveyLongTerm	No	string	Indicate whether this is a long-term user of the arming group: 1=Yes, 0=No
- surveyGroupIds	No	array	Arming group ID
- entranceGroups	No	array	Entrance and exit group
- - parkingLotId	No	string	Parking lot ID, required if "entranceGroups" is not empty
- - entranceLongTerm	No	string	Indicate whether this is a long-term user of the entrance and exit group: 1=Yes, 0=No
- - entranceStartTime	No	string	The start time of the validity period of the vehicle group (in seconds). Enter 1 when it is empty.
- - entranceEndTime	No	string	The end time of the validity period of the vehicle group (in seconds). Enter 1 when it is empty.
- - entranceGroupIds	No	array	ID of the entrance and exit group

Request Example

```
POST /ipms/api/v1.1/vehicle/save/batch HTTP/1.1
```

Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0a2cc19af0b2486f98912d1d4d5279c1

```
{
  "person": {
    "companyName": "",
    "tel": "",
    "email": "",
    "remark": "",
    "personId": ""
  },
  "vehicles": [
    {
      "id": "",
      "plateNo": "A61U3S",
      "vehicleColor": "100",
      "vehicleBrand": "-1",
      "orgCode": "001",
      "orgCodes": [
        "001"
      ],
      "remark": "cs",
      "surveyGroupIds": [],
      "surveyStartTime": "-1",
      "bindingDeviceCode": "",
      "companyTelephone": "",
      "companyOwner": "",
      "affiliatedCompany": "",
      "surveyEndTime": "-1",
      "surveyLongTerm": "0",
      "entranceGroups": [
        {
          "parkingLotId": "2",
          "entranceLongTerm": "1",
          "entranceStartTime": "-1",
          "entranceEndTime": "-1",
          "entranceGroupIds": [
            "10",
            "11"
          ]
        }
      ]
    }
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- existingPlateNos	array	Repeated license plate
- visiterExistingPlateNos	array	Repeated license plate in the visitor group
- groupExistingPlateNos	array	Repeated license plate in the entrance and exit group or the arming group
- overstepPlateNos	array	The plate number exceeding the limit

Return Example

```
{
  "code": 10004,
  "desc": "The carNo. is already exists or repeat!",
  "data": {
    "repeatPlateNos": [
      "A61U3S"
    ],
    "visiterExistingPlateNos": [],
    "groupExistingPlateNos": [],
    "overstepPlateNos": []
  }
}
```

3.11.2.2 Delete Vehicle Information

Brief description

- Delete vehicles

Request Method

- DELETE

Request URL

- /ipms/api/v1.1/vehicle?ids={ids}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
ids	Yes	string	Vehicle ID. Multiple IDs can be separated by commas.

Request Example

```
DELETE /ipms/api/v1.1/vehicle?ids=1 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.11.2.3 Get the List of Vehicles in Pages

Brief Description

- Get list of vehicles in pages

Request Method

- GET

Request URL

- /ipms/api/v1.1/vehicle/page?page={page}&pageSize={pageSize}&keyword={keyword}&plateNo={plateNo}&personId={personId}&personName={personName}&vehicleColor={vehicleColor}&vehicleBrand={vehicleBrand}&orgCode={orgCode}&containChild={containChild}&surveyGroupId={surveyGroupId}&entranceGroupId={entranceGroupId}&groupType={groupType}&companyName={companyName}

&entranceEffectiveStatus={entranceEffectiveStatus}&surveyEffectiveStatus={surveyEffectiveStatus}&entranceEffectiveStartTime={entranceEffectiveStartTime}&entranceEffectiveEndTime={entranceEffectiveEndTime}&entranceFailureStartTime={entranceFailureStartTime}&entranceFailureEndTime={entranceFailureEndTime}&surveyEffectiveStartTime={surveyEffectiveStartTime}&surveyEffectiveEndTime={surveyEffectiveEndTime}&surveyFailureStartTime={surveyFailureStartTime}&surveyFailureEndTime={surveyFailureEndTime}&containVisitorVehicle={containVisitorVehicle}&filterAuthorityOrg={filterAuthorityOrg}&parkingLotId={parkingLotId}&entranceGroupIds={entranceGroupIds}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
keyword	No	string	Keywords, matching name, plate number, group name
personId	No	string	Person ID
personName	No	string	Vehicle owner name
plateNo	No	string	Plate number
vehicleColor	No	string	Vehicle color, dictionary type. See [6.1.10 Color of motor vehicles]
vehicleBrand	No	string	Vehicle brand, dictionary type. See [6.1.12 Brand of motor vehicles]
surveyGroupId	No	string	Arming group ID
entranceGroupId	No	string	Vehicle group ID
groupType	No	string	Group type: 2 = Vehicle group; 3 = Arming group
orgCode	Yes	string	Organization code
containChild	Yes	string	Whether to associate with subordinate organizations, 0=No, 1=Yes
companyName	No	string	Company name
entranceEffectiveStatus	No	string	Whether the entrance and exit group is in effect; 0 = Not in effect; 1 = In effect; 2 = Invalidated; null means all

Parameter	Required	Type	Description
surveyEffectiveStatus	No	string	Whether the arming group is in effect; 0 = Not in effect; 1 = In effect; 2 = Invalidated; null means all
entranceEffectiveStartTime	No	string	Start time of entrance and exit group taking effect (second)
entranceEffectiveEndTime	No	string	End time of entrance and exit group taking effect (second)
entranceFailureStartTime	No	string	Start time of entrance and exit group losing effect (second)
entranceFailureEndTime	No	string	End time of entrance and exit group losing effect (second)
surveyEffectiveStartTime	No	string	Start time of arming group taking effect (second)
surveyEffectiveEndTime	No	string	End time of arming group taking effect (second)
surveyFailureStartTime	No	string	Start time of arming group losing effect (second)
surveyFailureEndTime	No	string	End time of arming group losing effect (second)
containVisitorVehicle	No	string	Visitor vehicle included or not: 1 = Yes; 0 = No; 0 by default
filterAuthorityOrg	No	string	Filter organizational permissions: 0=No, 1=Yes, default to 1
parkingLotId	No	string	Parking lot ID
entranceGroupIds	No	string	Entrance and exit group ID separated by commas

Request Example

```
GET /ipms/api/v1.1/vehicle/page?orgCode=001&companyName=&entranceGroupId=
&keyword=&personId=&plateNo=&containVisitorVehicle=0&surveyEffectiveStatus=&v
ehicleBrand=&entranceEffectiveStartTime=-1&surveyFailureEndTime=-1&entranceEffect
iveStatus=&vehicleColor=&surveyFailureStartTime=-1&surveyEffectiveStartTime=-1&pa
ge=1&surveyEffectiveEndTime=-1&entranceFailureStartTime=-1&entranceFailureEndTi
me=-1&personName=&entranceEffectiveEndTime=-1&orgName=&groupType=2&pag
eSize=20&surveyGroupId= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 307d68bd2c744865b91266dde93df21c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	Vehicle ID
- - entranceStartTime	string	Start time of entrance and exit group validity period (second)
- - entranceEndTime	string	End time of entrance and exit group validity period (second)
- - surveyStartTime	string	Start time of arming group validity period (second)
- - surveyEndTime	string	End time of arming group validity period (second)
- - plateNo	string	Plate number
- - vehicleColor	string	Vehicle color
- - vehicleColorName	string	Vehicle color name
- - vehicleBrand	string	Vehicle trademark
- - vehicleBrandName	string	Vehicle trademark name
- - bindingDeviceCode	string	Bind vehicle-mounted equipment code
- - companyTelephone	string	Company phone
- - companyOwner	string	Responsible person of the company
- - affiliatedCompany	string	Belonging company
- - orgCode	string	Person group code
- - orgName	string	Person group name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - remark	string	Remarks
- - entranceLongTerm	string	Long-term user of entrance and exit group: 1 = Yes; 0 = No

Parameter	Type	Description
-- surveyLongTerm	string	Long-term user of arming group: 1 = Yes; 0 = No
-- entranceRemaningTime	string	Remaining time of Vehicle database arming
-- entranceEffectiveStatus	string	Validity status of Vehicle database arming: 0 = Not in effect; 1 = In effect; 2 = Invalidated
-- surveyRemaningTime	string	Remaining time of Vehicle database arming
-- surveyEffectiveStatus	string	Validity status of Vehicle database arming: 0 = Not in effect; 1 = In effect; 2 = Invalidated
-- updateTime	int	Update Time (seconds)
-- entranceGroups	array	Vehicle group at entrance and exit
-- -- groupId	string	Arming group ID
-- -- groupName	string	Vehicle group name
-- -- groupColor	string	Group background color
-- -- remark	string	Remarks
-- -- level	string	Level
-- -- parkingLotId	string	Parking lot ID
-- -- parkingLotName	string	Parking lot name
-- -- entranceStartTime	string	Start time of entrance and exit group validity period (second)
-- -- entranceEndTime	string	End time of entrance and exit group validity period (second)
-- -- entranceLongTerm	string	Long-term user of entrance and exit group: 1 = Yes; 0 = No
-- surveyGroups	array	Arming group
-- -- groupId	string	Arming group ID
-- -- groupName	string	Vehicle group name
-- -- groupColor	string	Group background color
-- -- remark	string	Remarks
-- personInfo	object	Person information
-- -- personId	string	Vehicle ID
-- -- personName	string	Vehicle owner name
-- -- companyName	string	Company name
-- -- orgCode	string	Person group code, 001 by default
-- -- orgName	string	Person group name
-- -- orgInfos	array	Personnel multi organization infomation
-- -- -- orgCode	string	Person group code

Parameter	Type	Description
- - - orgName	string	Person group name
- - - hasAuthority	string	Has authority, 0=No; 1=Yes
- - - facePictures	array	Face picture URL, such as https://192.168.1.1:443/upload/person/xxx.jpg
- - - email	string	Email address
- - - tel	string	Phone number
- - - enableParkingSpaceQuota	string	Parking space enabled, 0 = No, 1 = Yes
- - - parkingSpaceQuota	string	Number of parking spaces, 0 by default
- - - remark	string	Person remarks

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "36",
    "pageData": [
      {
        "id": "63",
        "plateNo": "A888888",
        "vehicleBrand": "-1",
        "vehicleBrandName": "",
        "vehicleColor": "100",
        "vehicleColorName": "",
        "bindingDeviceCode": "",
        "companyTelephone": "",
        "companyOwner": "",
        "affiliatedCompany": "",
        "orgCode": "001",
        "orgName": "All Persons",
        "orgInfos": [
          {
            "orgCode": "001",
            "orgName": "All Persons"
          }
        ],
        "remark": "",
        "entranceStartTime": "0",
        "entranceEndTime": "0",
        "entranceLongTerm": "1",
        "surveyStartTime": "-1",
        "surveyEndTime": "-1",

```

```

"surveyLongTerm": "0",
"updateTime": 1748313883,
"personInfo": {
  "personId": "1",
  "personName": "1",
  "companyName": "",
  "orgCode": "001",
  "orgName": "All Persons",
  "orgInfos": [
    {
      "orgCode": "001",
      "orgName": "All Persons",
      "hasAuthority": "1"
    }
  ],
  "facePictures": [],
  "email": "",
  "tel": "",
  "enableParkingSpaceQuota": "1",
  "parkingSpaceQuota": "1",
  "remark": ""
},
"entranceGroups": [
  {
    "groupId": "15",
    "groupName": "123",
    "groupColor": "1",
    "level": "0",
    "remark": "",
    "parkingLotId": "18",
    "parkingLotName": "parking lot 1",
    "entranceStartTime": "1711184779",
    "entranceEndTime": "2026717579",
    "entranceLongTerm": "1"
  }
],
"surveyGroups": [],
"entranceRemaningTime": "0",
"entranceEffectiveStatus": "2",
"surveyRemaningTime": "0",
"surveyEffectiveStatus": "2"
}
]
}
}

```

3.11.2.4 Obtain Vehicle Details by License Plate Number

Brief Description

- Obtain vehicle details by license plate number

Request Method

- POST

Request URL

- /ipms/api/v1.1/vehicle/fetch-by-plate-no

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
plateNo	Yes	string	License plate number
filterAuthorityOrg	No	string	Whether to filter organization permissions: 0=no, 1=yes, default 1

Request Example

```
POST /ipms/api/v1.1/vehicle/fetch-by-plate-no HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ca19ae28474040f192b9fc8cc7b9c69a

{
  "plateNo": "A888888",
  "filterAuthorityOrg": "0"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- id	string	Vehicle ID
- surveyStartTime	string	Start time of validity period of survey group (seconds)
- surveyEndTime	string	End time of validity period of survey group (seconds)
- plateNo	string	License plate number

Parameter	Type	Description
- vehicleColor	string	Vehicle color
- vehicleBrand	string	Vehicle brand
- remark	string	Remark
- surveyLongTerm	string	Long-term users of survey group: 1=yes, 0=no
- entranceRemaningTime	string	Remaining time of vehicle entrance and exit group
- entranceEffectiveStatus	string	Entrance and exit group effective status: 0=not effective, 1=effective, 2=invalid
- surveyRemaningTime	string	Remaining time of vehicle survey group
- surveyEffectiveStatus	string	Survey group effective status: 0=not effective, 1=effective, 2=invalid
- entranceGroups	array	Entrance and exit groups
- - parkingLotId	string	Parking lot ID
- - parkingLotName	string	Parking lot name
- - groupId	string	Entrance and exit group ID
- - groupName	string	Entrance and exit group name
- - groupColor	string	Entrance and exit group background color
- - remark	string	Remark
- - level	string	Level
- - entranceStartTime	string	Start time of validity period of entrance and exit group (seconds)
- - entranceEndTime	string	End time of validity period of entrance and exit group (seconds)
- - entranceLongTerm	string	Long-term users of entrance and exit group: 1=yes, 0=no
- surveyGroups	array	Survey groups
- - groupId	string	Survey group ID
- - groupName	string	Survey group name
- - groupColor	string	Survey group background color
- - remark	string	Remark
- personInfo	object	Person infomation
- - personId	string	Person ID
- - personName	string	Name of vehicle owner
- - companyName	string	Company name
- - orgCode	string	Personnel group code, default 001
- - orgName	string	Personnel group name
- - orgInfos	array	Personnel group infomations

Parameter	Type	Description
- - - orgCode	string	Personnel group code
- - - orgName	string	Personnel group name
- - facePictures	array	Face image URL, for example: https://192.168.1.1:443/upload/person/xxx.jpg
- - email	string	Email address
- - tel	string	Mobile phone number
- - enableParkingSpaceQuota	string	Parking space enabled, 0 = No, 1 = Yes
- - parkingSpaceQuota	string	Number of parking spaces, 0 by default
- - remark	string	Person remark
- hasPermission	string	Have permission for this vehicle: 0=no, 1=yes

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "25",
    "plateNo": "A888888",
    "vehicleBrand": "-1",
    "vehicleColor": "1",
    "remark": "",
    "entranceStartTime": "1719215584",
    "entranceEndTime": "2034748384",
    "entranceLongTerm": "1",
    "surveyStartTime": "1719215459",
    "surveyEndTime": "2034748259",
    "surveyLongTerm": "1",
    "personInfo": {
      "personId": "",
      "personName": "",
      "companyName": "",
      "orgCode": "",
      "orgName": "",
      "facePictures": [],
      "email": "",
      "tel": "",
      "enableParkingSpaceQuota": "1",
      "parkingSpaceQuota": "1",
      "remark": ""
    },
    "entranceGroups": [
      {
```

```
        "groupId": "21",
        "groupName": "VIP",
        "groupColor": "4",
        "level": "2",
        "remark": "",
        "parkingLotId": "5",
        "parkingLotName": "lot1",
        "entranceStartTime": "1719215584",
        "entranceEndTime": "2034748384",
        "entranceLongTerm": "1"
    },
    ],
    "surveyGroups": [
        {
            "groupId": "8",
            "groupName": "1",
            "groupColor": "1",
            "remark": ""
        }
    ],
    "entranceRemaningTime": "3652",
    "entranceEffectiveStatus": "1",
    "surveyRemaningTime": "3652",
    "surveyEffectiveStatus": "1",
    "orgInfos": [
        {
            "orgCode": "001",
            "orgName": "All Persons and Vehicles"
        }
    ],
    "hasPermission": "1"
}
```

3.11.3 Vehicle Search

3.11.3.1 Get the Records of Entering Vehicles at the Entrance and Exit in Pages

Brief Description

- Get vehicle entrance/exit records

Request Method

- POST

Request URL

- /ipms/api/v1.1/entrance/vehicle-enter/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
positionIds	No	array	Position ID
startTime	Yes	string	Search start time, timestamp (seconds)
endTime	Yes	string	Search end time, timestamp (seconds)
vehicleType	No	string	Vehicle type: 1= visitor, 5= temporary car
entranceGroupId	No	string	Vehicle entrance and exit group ID
plateNo	No	string	Plate number, which supports fuzzy search
plateNoMatchMode	Yes	string	Plate number matching mode, 0 = Fuzzy matching, 1 = Exact matching
personName	No	string	Owner name, which supports fuzzy search
company	No	string	Company name, which supports fuzzy search
orgCode	No	string	Person group ID
vehicleColor	No	string	Vehicle color, dictionary type. See [API Reference > Dictionary > Vehicle Color].
vehicleModel	No	string	Vehicle type, dictionary type. See [API Reference > Dictionary > Vehicle Model]
vehicleBrand	No	string	Vehicle brand, dictionary type. See [API Reference > Dictionary > Vehicle Brand].
status	No	string	Exit status: 1 = Already left; 2 = Not left
NocardNo	No	string	Card number
cardPersonId	No	string	Card swiping personnel ID
cardPersonName	No	string	Card swiping personnel name

Parameter	Required	Type	Description
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster
issuePlace	No	string	Place of Issue
plateNoLength	No	string	License plate length scale: 1 = short, 2 = medium, 3 = long
plateNoType	No	string	License plate type, dictionary type 9048
plateNoAreaCode	No	string	License plate area code

Request Example

```
POST /ipms/api/v1.1/entrance/vehicle-enter/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: zh
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

```
{
  "plateNoMatchMode": "1",
  "company": "",
  "entranceGroupID": "",
  "cardPersonName": "",
  "plateNo": "",
  "positionIds": [],
  "startTime": "1701100800",
  "vehicleBrand": "O",
  "cardNo": "",
  "vehicleModel": "O",
  "vehicleColor": "O",
```

```

"endTime": "1701187199",
"page": "13",
"splitTime": "1701109256",
"splitId": "2023112700000000781",
"currentPage": "11",
"personName": "",
"status": "O",
"orgCode": "",
"cardPersonId": "",
"pageSize": "20",
"issuePlace": "",
"plateNoLength": "",
"plateNoType": "",
"plateNoAreaCode": ""
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	Page data
- - id	string	Record ID
- - plateNo	string	Plate number
- - originalPlateNo	string	Original license plate number
- - issuePlace	string	Place of Issue
- - plateNoLength	string	License plate length
- - plateNoAreaCode	string	License plate region code
- - extPlateNo	string	Expand license plates
- - plateNoType	string	License plate type

Parameter	Type	Description
-- plateNoTypeName	string	License plate type name
-- plateNoPictures	array	Entry license plate URL
-- fuzzyMatched	string	Has fuzzy matching been performed: 0=No, 1=Yes
-- accessAuthority	string	Access permissions: 1=allow passage, 2=prohibit passage, 3=verify passage
-- matchFlag	string	Matching identification: 0=Secondary camera image, 1=Main camera image
-- capturePictures	array	Snapshot URL, index[0] = Main snapshot, index[1] = Secondary snapshot. Append ?token={credential} after the URL to get the snapshot
-- enterTime	string	Entry time, timestamp (seconds)
-- parkingLotId	string	Parking lot ID
-- parkingLotName	string	Parking lot name
-- parentParkingLotId	string	Parent parking lot ID
-- parentParkingLotName	string	Parent parking lot name
-- positionId	string	Point ID
-- positionName	string	Point name
-- pointId	string	Gate ID
-- pointName	string	Gate name
-- personName	string	Owner name
-- company	string	Company name
-- orgCode	string	Person group ID
-- orgName	string	Person group name
-- vehicleBrand	string	Vehicle trademark
-- vehicleModel	string	Vehicle model
-- vehicleColor	string	Vehicle color
-- cardPersonId	string	Card swiping personnel ID
-- cardPersonName	string	Card swiping personnel name
-- cardNo	string	Swipe card number

Parameter	Type	Description
- - status	string	Exit status: 1 = Already left; 2 = Not left
- - vehicleType	string	Vehicle type: 1=Visitor, 5=Temporary vehicle
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - entranceGroups	array	Entrance and exit vehicle groups
- - - entranceGroupId	string	Vehicle entry and exit group ID
- - - entranceGroupName	string	Vehicle entry and exit group name
- - parkingLotOrgCode	string	Parking lot organization code
- - parkingLotOrgName	string	Parking lot organization name
- - enterMode	string	Open gate verification method
- - mergePictures	array	Composite image URL
- - vehiclePictures	array	Vehicle image URL

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-82",
    "pageData": [
      {
        "id": "2023112700000000760",
        "plateNo": "AB123",
        "originalPlateNo": "AB123",
        "issuePlace": "",
        "plateNoLength": "3",
        "plateNoAreaCode": "",
        "extPlateNo": "",
        "plateNoType": "15",
        "plateNoTypeName": "111",
        "fuzzyMatched": "0",

```

```
    "accessAuthority": "1",
    "plateNoPictures": [
      "http://192.168.1.1:80/v1/s/192.168.1.1/9901/"
    ],
    "capturePictures": [
      "http://192.168.1.1:80/v1/s/192.168.1.1/9901/"
    ],
    "matchFlag": "1",
    "enterTime": "1701107438",
    "parkingLotId": "1",
    "parkingLotName": "",
    "parentParkingLotId": "01",
    "parentParkingLotName": "123",
    "positionId": "1",
    "positionName": "1",
    "pointId": "1",
    "pointName": "1",
    "personName": "",
    "company": "",
    "orgCode": "",
    "mergePictures": [

    ],
    "vehiclePictures": [

    ],
    "enterMode": "Swiping Card",
    "orgName": "",
    "vehicleBrand": "-1",
    "vehicleColor": "99",
    "vehicleModel": "0",
    "cardPersonId": "",
    "cardPersonName": "",
    "cardNo": "",
    "status": "1",
    "vehicleType": "5",
    "orgInfos": [

    ],
    "entranceGroups": [

    ],
    "parkingLotOrgCode": "001001",
    "parkingLotOrgName": "test"
  }
}
}
```

3.11.3.2 Get the Record Details of Entering Vehicles at the Entrance and Exit

Brief Description

- Get vehicle entering records at entrance and exit. You can filter out the vehicles present by status=2 in the return value.

Request Method

- GET

Request URL

- /ipms/api/v1.1/entrance/vehicle-enter/record/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	Record ID

Request Example

```
GET /ipms/api/v1.1/entrance/vehicle-enter/record/1 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- id	string	Entering record ID
- status	string	Exit status: 1 = Already left; 2 = Not left
- plateNo	string	Plate number
- originalPlateNo	string	Original license plate number
- issuePlace	string	Place of Issue

Parameter	Type	Description
- plateNoLength	string	License plate length
- plateNoAreaCode	string	License plate region code
- extPlateNo	string	Expand license plates
- plateNoType	string	License plate type
- plateNoTypeName	string	License plate type name
- fuzzyMatched	string	Has fuzzy matching been performed: 0=No, 1=Yes
- accessAuthority	string	Access permissions: 1=allow passage, 2=prohibit passage, 3=verify passage
- parkingLotId	string	Parking lot ID
- parkingLotName	string	Parking lot name
- parentParkingLotId	string	Parent parking lot ID
- parentParkingLotName	string	Parent parking lot name
- positionId	string	Point ID
- positionName	string	Point name
- pointId	string	Gate ID
- pointName	string	Gate name
- enterTime	string	Entering time, timestamp (second)
- channels	array	Entering channel information
- - channelId	string	Channel code
- - channelName	string	Channel name
- - bindingChannelIds	array	Bound channel encoding
- layerInfo	object	
- -- layerId	string	layerId
- -- layerName	string	layerName
- -- locationX	string	locationX
- - -locationY	string	locationY
- - -height	string	Icon height
- - -width	string	Icon width
- - -angle	string	Icon Angle
- - -positionId	string	positionId

Parameter	Type	Description
- -- positionName	string	positionName
- -- pictureUrl	string	pictureUrl
- capturePictures	array	Snapshot URL, index[0] = Main snapshot; index[1] = Secondary snapshot. Append ?token={credential} after the URL to get the snapshot
- personName	string	Vehicle owner name
- company	string	Company name
- personPicture	string	Person avatar URL
- orgCode	string	Person group code
- orgName	string	Person group name
- orgInfos	array	Personnel multi organization information
- - orgCode	string	Person group code
- - orgName	string	Person group name
- vehicleBrand	string	Vehicle brand
- vehicleColor	string	Vehicle color
- entranceGroups	array	Entrance and exit group information of entering vehicles
- - entranceGroupId	string	Vehicle entrance and exit group ID
- - entranceGroupName	string	Vehicle entrance and exit group name
- exitInfo	object	Vehicle leaving information
- - id	string	Exit record ID
- - plateNo	string	Plate number
- - originalPlateNo	string	Original license plate number
- - issuePlace	string	Place of Issue
- - plateNoLength	string	License plate length
- - plateNoAreaCode	string	License plate region code
- - extPlateNo	string	Expand license plates
- - plateNoType	string	License plate type
- - plateNoTypeName	string	License plate type name
- - parkingLotId	string	Parking lot ID

Parameter	Type	Description
- - parkingLotName	string	Parking lot name
- - positionId	string	Point ID
- - positionName	string	Point name
- - pointId	string	Gate ID
- - pointName	string	Gate name
- - exitTime	string	Exit time, timestamp (seconds)
- - channels	array	Exit channel information
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - - bindingChannelIds	array	Bound channel encoding
- - capturePictures	array	Snapshot URL, index[0] = Main snapshot, index[1] = Secondary snapshot. Append ?token={credential} after the URL to get the snapshot
- - personName	string	Owner name
- - company	string	Company name
- - orgCode	string	Person group ID
- - orgName	string	Person group name
- - vehicleBrand	string	Vehicle trademark
- - vehicleColor	string	Vehicle color
- - entranceGroups	array	Entrance and exit group information of entering vehicles
- - - entranceGroupId	string	Vehicle entrance and exit group ID
- - - entranceGroupName	string	Vehicle entrance and exit group name
- - layerInfo	object	
- - - layerId	string	layerId
- - - layerName	string	layerName
- - - locationX	string	locationX
- - - locationY	string	locationY
- - - height	string	Icon height
- - - width	string	Icon width
- - - angle	string	Icon Angle
- - - positionId	string	positionId
- - - positionName	string	positionName
- - - pictureUrl	string	pictureUrl

Parameter	Type	Description
- originalPlateNo	string	Match the license plate number
- vehicleType	string	Vehicle type: 1= visitor, 2= allowlist, 4= blocklist, 5= temporary car
- matchFlag	string	Matching identifier: 0= auxiliary camera picture, 1= primary camera picture
- parkingRecordIds	array	parkingRecordIds
- parkingLotOrgCode	string	Parking lot organization code
- parkingLotOrgName	string	Name of car park organization

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "1441672",
    "status": "2",
    "plateNo": "A12A26",
    "originalPlateNo": "A12A26",
    "issuePlace": "SAU",
    "plateNoLength": "3",
    "plateNoAreaCode": "P",
    "extPlateNo": "ABC",
    "plateNoType": "4",
    "plateNoTypeName": "Police",
    "fuzzyMatched": "0",
    "accessAuthority": "1",
    "parkingLotId": "3",
    "parkingLotName": "Los ParkingLot",
    "positionId": "3",
    "positionName": "western gate",
    "pointId": "1675",
    "pointName": "entrance1",
    "enterTime": "1683682785",
    "channels": [{
      "channelId": "1000102$1$0$0",
      "channelName": "Channel0",
      "bindingChannelIds": [""]
    }],
    "capturePictures": ["http://192.168.1.1:80/v1/s/192.168.1.1/9901/c4ebe5bb-b22f-11ec-b244-2cf05dcafd2/20220408/3/dsf_a5e1bfd7-b6d8-11ec-8119-2cf05dcafd2_12466255_12545134.jpg"],
    "plateNoPictures": ["http://192.168.1.1:80/v1/s/192.168.1.1/9901/c4ebe5bb-b22f-11ec-b244-2cf05dcafd2/20220408/3/dsf_a5e1bfd7-b6d8-11ec-8119-2cf05dcafd2_12624013_12628458.jpg"],
  }
}
```

```
"layerInfo": {
  "layerId": "",
  "layerName": "",
  "locationX": "",
  "locationY": "",
  "height": "",
  "width": "",
  "angle": "",
  "positionId": "",
  "positionName": "",
  "pictureUrl": ""
},
"personName": "peter",
"company": "",
"cardNo": "",
"cardPersonId": "",
"cardPersonName": "",
"orgCode": "001",
"orgName": "All Persons and Vehicles",
"orgInfos": [
  {
    "orgCode": "001",
    "orgName": "All Persons and Vehicles"
  }
],
"vehicleBrand": "1",
"vehicleColor": "2",
"vehicleModel": "27",
"inputVehicleBrand": "-1",
"inputVehicleColor": "100",
"entranceGroups": [{
  "entranceGroupId": "3",
  "entranceGroupName": "VIP"
}],
"vehicleType": "2",
"matchFlag": "1",
"exitInfo": {
  "id": "",
  "plateNo": "",
  "originalPlateNo": "",
  "issuePlace": "SAU",
  "plateNoLength": "3",
  "plateNoAreaCode": "P",
  "extPlateNo": "ABC",
  "plateNoType": "4",
  "plateNoTypeName": "Police",
  "parkingLotId": "",
  "parkingLotName": "",
  "positionId": "",
```

```

    "positionName": "",
    "pointId": "",
    "pointName": "",
    "exitTime": "",
    "parkingTime": "",
    "fuzzyMatched": "",
    "accessAuthority": "",
    "channels": [{
        "channelId": "",
        "channelName": "",
        "bindingChannelIds" : [""]
    }],
    "capturePictures": [""],
    "plateNoPictures": [""],
    "layerInfo": {
        "layerId": "",
        "layerName": "",
        "locationX": "",
        "locationY": "",
        "height": "",
        "width": "",
        "angle": "",
        "positionId": "",
        "positionName": "",
        "pictureUrl": ""
    },
    "personName": "",
    "cardNo": "",
    "cardPersonId": "",
    "cardPersonName": "",
    "company": "",
    "personPicture": "",
    "orgCode": "",
    "orgName": "",
    "vehicleBrand": "",
    "vehicleColor": "",
    "vehicleModel": "",
    "inputVehicleBrand": "",
    "inputVehicleColor": "",
    "entranceGroups": [],
    "vehicleType": "",
    "matchFlag": ""
},
"parkingRecordIds": [],
"parkingLotOrgCode": "001001",
"parkingLotOrgName": "test"
}

```

3.11.3.3 Get the Records of Exiting Vehicles at the Entrance and Exit in Pages

Brief Description

- Get vehicle exit records at the entrance and exit in pages

Request Method

- POST

Request URL

- /ipms/api/v1.1/entrance/vehicle-exit/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
company	No	string	Company name, which supports fuzzy search
vehicleColor	No	string	Vehicle color, dictionary type. See [API Reference > Dictionary > Vehicle Color].
endTime	Yes	string	Search end time, timestamp (seconds)
entranceGroupId	No	string	Vehicle entrance and exit group ID
personName	No	string	Owner name, which supports fuzzy search
plateNo	No	string	Plate number, which supports fuzzy search
orgCode	No	string	Person group ID
positionIds	No	array	Position ID
startTime	Yes	string	Search start time, timestamp (seconds)
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
vehicleType	No	string	Vehicle type: 1= visitor, 5= temporary car
vehicleBrand	No	string	Vehicle brand, dictionary type. See [API Reference > Dictionary > Vehicle Brand].
vehicleModel	No	string	Vehicle type, dictionary type. See [API Reference > Dictionary > Vehicle Model]
plateNoMatchMode	Yes	string	Plate number matching mode, 0 = Fuzzy matching, 1 = Exact matching

Parameter	Required	Type	Description
cardNo	No	string	Card number
cardPersonId	No	string	Card swiping personnel ID
cardPersonName	No	string	Card swiping personnel name
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster
issuePlace	No	string	Place of Issue
plateNoLength	No	string	License plate length scale: 1 = short, 2 = medium, 3 = long
plateNoType	No	string	License plate type, dictionary type 9048
plateNoAreaCode	No	string	License plate area code

Request Example

```
POST /ipms/api/v1.1/entrance/vehicle-exit/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: b9d6f432b3184903a549a99919b2935e
```

```
{
  "company": "",
  "vehicleColor": "1",
  "endTime": "1615651199",
  "entranceGroupId": "",
  "page": "1",
  "personName": "",
  "plateNo": "",
  "orgCode": "",
  "positionIds": [],
  "startTime": "1615564800",
  "pageSize": "20",
```

```

"vehicleBrand": "1",
"vehicleModel": "1",
"plateNoMatchMode": "0",
"cardNo": "",
"cardPersonId": "",
"cardPersonName": "",
"splitTime": "",
"splitId": "",
"currentPage": "",
"issuePlace": "",
"plateNoLength": "",
"plateNoType": "",
"plateNoAreaCode": ""
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	Page data
- - id	string	Exit record ID
- - exitPlateNo	string	Exit plate number
- - originalExitPlateNo	string	Original exit license plate number
- - issuePlace	string	Place of Issue
- - plateNoLength	string	License plate length
- - plateNoAreaCode	string	License plate region code
- - extExitPlateNo	string	Expand license plates
- - plateNoType	string	License plate type
- - plateNoTypeName	string	License plate type name
- - exitMergePictures	array	Exit composite image URL
- - exitVehiclePictures	array	Exit vehicle diagram URL
- - enterMode	string	Open gate verification method

Parameter	Type	Description
-- enterMergePictures	array	Enter composite image URL
-- enterVehiclePictures	array	Enter vehicle diagram URL
-- exitPlateNoPictures	array	Exit license plate URL
-- fuzzyMatched	string	Has fuzzy matching been performed: 0=No, 1=Yes
-- originalEnterPlateNo	string	Original entry license plate number
-- enterExtPlateNo	string	Extension of license plates upon entry
-- accessAuthority	string	Access permissions: 1=allow passage, 2=prohibit passage, 3=verify passage
-- exitMatchFlag	string	Matching identification: 0=Secondary camera image, 1=Main camera image
-- exitCapturePictures	array	Exit snapshot URL, index[0] = Main image, index[1] = Auxiliary image. Append ?token={credential} to the URL to get images
-- exitTime	string	Exit time, timestamp (seconds)
-- exitParkingLotId	string	Exit parking lot ID
-- exitParkingLotName	string	Exit parking lot name
-- parentParkingLotId	string	Parent parking lot ID
-- parentParkingLotName	string	Parent parking lot name
-- exitPositionId	string	Exit position ID
-- exitPositionName	string	Exit position name
-- exitPositionId	string	Exit gate ID
-- exitPositionName	string	Exit gate name
-- parkingTime	string	Parking duration, in seconds
-- enterPlateNo	string	Entry vehicle license plate
-- enterCapturePictures	array	Entry snapshot URL, index[0] = Main image, index[1] = Auxiliary image. Append ?token={credential} to the URL to get images
-- enterTime	string	Entry time, timestamp (seconds)
-- enterParkingLotId	string	Entry parking lot ID
-- enterParkingLotName	string	Entry parking lot name
-- enterPositionId	string	Entry position ID
-- enterPositionName	string	Entry position name
-- enterPositionId	string	Entry gate ID
-- enterPositionName	string	Entry gate name

Parameter	Type	Description
- - personName	string	Owner name
- - company	string	Company name
- - orgCode	string	Person group ID
- - orgName	string	Person group name
- - vehicleBrand	string	Vehicle trademark
- - vehicleModel	string	Vehicle model
- - vehicleColor	string	Vehicle color
- - cardPersonId	string	Card swiping personnel ID
- - cardPersonName	string	Card swiping personnel name
- - cardNo	string	Swipe card number
- - vehicleType	string	Vehicle type: 1=Visitor, 5=Temporary vehicle
- - orgInfos	array	Personnel multi organization infomation
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - entranceGroups	array	Entrance and exit vehicle groups
- - - entranceGroupId	string	Vehicle entry and exit group ID
- - - entranceGroupName	string	Vehicle entry and exit group name
- - parkingLotOrgCode	string	Parking lot organization code
- - parkingLotOrgName	string	Parking lot organization name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-140",
    "pageData": [
      {
        "id": "2023110800000001564",
        "exitPlateNo": "AB123",

```

```
"originalExitPlateNo": "AB123",
"issuePlace": "",
"originalEnterPlateNo": "",
"plateNoLength": "3",
"plateNoAreaCode": "",
"exitExtPlateNo": "",
"exitMergePictures": [

],
"exitVehiclePictures": [

],
"enterMode": "2",
"enterMergePictures": [

],
"enterVehiclePictures": [

],
"plateNoType": "0",
"plateNoTypeName": "111",
"enterExtPlateNo": "",
"fuzzyMatched": "0",
"accessAuthority": "1",
"exitCapturePictures": [
    "http://192.168.1.1:80/v1/s/192.168.1.1/9901/"
],
"exitPlateNoPictures": [
    "http://192.168.1.1:80/v1/s/192.168.1.1/9901/"
],
"exitMatchFlag": "1",
"exitTime": "1699454311",
"exitParkingLotId": "1",
"exitParkingLotName": "AB123",
"exitPositionId": "1",
"exitPositionName": "1",
"exitPointId": "3",
"exitPointName": "2",
"parkingTime": "",
"enterPlateNo": "",
"enterCapturePictures": [
    ""
],
"enterPlateNoPictures": [
    ""
],
"enterMatchFlag": "",
"enterTime": "",
"enterParkingLotId": "",
```

```

        "enterParkingLotName": "",
        "enterPositionId": "",
        "enterPositionName": "",
        "enterPointId": "",
        "enterPointName": "",
        "personName": "",
        "company": "",
        "orgCode": "",
        "orgName": "",
        "vehicleColor": "2",
        "vehicleBrand": "1",
        "vehicleModel": "-1",
        "cardPersonId": "",
        "cardPersonName": "",
        "cardNo": "",
        "entranceGroups": [
            ],
        "vehicleType": "5",
        "orgInfos": [
            ],
        "parkingLotOrgCode": "001001",
        "parkingLotOrgName": "test"
    }
}

```

3.11.3.4 Get the Record Details of Exiting Vehicles at the Entrance and Exit

Brief Description

- Get the record details about exiting vehicles at the entrance and exit

Request Method

- GET

Request URL

- /ipms/api/v1.1/entrance/vehicle-exit/record/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	Record ID

Request Example

```
GET /ipms/api/v1.1/entrance/vehicle-exit/record/11506 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: cdbd9cb9315c4bfeb5cbef2b5c20156c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- id	string	Entering record ID
- status	string	Exit status: 1 = Already left; 2 = Not left
- plateNo	string	Plate number
- parkingLotId	string	Parking lot ID
- parkingLotName	string	Parking lot name
- parentParkingLotId	string	Parent parking lot ID
- parentParkingLotName	string	Parent parking lot name
- positionId	string	Point ID
- mergePictures	array	Composite image URL
- positionName	string	Point name
- pointId	string	Gate ID
- pointName	string	Gate name
- exitTime	string	Exit time, timestamp, in seconds
- channels	array	Entering channel information
- - channelId	string	Channel code
- - channelName	string	Channel name
- - bindingChannelIds	array	Bound channel encoding
- originalPlateNo	string	Original license plate number
- issuePlace	string	Place of Issue

Parameter	Type	Description
- fuzzyMatched	string	Has fuzzy matching been performed: 0=No, 1=Yes
- accessAuthority	string	Access permissions: 1=allow access, 2=prohibit access, 3=verify access, 4=visitor is not within the validity period, 5=insufficient space in the venue, 6=user has no available parking spaces, 7=vehicle group has no available parking spaces, 8=vehicle color does not match, 9=timeout vehicle is prohibited from access
- matchFlag	string	Matching identification: 0=Secondary camera image, 1=Main camera image
- parkingTime	string	Parking time, seconds
- plateNoPictures	array	Entry license plate URL
- capturePictures	array	Snapshot URL, index[0] = Main snapshot; index[1] = Secondary snapshot. Append ?token={credential} after the URL to get the snapshot
- personName	string	Vehicle owner name
- personPicture	string	Person avatar URL
- company	string	Company name
- orgCode	string	Person group code
- orgName	string	Person group name
- vehicleBrand	string	Vehicle brand
- vehicleColor	string	Vehicle color
- entranceGroups	array	Entrance and exit group information of entering vehicles
- - entranceGroupId	string	Vehicle entrance and exit group ID
- - entranceGroupName	string	Vehicle entrance and exit group name
- enterInfo	object	Vehicles entry information
- - id	string	Exit record ID
- - plateNo	string	Plate number
- - parkingLotId	string	Parking lot ID
- - parkingLotName	string	Parking lot name
- - positionId	string	Point ID
- - mergePictures	array	Composite image URL
- - positionName	string	Point name
- - pointId	string	Gate ID

Parameter	Type	Description
- - pointName	string	Gate name
- - enterTime	string	Entry time, timestamp (seconds)
- - channels	array	Exit channel information
- - - channelId	string	Channel code
- - - channelName	string	Channel name
- - - bindingChannelIds	array	Bound channel encoding
- - - plateNoPictures	array	Entry license plate URL
- - capturePictures	array	Snapshot URL, index[0] = Main snapshot, index[1] = Secondary snapshot. Append ?token={credential} after the URL to get the snapshot
- - layerInfo	object	Layer information
- - - layerId	string	Layer ID
- - - layerName	string	Layer name
- - - locationX	string	X-axis coordinates
- - - locationY	string	Y-axis coordinates
- - - height	string	Icon height
- - - width	string	Icon width
- - - angle	string	Icon angle
- - - positionId	string	Point ID
- - - positionName	string	Point name
- - - pictureUrl	string	Bottom image URL
- - personName	string	Owner name
- - company	string	Company name
- - orgCode	string	Person group ID
- - orgName	string	Person group name
- - orgInfos	array	Personnel multi organization information
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - vehicleBrand	string	Vehicle trademark
- - vehicleColor	string	Vehicle color
- - vehicleBrand	string	Vehicle brand
- - vehicleColor	string	Vehicle color
- - vehicleModel	string	Vhicle model
- - inputVehicleBrand	string	Vehicle brand(input)

Parameter	Type	Description
- - inputVehicleColor	string	Vehicle color(input)
- - vehicleType	string	Vehicle types: 1=visitor, 2=allowlist, 4=blocklist, 5=temporary vehicle
- - matchFlag	string	Matching identification: 0=Secondary camera image, 1=Main camera image
- - entranceGroups	object	Entrance and exit group information of entering vehicles
- - - entranceGroupId	string	Vehicle entrance and exit group ID
- - - entranceGroupName	string	Vehicle entrance and exit group name
- parkingRecordIds	array	Parking record ID
- parkingLotOrgCode	string	Parking lot organization code
- parkingLotOrgName	string	Name of car park organization

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "1",
    "status": "1",
    "plateNo": "EA12A26",
    "parkingLotId": "2",
    "parkingLotName": "ttt",
    "parentParkingLotId": "1",
    "parentParkingLotName": "",
    "positionId": "376",
    "positionName": "12",
    "mergePictures": [
    ],
    "pointId": "948",
    "pointName": "2008",
    "exitTime": "1615650191",
    "channels": [
      {
        "channelId": "1000049$1$0$0",
        "channelName": "Channel0",
        "bindingChannelIds": [
        ]
      }
    ]
  },
  "originalPlateNo": "A12A26",
}
```

```
"issuePlace": "ABC",
"fuzzyMatched": "0",
"accessAuthority": "1",
"matchFlag": "1",
"parkingTime": "2200",
"plateNoPictures": [
  "http://192.168.1.1:80/v1/s/192.168.1.1/9901/6f059d84-ee63-11ed-ae01-
d094665d8de6/20230510/1/dsf_ba340d83-eece-11ed-b2d5-
d094665d8de6_1442047_1446492.jpg"
],
"capturePictures": [
  "https://192.168.1.1:8082/d/1000049$1$0$0/20210313/23/4326-44742-0-
Peccancy.jpg?streamID:2219@capTime:1615650191"
],
"personName": "",
"personPicture": "",
"company": "",
"orgCode": "",
"orgName": "",
"vehicleBrand": "-1",
"vehicleColor": "100",
"orgInfos": [

],
"entranceGroups": [
  {
    "entranceGroupId": "2",
    "entranceGroupName": "General"
  },
  {
    "entranceGroupId": "3",
    "entranceGroupName": "VIP"
  }
],
"enterInfo": {
  "id": "",
  "plateNo": "",
  "parkingLotId": "",
  "parkingLotName": "",
  "positionId": "",
  "positionName": "",
  "mergePictures": [

  ],
  "pointId": "",
  "pointName": "",
  "enterTime": "",
  "channels": [
    {
```



```
        "channelId": "",
        "channelName": "",
        "bindingChannelIds": [

            ]
    },
    ],
    "plateNoPictures": [

    ],
    "capturePictures": [
        ""
    ],
    "layerInfo": {
        "layerId": "",
        "layerName": "",
        "locationX": "",
        "locationY": "",
        "height": "",
        "width": "",
        "angle": "",
        "positionId": "",
        "positionName": "",
        "pictureUrl": ""
    },
    "personName": "",
    "company": "",
    "orgCode": "",
    "orgName": "",
    "orgInfos": [
        {
            "orgCode": "001",
            "orgName": "All Persons and Vehicles"
        }
    ],
    "vehicleBrand": "",
    "vehicleColor": "",
    "vehicleModel": "27",
    "inputVehicleBrand": "-1",
    "inputVehicleColor": "100",
    "vehicleType": "5",
    "entranceGroups": [

    ],
    "matchFlag": "1"
},
"parkingRecordIds": [

]
]
```

```
}
}
```

3.11.3.5 Get the Snapshot Records of Vehicles at the Entrance and Exit in Pages

Brief Description

- Get vehicle snapshot records at entrance and exit in pages

Request Method

- POST

Request URL

- /ipms/api/v1.1/entrance/vehicle-capture/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
plateNoMatchMode	Yes	string	Plate number matching mode, 0 = Fuzzy matching, 1 = Exact matching
plateNos	No	array	Plate number, fuzzy search supported
company	No	string	Company name, which supports fuzzy search
vehicleType	No	string	Vehicle type: 1= visitor, 5= temporary car
vehicleColor	No	string	Vehicle color, dictionary type, see [API Reference > Dictionary > Vehicle Color]
vehicleModel	No	string	Vehicle model, dictionary type, see [API Reference > Dictionary > Vehicle Model]
endTime	Yes	string	Search end time, timestamp (seconds)
entranceGroupId	No	string	Vehicle entrance and exit group ID
personName	No	string	Owner name, which supports fuzzy search
channelIds	No	array	Channel code
orgCode	No	string	Person group ID
startTime	Yes	string	Search start time, timestamp (seconds)
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647

Parameter	Required	Type	Description
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
vehicleBrand	No	string	Vehicle brand, dictionary type, see [API Reference > Dictionary > Vehicle Brand]
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster

Request Example

```
POST /ipms/api/v1.1/entrance/vehicle-capture/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: zh
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: aabe70b944d04218ba3dd78f9ede4b2c
```

```
{
  "plateNoMatchMode": "0",
  "plateNos": [
    ""
  ],
  "company": "",
  "vehicleColor": "1",
  "vehicleModel": "1",
  "endTime": "1615823999",
  "entranceGroupId": "",
  "page": "1",
  "personName": "",
  "channelIds": [],
  "orgCode": "001",
  "startTime": "1615737600",
  "pageSize": "20",
  "vehicleBrand": "1",
  "splitTime": "",
```

```

    "splitId": "",
    "currentPage": ""
  }

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	Page data
- - id	string	Record ID
- - channelId	string	Channel code
- - channelName	string	Channel name
- - captureTime	string	Capture time, timestamp (second)
- - plateNo	string	Plate number
- - issuePlace	string	Place of Issue
- - plateNoLength	string	License plate length
- - plateNoAreaCode	string	License plate region code
- - extPlateNo	string	Expand license plates
- - plateNoType	string	License plate type
- - plateNoTypeName	string	License plate type name
- - inputVehicleBrand	string	Vehicle brand(input)
- - inputVehicleColor	string	Vehicle color(input)
- - vehicleType	string	The vehicle type: 1 = Vehicle of visitors, 2 = Vehicle in the trusted list, 4 = Vehicle in the restricted list, 5 = Temporary vehicle
- - capturePicture	string	Snapshot URL, append ?token={credential} after the URL to get the snapshot
- - plateNoPicture	string	Plate picture URL, append ?token={credential} after the URL to get the picture
- - parkingLotId	string	Parking lot ID

Parameter	Type	Description
-- parkingLotName	string	Parking lot name
-- parentParkingLotId	string	Parent parking lot ID
-- parentParkingLotName	string	Parent parking lot name
-- travelDirect	string	Driving direction: 8 = Driving in; 9 = Driving out
-- travelDirectName	string	Name of driving direction
-- vehicleBrand	string	Vehicle trademark
-- vehicleColor	string	Vehicle color
-- vehicleModel	string	Vehicle model
-- bindingChannelIds	array	Bound channel encoding
-- personName	string	Owner name
-- personPicture	string	Person avatar URL
-- company	string	Company name
-- orgCode	string	Person group ID
-- orgName	string	Person group name
-- orgInfos	array	Personnel multi organization information
-- - orgCode	string	Person group code
-- - orgName	string	Person group name
-- entranceGroups	array	Entrance and exit group information of entering vehicles
-- - entranceGroupId	string	Vehicle entrance and exit group ID
-- - entranceGroupName	string	Vehicle entrance and exit group name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-121",
    "pageData": [
      {
        "id": "1",
        "channelId": "1000023$1$0$9",
        "channelName": "Test Chn",
        "captureTime": "1615807058",
        "plateNo": "UKERT234",
        "issuePlace": "SAU",

```

```

        "plateNoLength": "3",
        "plateNoAreaCode": "P",
        "extPlateNo": "ABC",
        "plateNoType": "4",
        "plateNoTypeName": "Police",
        "plateNoPicture": "",
        "inputVehicleBrand": "",
        "inputVehicleColor": "",
        "vehicleType": "5",
        "capturePicture":
        "https://192.168.1.1:8082/d/1000023$1$0$9/20210315/19/1741-320730-
        0.jpg?streamID:659@capTime:1615807058",
        "parkingLotId": "87",
        "parkingLotName": "",
        "parentParkingLotId": "1",
        "parentParkingLotName": "",
        "travelDirect": "8",
        "travelDirectName": "IN",
        "vehicleBrand": "1",
        "vehicleColor": "1",
        "vehicleModel": "1",
        "bindingChannelIds": []
        "personName": "",
        "personPicture": "",
        "company": "",
        "orgCode": "",
        "orgName": "",
        "orgInfos": [],
        "entranceGroups": [
            {
                "entranceGroupId": "2",
                "entranceGroupName": "General"
            }
        ]
    }
}

```

3.11.3.6 Obtaining the Parking Records at the Vehicle Entrance/Exit by Page

Brief Description

- Obtain the parking records at the vehicle entrance/exit by page.

Request Method

- POST

Request URL

- /ipms/api/v1.1/entrance/vehicle-parking/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
parkingSpaceIds	No	array	The ID of the parking space.
parkingSpaceStatus	No	string	The status of the parking space: 0 = All, 1 = Occupied.
startTime	Yes	string	The start time of the query, in seconds.
endTime	Yes	string	The end time of the query, in seconds.
entranceGroupId	No	string	The ID of the vehicle group. Set this parameter only when the vehicle group has an ID.
vehicleType	No	string	The vehicle type: 1 = Vehicle of visitors, 5 = Temporary vehicle. Set this parameter only when the vehicles to be queried are vehicles of visitors or temporary vehicles.
plateNo	No	string	The number of the license plate. Fuzzy search is supported.
personName	No	string	The name of the vehicle owner. Fuzzy search is supported.
company	No	string	The company name. Fuzzy search is supported.
orgCode	No	string	The code of the personnel group.
vehicleColor	No	string	The vehicle color, which is a dictionary. For details, see [API Reference > Dictionary > Vehicle Color].

Parameter	Required	Type	Description
vehicleBrand	No	string	The vehicle trademark, which is a dictionary. For details, see [API Reference > Dictionary > Vehicle Brand].
plateNoMatchMode	Yes	string	The matching mode for license plate numbers: 0 = Fuzzy matching, 1 = Exact matching.
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster

Request Example


```
POST /ipms/api/v1.1/entrance/vehicle-parking/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 2f159ca6c41740f9b0acec78842bc85f
X-Subject-Sequence: 1649470123
```

```
{
  "plateNoMatchMode": "1",
  "company": "",
  "vehicleColor": "O",
  "endTime": "1649779199",
  "entranceGroupld": "",
  "vehicleType": "",
  "page": "1",
  "vehicleBrand": "O",
  "orgCode": "",
  "parkingSpaceStatus": "0",
  "plateNo": "",
  "personName": "",
  "startTime": "1649692800",
  "pageSize": "20",
  "parkingSpaceIds": [],
  "splitTime": "",
  "splitld": "",
  "currentPage": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	The page data.
- - id	string	The record ID.
- - plateNo	string	The number of the license plate.

Parameter	Type	Description
-- parkingSpaceId	string	The ID of the parking space.
-- parkingSpaceName	string	The number of the parking space.
-- parkingSpaceStatus	string	The status of the parking space: 0 = Unoccupied, 1 = Occupied.
-- plateNoPictures	array	The URL of the plate image: index[0] = Plate image in which the vehicle is pulling into the parking space, index[1] = Plate image in which the vehicle is pulling out of the parking space. To obtain the image, append ?token={credential} to the image URL.
-- driveInTime	string	The time when the vehicle pulls into the parking space, in seconds.
-- driveOutTime	string	The time when the vehicle pulls out of the parking space, in seconds.
-- parkingTime	string	The parking duration, in seconds.
-- parkingLotId	string	The ID of the parking lot.
-- parkingLotName	string	The name of the parking lot.
-- parentParkingLotId	string	Parent parking lot ID
-- parentParkingLotName	string	Parent parking lot name
-- positionId	string	The position ID.
-- mergePictures	array	Composite URL, the first one is for entering, the second one is for exiting
-- vehiclePictures	array	Vehicle body picture URL, the first one is for entering, the second one if for exiting
-- positionName	string	The position name.
-- pointId	string	The gate ID.
-- pointName	string	The gate name.
-- exitPointId	string	Exit ID
-- exitPointName	string	Exit name
-- exitPositionId	string	Exit point ID
-- exitPositionName	string	Exit point name
-- personName	string	The name of the vehicle owner.
-- company	string	The company name.
-- orgCode	string	The code of the personnel group.
-- orgName	string	The name of the personnel group.

Parameter	Type	Description
- - vehicleBrand	string	The vehicle trademark, which is a dictionary. For details, see [API Reference > Dictionary > Vehicle Brand].
- - vehicleColor	string	The vehicle color, which is a dictionary. For details, see [API Reference > Dictionary > Vehicle Color].
- - orgInfos	array	Personnel multi organization infomation
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- - entranceGroups	array	Vehicle Group
- - - entranceGroupId	string	The ID of the vehicle group.
- - - entranceGroupName	string	The name of the vehicle group.
- - vehicleType	string	The vehicle type: 1 = Vehicle of visitors, 2 = Vehicle in the trusted list, 4 = Vehicle in the restricted list, 5 = Temporary vehicle.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-21",
    "pageData": [
      {
        "id": "1029",
        "plateNo": "A12A28",
        "parkingSpaceId": "15",
        "parkingSpaceName": "space3",
        "parkingSpaceStatus": "0",
        "plateNoPictures": [
          ""
        ],
        "driveInTime": "1649759226",
        "driveOutTime": "1649759257",
        "parkingTime": "31",
        "parkingLotId": "3",
        "parkingLotName": "parkingLot2",
        "parentParkingLotId": "1",
        "parentParkingLotName": "",
        "positionId": "3",

```

```

        "positionName": "position3",
        "pointId": "2388",
        "pointName": "point8",
        "exitPointId": "11",
        "exitPointName": "111",
        "exitPositionId": "22",
        "exitPositionName": "222",
        "personName": "john",
        "company": "VMS",
        "orgCode": "001",
        "orgName": "All Persons",
        "vehicleBrand": "-1",
        "vehicleColor": "1",
        "orgInfos": [
        ],
        "entranceGroups": [
            {
                "entranceGroupId": "3",
                "entranceGroupName": "VIP"
            }
        ],
        "vehicleType": "2",
        "mergePictures": [
        ],
        "vehiclePictures": [
        ]
    }
}

```

3.11.3.7 Obtaining the Parking Records at the Vehicle Entrance/Exit in Batches

Brief Description

- Obtain the parking records at the vehicle entrance/exit in batches.

Request Method

- GET

Request URL

- /ipms/api/v1.1/entrance/vehicle-parking/record/list?ids={ids}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
ids	Yes	string	The record ID. Separate multiple records with commas.

Request Example

```
GET /ipms/api/v1.1/entrance/vehicle-parking/record/list?ids=1029 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: f4982e72223443ceb33aa08901b45f07
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- id	string	The ID of the parking record.
- plateNo	string	The number of the license plate.
- driveInPicture	string	The URL of the vehicle image in which the vehicle is pulling into the parking space.
- driveOutPicture	string	The URL of the vehicle image in which the vehicle is pulling out of the parking space.
- driveInPlateNoPicture	string	The URL of the plate image in which the vehicle is pulling into the parking space.
- driveOutPlateNoPicture	string	The URL of the plate image in which the vehicle is pulling out of the parking space.
- driveInMergePicture	string	Vehicle entering composite image
- driveOutMergePicture	string	Vehicle exit composite image
- driveInVehiclePicture	string	Vehicle entering car image
- driveOutVehiclePicture	string	Vehicle exit car image
- parkingSpaceId	string	The ID of the parking space.
- parkingSpaceName	string	The number of the parking space.

Parameter	Type	Description
- channelId	string	The channel code.
- channelName	string	The channel name.
- deviceCode	string	The device code.
- deviceName	string	The device name.
- driveInTime	string	The time when the vehicle pulls into the parking space, in seconds.
- driveOutTime	string	The time when the vehicle pulls out of the parking space, in seconds.
- parkingTime	string	The parking duration, in seconds.
- parkingLotId	string	The ID of the parking lot.
- parkingLotName	string	The name of the parking lot.
- parentParkingLotId	string	The ID of the parent parking lot.
- parentParkingLotName	string	The name of the parent parking lot.
- personName	string	The name of the vehicle owner.
- personPicture	string	Person avatar URL
- company	string	The company name.
- orgCode	string	The code of the personnel group.
- orgName	string	The name of the personnel group.
- vehicleBrand	string	The vehicle trademark.
- vehicleColor	string	The vehicle color.
- inputVehicleBrand	string	The input vehicle trademark.
- inputVehicleColor	string	The input vehicle color.
- layerInfo	object	The related layer information.
- - layerId	string	The layer ID.
- - layerName	string	The layer name.
- - locationX	string	The coordinate value of the x-axis.
- - locationY	string	The coordinate value of the y-axis.
- - height	string	The icon height.
- - width	string	The icon width.
- - angle	string	The icon angle.
- - parkingSpaceId	string	The ID of the parking space.
- - parkingSpaceName	string	The number of the parking space.
- - pictureUrl	string	The URL of the bottom layer.
- - orgInfos	array	Personnel multi organization information

Parameter	Type	Description
- - - orgCode	string	Person group code
- - - orgName	string	Person group name
- entranceGroups	object	The information of the vehicle group.
- - entranceGroupId	string	The ID of the vehicle group.
- - entranceGroupName	string	The name of the vehicle group.
- vehicleType	string	The vehicle type: 1 = Vehicle of visitors, 2 = Vehicle in the trusted list, 4 = Vehicle in the restricted list, 5 = Temporary vehicle.
- enterRecordId	string	The ID of the related entrance record.
- exitRecordId	string	The ID of the related exit record.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "id": "1029",
        "plateNo": "A12A28",
        "driveInPicture": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/d2ed1e6a-a8ec-11ec-abbf-2cf05dcafd2/20220322/1/dsf_cb617505-a9d1-11ec-bd74-2cf05dcafd2_1211310_1624116.jpg",
        "driveOutPicture": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/d2ed1e6a-a8ec-11ec-abbf-2cf05dcafd2/20220322/1/dsf_cb617505-a9d1-11ec-bd74-2cf05dcafd2_1211310_1624116.jpg",
        "driveInPlateNoPicture": "",
        "driveOutPlateNoPicture": "",
        "parkingSpaceId": "1000107$20$2$0",
        "parkingSpaceName": "space3",
        "channelId": "1000107$1$0$0",
        "channelName": "channel1",
        "deviceCode": "1000107",
        "deviceName": "Detector",
        "driveInTime": "1649759226",
        "driveOutTime": "1649759257",
        "parkingTime": "31",
        "parkingLotId": "3",
        "parkingLotName": "parkingLot2",
        "personName": "test",
        "personPicture": ""
      }
    ]
  }
}
```

```

"company": "",
"orgCode": "001",
"orgName": "All Persons and Vehicles",
"orgInfos": [
  {
    "orgCode": "001",
    "orgName": "All Persons and Vehicles"
  }
],
"vehicleBrand": "-1",
"vehicleColor": "99",
"inputVehicleBrand": "",
"inputVehicleColor": "",
"layerInfo": {
  "layerId": "",
  "layerName": "",
  "locationX": "",
  "locationY": "",
  "height": "",
  "width": "",
  "angle": "",
  "parkingSpaceId": "",
  "parkingSpaceName": "",
  "pictureUrl": ""
},
"entranceGroups": [
  {
    "entranceGroupId": "3",
    "entranceGroupName": "VIP"
  }
],
"vehicleType": "2",
"enterRecordId": "1441678",
"exitRecordId": "1926820",
"parkingLotOrgCode": "001001",
"parkingLotOrgName": "test",
"driveInMergePicture": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/d2afdf2_1211310_1624116.jpg",
"driveOutMergePicture": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/d2afdf2_1211310_1624116.jpg",
"driveInVehiclePicture": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/d2afdf2_1211310_1624116.jpg",
"driveOutVehiclePicture": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/d2afdf2_1211310_1624116.jpg"
}
]
}
}

```


3.11.3.8 Get the Historical Information of the Parking Lot

Brief Description

- Get the historical Information of the parking lot

Request Method

- POST

Request URL

- /ipms/api/v1.1/parking-lot/statistics/history

Request Parameters

- Body Parameters

Parameter	Required	Type	Description
countType	Yes	string	Statistics Type: 1 = day, 2 = week, 3 = month, Required. When countType is 2, the start date of each week will be entered (that is the date of Sunday), such as: 2022-09-11, when countType is 3, only month will be entered, such as: 2022-09
countDate	No	string	Date of the counting, format: YYYY-MM-DD, Required
startDate	No	string	Statistics start date, format: YYYY-MM-DD, required when countType = 4
endDate	No	string	Statistics end date, format: YYYY-MM-DD, required when countType = 4
parkingLotIds	No	array	Parking lot ID, empty means all the parking lots

Request Example

```
POST /ipms/api/v1.1/parking-lot/statistics/history HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: e935049cad0e436c88f0a7ca0b8d5b3c

{
  "countType": "1",
```

```

    "countDate": "2022-09-13",
    "parkingLotIds": [
        "1"
    ],
    "startDate": "",
    "endDate": ""
}

```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- enterVehicleCounts	array	Y axis: The number of the entered vehicles
- exitVehicleCounts	array	Y axis: The number of the leaving vehicles
- presentVehicleCounts	array	Y axis: The number of the present vehicles
- parkingTurnoverRates	array	Y axis, Parking turnover rates
- parkingUtilizationRates	array	Y axis, Parking utilization rates
- periods	array	X axis, Periods

Return Example

```

{
  "code": 1000,
  "desc": "Success",
  "data": {
    "presentVehicleCounts": [
      "0", "0", "0", "0", "0", "0", "0", "0",
      "0", "0", "0", "0", "0", "0", "0", "0",
      "0", "0", "0", "0", "0", "0", "0", "0"
    ],
    "enterVehicleCounts": [
      "0", "0", "0", "0", "0", "0", "0", "0",
      "0", "0", "0", "0", "0", "0", "0", "0",
      "0", "0", "0", "0", "0", "0", "0", "0"
    ],
    "exitVehicleCounts": [
      "0", "0", "0", "0", "0", "0", "0", "0",
      "0", "0", "0", "0", "0", "0", "0", "0",
      "0", "0", "0", "0", "0", "0", "0", "0"
    ],
    "parkingTurnoverRates": [
      "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0",
      "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0"
    ]
  }
}

```

```

        "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0"
    ],
    "parkingUtilizationRates": [
        "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0",
        "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0",
        "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0", "0.0"
    ],
    "periods": [
        "01:00", "02:00", "03:00", "04:00", "05:00", "06:00", "07:00", "08:00",
        "09:00", "10:00", "11:00", "12:00", "13:00", "14:00", "15:00", "16:00",
        "17:00", "18:00", "19:00", "20:00", "21:00", "22:00", "23:00", "00:00"
    ]
}
}

```

3. 11. 3. 9 Get the Real-time Information of the Parking Lot

Brief Description

- Get the real-time information of the parking lot

Request Method

- POST

Request URL

- /ipms/api/v1.1/parking-lot/statistics/real-time

Request Parameters

- Body Parameters

Parameter	Required	Type	Description
parkingLotIds	true	array	Parking lot ID

Request Example

```

POST /ipms/api/v1.1/parking-lot/statistics/real-time HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
X-Subject-Token: 9BE5357CAF244760822cf61ae689cad2
Accept-Language: zh
Time-Zone: Asia/Shanghai
Content-Type: application/json

{

```

```

    "parkingLotIds": [
      "1",
      "2",
      "3",
      "5",
      "6",
      "9"
    ]
  }

```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	
- totalParkingSpaceCount	string	Total number of parking spaces
- idleParkingSpaceCount	string	Number of free parking spaces
- idleParkingSpaceRate	string	Parking space free rate
- presentVehicleCount	string	Number of vehicles present
- enterVehicleCount	string	Number of incoming vehicles
- exitVehicleCount	string	Number of exit vehicles
- addVehicleCount	string	Number of new vehicles
- parkingTurnover	string	Parking turnover rates
- parkingUtilizationRate	string	Parking utilization rates

Return Example

```

{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalParkingSpaceCount": "13",
    "idleParkingSpaceCount": "11",
    "idleParkingSpaceRate": "0.846",
    "presentVehicleCount": "2",
    "enterVehicleCount": "2",
    "exitVehicleCount": "2",
    "parkingTurnover": "0.308",
    "parkingUtilizationRate": "0.163",
    "addVehicleCount": "0"
  }
}

```

3.11.4 Parking Lot

3.11.4.1 Get Parking Resource Tree

Brief Description

- Get parking resource tree (Filter device permissions)

Request Method

- GET

Request URL

- /ipms/api/v1.1/parking-lot/summary/list?parkingLotIds={parkingLotIds}&mode={mode}&orgCode={orgCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
parkingLotIds	No	string	ID of the parking lot, separated by multiple commas. A blank value indicates all of them.
mode	No	string	Mode: 1 = configuration mode, 2 = application mode, 3 = recording mode, default 1
orgCode	No	string	Parking lot organization code

Request Example

```
GET /ipms/api/v1.1/parking-lot/summary/list?parkingLotIds=&mode=1&orgCode= HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 4572d75b9c34431bbd51696ee953e8be
```

Return Parameter

Parameter	Type	Description
code	int	Error code

Parameter	Type	Description
desc	string	Result description
data	object	Data
- results	array	Result
- - id	string	Parking lot ID
- - orgCode	string	Parking lot organization code
- - orgName	string	Parking lot organization name
- - parkingLotName	string	Parking lot name
- - parentParkingLotId	string	Parent parking lot ID
- - parentParkingLotName	string	Parent parking lot name
- - parkingLotMode	string	Parking lot field mode: 1 = entrance and exit mode, 2 = no entrance and exit mode
- - enableParkingSpaceStatistics	string	Start parking statistics: 1 = Yes, 0 = No
- - parkingSpaceStatisticsType	string	Parking space statistics mode: 1 = residual position mode, 2 = vehicle detector mode
- - positions	array	Parking point
- - - id	string	Parking point ID
- - - positionName	string	Parking point name
- - - remark	string	Remark
- - - points	array	Parking lot gate
- - - - id	string	Parking lot gate ID
- - - - pointName	string	Parking lot gate name
- - - - direction	string	Driving direction: 8 = in, 9 = out
- - - - remark	string	Remark
- - - - mode	string	Gate mode: 1 = with gate, 2 = without gate
- - - - refPointId	string	Reuse gate ID, 0 or empty means not reused

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
```

```

    "id": "208",
    "orgCode": "001001",
    "orgName": "test",
    "parentParkingLotId": "0",
    "parentParkingLotName": "",
    "parkingLotMode": "1",
    "parkingLotName": "superMarket",
    "parkingSpaceStatisticsType": "1",
    "enableParkingSpaceStatistics": "1",
    "positions": [
      {
        "id": "256",
        "points": [
          {
            "direction": "8",
            "id": "497",
            "mode": "1",
            "pointName": "east-enter",
            "refPointId": "0",
            "remark": ""
          }
        ],
        "positionName": "east",
        "remark": ""
      }
    ]
  }
}

```

3.11.4.2 Add a Parking Lot

Brief Description

- Add parking lot

Request Method

- POST

Request URL

- /ipms/api/v1.1/parking-lot

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
parentParkingLotId	No	string	Parent zone ID. 0 or empty indicates the current parent zone
orgCode	No	string	Parking organization code, default 001
parkingLotName	Yes	string	parkingLotName
parkingLotMode	No	string	Parking lot mode: 1= access mode, 2= no access mode, default 1
enableParkingSpaceStatistics	No	string	Whether to enable parking statistics. 1= yes, 0= no, default 0
enableParkingOverTimePass	No	string	Whether to enable parking timeout to allow passage, 1 = yes, 0 = no, default 0
parkingSpaceStatisticsType	No	string	Parking statistical mode: 1= residual mode, 2= vehicle detector mode
totalParkingSpaceCount	No	string	Total number of parking spaces, up to 10,000
idleParkingSpaceCount	No	string	Number of empty parking spots
enableIdleReset	No	string	Whether to enable residual reset. 1= yes, 0= no, 0 by default
idleResetType	No	string	Remaining position reset type, 1 = reset to the total parking space, 2 = reset the remaining position according to the vehicle present, 3 = reset to the specified remaining position, default 1
idleResetTime	No	string	Rest reset time. The format is HH:mm:ss
idleResetToSpaceCount	No	string	Specify the number of reset bits
enableForceExitRecordIdleReset	No	string	Whether to force the vehicle out of the field when the rest bit is reset: 1= yes, 0= no, 0 by default
enableRecordCheck	No	string	Whether record verification is enabled: 1=yes, 0=no
recordCheckIntervalTime	No	string	Record inspection time interval (the same license plate is recognized again within the configured time and the entry and exit directions are inconsistent, and it is judged as

Parameter	Required	Type	Description
			repeated reporting and no processing), range 1-20, unit second
parkingRuleType	No	string	Parking rule type: 1=default, exit from the parking lot or parking space is counted as exit; 2=user-defined, the same record is counted as the same record when the license plate enters the parking space again within the user-defined interval
parkingRuleIntervalTime	No	string	The interval time configured by the user-defined parking rule (re-entering the parking space in the same license plate interval time, counting the same record), range 1-60, unit second
parkingSpaceStatisticsRuleType	No	string	Type of residual statistics rule: 1= Residual for all vehicles, 2= residual for unregistered vehicles, 3= Custom, default 1, condition: enableParkingSpaceStatistics=1、parkingSpaceStatisticsType=1、enterRuleType=3, condition: enableParkingSpaceStatistics=1, parkingSpaceStatisticsType=1, enterRuleType=3
enableFuzzyMatch	No	string	Whether to enable fuzzy matching: 1= yes, 0= no, 0 by default
enableFirstPlaceOneMore	No	string	More than one digit is allowed: 1= yes, 0= no, 0 is the default
enableFirstPlaceOneLess	No	string	The first digit is allowed to be less: 1= yes, 0= no, 0 is the default
enableLastPlaceOneMore	No	string	More bits are allowed at the end: 1= yes, 0= no, 0 is the default
enableLastPlaceOneLess	No	string	One bit less is allowed at the end: 1= yes, 0= no, 0 is the default
fuzzyMatchBitCount	No	string	Fuzzy matching of error bits is allowed. Only 1 is supported for the time being

Parameter	Required	Type	Description
enableOverrideEnterRepeatVehicle	Yes	string	Whether the approach record is overwritten: 1= yes, 0= no, default 0
issueVehicles	Yes	string	Whether to deliver to the garage: 1= yes, 0= no, default 0
enterRuleType	No	string	Entry rules configuration types: 1= registered vehicles, 2= all vehicles, 3= Custom, default 1
enableEnterAutoPassUnRegisterVehicle	No	string	Whether the temporary car is automatically released: 1= yes, 0= no, default 0, condition: enterRuleType=2 or 3
enterTimeTemplateId	No	string	Automatic admission time template ID, default 1
enableEnterAutoPassBlockVehicle	No	string	Whether the blocklist is automatically released: 1= yes, 0= no, 0 is the default, condition: enterRuleType=2
enableEnterAutoPassUnlicensedVehicle	No	string	Whether the unlicensed car is automatically released: 1= yes, 0= no, default 1, condition: enterRuleType=2
enableEmptyParkingSpaceEnterRule	No	string	Whether to enable rule configuration when the empty parking spot is 0: 1 = Yes, 0 = No
enableEnterCheckAuth	No	string	Whether to enable the approach verification switch: 1= yes, 0= no, default 0. (Abandoned)
enableEnterCheckAuthByCard	No	string	Check whether swipe verification is enabled: 1= yes, 0= no, default 0. (Abandoned)
enableEnterCardAuth	No	string	Enter to enable swipe card opening: 1= yes, 0= no, default 0. (Abandoned)
exitRuleType	No	string	Exit rules Configuration types: 1= registered vehicles, 2= all vehicles, 3= Custom, default 1
enableExitAutoPassUnRegisterVehicle	No	string	Whether the temporary car is automatically released: 1= yes, 0= no, default 0, condition: exitRuleType=2 or 3

Parameter	Required	Type	Description
exitTimeTemplateId	No	string	ID of the automatic release time template
enableExitAutoPassBlockVehicle	No	string	Whether the blocklist is automatically allowed: 1= yes, 0= no, 0 is the default, condition: exitRuleType=2
enableExitAutoPassUnlicensedVehicle	No	string	Whether the unlicensed car is automatically released: 1= yes, 0= no, default 1, condition: exitRuleType=2
enableExitCheckAuth	No	string	Whether to enable the appearance verification switch: 1= yes, 0= no, 0 is the default. (Abandoned)
enableExitCheckAuthByCard	No	string	Check whether credit card verification is enabled: 1= yes, 0= no, default 0. (Abandoned)
enableExitCardAuth	No	string	Activate the swipe card switch: 1= yes, 0= no, default 0. Abandoned
enterReleaseRuleCheckType	No	string	Vehicle enter release rule verification type: 1=do not enable validation, 2=open the gate manually, 3=open the gate directly through validation, 4=open the gate through double validation
enterReleaseRuleDoubleCheckType	No	string	Vehicle enter double verification opening type: 1=owner verification (consistent with people and vehicles), 2=body color verification
enableEnterUnrecognizedVehicleColorRelease	No	string	When the body color is not recognized during the incoming body color verification, is it released: 1=yes, 0=no
exitReleaseRuleCheckType	No	string	Vehicle exit double verification opening type: 1=do not enable validation, 2=open the gate manually, 3=open the gate directly through validation, 4=open the gate through double validation

Parameter	Required	Type	Description
exitReleaseRuleDoubleCheckType	No	string	Double verification and opening type: 1=owner verification (consistent with people and vehicles), 2=body color verification
enableExitUnrecognizedVehicleColorRelease	No	string	When the body color is not recognized during the outgoing body color verification, is it released: 1=yes, 0=no
bindingDetectors	No	array	Vehicle inspection device code. (Abandoned)
bindingLeds	No	array	LED device coding. (Abandoned)
bindingDetectorSpacelds	No	array	Parking ID
bindingLedPartitionIds	No	array	Split screen ID
bindingLedPartitions	Yes	array	
- ledPartitionId	Yes	string	Screen ID
- dataSource	No	string	Type of data statistics: 1= vehicle entry and exit statistics, 2= vehicle detector statistics
- bindingAllDetectors	No	string	Whether to bind all auto detectors, 1= yes, 0= no
- parkingSpacelds	No	array	Parking ID
- parkingLotIds	No	array	Parking ID
- parkingSpaceGroupIds	No	array	Parking space type ID
- bindingType	No	string	Binding type: 1= parking space, 2= parking lot, 3=current area, 4=parking space type,default 2
bindingRemainLedChannels	Yes	array	(Abandoned)
- channelId	Yes	string	LED channel coding
- displayColor	Yes	string	Display text colors: 1= red, 2= green, 3= yellow
- noParkingSpaceDisplayContent	Yes	string	When there is no empty parking spot, the display content is: 0000, FULL
parkingSpaceStatisticsRules	Yes	array	condition: EnableParkingSpaceStatistics=1, parkingSpaceStatisticsType=1, enterRuleType=3
- groupId	Yes	string	Entrance and exit group ID (blocklist cannot be configured)

Parameter	Required	Type	Description
- statisticsFlag	Yes	string	Statistical identification: 1= calculated, 0= not calculated
positions	Yes	array	
- positionName	Yes	string	positionName
- remark	No	string	remark
- passRules	Yes	array	
- - groupId	Yes	string	Entrance and exit group ID (blocklist cannot be configured): 2= normal group, 3=VIP
- - timeTemplateId	Yes	string	Time template ID, condition: enterRuleType=3 or exitRuleType=3
- - direction	Yes	string	In and out directions: 1= in, 2= out, condition: enterRuleType=3 or exitRuleType=3
- - enableEmptyParkingSpaceEnterRule	No	string	Whether the rules take effect when the empty parking spot is 0: 1 = Yes, 0 = No
- issueVehicleRules	Yes	array	condition: issueVehicles=1
- - groupId	Yes	string	ID of an import or export group (with blocklist) : -2= common group, -3=VIP, and -4= blocklist
- - direction	Yes	string	In and out directions: 1= in, 2= out
- points	Yes	array	
- - pointName	Yes	string	pointName
- - direction	Yes	string	Driving direction: 8= in, 9= out
- - remark	No	string	remark

Parameter	Required	Type	Description
- - refPointId	No	string	Reference gate ID. 0 or empty indicates no reference
- - multipleCameras	No	string	Multi-camera identification: 0 = Single camera, 1 = Dual camera
- - coordinateTime	No	string	Dual camera collaboration time
- - pointMode	No	string	Gate mode: 1= one gate, 2= no gate, default 1
- - bindingLtcChannels	Yes	array	
- - - channelId	Yes	string	channelId
- - - channelType	Yes	string	Channel type: 1 = Main camera, 2 = Auxiliary camera, condition: when channelType=2, multipleCameras=1
- - bindingVtoChannels	Yes	array	
- - - channelId	Yes	string	channelId
- - bindingCommonLedChannels	Yes	array	(Abandoned)
- - - channelId	Yes	string	channelId
- - - displayColor	Yes	string	Display text colors: 1= red, 2= green, 3= yellow
- - - displayRows	No	string	Display line count
- - - screenHeight	No	string	LED screen height
- - - screenWidth	No	string	LED screen width
- - - rollType	Yes	string	Content switching mode: 1= scroll, 2= cut screen
- - - rollSpeed	Yes	string	Rolling speed, range: 1-100

Parameter	Required	Type	Description
--idleContentType	Yes	string	When idle, display content types: 0= empty, 1= residual bits, 2= Custom
--idleUserDefinedContent	No	string	Displays custom content when idle
--displayContents	Yes	array	
---id	Yes	string	Serial number, starting from 0
----content	Yes	string	Content number: 1= license plate number, 2= entry time,...
----userDefined	No	string	User-defined content or not: 1= yes, 0= no, 0 by default

Request Example

```
POST /ipms/api/v1.1/parking-lot HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 0975fa3db69c495a860c98013ca6f0fa
```

```
{
  "bindingDetectorSpaceIds": [
    "1000040$20$0$0",
    "1000040$20$0$1"
  ],
  "bindingDetectors": [],
  "bindingLedPartitionIds": [
    "1000041$19$0$0",
    "1000041$19$0$1"
  ],
  "bindingLedPartitions": [
    {
      "bindingAllDetectors": "0",
      "bindingType": "3",
      "dataSource": "1",
      "ledPartitionId": "1000041$19$0$0",
      "parkingLotIds": [],
      "parkingSpaceIds": [],
      "parkingSpaceGroupIds": []
    }
  ]
}
```

```
    },
    {
        "bindingAllDetectors": "0",
        "bindingType": "1",
        "dataSource": "2",
        "ledPartitionId": "1000041$19$0$1",
        "parkingLotIds": [],
        "parkingSpaceIds": [
            "1000040$20$0$0",
            "1000040$20$0$1"
        ],
        "parkingSpaceGroupIds": []
    }
],
"bindingLeds": [],
"bindingRemainLedChannels": [],
"enableEmptyParkingSpaceEnterRule": "0",
"enableEnterAutoPassBlockVehicle": "0",
"enableEnterAutoPassUnRegisterVehicle": "0",
"enableEnterAutoPassUnlicensedVehicle": "0",
"enableEnterCardAuth": "0",
"enableEnterCheckAuth": "0",
"enableEnterCheckAuthByCard": "0",
"enableExitAutoPassBlockVehicle": "0",
"enableExitAutoPassUnRegisterVehicle": "0",
"enableExitAutoPassUnlicensedVehicle": "0",
"enableExitCardAuth": "0",
"enterReleaseRuleCheckType": "",
"enterReleaseRuleDoubleCheckType": "",
"enableEnterUnrecognizedVehicleColorRelease": "",
"exitReleaseRuleCheckType": "",
"exitReleaseRuleDoubleCheckType": "",
"enableExitUnrecognizedVehicleColorRelease": "",
"enableExitCheckAuth": "0",
"enableExitCheckAuthByCard": "0",
"enableFirstPlaceOneLess": "0",
"enableFirstPlaceOneMore": "0",
"enableForceExitRecordIdleReset": "0",
"enableRecordCheck": "",
"recordCheckIntervalTime": "",
"parkingRuleType": "",
"parkingRuleIntervalTime": "",
"enableFuzzyMatch": "0",
"enableIdleReset": "0",
"idleResetType": "1",
"idleResetToSpaceCount": "",
"enableLastPlaceOneLess": "0",
"enableLastPlaceOneMore": "0",
"enableOverrideEnterRepeatVehicle": "0",
```



```
"enableParkingSpaceStatistics": "1",
"enableParkingOverTimePass": "0",
"enterRuleType": "1",
"enterTimeTemplateId": "1",
"exitRuleType": "1",
"exitTimeTemplateId": "1",
"fuzzyMatchBitCount": "0",
"idleParkingSpaceCount": "1000",
"idleResetTime": "00:00:00",
"issueVehicles": "0",
"parentParkingLotId": "",
"orgCode": "001",
"parkingLotMode": "1",
"parkingLotName": "1399",
"parkingSpaceStatisticsRuleType": "0",
"parkingSpaceStatisticsRules": [],
"parkingSpaceStatisticsType": "1",
"positions": [
  {
    "id": "",
    "issueVehicleRules": [],
    "passRules": [],
    "points": [
      {
        "bindingCommonLedChannels": [],
        "bindingItcChannels": [
          {
            "channelId": "1000008$1$0$0",
            "channelType": "1"
          }
        ],
        "bindingVtoChannels": [],
        "coordinateTime": "3",
        "direction": "8",
        "id": "",
        "multipleCameras": "0",
        "pointMode": "1",
        "pointName": "enter",
        "remark": ""
      },
      {
        "bindingCommonLedChannels": [],
        "bindingItcChannels": [
          {
            "channelId": "1000008$1$0$4",
            "channelType": "1"
          }
        ],
        "bindingVtoChannels": [],
```

```

        "coordinateTime": "3",
        "direction": "9",
        "id": "",
        "multipleCameras": "0",
        "pointMode": "1",
        "pointName": "exit",
        "remark": ""
    }
    ],
    "positionName": "west",
    "remark": ""
}
],
"totalParkingSpaceCount": "1000"
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- id	string	Parking lot ID

Return Example

```

{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "1"
  }
}

```

3.11.4.3 Update a Parking Lot

Brief Description

- Update parking lot

Request Method

- PUT

Request URL

- /ipms/api/v1.1/parking-lot/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	Parking lot ID

- Body Parameters

Parameter	Required	Type	Description
mode	No	string	Update mode: 1=normal mode, 2=forced mode, default 2
orgCode	No	string	Parking organization code, default 001
parkingLotName	Yes	string	Parking lot name
parkingLotMode	No	string	Parking lot mode: 1= access mode, 2= no access mode, default 1
enableParkingSpaceStatistics	No	string	Whether to enable parking statistics. 1= yes, 0= no, default 0
enableParkingOverTimePass	No	string	Whether to enable parking timeout to allow passage, 1 = yes, 0 = no, default 0
parkingSpaceStatisticsType	No	string	Parking statistical mode: 1= residual mode, 2= vehicle detector mode
totalParkingSpaceCount	No	string	Total number of parking spaces, up to 10,000
idleParkingSpaceCount	No	string	Number of empty parking spots
enableIdleReset	No	string	Whether to enable residual reset. 1= yes, 0= no, 0 by default

Parameter	Required	Type	Description
idleResetType	No	string	Remaining position reset type, 1 = reset to the total parking space, 2 = reset the remaining position according to the vehicle present, 3 = reset to the specified remaining position, default 1
idleResetTime	No	string	Rest reset time. The format is HH:mm:ss
idleResetToSpaceCount	No	string	Specify the number of reset bits
enableForceExitRecordIdleReset	No	string	Whether to force the vehicle out of the field when the rest bit is reset: 1= yes, 0= no, 0 by default
enableRecordCheck	No	string	Whether record verification is enabled: 1=yes, 0=no
recordCheckIntervalTime	No	string	Record inspection time interval (the same license plate is recognized again within the configured time and the entry and exit directions are inconsistent, and it is judged as repeated reporting and no processing), range 1-20, unit second
parkingRuleType	No	string	Parking rule type: 1=default, exit from the parking lot or parking space is counted as exit; 2=user-defined, the same record is counted as the same record when the

Parameter	Required	Type	Description
			license plate enters the parking space again within the user-defined interval
parkingRuleIntervalTime	No	string	The interval time configured by the user-defined parking rule (re-entering the parking space in the same license plate interval time, counting the same record), range 1-60, unit second
parkingSpaceStatisticsRuleType	No	string	Type of residual statistics rule: 1= Residual for all vehicles, 2= residual for unregistered vehicles, 3= Custom, default 1, condition: enableParkingSpaceStatistics=1, parkingSpaceStatisticsType=1, enterRuleType=3
enableFuzzyMatch	No	string	Whether to enable fuzzy matching: 1= yes, 0= no, 0 by default
enableFirstPlaceOneMore	No	string	More than one digit is allowed: 1= yes, 0= no, 0 is the default
enableFirstPlaceOneLess	No	string	The first digit is allowed to be less: 1= yes, 0= no, 0 is the default
enableLastPlaceOneMore	No	string	More bits are allowed at the end: 1= yes, 0= no, 0 is the default
enableLastPlaceOneLess	No	string	One bit less is allowed at the end:

Parameter	Required	Type	Description
			1= yes, 0= no, 0 is the default
fuzzyMatchBitCount	No	string	Fuzzy matching of error bits is allowed. Only 1 is supported for the time being
enableOverrideEnterRepeatVehicle	Yes	string	Whether the approach record is overwritten: 1= yes, 0= no, default 0
issueVehicles	Yes	string	Whether to deliver to the garage: 1= yes, 0= no, default 0
enterRuleType	No	string	Entry rules configuration types: 1= registered vehicles, 2= all vehicles, 3= Custom, default 1
enableEnterAutoPassUnRegisterVehicle	No	string	Whether the temporary car is automatically released: 1= yes, 0= no, default 0, condition: enterRuleType=2 or 3
enterTimeTemplateId	No	string	Automatic admission time template ID, default 1
enableEnterAutoPassBlockVehicle	No	string	Whether the blocklist is automatically released: 1= yes, 0= no, 0 is the default, condition: enterRuleType=2
enableEnterAutoPassUnlicensedVehicle	No	string	Whether the unlicensed car is automatically released: 1= yes, 0= no, default 1, condition: enterRuleType=2

Parameter	Required	Type	Description
enableEmptyParkingSpaceEnterRule	No	string	Whether to enable rule configuration when the empty parking spot is 0: 1 = Yes, 0 = No
enableEnterCheckAuth	No	string	Whether to enable the approach verification switch: 1= yes, 0= no, default 0(Abandoned)
enableEnterCheckAuthByCard	No	string	Check whether swipe verification is enabled: 1= yes, 0= no, default 0(Abandoned)
enableEnterCardAuth	No	string	Enter to enable swipe card opening: 1= yes, 0= no, default 0(Abandoned)
exitRuleType	No	string	Exit rules Configuration types: 1= registered vehicles, 2= all vehicles, 3= Custom, default 1
enableExitAutoPassUnRegisterVehicle	No	string	Whether the temporary car is automatically released: 1= yes, 0= no, default 0, condition: exitRuleType=2 or 3
exitTimeTemplateId	No	string	ID of the automatic release time template
enableExitAutoPassBlockVehicle	No	string	Whether the blocklist is automatically allowed: 1= yes, 0= no, 0 is the default, condition: exitRuleType=2

Parameter	Required	Type	Description
enableExitAutoPassUnlicensedVehicle	No	string	Whether the unlicensed car is automatically released: 1= yes, 0= no, default 1, condition: exitRuleType=2
enableExitCheckAuth	No	string	Whether to enable the appearance verification switch: 1= yes, 0= no, 0 is the default(Abandoned)
enableExitCheckAuthByCard	No	string	Check whether credit card verification is enabled: 1= yes, 0= no, default 0(Abandoned)
enableExitCardAuth	No	string	Activate the swipe card switch: 1= yes, 0= no, default 0(Abandoned)
enterReleaseRuleCheckType	No	string	Vehicle enter release rule verification type: 1=do not enable validation, 2=open the gate manually, 3=open the gate directly through validation, 4=open the gate through double validation
enterReleaseRuleDoubleCheckType	No	string	Vehicle enter double verification opening type: 1=owner verification (consistent with people and vehicles), 2=body color verification
enableEnterUnrecognizedVehicleColorRelease	No	string	When the body color is not recognized during the incoming body color verification, is it

Parameter	Required	Type	Description
			released: 1=yes, 0=no
exitReleaseRuleCheckType	No	string	Vehicle exit double verification opening type: 1=do not enable validation, 2=open the gate manually, 3=open the gate directly through validation, 4=open the gate through double validation
exitReleaseRuleDoubleCheckType	No	string	Double verification and opening type: 1=owner verification (consistent with people and vehicles), 2=body color verification
enableExitUnrecognizedVehicleColorRelease	No	string	When the body color is not recognized during the outcoming body color verification, is it released: 1=yes, 0=no
bindingDetectors	No	array	Vehicle inspection device code(Abandoned)
bindingLeds	No	array	LED device coding(Abandoned)
bindingDetectorSpaceIds	No	array	Parking ID
bindingLedPartitionIds	No	array	Split screen ID
bindingLedPartitions	Yes	array	
- ledPartitionId	Yes	string	Screen ID
- dataSource	No	string	Type of data statistics: 1= vehicle entry and exit statistics, 2= vehicle detector statistics

Parameter	Required	Type	Description
- bindingAllDetectors	No	string	Whether to bind all auto detectors, 1= yes, 0= no
- parkingSpaceIds	No	array	Parking ID
- parkingLotIds	No	array	Parking ID
- bindingType	No	string	Binding type: 1= parking space, 2= parking lot, default 2
bindingRemainLedChannels	Yes	array	(Abandoned)
- channelId	Yes	string	LED channel coding
- displayColor	Yes	string	Display text colors: 1= red, 2= green, 3= yellow
- noParkingSpaceDisplayContent	Yes	string	When there is no empty parking spot, the display content is: 0000, FULL
parkingSpaceStatisticsRules	Yes	array	condition: EnableParkingSpaceStatistics=1, parkingSpaceStatisticType=1, enterRuleType=3
- groupId	Yes	string	Entrance and exit group ID (blocklist cannot be configured)
- statisticsFlag	Yes	string	Statistical identification: 1= calculated, 0= not calculated
positions	Yes	array	
- positionName	Yes	string	positionName
- remark	No	string	remark
- passRules	Yes	array	
- - groupId	Yes	string	Entrance and exit group ID (blocklist cannot be configured): 2= normal group, 3=VIP

Parameter	Required	Type	Description
- - timeTemplateId	Yes	string	Time template ID, condition: enterRuleType=3 or exitRuleType=3
- - direction	Yes	string	In and out directions: 1= in, 2= out, condition: enterRuleType=3 or exitRuleType=3
- - enableEmptyParkingSpaceEnterRule	No	string	Whether the rules take effect when the empty parking spot is 0: 1 = Yes, 0 = No
- issueVehicleRules	Yes	array	condition: issueVehicles=1
- - groupId	Yes	string	ID of an import or export group (with blocklist) : -2= common group, -3=VIP, and -4= blocklist
- - direction	Yes	string	In and out directions: 1= in, 2= out
- points	Yes	array	
- - pointName	Yes	string	pointName
- - direction	Yes	string	Driving direction: 8= in, 9= out
- - remark	No	string	remark
- - refPointId	No	string	Reference gate ID. 0 or empty indicates no reference
- - multipleCameras	No	string	Multi-camera identification: 0 = Single camera, 1 = Dual camera
- - coordinateTime	No	string	Dual camera collaboration time
- - pointMode	No	string	Gate mode: 1= one gate, 2= no gate, default 1
- - bindingItcChannels	Yes	array	

Parameter	Required	Type	Description
- - - channelId	Yes	string	channelId
- - - channelType	Yes	string	Channel type: 1 = Main camera, 2 = Auxiliary camera, condition: when channelType=2, multipleCameras=1
- - bindingVtoChannels	Yes	array	
- - - channelId	Yes	string	channelId
- - bindingCommonLedChannels	Yes	array	(Abandoned)
- - - channelId	Yes	string	channelId
- - - displayColor	Yes	string	Display text colors: 1= red, 2= green, 3= yellow
- - - displayRows	No	string	Display line count
- - - screenHeight	No	string	LED screen height
- - - screenWidth	No	string	LED screen width
- - - rollType	Yes	string	Content switching mode: 1= scroll, 2= cut screen
- - - rollSpeed	Yes	string	Rolling speed, range: 1-100
- - - idleContentType	Yes	string	When idle, display content types: 0= empty, 1= residual bits, 2= Custom
- - - idleUserDefinedContent	No	string	Displays custom content when idle
- - - displayContents	Yes	array	
- - - - id	Yes	string	Serial number, starting from 0
- - - - content	Yes	string	Content number: 1= license plate number, 2= entry time,...
- - - - userDefined	No	string	User-defined content or not: 1= yes, 0= no, 0 by default

Request Example

PUT /ipms/api/v1.1/parking-lot/73 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 28a54240216846179a214b4e04cb674d

```
{
  "idleParkingSpaceCount": "900",
  "enableEnterAutoPassUnRegisterVehicle": "1",
  "exitTimeTemplateId": "0",
  "enterTimeTemplateId": "1",
  "enableExitAutoPassUnRegisterVehicle": "0",
  "enableOverrideEnterRepeatVehicle": "0",
  "enableEmptyParkingSpaceEnterRule": "1",
  "idleResetType": "1",
  "idleResetToSpaceCount": "",
  "positions": [
    {
      "points": [
        {
          "pointName": "e1",
          "direction": "8",
          "bindingVtoChannels": [],
          "multipleCameras": "0",
          "bindingItcChannels": [
            {
              "channelType": "1",
              "channelId": "1000143$1$0$3"
            }
          ],
          "coordinateTime": "0",
          "bindingCommonLedChannels": [],
          "remark": ""
        }
      ],
      "positionName": "p1",
      "passRules": [],
      "remark": ""
    }
  ],
  "exitRuleType": "3",
  "issueVehicles": "0",
  "parkingLotName": "parking lot 1",
  "enableParkingSpaceStatistics": "1",
  "enableParkingOverTimePass": "0",
  "enableRecordCheck": "",
  "recordCheckIntervalTime": ""
}
```

```
{
  "parkingRuleType": "",
  "parkingRuleIntervalTime": "",
  "enterReleaseRuleCheckType": "",
  "enterReleaseRuleDoubleCheckType": "",
  "enableEnterUnrecognizedVehicleColorRelease": "",
  "exitReleaseRuleCheckType": "",
  "exitReleaseRuleDoubleCheckType": "",
  "enableExitUnrecognizedVehicleColorRelease": "",
  "bindingRemainLedChannels": [
    {
      "color": "2",
      "channelId": "1000166$15$0$0",
      "emptyParkingSpaceContent": "0"
    }
  ],
  "totalParkingSpaceCount": "1000"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data
- prompt	string	Prompt information code: 150106 = actual presence vehicles are greater than configured presence vehicles, 150107 = actual presence vehicles are less than configured presence vehicles

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "prompt": "150106"
  }
}
```

3.11.4.4 Batch Delete Parking Lots

Brief Description

- Batch delete parking lots

Request Method

- DELETE

Request URL

- /ipms/api/v1.1/parking-lot?ids={ids}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
ids	Yes	string	Parking lot ID; multiple IDs are separated with commas

Request Example

```
DELETE /ipms/api/v1.1/parking-lot?ids=1 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8b8a07b9d6944ba1b62b84774b175463
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.11.4.5 Get Details of the Parking Lot

Brief Description

- Get details of the parking lot

Request Method

- GET

Request URL

- /ipms/api/v1.1/parking-lot/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	Parking lot ID

Request Example

```
GET /ipms/api/v1.1/parking-lot/{id} HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ca19ae28474040f192b9fc8cc7b9c69a
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- id	string	Parking lot ID
- orgCode	string	Parking lot organization code
- orgName	string	Parking lot organization name
- parkingLotName	string	Parking lot name
- parentParkingLotId	string	Parent parking lot ID
- parkingLotMode	string	Parking lot mode: 1= access mode, 2= no access mode, default 1
- enableParkingSpaceStatistics	string	Whether to enable parking statistics. 1= yes, 0= no, default 0
- enableParkingOverTimePass	string	Whether to enable parking timeout to allow passage, 1 = yes, 0 = no, default 0
- parkingSpaceStatisticsType	string	Parking statistical mode: 1= residual mode, 2= vehicle detector mode
- totalParkingSpaceCount	string	Total number of parking spaces, up to 10,000

Parameter	Type	Description
- idleParkingSpaceCount	string	Number of empty parking spots
- enableIdleReset	string	Whether to enable residual reset. 1= yes, 0= no
- idleResetType	string	Remaining position reset type, 1 = reset to the total parking space, 2 = reset the remaining position according to the vehicle present, 3 = reset to the specified remaining position, default 1
- idleResetTime	string	Rest reset time. The format is HH:mm:ss
- idleResetToSpaceCount	string	Specify the number of reset bits
- enableForceExitRecordIdleReset	string	Whether to force the vehicle out of the field when the rest bit is reset: 1= yes, 0= no, 0 by default
- enableRecordCheck	string	Whether record verification is enabled: 1=yes, 0=no
- recordCheckIntervalTime	string	Record inspection time interval (the same license plate is recognized again within the configured time and the entry and exit directions are inconsistent, and it is judged as repeated reporting and no processing), range 1-20, unit second
- parkingRuleType	string	Parking rule type: 1=default, exit from the parking lot or parking space is counted as exit; 2=user-defined, the same record is counted as the same record when the license plate enters the parking space again within the user-defined interval
- parkingRuleIntervalTime	string	The interval time configured by the user-defined parking rule (re-entering the parking space in the same license plate interval time, counting the same record), range 1-60, unit second
- parkingSpaceStatisticsRuleType	string	Type of residual statistics rule: 1= Residual for all vehicles, 2= residual for unregistered vehicles, 3= Custom, default 1
- enableFuzzyMatch	string	Whether to enable fuzzy matching: 1= yes, 0= no, 0 by default
- enableFirstPlaceOneMore	string	More than one digit is allowed: 1= yes, 0= no, 0 is the default
- enableFirstPlaceOneLess	string	The first digit is allowed to be less: 1= yes, 0= no, 0 is the default

Parameter	Type	Description
- enableLastPlaceOneMore	string	More bits are allowed at the end: 1= yes, 0= no, 0 is the default
- enableLastPlaceOneLess	string	One bit less is allowed at the end: 1= yes, 0= no, 0 is the default
- fuzzyMatchBitCount	string	Fuzzy matching of error bits is allowed. Only 1 is supported for the time being
- enableOverrideEnterRepeatVehicle	string	Whether the approach record is overwritten: 1= yes, 0= no, default 0
- issueVehicles	string	Whether to deliver to the garage: 1= yes, 0= no
- enterRuleType	string	Entry rules configuration types: 1= registered vehicles, 2= all vehicles, 3= Custom
- enableEnterAutoPassUnRegisterVehicle	string	Whether the temporary car is automatically released: 1= yes, 0= no, default 0
- enterTimeTemplateId	string	Automatic admission time template ID
- enableEnterAutoPassBlockVehicle	string	Whether the blocklist is automatically released: 1= yes, 0= no, 0 is the default
- enableEnterAutoPassUnlicensedVehicle	string	Whether the unlicensed car is automatically released: 1= yes, 0= no, default 1
- enableEmptyParkingSpaceEnterRule	string	Whether to enable rule configuration when the empty parking spot is 0: 1 = Yes, 0 = No
- enableEnterCheckAuth	string	Whether to enable the approach verification switch: 1= yes, 0= no, default 0(Abandoned)
- enableEnterCheckAuthByCard	string	Check whether swipe verification is enabled: 1= yes, 0= no, default 0(Abandoned)
- enableEnterCardAuth	string	Enter to enable swipe card opening: 1= yes, 0= no, default 0(Abandoned)
- exitRuleType	string	Exit rules Configuration types: 1= registered vehicles, 2= all vehicles, 3= Custom
- enableExitAutoPassUnRegisterVehicle	string	Whether the temporary car is automatically released: 1= yes, 0= no, default 0
- exitTimeTemplateId	string	ID of the automatic release time template
- enableExitAutoPassBlockVehicle	string	Whether the blocklist is automatically allowed: 1= yes, 0= no, 0 is the default

Parameter	Type	Description
- enableExitAutoPassUnlicensedVehicle	string	Whether the unlicensed car is automatically released: 1= yes, 0= no, default 1
- enableExitCheckAuth	string	Whether to enable the appearance verification switch: 1= yes, 0= no, 0 is the default(Abandoned)
- enableExitCheckAuthByCard	string	Check whether credit card verification is enabled: 1= yes, 0= no, default 0(Abandoned)
- enableExitCardAuth	string	Activate the swipe card switch: 1= yes, 0= no, default 0(Abandoned)
- enterReleaseRuleCheckType	string	Vehicle enter release rule verification type: 1=do not enable validation, 2=open the gate manually, 3=open the gate directly through validation, 4=open the gate through double validation
- enterReleaseRuleDoubleCheckType	string	Vehicle enter double verification opening type: 1=owner verification (consistent with people and vehicles), 2=body color verification
- enableEnterUnrecognizedVehicleColorRelease	string	When the body color is not recognized during the incoming body color verification, is it released: 1=yes, 0=no
- exitReleaseRuleCheckType	string	Vehicle exit double verification opening type: 1=do not enable validation, 2=open the gate manually, 3=open the gate directly through validation, 4=open the gate through double validation
- exitReleaseRuleDoubleCheckType	string	Double verification and opening type: 1=owner verification (consistent with people and vehicles), 2=body color verification
- enableExitUnrecognizedVehicleColorRelease	string	When the body color is not recognized during the outgoing body color verification, is it released: 1=yes, 0=no
- issueStatus	string	Vehicle library distribution status: 1=being distributed, 2=successfully distributed or no distributed task, 3=partially or completely distributed failed
- bindingDetectors	array	Vehicle detector device code(Abandoned)
- - deviceCode	string	Device code
- - deviceName	string	Device name

Parameter	Type	Description
- - status	string	Online Status: 0=offline, 1=online
- - orgCode	string	Organization code
- - orgName	string	Organization name
- - deviceIp	string	Device IP
- - devicePort	string	Device port
- - deviceManufacturer	string	Device manufacturer
- - deviceCategory	string	Device category
- - deviceType	string	Device type
- - serverCode	string	Code of the server where the device is located
- - serverIp	string	IP of the server where the device is located
- - serverName	string	Name of the server where the device is located
- - parkingSpaceCount	string	Parking space count
- - detectorTypes	array	Vehicle detector type: 1=indoor, 2=outdoor
- - parkingSpaces	array	Parking space
- - - parkingSpaceId	string	Parking space ID
- - - parkingSpaceName	string	Parking space name
- - - detectorType	string	Vehicle detector type: 1=indoor, 2=outdoor
- bindingDetectorSpaces	array	
- - parkingSpaceId	string	Parking space ID
- - parkingSpaceName	string	Parking space name
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - deviceIp	string	Device IP
- - deviceCategory	string	Device category
- - deviceType	string	Device type
- - parkingSpaceGroupId	string	Parking space type ID
- - parkingSpaceGroupName	string	Parking space type name
- - parkingSpaceGroupColor	string	Parking space type color
- - status	string	Online Status: 0=offline, 1=online
- - orgCode	string	Organization code
- - orgName	string	Organization name
- - detectorType	string	Parking space type: 1=indoor parking space detector, 2=outdoor parking space detector

Parameter	Type	Description
- bindingLedPartitionInfos	array	
- - ledPartitionId	string	Screen ID
- - ledPartitionName	string	Screen name
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - deviceIp	string	Device IP
- - deviceCategory	string	Device category
- - deviceType	string	Device type
- - status	string	Online Status: 0=offline, 1=online
- - orgCode	string	Organization code
- - orgName	string	Organization name
- bindingLeds	array	LED device code(Abandoned)
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - status	string	Online Status: 0=offline, 1=online
- - orgCode	string	Organization code
- - orgName	string	Organization name
- - deviceIp	string	Device IP
- - devicePort	string	Device port
- - deviceManufacturer	string	Device manufacturer
- - deviceCategory	string	Device category
- - deviceType	string	Device type
- - serverCode	string	Code of the server where the device is located
- - serverIp	string	IP of the server where the device is located
- - serverName	string	Name of the server where the device is located
- - ledPartitionCount	string	Screen count
- bindingLedPartitions	array	
- - ledPartitionId	string	Screen ID
- - ledDeviceCode	string	Remaining screen device code
- - dataSource	string	Type of data statistics: 1= vehicle entry and exit statistics, 2= vehicle detector statistics(Abandoned)
- - bindingAllDetectors	string	Whether to bind all auto detectors, 1= yes, 0= no (Abandoned)

Parameter	Type	Description
- - bindingType	string	Binding type: 1= parking space, 2= parking lot, default 2
- - parkingSpaces	array	
- - - parkingSpaceId	string	Parking space ID
- - - parkingSpaceName	string	Parking space name
- - - detectorDeviceCode	string	Vehicle detector device code
- - - detectorDeviceName	string	Vehicle detector device name
- - - parkingSpaceGroupId	string	Parking space type ID
- - - parkingSpaceGroupName	string	Parking space type name
- - - parkingSpaceGroupColor	string	Parking space type color
- - - orgCode	string	Organization code
- - - orgName	string	Organization name
- - parkingLots	array	Parking lot
- - - parkingLotId	string	Parking lot ID
- - - parkingLotName	string	Parking lot name
- - parkingSpaceGroups	array	Parking space type
- - - parkingSpaceGroupId	string	Parking space type ID
- - - parkingSpaceGroupName	string	Parking space type name
- - - parkingSpaceGroupColor	string	Parking space type color
- bindingRemainLedChannels	array	(Abandoned)
- - channelId	string	Channel code
- - channelName	string	Channel name
- - status	string	Online Status: 0=offline, 1=online
- - orgCode	string	Organization code
- - orgName	string	Organization name
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - deviceIp	string	Device IP
- - devicePort	string	Device port
- - deviceManufacturer	string	Device manufacturer
- - deviceCategory	string	Device category
- - deviceType	string	Device type
- - serverCode	string	Code of the server where the device is located
- - serverIp	string	IP of the server where the device is located

Parameter	Type	Description
-- serverName	string	Name of the server where the device is located
-- displayColor	string	Display text colors: 1= red, 2= green, 3= yellow
-- noParkingSpaceDisplayContent	string	When there is no empty parking spot, the display content is: 0000, FULL
- parkingSpaceStatisticsRules	array	
-- groupId	string	Enter/exit group ID (not blocklisted)
-- groupName	string	Enter/exit group name
-- statisticsFlag	string	Statistical identification: 1=calculated, 0=not calculated
- positions	array	
-- id	string	Point ID
-- positionName	string	Point name
-- remark	string	Remark
-- passRules	array	Rules of passage
-- -- groupId	string	Enter/exit group ID (not blocklisted)
-- -- groupName	string	Enter/exit group name
-- -- timeTemplateId	string	Time template ID
-- -- timeTemplateName	string	Time template name
-- -- direction	string	In and out direction: 1=in, 2=out
-- -- enableEmptyParkingSpaceEnterRule	string	Whether the rules take effect when the empty parking spot is 0: 1 = Yes, 0 = No
-- issueVehicleRules	array	
-- -- groupId	string	Enter/exit group ID (not blocklisted)
-- -- groupName	string	Enter/exit group name
-- -- direction	string	In and out direction: 1=in, 2=out
-- points	array	
-- -- id	string	Gate ID
-- -- pointName	string	Gate name
-- -- direction	string	Driving direction: 8= in, 9= out
-- -- remark	string	Remark
-- -- refPointId	string	Reference gate ID
-- -- refPointName	string	Reference gate name
-- -- refPositionId	string	Reference gate point ID

Parameter	Type	Description
- - - refPositionName	string	Reference gate point name
- - - refParkingLotId	string	Reference gate parking lot ID
- - - refParkingLotName	string	Reference gate parking lot name
- - - multipleCameras	string	Multi-camera identification: 0=Single camera, =Dual camera
- - - coordinateTime	string	Dual camera collaboration time
- - - pointMode	string	Gate mode: 1= one gate, 2= no gate
- - - bindingIltcChannels	array	
- - - - channelId	string	Channel code
- - - - channelType	string	Channel type: 1 = Main camera, 2 = Auxiliary camera
- - - - channelName	string	Channel name
- - - - status	string	Online Status: 0=offline, 1=online
- - - - orgCode	string	Organization code
- - - - orgName	string	Organization name
- - - - deviceCode	string	Device code
- - - - deviceName	string	Device name
- - - - deviceIp	string	Device IP
- - - - devicePort	string	Device port
- - - - deviceManufacturer	string	Device manufacturer
- - - - deviceCategory	string	Device category
- - - - deviceType	string	Device type
- - - - serverCode	string	Code of the server where the device is located
- - - - serverIp	string	IP of the server where the device is located
- - - - serverName	string	Name of the server where the device is located
- - - bindingVtoChannels	array	
- - - - channelId	string	Channel code
- - - - channelName	string	Channel name
- - - - status	string	Online Status: 0=offline, 1=online
- - - - orgCode	string	Organization code
- - - - orgName	string	Organization name
- - - - deviceCode	string	Device code
- - - - deviceName	string	Device name

Parameter	Type	Description
- - - - deviceId	string	Device IP
- - - - devicePort	string	Device port
- - - - deviceManufacturer	string	Device manufacturer
- - - - deviceCategory	string	Device category
- - - - deviceType	string	Device type
- - - - serverCode	string	Code of the server where the device is located
- - - - serverIp	string	IP of the server where the device is located
- - - - serverName	string	Name of the server where the device is located
- - - bindingCommonLedChannels	array	Common screen(Abandoned)
- - - - channelId	string	Channel code
- - - - channelName	string	Channel name
- - - - status	string	Online Status: 0=offline, 1=online
- - - - orgCode	string	Organization code
- - - - orgName	string	Organization name
- - - - deviceCode	string	Device code
- - - - deviceName	string	Device name
- - - - deviceId	string	Device IP
- - - - devicePort	string	Device port
- - - - deviceManufacturer	string	Device manufacturer
- - - - deviceCategory	string	Device category
- - - - deviceType	string	Device type
- - - - serverCode	string	Code of the server where the device is located
- - - - serverIp	string	IP of the server where the device is located
- - - - serverName	string	Name of the server where the device is located
- - - - displayColor	string	Display text colors: 1= red, 2= green, 3= yellow
- - - - displayRows	string	Display line count
- - - - screenHeight	string	LED screen height
- - - - screenWidth	string	LED screen width
- - - - rollType	string	Content switching mode: 1= scroll, 2= cut screen
- - - - rollSpeed	string	Rolling speed, range: 1-100

Parameter	Type	Description
----- idleContentType	string	When idle, display content types: 0= empty, 1= residual bits, 2= Custom
----- idleUserDefinedContent	string	Displays custom content when idle
----- displayContents	array	
----- id	string	Serial number, starting from 0
----- content	string	Content number: 1= license plate number, 2= entry time,...
----- userDefined	string	User-defined content or not: 1= yes, 0= no, 0 by default

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": "1",
    "parkingLotName": "test",
    "enableParkingSpaceStatistics": "1",
    "enableParkingOverTimePass": "0",
    "parkingSpaceStatisticsType": "2",
    "totalParkingSpaceCount": "0",
    "idleParkingSpaceCount": "0",
    "parkingSpaceStatisticsRuleType": "1",
    "enableFuzzyMatch": "0",
    "enableFirstPlaceOneMore": "0",
    "enableFirstPlaceOneLess": "0",
    "enableLastPlaceOneMore": "0",
    "enableLastPlaceOneLess": "0",
    "fuzzyMatchBitCount": "0",
    "enableOverrideEnterRepeatVehicle": "0",
    "issueVehicles": "0",
    "issueStatus": "2",
    "enterRuleType": "1",
    "enableEnterAutoPassUnRegisterVehicle": "0",
    "enterTimeTemplateId": "1",
    "enableEnterAutoPassBlockVehicle": "0",
    "enableEnterAutoPassUnlicensedVehicle": "0",
    "enableEmptyParkingSpaceEnterRule": "0",
    "enableEnterCheckAuth": "0",
    "enableEnterCheckAuthByCard": "0",
    "enableEnterCardAuth": "0",
    "exitRuleType": "1",
    "enableExitAutoPassUnRegisterVehicle": "0",
    "exitTimeTemplateId": "1",
  }
}
```

```
"enableExitAutoPassBlockVehicle": "0",
"enableExitAutoPassUnlicensedVehicle": "0",
"enableExitCheckAuth": "0",
"enableExitCheckAuthByCard": "0",
"enableExitCardAuth": "0",
"parentParkingLotId": "0",
"parkingLotMode": "1",
"enableIdleReset": "0",
"idleResetTime": "00:00:00",
"enableForceExitRecordIdleReset": "0",
"enableRecordCheck": "1",
"recordCheckIntervalTime": "10",
"parkingRuleType": "1",
"parkingRuleIntervalTime": "1",
"idleResetType": "1",
"idleResetToSpaceCount": "0",
"orgCode": "001",
"orgName": "Current Site",
"bindingRemainLedChannels": [],
"enterReleaseRuleCheckType": "1",
"enterReleaseRuleDoubleCheckType": "1",
"enableEnterUnrecognizedVehicleColorRelease": "1",
"exitReleaseRuleCheckType": "1",
"exitReleaseRuleDoubleCheckType": "1",
"enableExitUnrecognizedVehicleColorRelease": "1",
"positions": [
  {
    "id": "44",
    "positionName": "aaaa",
    "parkingLotId": "68",
    "remark": "",
    "passRules": [],
    "issueVehicleRules": [],
    "points": [
      {
        "id": "62",
        "pointName": "aa",
        "positionId": "44",
        "multipleCameras": "1",
        "coordinateTime": "3",
        "direction": "8",
        "remark": "",
        "refPointId": "0",
        "refPointName": "",
        "refPositionId": "",
        "refPositionName": "",
        "refParkingLotId": "",
        "refParkingLotName": "",
        "pointMode": "1",
```


- PUT

Request URL

- /ipms/api/v1.0/entrance/channel/remote-open-sluice/{channelId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
channelId	Yes	string	Channel code

- Body Parameters

Parameter	Required	Type	Description
correctCarNo	No	string	When the identified license plate is found to be incorrect, correct the license plate and the platform will synchronously modify the entry or exit records
correctTime	No	string	Time correction (second)
forceRecapture	No	string	Is it mandatory to retake the image when the identified license plate is found to be incorrect: 1=Yes, 0=No, default to 0
parkingSpaceStatisticsType	No	string	Remainder calculation rules: 1=calculated, 2=not calculated, 3=calculated according to configuration rules, default to 3
saveRecord	No	string	Save entry/exit records: 1=Yes, 0=No, default to 1

Request Example

```
PUT /ipms/api/v1.0/entrance/channel/remote-open-sluice/1000024$1$0$1 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 8adeb2e9bfba4c09b2a3d6857206352c

{
  "correctCarNo": "",
```

```
"correctTime": "0",  
"forceRecapture": "0",  
"parkingSpaceStatisticsType": "1",  
"saveRecord": "0"  
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{  
  "code": 1000,  
  "desc": "Success",  
  "data": null  
}
```

3.12 Face

3.12.1 Face Comparison Group

3.12.1.1 Add a Face Comparison Group

Brief Description

- Add face comparison group

Request Method

- POST

Request URL

- /brms/api/v1.1/face/repository

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
repositoryName	Yes	string	Face comparison group, excluding special character < >, with length no more than 50
color	Yes	string	Face comparison group color, 1 = gray, 2 = orange, 3 = yellow, 4 = green, 5 = cyan, 6 = blue, 7 = purple, 8 = red
memo	No	string	Face comparison group remarks, without special character < >, length no more than 255
roleIds	Yes	array	Role ID
source	No	string	Source: 1 = platform, 2 = education plug-in, default 1
surveyType	No	string	Type of personnel control library: 1 = general control, 2 = mobile control, default 1

Request Example

```
POST /brms/api/v1.1/face/repository HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
Accept-Language: en
```

```
{
  "repositoryName": "face_test",
  "color": "1",
  "memo": "test",
  "roleIds": [
    "1"
  ],
  "source": "1",
  "surveyType": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information

Parameter	Type	Description
- id	long	Face comparison group ID

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "id": 12345
  }
}
```

3.12.1.2 Update a Face Comparison Group

Brief Description

- Update the information about the face comparison group

Request Method

- PUT

Request URL

- /brms/api/v1.1/face/repository/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	ID of face comparison group. Required.

- Body Parameters

Parameter	Required	Type	Description
repositoryName	Yes	string	Face comparison group, excluding special character < >, with length no more than 50
color	Yes	string	Face comparison group color: 1 = gray, 2 = orange, 3 = yellow, 4 = green, 5 = cyan, 6 = blue, 7 = purple, 8 = red
memo	No	string	Face comparison group remarks, without special character < >, length no more than 255

Parameter	Required	Type	Description
roleIds	Yes	array	Role ID, required
source	No	string	Source: 1 = platform, 2 = education plug-in, default 1

Request Example

```
PUT /brms/api/v1.1/face/repository/11 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
Accept-Language: en

{
  "color": "2",
  "repositoryName": "face001",
  "memo": "face detection",
  "roleIds": ["1"],
  "source": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return Message

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.12.1.3 Delete a Face Comparison Group

Brief Description

- Delete the face comparison group

Request Method

- DELETE

Request URL

- /brms/api/v1.1/face/repository/{id}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
id	Yes	string	ID of face comparison group. Required.

Request Example

```
DELETE /brms/api/v1.1/face/repository/3 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
Accept-Language: en
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Returned information
- - failedDevices	array	List of devices that failed to be deleted
- - - deviceCode	string	Device code
- - - deviceName	string	Device name
- - failedChannels	array	List of channels that failed to be deleted
- - - channelId	string	Channel code
- - - channelName	string	Channel name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "failedDevices": [
      {
        "deviceCode": "1000001",
        "deviceName": "Face IVSS"
      }
    ]
  }
}
```

```

    ],
    "failedChannels": [
      {
        "channelId": "1000043&1&0&0",
        "channelName": "EVS"
      }
    ]
  }
}

```

3.12.1.4 Get Face Comparison Group List in pages

Brief Description

- Get face comparison group List in pages

Request Method

- GET

Request URL

- /brms/api/v1.1/face/application/control/repository/page?page={page}&pageSize={pageSize}&orderType={orderType}&orderDirection={orderDirection}&keyword={keyword}&surveyType={surveyType}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orderType	No	string	Sorting type
orderDirection	No	string	Sorting direction: 0 = Ascending sequence; 1 = Descending sequence
keyword	No	string	Keywords
source	No	string	Source: 1 = Platform, 2 = Education plug-in
surveyType	No	string	Type of personnel control library: 1 = general control,

Parameter	Required	Type	Description
			2 = mobile control, default 1

Request Example

```
GET /brms/api/v1.1/face/application/control/repository/page?orderType=0&orderDirection=0&keyword=&source=&page=1&pageSize=20&surveyType= HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 703301f9ca3e422da5cfc49ea86a563a
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Returned information
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - repositoryId	string	Face watch list ID
- - color	string	Repository name, 1=Gray, 2=Red, 3=Orange, 4=Yellow, 5=Green, 6=Cyan, 7=Blue, 8=Purple
- - colorName	string	Face comparison group color name
- - repositoryName	string	Face comparison group, excluding special character < >, with length no more than 50
- - memo	string	Face comparison group remarks, without special character < >, length no more than 255
- - source	string	Source: 1 = platform, 2 = education plug-in
- - createTime	string	Create time, timestamp, in seconds

Parameter	Type	Description
- - updateTime	string	Update time,timestamp, in seconds
- - facePersonCount	string	Face count,50
- - isFailed	string	Whether there is a person or channel that fails to watch,0=No, 1=Yes
- - thumbnail	string	Thumbnail
- - surveyType	string	Type of personnel control library: 1 = general control, 2 = mobile control
- - process	string	Release progress, for example: 80
- - channelInfos	array	Channel infos
- - - cameraType	string	Camera type,1= Bullet Camera,2= PTZ Camera,3= Dome Camera
- - - channelId	string	Channel code
- - - channelName	string	Channel name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "repositoryId": "1",
        "repositoryName": "7788",
        "color": "7",
        "colorName": "",
        "memo": "11223",
        "source": "1",
        "createTime": "1594106953",
        "updateTime": "1594106953",
        "facePersonCount": "15",
        "isFailed": "1",
        "thumbnail": null,
        "surveyType": "1",
        "process": "80",
        "channelInfos": [
          {
            "cameraType": "1",
            "channelName": "192.168.1.1",
            "channelId": "1000747$1$0$0"
          }
        ]
      }
    ]
  }
}
```

```
}
  ]
}
}
```

3.12.1.5 Issuing the Face Comparison Group

Brief Description

- Issue the face comparison group.

Request Method

- POST

Request URL

- /brms/api/v1.1/face/application/issue/repository/{repositoryId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
repositoryId	Yes	string	The ID of the face database.

- Body Parameters

Parameter	Required	Type	Description
operateType	Yes	string	The issue status: 0 = Abnormal, 1 = Normal.
deviceCodes	No	array	The device code, required when channelIds is blank.
channelIds	No	array	The channel code, required when deviceCodes is blank.

Request example

```
POST /brms/api/v1.1/face/application/issue/repository/22 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075

{
  "operateType": "1",
```

```

    "deviceCodes": [
      "1000051"
    ],
    "channelIds": [
      "1000051$1$0$0"
    ]
  }

```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	Return Message

Return Example

```

{
  "code": 1000,
  "desc": "Success",
  "data": null
}

```

3.12.1.6 Cancel the Issue of the Face Comparison Group

Brief Description

- Cancel the issue of the face comparison group.

Request Method

- POST

Request URL

- /brms/api/v1.1/face/application/issue/repository/{repositoryId}/cancel

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
repositoryId	Yes	string	The ID of the face database.

- Body Parameters

Parameter	Required	Type	Description
deviceCodes	No	array	The device code, required when channelIds is blank.
channelIds	No	array	The channel code, required when deviceCodes is blank.

Request Example

```
POST /brms/api/v1.1/face/application/issue/repository/22/cancel HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075

{
  "deviceCodes": ["1000051"],
  "channelIds": ["1000043$1$0$0"]
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	The returned information.
- failedResources	array	None
- - deviceCode	string	The device code.
- - deviceName	string	The device name.
- - channelId	string	The channel code.
- - channelName	string	The channel name.
- - resourceType	string	The resource type: 0 = Device, 1 = Channel.
- - errorCode	string	The error code.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "failedResources": [
      {
        "deviceCode": "1000043",
```



```

        "deviceName": "EVS",
        "channelId": "1000043$1$0$0",
        "channelName": "IPC",
        "resourceType": "1",
        "errorCode": "110708"
    }
}
]
}
}

```

3.12.1.7 Arming the Face Comparison Group

Brief Description

- Arm the face comparison group.

Request Method

- POST

Request URL

- /brms/api/v1.1/face/application/control/repository/{repositoryId}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
repositoryId	Yes	string	The ID of the face database.

- Body Parameters

Parameter	Required	Type	Description
channels	Yes	array	The armed channel.
- - channelId	Yes	string	The channel code.
- - similarity	Yes	string	The similarity, value range [1-100]

Request Example

```

POST /brms/api/v1.1/face/application/control/repository/22 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075

```

```
{
  "channels":
  [
    {
      "channelId": "1000051$1$0$0",
      "similarity": "80"
    }
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	Return Message

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.12.1.8 Canceling the Arming of the Face Comparison Group

Brief Description

- Cancel the arming of the face comparison group.

Request Method

- POST

Request URL

- /brms/api/v1.1/face/application/control/repository/{repositoryId}/cancel

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
repositoryId	Yes	String	The ID of the face database.

- Body Parameters

Parameter	Required	Type	Description
channelId	Yes	array	The channel code.

Request Example

```
POST /brms/api/v1.1/face/application/control/repository/22/cancel HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
```

```
{
  "channelIds": [
    "1000043$1$0$0"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	Int	The error code.
desc	string	The result description.
data	object	The returned information.
- failedChannels	array	None
- - deviceCode	string	The device code.
- - deviceName	string	The device name.
- - channelId	string	The channel code.
- - channelName	string	The channel name.
- - errorCode	string	The error code.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "failedChannels":
    [
      {
        "deviceCode": "1000043",
        "deviceName": "IPC",
        "channelId": "1000043$1$0$0",

```

```

        "channelName": "IPC",
        "errorCode": "110708"
      }
    ]
  }
}

```

3.12.2 Face Search

3.12.2.1 Starting Search the Face Snapshot Records Asynchronously by Searching by Image

Brief Description

- Start to search by image, and search the face snapshot records asynchronously. For the searching progress, see MQ messages: [4.7.3 Notification of Face Search by Image Result Progress].

Request Method

- POST

Request URL

- /brms/api/v1.1/face/detection/record/search-by-picture/start

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
beginTime	Yes	string	Start time, timestamp, in seconds
endTime	Yes	string	End time, timestamp, in seconds
similarity	Yes	string	The similarity, value range [1-100]
channelIds	Yes	array	Channel code
recordSource	No	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
taskIds	No	array	Analysis task ID
faceImageData	Yes	string	Face image (base64) in the .jpg format
analyseMode	Yes	string	Comparison mode: 1 = device comparison, 2 = central intelligence comparison
searchCount	No	string	Total number of searches, value range [1-1000]

Request Example

```
POST /brms/api/v1.1/face/detection/record/search-by-picture/start HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
Accept-Language: en

{
  "beginTime": "1609257600",
  "endTime": "1609343999",
  "similarity": "92",
  "channelIds": [
    "1000000$1$0$0",
    "1000000$1$0$1",
    "1000000$1$0$6",
    "1000000$1$0$7"
  ],
  "recordSource": "0",
  "taskIds": [
    "1"
  ],
  "faceImageData": "iVBORw0KGgoAAAANSUheEU...TeHTY5gAAAABJR5ErkJggg==",
  "analyseMode": "2",
  "searchCount": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- session	string	Session mark

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "session": "2acd14e7290048d88611101e779543fb"
  }
}
```

3.12.2.2 Stopping Asynchronously Querying Face Capture Records in Reverse Image Search

Brief Description

- Stop asynchronously querying face capture records in reverse image search.

Request Method

- POST

Request URL

- / brms /api/v1.1/face/detection/record/search-by-picture/stop

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
session	Yes	string	The session ID.

Request Example

```
POST /brms/api/v1.1/face/detection/record/search-by-picture/stop HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "session": "2acd14e7290048d88611101e779543fb"
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	Return Message

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.12.2.3 Search the Face Snapshot Records in Pages by Searching by Image

Brief Description

- Search face snapshot records in pages through Search by Picture

Request Method

- POST

Request URL

- /brms/api/v1.1/face/detection/record/search-by-picture/{session}/page

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
session	Yes	string	Session mark

- Body Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orderType	Yes	string	Sorting type, 1 = Time; 2 = Similarity
orderDirection	Yes	string	Sorting direction, 0 = Ascending order, 1 = Descending order
beginAge	No	string	Minimum age, value range [-1,255]
endAge	No	string	Maximum age, value range [-1, integer maximum]
gender	No	string	Gender: 0 = unspecified, 1 = male, 2 = female
mask	No	string	Mask: 1 = unknown, not recognized, 2 = no mask, 3 = mask, empty = all

Parameter	Required	Type	Description
beard	No	string	Beard: 1 = Unknown, Unrecognized, 2 = No Beard, 3 = Beard, Empty = All
glasses	No	string	Glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses, Empty = All
emotion	No	string	Emotion: -1 = unknown, 0 = smiling, 1 = angry, 2 = sad, 3 = disgusted, 4 = scared, 5 = surprised, 6 = normal, 7 = laughing, 8 = happy, 9 = confused, 10 = screaming, empty = all
recordType	No	string	Record type: 1 = capture record, 2 = recognition record, blank = all

Request Example

```

POST /brms/api/v1.1/face/detection/record/search-by-picture/2acd14e729543fb/page
HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "page": "1",
  "pageSize": "20",
  "orderType": "1",
  "orderDirection": "1",
  "beginAge": "",
  "endAge": "",
  "gender": null,
  "mask": null,
  "beard": null,
  "glasses": null,
  "emotion": null,
  "recordType": null
}

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	Record ID
- - channelId	string	Channel code
- - channelName	string	Channel name
- - recordSource	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
- - facelImageUrl	string	Face snapshot
- - pictureUrl	string	Scenario picture
- - age	string	Age
- - gender	string	Gender: 0 = Unidentified, 1 = Male, 2 = Female
- - fringe	string	Fringe: 0 = No, 1 = Yes
- - eye	string	Eyes: 1 = Unidentified, 2 = Closed, 3 = Open
- - mouth	string	Mouth: 1 = Unidentified, 2 = Closed, 3 = Open
- - mask	string	Mask: 0 = Unknown (SDK), 1 = Unrecognized; 2 = No mask, 3 = Wear mask
- - beard	string	Beard: 0 = Unknown (SDK), 1 = Unrecognized, 2 = No beard, 3 = With beard
- - glasses	string	Glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
- - emotion	string	Expressions: 0 = Smile, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Scared, 5 = Surprised, 6 = Normal, 7 = Laugh, 8 = Happy, 9 = Confused, 10 = Scream
- - captureTime	string	Capture time, timestamp, in seconds
- - similarity	string	The Similarity, Value range [1-100]
- - personId	string	No person is recognized if the person ID that the face snapshot matches is null
- - personName	string	Person name

Parameter	Type	Description
-- personSimilarity	string	No similarity is recognized if the similarity between the face snapshot and the recognized face from the face database, Value range [1-100]
-- dataSource	string	Data source: 1=device, 2=platform
-- repositoryName	string	Name of facial comparison group
-- eventCode	string	Unique identification of an event used to determine the original event information of the captured image
-- gpsX	string	Longitude
-- gpsY	string	Latitude
-- mapId	string	Map id

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-140",
    "pageData": [
      {
        "id": "2021040220000523001",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "recordSource": "0",
        "faceImageUrl": "https://192.168.1.1:9901/e3e8bfda-656_373.jpg",
        "pictureUrl": "https://192.168.1.1:9901/e3e8bfda-65c_376.jpg",
        "age": "35",
        "gender": "1",
        "fringe": "0",
        "eye": "3",
        "mouth": "2",
        "mask": "2",
        "beard": "2",
        "glasses": "0",
        "emotion": "6",
        "captureTime": "1612334628",
        "similarity": "34.74",
        "personId": "1234",
        "personName": "zhangsan",
        "personSimilarity": "95",
        "dataSource": "2",
        "repositoryName": "g1",
        "eventCode": "23aeaq43",
      }
    ]
  }
}
```

```

        "gpsX": "23.454656",
        "gpsY": "56.23232",
        "mapId": "2"
      }
    ]
  }
}

```

3.12.2.4 Face Snapshot Records — Search by Image (Download Image)

Brief Description

- Face snapshot record—Search by picture (download picture). The picture URL obtained from the original IVSS device is a relative path on the device, and you should manually download the picture (up to 10 pictures in each request) to store it into the OSS. Download is not required subsequently.

Request Method

- POST

Request URL

- /brms/api/v1.1/face/detection/record/device/picture/download

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
session	Yes	string	Search by picture, session mark
deviceCode	Yes	string	Device number
urls	Yes	object	None
- id	Yes	string	Record ID
- type	Yes	string	Record picture type: 1 = Face picture; 2 = Scenario picture
- url	Yes	string	Record picture URL

Request Example

```

POST /brms/api/v1.1/face/detection/record/device/picture/download HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
Accept-Language: en

```

```
{
  "session": "30456737dd3845cf89929c1968b52265",
  "deviceCode": "1000000",
  "urls": [
    {
      "id": "2",
      "type": "1",
      "url": "/picid/2789283.jpg"
    }
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return information
- results	object	None
- - id	string	Record ID
- - type	string	Record picture type: 1 = Face picture; 2 = Scenario picture
- - url	string	Record picture URL.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "id": "2",
        "type": "1",
        "url": "https://192.168.1.1:9901/cfd4...278_11787969.jpg"
      }
    ]
  }
}
```

3.12.2.5 Search the Face Snapshot Records in Pages Based on Features

Brief Description

- Search face snapshot records in pages based on features.

Request Method

- POST

Request URL

- /iams/api/v1.1/face/detection/record/search-by-feature/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
channelIds	No	array	Channel code
recordSource	No	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
taskIds	No	array	Analysis task ID
beginTime	Yes	string	Start time, timestamp, in seconds
endTime	Yes	string	End time, timestamp, in seconds
beginAge	No	string	Minimum age, Value range [1-100]
endAge	No	string	Maximum age, Value range [-1, integer maximum]
gender	No	string	Gender: 1 = Male, 2 = Female
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
currentPage	Yes	string	Current page number, Greater than or equal to 1
orderType	No	string	Sort type: 1 = Time
orderDirection	Yes	string	Sorting direction: 0 = Ascending order, 1 = Descending order

Parameter	Required	Type	Description
splitTime	No	string	Time of the first record when paging up and time of the last record when paging down, timestamp, in seconds
splitId	No	string	ID of the first record when paging up and ID of the last record when paging down
glasses	No	string	Glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses, empty=all
mask	No	string	Mask: 1=unknown, not recognized, 2=no mask, 3=wearing mask, empty=all
beard	No	string	Beard: 1=unknown, unrecognized, 2=no beard, 3=bearded, empty=all
recordType	No	string	Record types: 1=capture record, 2=identification record, empty=all

Request Example

```
POST /iams/api/v1.1/face/detection/record/search-by-feature/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "channelIds": [
    "1000000$1$0$0"
  ],
  "recordSource": "0",
  "taskIds": [
    "1"
  ],
  "beginTime": "1614219854",
  "endTime": "1614637411",
  "beginAge": "12",
  "endAge": "50",
  "gender": ""
```

```

    "page": "1",
    "pageSize": "20",
    "currentPage": "1",
    "orderType": "1",
    "orderDirection": "1",
    "splitTime": "",
    "splitId": "",
    "glasses": "",
    "mask": "",
    "beard": "",
    "recordType": "1"
  }

```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	Record ID
- - channelId	string	Channel code
- - channelName	string	Channel name
- - recordSource	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
- - faceImageUrl	string	Add the token when accessing the face snapshot URL, such as: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - pictureUrl	string	Add the token when accessing the scenario image URL, such as: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - age	string	Age

Parameter	Type	Description
-- gender	string	Gender: 0 = Unidentified, 1 = Male, 2 = Female
-- fringe	string	Fringe: 0 = No, 1 = Yes
-- eye	string	Eyes: 1 = Unidentified, 2 = Closed, 3 = Open
-- mouth	string	Mouth: 1 = Unidentified, 2 = Closed, 3 = Open
-- mask	string	Mask: 0 = Unknown, 1 = Unidentified, 2 = No mask, 3 = Wear mask
-- beard	string	Beard: 0 = Unknown, 1 = Unidentified, 2 = No beard, 3 = With beard
-- glasses	string	Glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
-- emotion	string	Expressions: 0 = Smile, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Scared, 5 = Surprised, 6 = Normal, 7 = Laugh, 8 = Happy, 9 = Confused, 10 = Scream
-- captureTime	string	Capture time, timestamp, in seconds
-- personId	string	No person is recognized if the person ID that the face snapshot matches is null
-- personName	string	Person name
-- personSimilarity	string	No similarity is recognized if the similarity between the face snapshot and the recognized face from the face database
-- repositoryName	string	Face comparison group name

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.

Parameter	Type	Description
- pageData	object	None
- - id	string	Record ID
- - channelId	string	Channel code
- - channelName	string	Channel name
- - faceImageUrl	string	Add the token when accessing the face snapshot URL, such as: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - pictureUrl	string	Add the token when accessing the scenario image URL, such as: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - age	string	Age
- - gender	string	Gender: 0 = Unidentified, 1 = Male, 2 = Female
- - fringe	string	Fringe: 0 = No, 1 = Yes
- - eye	string	Eyes: 1 = Unidentified, 2 = Closed, 3 = Open
- - mouth	string	Mouth: 1 = Unidentified, 2 = Closed, 3 = Open
- - mask	string	Mask: 0 = Unknown, 1 = Unidentified, 2 = No mask, 3 = Wear mask
- - beard	string	Beard: 0 = Unknown, 1 = Unidentified, 2 = No beard, 3 = With beard
- - glasses	string	Glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
- - emotion	string	Expressions: 0 = Smile, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Scared, 5 = Surprised, 6 = Normal, 7 = Laugh, 8 = Happy, 9 = Confused, 10 = Scream
- - captureTime	string	Capture time, timestamp, in seconds
- - personId	string	No person is recognized if the person ID that the face snapshot matches is null
- - personName	string	Person name
- - personSimilarity	string	No similarity is recognized if the similarity between the face snapshot and the recognized face from the face database, value range [1-100]
- - repositoryName	string	Name of facial comparison group

Parameter	Type	Description
-- gpsX	string	Latitude
-- gpsY	string	Longitude
-- mapId	string	Map id

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": -140,
    "pageData": [
      {
        "id": "2021040220000254123",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "recordSource": "0",
        "faceImageUrl": "https://192.168.1.1:9901/e3e8bfda-656_373.jpg",
        "pictureUrl": "https://192.168.1.1:9901/e3e8bfda-65c_376.jpg",
        "age": "35",
        "gender": "1",
        "fringe": "0",
        "eye": "3",
        "mouth": "2",
        "mask": "2",
        "beard": "2",
        "glasses": "0",
        "emotion": "6",
        "captureTime": "1612334628",
        "personId": "1234",
        "personName": "zhangsan",
        "personSimilarity": "95",
        "repositoryName": "Lib 1"
      },
      {
        "id": "2021040220000254156",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "recordSource": "0",
        "faceImageUrl": "https://192.168.1.1:9901/e3e8bfda-65c-471.jpg",
        "pictureUrl": "https://192.168.1.1:9901/e3e8bfda-65c_425.jpg",
        "age": "30",
        "gender": "1",
        "fringe": "0",
        "eye": "3",
        "mouth": "3",

```

```

        "mask": "2",
        "beard": "2",
        "glasses": "0",
        "emotion": "8",
        "captureTime": "1612334594",
        "personId": null,
        "personName": "",
        "personSimilarity": null,
        "repositoryName": "g1",
        "gpsX": "23.232323",
        "gpsY": "45.454545",
        "mapId": "12"
    }
}
}

```

3.12.2.6 Search the Face Recognition Records in Pages Based on Features

Brief Description

- Search face recognition record in pages based on features

Request Method

- POST

Request URL

- /iams/api/v1.1/face/recognition/record/search-by-feature/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
channelIds	No	array	Channel code
beginTime	Yes	string	Start time, timestamp, in seconds
endTime	Yes	string	End time, timestamp, in seconds
personId	No	string	Person ID
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647

Parameter	Required	Type	Description
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
currentPage	Yes	string	Current page number, greater than or equal to 1
splitTime	No	string	Time of the first record when paging up and time of the last record when paging down, timestamp, in seconds
splitId	No	string	ID of the first record when paging up and ID of the last record when paging down

Request Example

```
POST /iams/api/v1.1/face/recognition/record/search-by-feature/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "channelIds": [
    "1000000$1$0$0"
  ],
  "beginTime": "1614219854",
  "endTime": "1614637411",
  "personId": "",
  "page": "2",
  "pageSize": "20",
  "currentPage": "1",
  "splitTime": "1617297162",
  "splitId": "2021040220000400001"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of

Parameter	Type	Description
		pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	Record ID
- - personId	string	Person ID
- - personName	string	Person name in face database
- - certNo	string	Person ID number in face database
- - personGender	string	Gender of persons in face database
- - personBirthday	string	Date of birth of persons in face database
- - personNationality	string	Country or region of persons in face database
- - memo	string	Description
- - repositoryId	string	Face database ID
- - repositoryName	string	Face database name
- - channelId	string	Channel code
- - channelName	string	Channel name
- - captureTime	string	Capture time, timestamp, in seconds
- - personFacelImageUrl	string	Face picture URL of persons in face database. Append a token when accessing the URL. For example: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - facelImageUrl	string	Add the token when accessing the face snapshot URL, such as: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - pictureUrl	string	Add the token when accessing the scenario image URL, such as: https://192.168.1.1:9901/...e3a-65c-433.jpg?token=12837734
- - similarity	string	Similarity
- - age	string	Age
- - gender	string	Gender: 0 = Unidentified, 1 = Male, 2 = Female
- - fringe	string	Fringe: 0 = No, 1 = Yes

Parameter	Type	Description
-- eye	string	Eyes: 1 = Unidentified, 2 = Closed, 3 = Open
-- mouth	string	Mouth: 1 = Unidentified, 2 = Closed, 3 = Open
-- mask	string	Mask: 0 = Unknown (SDK); 1 = Unrecognized; 2 = No mask; 3 = Wear mask
-- beard	string	Beard: 0 = Unknown (SDK); 1 = Unrecognized; 2 = No beard; 3 = With beard
-- glasses	string	Glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
-- emotion	string	Expressions: 0 = Smile, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Scared, 5 = Surprised, 6 = Normal, 7 = Laugh, 8 = Happy, 9 = Confused, 10 = Scream

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-10",
    "pageData": [
      {
        "id": "2021040220000433001",
        "personId": "12564",
        "personName": "zhangsan",
        "certNo": "56666",
        "personGender": "1",
        "personBirthday": "2000-12-15",
        "personNationality": "china",
        "memo": "test",
        "repositoryId": "1",
        "repositoryName": "face001",
        "channelId": "1000000$1$0$0",
        "channelName": "t002",
        "captureTime": "1614474762",
        "personFacelImageUrl": "https://192.168.1.1:9901/e3e8b...13846.jpg",
        "facelImageUrl": "https://192.168.1.1:9901/e3e8b...3744.jpg",
        "pictureUrl": "https://192.168.1.1:9901/e3e8b...97779.jpg",
        "similarity": "50",

```

```

    "age": "25",
    "gender": "1",
    "fringe": "0",
    "eye": "3",
    "mouth": "2",
    "mask": "2",
    "beard": "2",
    "glasses": "0",
    "emotion": "0"
  }
}
}

```

3.13 Metadata

3.13.1 Starting Asynchronously Querying Body Capture Records in Reverse Image Search

Brief Description

- Start asynchronously querying body capture records in reverse image search. Obtain the query progress from MQ. For details, see [4.8.4 Notifying the Progress of Searching for Human Snapshots by Images].

Request Method

- POST

Request URL

- /brms/api/v1.1/video-analyse/human/record/search-by-picture/start

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
startTime	Yes	string	The start time, in seconds.
endTime	Yes	string	The end time, in seconds.
similarity	Yes	string	The similarity.
channelIds	No	array	The channel code. The channels must belong to the same device.
imageData	Yes	string	The Base64-encoded face image, which is in the JPG format.

Parameter	Required	Type	Description
analyseMode	No	string	Comparison mode: 1 = equipment comparison, 2 = central intelligent comparison
searchCount	No	string	Total number of queries

Request Example

```
POST /brms/api/v1.1/video-analyse/human/record/search-by-picture/start HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075

{
  "startTime": "1609257600",
  "endTime": "1609343999",
  "similarity": "92",
  "analyseMode": "1",
  "channelIds": [
    "1000000$1$0$0",
    "1000000$1$0$1",
    "1000000$1$0$6",
    "1000000$1$0$7"
  ],
  "imageData": "iVBORw0KGgoAAAANSUHEU...TeHTY5gAAAABJRU5ErkJggg==",
  "searchCount": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- session	string	The session ID.
- searchingDevices	array	Searching for devices
- deviceCode	string	Device code
- deviceName	string	Device name

Return Example

```
{
  "code": 1000,
```



```
"desc": "Success",
"data": {
  "session": "2acd14e7290048d88611101e779543fb",
  "searchingDevices": [
    {
      "deviceCode": "1000001",
      "deviceName": "device1"
    }
  ]
}
```

3.13.2 Stopping Asynchronously Querying Body Capture Records in Reverse Image Search

Brief Description

- Stop asynchronously querying body capture records in reverse image search.

Request Method

- POST

Request URL

- /brms/api/v1.1/video-analyse/human/record/search-by-picture/stop

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
session	Yes	string	The session ID.

Request Example

```
POST /brms/api/v1.1/video-analyse/human/record/search-by-picture/stop HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "session": "2acd14e7290048d88611101e779543fb"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.13.3 Query Body Capture Records in Reverse Image Search by Page

Brief Description

- Query body capture records in reverse image search by page.

Request Method

- POST

Request URL

- /brms/api/v1.1/video-analyse/human/record/search-by-picture/{session}/page

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
session	Yes	string	The session ID, 32-Bit string

- Body Parameters

Parameter	Required	Type	Description
gender	No	string	Gender, 0 = unknown, 1 = male, 2 = female, empty = all
coat	No	string	Upper body clothes, 0 = unknown, 1 = long sleeve, 2 = short sleeve, 3 = sleeveless, empty = all

Parameter	Required	Type	Description
coatColors	No	array	Color of upper body clothes, 0 = unknown, 1 = white, 2 = orange, 3 = pink, 4 = black, 5 = red, 6 = yellow, 7 = gray, 8 = blue, 9 = green, 10 = purple, 11 = brown, 12 = silver, 13 = gold, 14 = green, empty = all
trouser	No	string	Underwear, 0 = unknown, 1 = pants, 2 = shorts, 3 = skirt, empty = all
trouserColors	No	array	Undercoat color, 0 = unknown, 1 = white, 2 = orange, 3 = pink, 4 = black, 5 = red, 6 = yellow, 7 = gray, 8 = blue, 9 = green, 10 = purple, 11 = brown, 12 = silver, 13 = gold, 14 = green, empty = all
hat	No	string	Hat, 0 = unknown, 1 = no, 2 = bands, empty = all
hatType	No	string	Hat type, 0 = unknown, 1 = normal hat, 2 = helmet, 3 = hard hat, 10 = no hat, empty = all
bag	No	string	Package, 0 = unknown, 1 = no package, 2 = with package, empty = all
bagType	No	string	Bag type, 0 = unknown, 1 = handbag, 2 = shoulder bag, 3 = backpack, 4 = trolley case, 5 = Fanny pack, 6 = no bag, empty = all
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orderType	No	string	The sort type: 1 = Time, 2 = Similarity, default 1
orderDirection	No	string	The sort order: 0 = Ascending order, 1 = Descending order, default 1

Request Example

```
POST /brms/api/v1.1/video-analyse/human/record/search-by-picture/2D693809559A4B86b4b31a01d6689576/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "gender": "1",
  "coat": "",
  "coatColors": [],
  "trouser": "",
  "trouserColors": [],
  "hat": "1",
  "hatType": "",
  "bag": "1",
  "bagType": "",
  "orderType": "1",
  "orderDirection": "1",
  "page": "1",
  "pageSize": "20"
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	The record ID.
- - channelId	string	The channel code.
- - channelName	string	The channel name.
- - humanImageUrl	string	The body image URL.
- - pictureUrl	string	The scene image URL.
- - facelImageUrl	string	The face image URL.
- - gender	string	The gender: 0 = Unrecognized, 1 = Male, 2 = Female.
- - coat	string	The tops: 0 = Unknown, 1 = Long sleeve, 2 = Short sleeve.
- - coatColor	string	The color of the tops: 0 = Unknown, 1 = White, 2 = Orange, 3 = Pink, 4 = Black, 5 =

Parameter	Type	Description
		Red, 6 = Yellow, 7 = Gray, 8 = Blue, 9 = Green, 10 = Purple, 11 = Brown.
- - coatColorName	string	The internationalized color of the tops.
- - trouser	string	The bottoms: 0 = Unknown, 1 = Pants, 2 = Shorts, 3 = Skirts.
- - trouserColor	string	The color of the bottoms: 0 = Unknown, 1 = White, 2 = Orange, 3 = Pink, 4 = Black, 5 = Red, 6 = Yellow, 7 = Gray, 8 = Blue, 9 = Green, 10 = Purple, 11 = Brown.
- - trouserColorName	string	The internationalized color of the bottoms.
- - hat	string	The hat: 0 = Unknown, 1 = No hat, 2 = With hat.
- - bag	string	The bag: 0 = Unknown, 1 = No bag, 2 = With bag.
- - emotion	string	The emotion: -1 = Unknown, 0 = Smiling, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Feared, 5 = Surprised, 6 = Normal, 7 = Laughing, 8 = Happy, 9 = Confused, 10 = Screaming.
- - glasses	string	The glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
- - beard	string	The beard: 0 = Unknown, 1 = Unrecognized, 2 = No beard, 3 = With beard.
- - mask	string	The mask: 0 = Unknown, 1 = Unrecognized, 2 = No mask, 3 = Wearing a mask.
- - age	string	The age: -1 = Unknown.
- - captureTime	string	The capture time, in seconds.
- - similarity	string	The similarity. Example: 80.
- - dataSource	string	The data source: 1 = Device, 2 = Platform.
- - eventCode	string	Unique identification of an event used to determine the original event information of the captured image
- - gpsX	string	Latitude
- - gpsY	string	Longitude
- - mapId	string	Map id

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-140",
    "pageData": [
      {
        "id": "111611394219827200",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "humanImageUrl": "...e3e8bfda-126_353.jpg",
        "pictureUrl": "...e3e8bfda-65c_376.jpg",
        "facelImageUrl": "https://192.168.1.1:9901/...e3e8bfda-656_373.jpg",
        "gender": "1",
        "coat": "1",
        "coatColor": "1",
        "coatColorName": "White",
        "trouser": "2",
        "trouserColor": "8",
        "trouserColorName": "Blue",
        "hat": "1",
        "bag": "2",
        "emotion": "6",
        "glasses": "0",
        "beard": "2",
        "mask": "2",
        "age": "35",
        "captureTime": "1612334628",
        "similarity": "34.74",
        "dataSource": "1"
      },
      {
        "id": "111611253383487488",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "humanImageUrl": "...e3e8bfda-656_123.jpg",
        "pictureUrl": "https://192.168.1.1:9901/e3e8bfda-65c_425.jpg",
        "facelImageUrl": "https://192.168.1.1:9901/e3e8bfda-65c-471.jpg",
        "gender": "1",
        "coat": "1",
        "coatColor": "1",
        "coatColorName": "White",
        "trouser": "2",
        "trouserColor": "8",
        "trouserColorName": "Blue",
        "hat": "0",
        "bag": "1",
        "emotion": "8",
        "glasses": "0",
```

```

        "beard": "2",
        "mask": "2",
        "age": "30",
        "captureTime": "1612334594",
        "similarity": "26.94",
        "dataSource": "1",
        "eventCode": "fqeqf23",
        "gpsX": "12.3456",
        "gpsY": "12.3456",
        "mapId": "23"
      }
    ]
  }
}

```

3.13.4 Downloading a Captured Body Image.

Brief Description

- Download a captured body image. If an image is obtained from the IVSS device of an earlier version, the image URL is a relative path. In this case, you must download the image again. Then, the image can be stored to OSS, and you do not need to download the image in the future.

Request Method

- POST

Request URL

- /brms/api/v1.1/video-analyse/human/record/device/picture/download

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
deviceCode	Yes	string	The device number, required.
urls	Yes	object	None
- id	Yes	string	The record ID, required.
- type	Yes	string	The type of the recorded image, required: 1 = Face image, 2 = Scene image, 3 = Body image.
- url	Yes	string	The URL of the recorded image, required.

Request Example

```
POST /brms/api/v1.1/video-analyse/human/record/device/picture/download HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 457cb8a2da6046c4afafb7d7a4ac4075
```

```
{
  "deviceCode": "1000000",
  "urls": [
    {
      "id": "2",
      "type": "1",
      "url": "/picid/2789283.jpg"
    }
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	The returned information.
- urls	array	None
- - id	string	The record ID.
- - type	string	The type of the recorded image: 1 = Face image, 2 = Scene image, 3 = Body image.
- - url	string	The URL of the recorded image.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "urls": [
      {
        "id": "2",
        "type": "1",
        "url": "https://192.168.1.1:9901/cfd4...278_11787969.jpg"
      }
    ]
  }
}
```



```
}
```

3.13.5 Querying Body Capture Records by Feature and Display the Records by Page

Brief Description

- Query body capture records by feature and Display the Records by Page.

Request Method

- POST

Request URL

- /iams/api/v1.1/video-analyse/human/record/search-by-feature/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
channelIds	No	object	The channel code.
recordSource	No	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
taskIds	No	array	Analysis task ID
startTime	Yes	string	The start time, in seconds.
endTime	Yes	string	The end time, in seconds.
gender	No	string	Gender, 0 = unknown, 1 = male, 2 = female, empty = all
coat	No	string	Upper body clothes, 0 = unknown, 1 = long sleeve, 2 = short sleeve, 3 = sleeveless, empty = all
coatColors	No	array	Color of upper body clothes, 0 = unknown, 1 = white, 2 = orange, 3 = pink, 4 = black, 5 = red, 6 = yellow, 7 = gray, 8 = blue, 9 = green, 10 = purple, 11 = brown, 12 = silver, 13 = gold, 14 = green, empty = all

Parameter	Required	Type	Description
trouser	No	string	Underwear, 0 = unknown, 1 = pants, 2 = shorts, 3 = skirt, empty = all
trouserColors	No	array	Undercoat color, 0 = unknown, 1 = white, 2 = orange, 3 = pink, 4 = black, 5 = red, 6 = yellow, 7 = gray, 8 = blue, 9 = green, 10 = purple, 11 = brown, 12 = silver, 13 = gold, 14 = green, empty = all
hat	No	string	Hat, 0 = unknown, 1 = no, 2 = bands, empty = all
hatType	No	string	Hat type, 0=unknown, 1=normal hat, 2=helmet, 3=safety helmet, 10=no hat
bag	No	string	Package, 0 = unknown, 1 = no package, 2 = with package, empty = all
bagType	No	string	Bag type, 0=unknown, 1=handbag, 2=shoulder bag, 3=backpack, 4=trolley case, 5=waist bag, 6=no bag
orderType	Yes	string	The sort type: 1 = Time.
orderDirection	Yes	string	The sort order: 0 = Ascending order, 1 = Descending order.
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
currentPage	Yes	string	The current page number.
splitTime	No	string	The timestamp of the first record when you scroll up a page or the timestamp of the last record when you scroll down a page, in seconds. This parameter is required when you scroll up or down a page.
splitId	No	string	The ID of the first record when you scroll up a page or the ID of the last record when you scroll down a page. This parameter is

Parameter	Required	Type	Description
			required when you scroll up or down a page.
direction	No	string	Movement direction, 0 = unknown, 1 = A->B direction, 2 = B->A direction, 3 = A->B and B->A direction

Request Example

POST /iams/api/v1.1/video-analyse/human/record/search-by-feature/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

```
{
  "channelIds": [
    "1000000$1$0$0"
  ],
  "recordSource": "0",
  "taskIds": [
    "1"
  ],
  "startTime": "1629648000",
  "endTime": "1629734399",
  "currentPage": "1",
  "splitTime": "1629734395",
  "splitId": "2021082300000015880",
  "gender": "1",
  "coat": "1",
  "coatColors": [
    "2",
    "4"
  ],
  "trouser": "2",
  "trouserColors": [
    "2",
    "8"
  ],
  "hat": "2",
  "hatType": "1",
  "bag": "2",
  "bagType": "1",
  "orderType": "1",
  "orderDirection": "1",
  "pageSize": "20",
}
```

```
"page": "1",
"direction": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	N/A
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	The record ID.
- - channelId	string	The channel code.
- - channelName	string	The channel name.
- - recordSource	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
- - humanImageUrl	string	The URL of the captured body image.
- - pictureUrl	string	The URL of the scene image.
- - faceImageUrl	string	The URL of the captured face image.
- - gender	string	The gender: 0 = Unrecognized, 1 = Male, 2 = Female.
- - coat	string	The tops: 0 = Unknown, 1 = Long sleeve, 2 = Short sleeve.
- - coatColor	string	The color of the tops: 0 = Unknown, 1 = White, 2 = Orange, 3 = Pink, 4 = Black, 5 = Red, 6 = Yellow, 7 = Gray, 8 = Blue, 9 = Green, 10 = Purple, 11 = Brown.
- - coatColorName	string	The internationalized color of the tops.
- - trouser	string	The bottoms: 0 = Unknown, 1 = Pants, 2 = Shorts, 3 = Skirts.

Parameter	Type	Description
- - trouserColor	string	The color of the bottoms: 0 = Unknown, 1 = White, 2 = Orange, 3 = Pink, 4 = Black, 5 = Red, 6 = Yellow, 7 = Gray, 8 = Blue, 9 = Green, 10 = Purple, 11 = Brown.
- - trouserColorName	string	The internationalized color of the bottoms.
- - hat	string	The hat: 0 = Unknown, 1 = No hat, 2 = With hat.
- - hatType	string	Hat type, 0=unknown, 1=normal hat, 2=helmet, 3=safety helmet, 10=no hat
- - bag	string	The bag: 0 = Unknown, 1 = No bag, 2 = With bag.
- - bagType	string	Bag type, 0=unknown, 1=handbag, 2=shoulder bag, 3=backpack, 4=trolley case, 5=waist bag, 6=no bag.
- - direction	string	Movement direction, 0 = unknown, 1 = A->B direction, 2 = B->A direction
- - emotion	string	The emotion: -1 = Unknown, 0 = Smiling, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Feared, 5 = Surprised, 6 = Normal, 7 = Laughing, 8 = Happy, 9 = Confused, 10 = Screaming.
- - glasses	string	The glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses.
- - beard	string	The beard: 0 = Unknown, 1 = Unrecognized, 2 = No beard, 3 = With beard.
- - mask	string	The mask: 0 = Unknown, 1 = Unrecognized, 2 = No mask, 3 = Wearing a mask.
- - age	string	The age: -1 = Unknown.
- - captureTime	string	The capture time, in seconds.
- - gpsX	string	Latitude
- - gpsY	string	Longitude
- - mapId	string	Map id

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": -140,
    "pageData": [
      {
        "id": "111611394219827200",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "recordSource": "0",
        "humanImageUrl": "https://192.168.1.1:9901/...e3e8bfda-656_355.jpg",
        "pictureUrl": "https://192.168.1.1:9901/...e3e8bfda-65c_376.jpg",
        "faceImageUrl": "https://192.168.1.1:9901/...e3e8bfda-656_373.jpg",
        "gender": "1",
        "coat": "1",
        "coatColor": "2",
        "coatColorName": "Orange",
        "trouser": "2",
        "trouserColor": "8",
        "trouserColorName": "Blue",
        "hat": "2",
        "hatType": "1",
        "bag": "2",
        "bagType": "1",
        "direction": "1",
        "emotion": "6",
        "glasses": "0",
        "beard": "2",
        "mask": "2",
        "age": "35",
        "captureTime": "1612334628",
        "gpsX": "12.121212",
        "gpsY": "34.343434",
        "mapId": "12"
      }
    ]
  }
}
```

3.13.6 Obtaining the Records of Capturing Motor Vehicles by Page

Brief Description

- Obtaining the records of capturing motor vehicles by page. Integrates the capture records of all motor vehicles on the platform, including metadata, entrance and exit, traffic businesses.

Request Method

- POST

Request URL

- /ipms/api/v1.1/fusion/vehicle-capture/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
currentPage	Yes	string	The current page number.
splitTime	No	string	The timestamp of the first record when you scroll up a page or the timestamp of the last record when you scroll down a page, in seconds. This parameter is required when you scroll up or down a page.
splitId	No	string	The ID of the first record when you scroll up a page or the ID of the last record when you scroll down a page. This parameter is required when you scroll up or down a page.
orderType	No	string	The sort type: 1 = Capture time.
orderDirection	No	string	The sort order: 0 = Descending order: 1 = Ascending order.
channelIds	No	array	The channel code.
recordSource	No	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
taskId	No	array	Analysis task ID
startTime	Yes	string	The start time of the query, in seconds.

Parameter	Required	Type	Description
endTime	Yes	string	The end time of the query, in seconds.
vehicleColor	No	string	The vehicle color, which is a dictionary. For details, see [6.1.10 Color of motor vehicles].
vehicleBrand	No	string	The vehicle trademark, which is a dictionary. For details, see [6.1.12 Brand of motor vehicles].
vehicleModel	No	string	Vehicle type, dictionary type. See [6.1.11 Model of motor vehicles]
plateNoMatchMode	No	string	The matching mode for license plate numbers: 0 = Fuzzy matching, 1 = Exact matching.
inVehicleGroup	No	string	Whether the vehicle is registered: 0 = No, 1: Yes.
plateNos	No	array	The number of the license plate.
direction	No	string	Movement direction, 0 = unknown, 1 = A->B direction, 2 = B->A direction, 3 = A->B and B->A direction
issuePlace	No	string	Place of Issue
plateNoLength	No	string	License plate length scale: 1 = short, 2 = medium, 3 = long
plateNoType	No	string	License plate type, dictionary type 9048
plateNoAreaCode	No	string	License plate area code
hasPlateNo	No	string	Is unlicensed car: 0=Yes, 1=No
confidence	No	string	Confidence level
minSpeed	No	string	Minimum vehicle speed
maxSpeed	No	string	Maximum vehicle speed

Request Example

```
POST /ipms/api/v1.1/fusion/vehicle-capture/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 187f7da5f6044d27bde9c1223531bcc1

{
```



```

"page": "1",
"pageSize": "20",
"currentPage": "1",
"splitTime": "1629734395",
"splitId": "2021082300000015880",
"orderType": "1",
"orderDirection": "0",
"channelIds": [
    "1000051$1$0$0"
],
"recordSource": "0",
"taskIds": [
    "1"
],
"startTime": "1629648000",
"endTime": "1629734399",
"vehicleColor": "1",
"vehicleBrand": "1",
"vehicleModel": "27",
"plateNoMatchMode": "0",
"inVehicleGroup": "1",
"plateNos": [

],
"direction": "1",
"issuePlace": "",
"plateNoLength": "",
"plateNoType": "",
"plateNoAreaCode": "",
"hasPlateNo": "0",
"minSpeed": "1",
"maxSpeed": "100",
"confidence": "1"
}

```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.

Parameter	Type	Description
- pageData	array	The page data.
- - id	string	The primary key ID.
- - channelId	string	The channel code.
- - channelName	string	The channel name.
- - recordSource	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
- - captureTime	string	The capture time, in seconds.
- - plateNo	string	The number of the license plate.
- - issuePlace	string	Place of Issue
- - plateNoLength	string	License plate length
- - plateNoAreaCode	string	License plate region code
- - extPlateNo	string	Expand license plates
- - plateNoType	string	License plate type
- - plateNoTypeName	string	License plate type name
- - capturePicture	string	The address of the captured image.
- - inVehicleGroup	string	Whether the vehicle is registered: 0 = No, 1: Yes.
- - vehicleBrand	string	The vehicle trademark, which is a dictionary. For details, see [6.1.12 Brand of motor vehicles].
- - vehicleColor	string	The vehicle color, which is a dictionary. For details, see [6.1.10 Color of motor vehicles].
- - vehicleModel	string	Vehicle type, dictionary type. See [6.1.11 Model of motor vehicles]
- - direction	string	Movement direction, 0 = unknown, 1 = A->B direction, 2 = B->A direction
- - parkingLotId	string	Parking lot ID
- - parkingLotName	string	Parking lot name
- - confidence	string	Confidence level
- - mergePicture	string	Composite image URL
- - vehiclePicture	string	Vehicle image URL
- - plateNoPicture	string	License plate image URL
- - plateColor	string	License plate color
- - carSpeed	string	speed
- - roadway	string	Lane number
- - carType	string	Vehicle type

Parameter	Type	Description
-- gpsX	string	Coordinate information, longitude
-- gpsY	string	Coordinate information, latitude
-- mapId	string	Map id

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-102",
    "pageData": [
      {
        "id": "95",
        "channelId": "1000000$1$0$0",
        "channelName": "moniqi-lmx",
        "recordSource": "0",
        "plateNo": "EA2779335335F",
        "originalPlateNo": "H12553",
        "issuePlace": "SAU",
        "plateNoLength": "3",
        "plateNoAreaCode": "P",
        "extPlateNo": "ABC",
        "plateNoType": "4",
        "plateNoTypeName": "Police",
        "fuzzyMatched": "0",
        "accessAuthority": "1",
        "matchFlag": "1",
        "plateNoPictures": [
          "http://192.168.1.1:80/v1/s/192.168.1.1/9901/"
        ],
        "capturePictures": [
          "https://192.168.1.1:8082/1000054$1$0$0/20210206/23/5457-3196639-0.jpg?streamID:1260@capTime:1612684797"
        ],
        "enterTime": "1614195048",
        "direction": "1",
        "parkingLotId": "2",
        "parkingLotName": "",
        "parentParkingLotId": "1",

```

```
    "parentParkingLotName": "",
    "positionId": "1",
    "positionName": "test position",
    "pointId": "1",
    "pointName": "enter",
    "personName": "XingMing",
    "company": "",
    "orgCode": "001",
    "orgName": "All Person",
    "vehicleBrand": "1",
    "vehicleModel": "1",
    "vehicleColor": "1",
    "cardPersonId": "",
    "cardPersonName": "",
    "cardNo": "",
    "status": "1",
    "vehicleType": "5",
    "orgInfos": [

    ],
    "entranceGroups": [

    ],
    "confidence": "1",
    "mergePicture": "https://aaa/bb.jpg",
    "vehiclePicture": "https://aaa/bb.jpg",
    "plateNoPicture": "https://aaa/bb.jpg",
    "plateColor": "2",
    "carSpeed": "90",
    "roadway": "111",
    "carType": "23",
    "gpsX": "34.3434343",
    "gpsY": "23.22323",
    "mapId": "232"
  }
]
}
```

3.13.7 Obtaining the Records of Capturing Non-motor Vehicles in Object Detection by Page

Brief Description

- Obtain the records of capturing non-motor vehicles in object detection by page.

Request Method

- POST

Request URL

- /iams/api/v1.1/video-analyse/non-vehicle/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
channelIds	No	array	The channel code.
recordSource	No	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
taskIds	No	array	Analysis task ID
startTime	Yes	string	The start time, in seconds.
endTime	Yes	string	The end time, in seconds.
vehicleColor	No	string	The vehicle color, which is a dictionary. For details, see [6.1.13 Color of non motorized vehicles].
vehicleType	No	string	The vehicle type, which is a dictionary. For details, see [6.1.14 Model of non motorized vehicles].
riderNum	No	string	The number of riders.
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
currentPage	Yes	string	The current page number.
splitTime	No	string	The timestamp of the first record when you scroll up a page or the timestamp of the last record when you scroll

Parameter	Required	Type	Description
			down a page, in seconds. This parameter is required when you scroll up or down a page.
splitId	No	string	The ID of the first record when you scroll up a page or the ID of the last record when you scroll down a page. This parameter is required when you scroll up or down a page.
orderType	No	string	Sort type: 1 = by capture time
orderDirection	No	string	Sort direction: 0 = Ascending 1 = Descending
direction	No	string	Movement direction, 0 = unknown, 1 = A->B direction, 2 = B->A direction, 3 = A->B and B->A direction

Request Example

```
POST /iams/api/v1.1/video-analyse/non-vehicle/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "channelIds": [
    "1000000$1$0$0"
  ],
  "recordSource": "0",
  "taskIds": [
    "1"
  ],
  "startTime": "1629648000",
  "endTime": "1629734399",
  "vehicleType": "1",
  "vehicleColor": "10",
  "riderNum": "2",
  "page": "3",
  "pageSize": "20",
  "currentPage": "1",
  "splitTime": "1629734395",
  "splitId": "2021082300000015880",
```

```

"orderType": "1",
"orderDirection": "1",
"direction": "1"
}

```

Return Parameter

Parameter	Type	Description
code	int	The error code.
desc	string	The result description.
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - id	string	The record ID
- - channelId	string	The channel code
- - channelName	string	The channel name
- - recordSource	string	Record Source: 0=real-time reporting, 1=video analysis, 2=file analysis, default 0
- - pictureUrl	string	The URL of the scene image.
- - vehicleImageUrl	string	The vehicle image URL.
- - vehicleType	string	The vehicle type, which is a dictionary. For details, see [6.1.12 Brand of motor vehicles].
- - vehicleTypeName	string	The internationalized vehicle type. The internationalization relies on the request locale.
- - vehicleColor	string	The vehicle color, which is a dictionary. For details, see [6.1.13 Color of non motorized vehicles].
- - vehicleColorName	string	The internationalized vehicle color. The internationalization relies on the request locale.
- - riderNum	string	The number of riders.
- - deviceCode	string	Device code

Parameter	Type	Description
-- deviceName	string	Device name
-- gpsX	string	Latitude
-- gpsY	string	Longitude
-- mapId	string	Map id
-- direction	string	Movement direction, 0 = unknown, 1 = A->B direction, 2 = B->A direction
-- captureTime	string	The capture time, in seconds.
-- faceInfos	array	None
--- faceUrl	string	The face image URL.
--- gender	string	The gender: 0 = Unrecognized, 1 = Male, 2 = Female.
--- mouth	string	The mouth: 0 = Unknown, 1 = Unrecognized, 2 = Closed, 3 = Open.
--- mask	string	The mask: 0 = Unknown, 1 = Unrecognized, 2 = No mask, 3 = Wearing a mask.
--- beard	string	The beard: 0 = Unknown, 1 = Unrecognized, 2 = No beard, 3 = With beard.
--- glasses	string	The glasses: 0 = Unknown, 1 = No glasses, 2 = Regular glasses, 3 = Sunglasses, 4 = Black-rim glasses.
--- emotion	string	The emotion: -1 = Unknown, 0 = Smiling, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Feared, 5 = Surprised, 6 = Normal, 7 = Laughing, 8 = Happy, 9 = Confused, 10 = Screaming.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "140",
    "pageData": [
      {
        "id": "9467",
        "channelId": "1000009$1$0$0",
        "channelName": "2111111",
        "recordSource": "0",
        "pictureUrl": "https://192.168.1.1:9901/cb72...974_4654819.jpg",

```



```
{
  "vehicleImageUrl": "https://192.168.1.1:9901/cb72...819_4665977.jpg",
  "vehicleType": "2",
  "vehicleTypeName": "Two-wheeler",
  "vehicleColor": "2",
  "deviceCode": "100001",
  "deviceName": "ipc-1",
  "gpsX": "23.232323",
  "gpsY": "12.121212",
  "mapId": "12",
  "vehicleColorName": "Red",
  "riderNum": "1",
  "direction": "1",
  "captureTime": "1615788460",
  "faceInfos": [
    {
      "faceUrl": "https://192.168.1.1:9901/cb72...977_4667540.jpg",
      "gender": "1",
      "mouth": "2",
      "mask": "2",
      "beard": "0",
      "glasses": "1",
      "emotion": "-1"
    }
  ]
}
```

3.14 PTZ Control

3.14.1 Save tours

Brief description

- Save tours

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Cruise/SaveCruise

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- cruiseId	No	string	Tours ID
- data	Yes	string	Tours data
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Cruise/SaveCruise HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "cruiseId": "1",
    "data": "</Cruise&\n</CruiseConfig&\n",
    "channelId": "1000296$1$0$0"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.14.2 Obtain tours

Brief description

- Obtain tours

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Cruise/GetCruises

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Cruise/GetCruises HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "data": {
    "channelId": "1000296$1$0$0"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- cruises	object	None
- - channelId	string	Channel code
- - cruiseId	string	Tours ID, not used for now
- - data	string	Tours data

Return example

```
{
  "code": 1000,
  "desc": "Success",
}
```

```

    "data": {
      "cruises": [
        {
          "channelId": "1000296$1$0$0",
          "cruiseId": "0",
          "data": "</Cruise&\n  </Cruise&\n</CruiseConfig&\n"
        }
      ]
    }
  }
}

```

3.14.3 PTZ negotiation

Brief description

- PTZ negotiation

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/ArrangePtz

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- extend	No	string	Extended data, not used for now
- channelId	Yes	string	Channel code
- lockTime	Yes	string	Lock time (second), 0 means always locked until released or robbed
- operateType	Yes	string	Operation type: 1-locking the current camera; 2-unlocking the current camera; 5-search the lock status

Request example

```

POST /brms/api/v1.0/DMS/Ptz/ArrangePtz HTTP/1.1
Host: 192.168.1.1

```

Connection: close
 Content-Type: application/json;charset=UTF-8
 X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

```
{
  "data": {
    "extend": "",
    "channelId": "1000296$1$0$0",
    "lockTime": "0",
    "operateType": "1"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success; refer to LockUser when failed. When searching the locking status: 0-unlocked; 1-locked
- lockUser	object	None
- - userName	string	Name of the user locking the PTZ in a control failure
- - userLevel	string	Level of the user locking the PTZ in a control failure
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "0",
    "lockUser": {
      "userName": "system",
      "userLevel": "10",
      "lowerDomain": "192.168.1.1"
    }
  }
}
```

```
}  
}
```

3.14.4 Obtain presets

Brief description

- Obtain presets

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/GetPresetPoints

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Ptz/GetPresetPoints HTTP/1.1  
Host: 192.168.1.1  
Connection: close  
Content-Type: application/json;charset=UTF-8  
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c  
  
{  
  "data": {  
    "channelId": "1000296$1$0$0"  
  }  
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None

Parameter	Type	Description
- channelId	string	Channel code
- presetPoints	array	Preset information
- - channelId	string	Channel code
- - presetPointCode	string	Preset code
- - presetPointName	string	Preset name

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "channelId": "1000296$1$0$0",
    "presetPoints": [
      {
        "channelId": "1000296$1$0$0",
        "presetPointCode": "1",
        "presetPointName": "1"
      }
    ]
  }
}
```

3.14.5 PTZ lens control

Brief description

- PTZ lens control

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/OperateCamera

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- extend	No	string	Extended data

Parameter	Required	Type	Description
- direct	Yes	string	Direction: 1-increase; 2-decrease
- command	Yes	string	Command: 0-stop; 1-start
- channelId	Yes	string	Channel code
- step	Yes	string	speed
- operateType	Yes	string	Operation type: 1-zoom; 2-focus; 3-iris

Request example

```
POST /brms/api/v1.0/DMS/Ptz/OperateCamera HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "data": {
    "extend": "",
    "direct": "1",
    "command": "0",
    "channelId": "1000008$1$0$0",
    "step": "4",
    "operateType": "1"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success; refer to LockUser when failed
- lockUser	object	None
- - userName	string	Name of the user locking the PTZ
- - userLevel	string	Level of the user locking the PTZ
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

- Success

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "1",
    "lockUser": null
  }
}
```

- Fail (The PTZ is locked by some user)

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "0",
    "lockUser": {
      "userName": "system",
      "userLevel": "10",
      "lowerDomain": "192.168.1.1"
    }
  }
}
```

3.14.6 PTZ direction control

Brief description

- PTZ direction control

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/OperateDirect

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- stepY	Yes	string	speed in the vertical direction
- extend	No	string	Extended data
- direct	Yes	string	Direction: 1-up; 2-down; 3-left; 4-right; 5-upper left; 6-lower left; 7-upper right; 8-lower right
- channelId	Yes	string	Channel code
- stepX	Yes	string	speed in the horizontal direction
- command	Yes	string	Command: 0-stop; 1-start

Request example

```
POST /brms/api/v1.0/DMS/Ptz/OperateDirect HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "stepY": "6",
    "extend": "",
    "direct": "4",
    "channelId": "1000171$1$0$0",
    "stepX": "6",
    "command": "0"
  }
}
```

Return parameters

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success; refer to LockUser when failed
- lockUser	object	information of the user locking the PTZ
- - userName	string	Name of the user locking the PTZ
- - userLevel	string	Level of the user locking the PTZ

Parameter	Type	Description
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

The PTZ is not locked by any user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "1",
    "lockUser": null
  }
}
```

The PTZ is locked by some user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "0",
    "lockUser": {
      "userName": "system",
      "userLevel": "10",
      "lowerDomain": null
    }
  }
}
```

3.14.7 Presets control

Brief description

- Preset control (positioning/setting/deletion/work time)

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/OperatePresetPoint

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- presetPointCode	Yes	string	Preset code
- presetPointName	No	string	Preset name
- channelId	Yes	string	Channel code
- operateType	Yes	string	Operation type: 1-position; 2-set; 3-delete

Request example

```
POST /brms/api/v1.0/DMS/Ptz/OperatePresetPoint HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "presetPointCode": "1",
    "presetPointName": "prepoint1",
    "channelId": "1000171$1$0$0",
    "operateType": "1"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success; refer to LockUser when failed
- lockUser	object	information of the user locking the PTZ
- - userName	string	Name of the user locking the PTZ
- - userLevel	string	Level of the user locking the PTZ
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

The PTZ is not locked by any user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "1",
    "lockUser": null
  }
}
```

The PTZ is locked by some user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "0",
    "lockUser": {
      "userName": "system",
      "userLevel": "10",
      "lowerDomain": null
    }
  }
}
```

3.14.8 Obtain electric focus status

Brief description

- Obtain electric focus status

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/GetFocusStatus

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelId	Yes	string	Channel code
- operateType	No	string	Operation type
- zoom	No	string	Zoom times
- focus	No	string	Focal length
- lensType	No	string	LensType

Request example

```
POST /brms/api/v1.0/DMS/Ptz/GetFocusStatus HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "data": {
    "channelId": "1000003$1$0$1",
    "operateType": "2",
    "zoom": "",
    "focus": "",
    "lensType": ""
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	string	None
- channelId	string	Channel code
- status	string	Focus status: 0-normal; 1-automatic focusing
- zoom	string	Zoom times
- focus	string	Focal length

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
```

```

    "channelId": "1000024$1$0$0",
    "status": "0",
    "zoom": "0.000000",
    "focus": "0.973770"
  }
}

```

3.14.9 Electric focus control

Brief description

- Electric focus control

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/OperateFocus

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- focus	Yes	string	Focal length
- channelId	Yes	string	Channel code
- zoom	Yes	string	Zoom times
- operateType	Yes	string	Operation type: 0-reset; 1-continuous focusing; 2-automatic focusing
- lensType	Yes	string	Lens type: 1-embedded vari-focal lens; 2-manual vari-focal lens; 3-external vari-focal lens

Request example

```

POST /brms/api/v1.0/DMS/Ptz/OperateFocus HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

```

```
{
  "data": {
    "focus": "0.97",
    "channelId": "1000024$1$0$0",
    "zoom": "0.05",
    "operateType": "0",
    "lensType": "1"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success
- lockUser	object	information of the user locking the PTZ
- - userName	string	Name of the user locking the PTZ
- - userLevel	string	Level of the user locking the PTZ
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "1",
    "lockUser": null
  }
}
```

3.14.10 3-dimensional positioning or radar and speed dome positioning

Brief description

- 3-dimensional positioning or radar and speed dome positioning

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/SitPosition

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- pointZ	Yes	string	Zoom times: -4 - 4
- extend	No	string	Extension data: 1- 3-dimensional common positioning; 2- 3-dimensional accurate positioning; 3-radar and speed dome positioning
- channelId	Yes	string	Channel code
- pointY	Yes	string	Vertical coordinate: -8192 - 8192
- pointX	Yes	string	Horizontal coordinate: -8192 - 8192

Request example

```
POST /brms/api/v1.0/DMS/Ptz/SitPosition HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "pointZ": "8",
    "extend": "1",
    "channelId": "1000296$1$0$0",
    "pointY": "326",
    "pointX": "-1039"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code

Parameter	Type	Description
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success; refer to LockUser when failed
- lockUser	object	Information locked by user
- - userName	string	Name of the user locking the PTZ
- - userLevel	string	Level of the user locking the PTZ
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

The PTZ is not locked by any user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "1",
    "lockUser": null
  }
}
```

The PTZ is locked by some user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "0",
    "lockUser": {
      "userName": "system",
      "userLevel": "10",
      "lowerDomain": "192.168.1.1"
    }
  }
}
```

3.14.11 Obtain the PTZ position information

Brief description

- Obtain the PTZ position information

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/GetPosition

Request parameters

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Ptz/GetPosition HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "channelId": "1000125$1$0$3"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- channelId	string	Channel code
- pointX	string	Horizontal coordinate: -8192 - 8192
- pointY	string	Vertical coordinate: -8192 - 8192
- pointZ	string	Zoom times: -4 - 4

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "channelId": "1000125$1$0$3",
    "pointX": "3105",
    "pointY": "387",
    "pointZ": "128"
  }
}
```

3.14.12 PTZ function control

Brief description

- PTZ function control

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/OperateFunction

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- extend	No	string	Extended data
- target	Yes	string	Operation target, valid only when operateType=[14,19,20,21], 0 in other conditions
- channelId	Yes	string	Channel code
- command	No	string	Command: 0-stop; 1-start; valid only when operateType=[11,12,13,14,15,18,19,20,21], 0 in other conditions
- operateType	Yes	string	See notes for detailed value description

Request example

```
POST /brms/api/v1.0/DMS/Ptz/OperateFunction HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "extend": "",
    "target": "0",
    "channelId": "1000171$1$0$0",
    "command": "0",
    "operateType": "25"
  }
}
```

Notes

Operation type: 11-light (target=0); 12- wiper (target=0); 13-infrared light (target=0); 14-tour (target=tour ID); 15-automatic rotation (target=0); 16-setting the left border for line scan (target=0); 17-setting the right border for line scan (target=0); 18-line scan (target=0); 19-setting tour track (target=tour track ID); 20-pattern (target=tour track ID); 21-auxiliary (target=auxiliary command); 22-opening the PTZ menu (target=0); 23-closing the PTZ menu (target=0); 24-confirming the PTZ menu (target=0); 25-moving the PTZ menu upwards (target=0); 26-moving the PTZ menu downwards (target=0); 27-moving the PTZ menu leftwards (target=0); 28-moving the PTZ menu rightwards (target=0)

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- result	string	Operation result: 0-failure; 1-success; refer to LockUser when failed
- lockUser	object	information of the user locking the PTZ
- - userName	string	Name of the user locking the PTZ
- - userLevel	string	Level of the user locking the PTZ
- - lowerDomain	string	Lower-level domain address; null or null character string means this level; if there is an address, it means the lower level

Return example

The PTZ is not locked by any user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "1",
    "lockUser": null
  }
}
```

The PTZ is locked by some user

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "result": "0",
    "lockUser": {
      "userName": "system",
      "userLevel": "10",
      "lowerDomain": null
    }
  }
}
```

3.14.13 Search the lens information

Brief description

- Search the lens information

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/GetLensInfo

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Ptz/GetLensInfo HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "channelId": "1000000$1$0$0"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- type	string	Lens type: 1-embedded vari-focal lens; 2-manual vari-focal lens; 3-external vari-focal lens

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "type": "1"
  }
}
```

3.14.14 Setting Visible Range

Brief description

- Set visible range

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/SetVisibleArea

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- azimuth	Yes	string	Azimuth
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Ptz/SetVisibleArea HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "azimuth": "",
    "channelId": "1000171$1$0$0"
  }
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.14.15 Getting Visible Range

Brief description

- Get visible range

Request Method

- POST

Request URL

- /brms/api/v1.0/DMS/Ptz/GetVisibleArea

Request parameter

- Body Parameters

Parameter	Required	Type	Description
data	Yes	object	None
- channelId	Yes	string	Channel code

Request example

```
POST /brms/api/v1.0/DMS/Ptz/GetVisibleArea HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "data": {
    "channelId": "1000171$1$0$0"
  }
}
```

Return parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- channelId	string	Channel code
- distance	string	Distance(radius)
- azimuth	string	Azimuth
- angle	string	Included angle
- inclination	string	Inclination angle

Return example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "channelId": "1000171$1$0$0",
    "distance": "5",
    "azimuth": "",
    "angle": "",
    "inclination": ""
  }
}
```

3.15 People Counting(Intelligent Analysis)

3.15.1 Counting History People Flow by Channel

Brief Description

- Counting history people flow by channel. To generate statistical charts.

Request Method

- POST

Request URL

- /iams/api/v1.1/intelligence-analyse/passenger-flow/channel/history/statistics

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
countType	Yes	string	Statistics type: 1 = day; 2 = week; 3 = month; 4 = year
countDate	Yes	string	Statistics date
channelIds	Yes	array	Channel number

Request Example

```
POST /iams/api/v1.1/intelligence-analyse/passenger-flow/channel/history/statistics HTTP/1.1
Host: 192.168.1.1
Connection: close
```

Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

```
{
  "countType": "1",
  "countDate": "2021-06-06",
  "channelIds": [
    "1000001$1$0$0",
    "1000002$1$0$0"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- enterNumbers	array	Y axis, number of people entering
- exitNumbers	array	Y axis, number of people exiting
- stayNumbers	array	Y axis, number of people staying
- passingNumbers	array	Y axis, number of people passing
- periods	array	X axis, period

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "enterNumbers": ["2", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "12", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0"],
    "exitNumbers": ["5", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "9", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0"],
    "stayNumbers": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0"],
    "passingNumbers": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0"],
    "periods": ["01:00", "02:00", "03:00", "04:00", "05:00", "06:00", "07:00", "08:00", "09:00", "10:00", "11:00", "12:00", "13:00", "14:00", "15:00", "16:00", "17:00", "18:00", "19:00", "20:00", "21:00", "22:00", "23:00", "00:00"]
  }
}
```

Remarks

- The number of people exiting and staying are available only on daily report.

3.15.2 Counting History People Flow by People Counting Group

Brief Description

- Counting history people flow by people counting group. To generate statistical charts.

Request Method

- POST

Request URL

- /iams/api/v1.1/intelligence-analyse/passenger-flow/group/history/statistics

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
countType	Yes	string	Statistics type: 1 = day; 2 = week; 3 = month; 4 = year
countDate	Yes	string	Statistics date
groupIds	Yes	array	People counting group ID

Request Example

```
POST /iams/api/v1.1/intelligence-analyse/passenger-flow/group/history/statistics HTTP
/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "countType": "1",
  "countDate": "2021-06-06",
  "groupIds": [
    "1",
    "2"
  ]
}
```

Return Parameter

- /iams/api/v1.1/intelligence-analyse/passenger-flow/area/history/statistics

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
countType	Yes	string	Statistics type: 1 = day, 2 = week, 3 = month
countDate	Yes	string	Statistics date
channelIds	Yes	array	Channel number
stayDuration	Yes	string	Stay duration, in second

Request Example

```
POST /iams/api/v1.1/intelligence-analyse/passenger-flow/area/history/statistics HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "countType": "1",
  "countDate": "2021-06-06",
  "channelIds": [
    "1000001$1$0$0",
    "1000002$1$0$0"
  ],
  "stayDuration": "5"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- lessStayDurationNumbers	array	Y axis, number of people staying shorter than the specified time
- moreStayDurationNumbers	array	Y axis, number of people staying longer than the specified time
- periods	array	X axis, period

Parameter	Required	Type	Description
displayDeviceTimezoneOffset	No	string	Enable device time zone display: 0=No, 1=Yes

Request Example

```
POST /iams/api/v1.1/intelligence-analyse/passenger-flow/channel/history/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "countType": "1",
  "page": "1",
  "pageSize": "15",
  "countDate": "2021-06-06",
  "channelIds": [
    "1000001$1$0$0",
    "1000002$1$0$0"
  ],
  "displayServerTimezoneOffset": "0",
  "displayDeviceTimezoneOffset": "0"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- pageData	object	Page data
- - enterNumbers	int	Number of people entering
- - exitNumbers	int	Number of people exiting
- - stayNumbers	int	Number of people staying (Difference between number of people entering and exiting in the period)
- - passingNumber	int	Number of people passing
- - source	string	Data sources: 1=regular passenger flow data, 2=facial report data
- - channelId	string	Channel number
- - channelName	string	Channel name

Parameter	Type	Description
- - startTime	string	Start time
- - endTime	string	End time
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "pageData": [
      {
        "enterNumber": 7,
        "exitNumber": 7,
        "stayNumber": 0,
        "passingNumber": 0,
        "source": "1",
        "channelId": "1000000$1$0$0",
        "channelName": "IPC-3",
        "startTime": "2021-06-24 00:00:00",
        "endTime": "2021-06-24 01:00:00"
      }
    ],
    "totalCount": "1"
  }
}
```

3.15.5 Obtaining History People Flow of People Counting Groups by Page

Brief Description

- Obtaining history people flow of people counting groups by page. To generate statistical reports.

Request Method

- POST

Request URL

- /iams/api/v1.1/intelligence-analyse/passenger-flow/group/history/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
countType	Yes	string	Statistics type: 1 = day; 2 = week; 3= month; 4= year
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
countDate	Yes	string	Statistics date
groupIds	Yes	array	People counting group ID
displayServerTimezoneOffset	Yes	string	Enable time zone display: 0=No, 1=Yes
displayDeviceTimezoneOffset	No	string	Enable device time zone display: 0=No, 1=Yes

Request Example

```
POST /iams/api/v1.1/intelligence-analyse/passenger-flow/group/history/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "countType": "1",
  "page": "1",
  "pageSize": "15",
  "countDate": "2021-06-06",
  "groupIds": [
    "1",
    "2"
  ],
  "displayServerTimezoneOffset": "0",
  "displayDeviceTimezoneOffset": "0"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- pageData	object	Page data
- - enterNumbers	int	Number of people entering
- - exitNumbers	int	Number of people exiting
- - stayNumbers	int	Number of people staying (Difference between number of people entering and exiting in the period)
- - passingNumber	int	Number of people passing
- - groupId	string	People counting group ID
- - groupName	string	People counting group name
- - startTime	string	Start time
- - endTime	string	End time
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "pageData": [
      {
        "enterNumber": 7,
        "exitNumber": 7,
        "stayNumber": 0,
        "passingNumber": 0,
        "groupId": "1",
        "groupName": "IPC-3",
        "startTime": "2021-06-24 00:00:00",
        "endTime": "2021-06-24 01:00:00"
      }
    ]
  },
}
```

```
"totalCount": "1"
}
```

3.15.6 Get Passenger Flow Analysis Group List

Brief Description

- Get passenger flow analysis group list

Request Method

- GET

Request URL

- /iams/api/v1.1/intelligence-analyse/passenger-flow/group/list

Request Parameter

None

Request Example

```
GET /iams/api/v1.1/intelligence-analyse/passenger-flow/group/list HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- results	array	Result
- - groupId	string	Passenger flow group ID
- - groupName	string	Passenger flow group name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
```

```
{
  "data": {
    "results": [
      {
        "groupId": "1",
        "groupName": "test"
      }
    ]
  }
}
```

3.16 Alarm Control Panel

3.16.1 Acquiring the Sub-system and Partition Lists of the Alarm Control Panel in Batches

Brief Description

- Acquiring sub-system and partition lists of the alarm control panel in batches

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/device/area/fetch/batch/list

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
deviceCodes	Yes	array	Device code

Request Example

```
POST /obms/api/v1.1/alarm-host/device/area/fetch/batch/list HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 6fffb31707843e8a3063167d007387e
```

```
{
  "deviceCodes": [
    "1000003"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- results	array	Return result
- - deviceCode	string	Device code
- - armStatus	string	Arming status: 0=Disarmed, 1=Armed, 2=P1 armed, 3=P2 armed, 4=P1 + P2 armed.
- - alarmStatus	string	Alarm status: 0=normal, 1=alarm
- - areald	string	Sub-system ID
- - areaName	string	Sub-system name
- - partitions	array	Partition
- - - partitionId	string	Partition ID
- - - partitionName	string	Partition name

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "results": [
      {
        "deviceCode": "1000003",
        "armStatus": "0",
        "alarmStatus": "0",
        "areald": "1000003$3$0$-1",
        "areaName": "Areas1",

```

```

    "partitions": [
      {
        "partitionId": "145",
        "partitionName": "Partition_1"
      }
    ]
  }
}

```

3.16.2 Acquiring the Protection Zone List of the Alarm Control Panel by page

Brief Description

- Exporting the protection zone list of the alarm control panel by page

Request Method

- GET

Request URL

- /obms/api/v1.1/alarm-host/zone/page?page={page}&pageSize={pageSize}&keyword={keyword}&deviceCode={deviceCode}&areald={areald}&bypassStatus={bypassStatus}&faultStatus={faultStatus}&realTimeStatus={realTimeStatus}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
keyword	No	string	Keyword, fuzzy match with zone name or partition name
deviceCode	No	string	Device code
areald	No	string	Sub-system ID
bypassStatus	No	string	Bypass status: 0 = Normal, 1 = Bypass, 2 = Isolated

Parameter	Required	Type	Description
faultStatus	No	string	Fault status: 0 = normal, 1 = tampering, 2 = short circuit, 3 = anti-masking, 4 = Main Power Failure, 5 = Storage Battery Missing, 6 = Low Battery, 7 = Not Connected
realTimeStatus	No	string	Realtime Status: 0 = Close, 1 = Open

Request Example

```
GET /obms/api/v1.1/alarm-
host/zone/page?pageSize=20&deviceCode=1000003&keyword=&page=1&areald=sub
system$1000003_0 HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: a1203ec07e2a463799525b8808564d34
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Record the total number. Negative numbers indicate no total number of pages to be paginated. They are used to predict the total number of the next 7 pages and assist in displaying paginated controls. Cannot be used for data statistics.
- pageData	array	None
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - areald	string	Sub-system ID
- - areaName	string	Sub-system name
- - zoneld	string	Protection zone ID
- - zoneName	string	Protection zone name
- - partitions	array	Partition
- - - partitionId	string	Partition ID
- - - partitionName	string	Partition name

Parameter	Type	Description
-- bypassStatus	string	Bypass status: 0 = Normal, 1 = Bypass, 2 = Isolated
-- faultStatus	string	Fault status: 0 = normal, 1 = tampering, 2 = short circuit, 3 = anti-masking, 4 = Main Power Failure, 5 = Storage Battery Missing, 6 = Low Battery, 7 = Not Connected
-- alarmStatus	string	Alarm status: 0=normal, 1=alarm
-- realTimeStatus	string	Realtime Status: 0 = Close, 1 = Open
-- supportSnapshot	string	Whether to support image capture: 0 = No, 1 = Yes
-- channelId	string	Channel code

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "1",
    "pageData": [
      {
        "deviceCode": "1000003",
        "deviceName": "Test Dev",
        "areaid": "subsystem$1000003_0",
        "areaName": "Areas1",
        "zoneId": "999",
        "zoneName": "WIRED ZONE1",
        "partitions": [
          {
            "partitionId": "177",
            "partitionName": "Partition_1"
          }
        ],
        "bypassStatus": "0",
        "faultStatus": "0",
        "alarmStatus": "0",
        "realTimeStatus": "0",
        "supportSnapshot": "0",
        "channelId": "1000003$3$0$0"
      }
    ]
  }
}
```

3.16.3 Synchronizing Alarm Control Panel Information (Asynchronization Notification)

Brief description

- Acquiring the synchronization results from MQ: [4.9.4 Information Synchronization Notification of Alarm Control Panel Devices]

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/device/{deviceCode}/info/synchronize

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
deviceCode	Yes	string	Device code

Request Example

```
POST /obms/api/v1.1/alarm-host/device/1000001/info/synchronize HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: a1203ec07e2a463799525b8808564d34
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {}
}
```

```
"data": null
}
```

3.16.4 Operating Alarm Control Panel Devices in Batches

Brief description

- Operating alarm control panel devices in batches

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/device/operate/batch

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
forceMode	Yes	string	Enforcement mode: 0 = No, 1 = Yes
operateType	Yes	string	Operation type: 1 = Arming, 5 = Disarming
deviceCodes	Yes	array	Device code

Request Example

```
POST /obms/api/v1.1/alarm-host/device/operate/batch HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: a1203ec07e2a463799525b8808564d34
```

```
{
  "forceMode": "0",
  "deviceCodes": [
    "1000001"
  ],
  "operateType": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return result
- failedResults	array	Return result
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - supportForceMode	string	Enforcement mode: 0 = No, 1 = Yes
- - errors	object	Error information
- - - errorType	string	Error type: 1 = Device, 2 = Module, 3 = Protection zone
- - - errorMsg	string	Reasons for error

Return Example

```
{
  "code": 1001,
  "desc": "Failed",
  "data": {
    "failedResults": [
      {
        "deviceCode": "1000001",
        "deviceName": "172.3.0.77",
        "supportForceMode": "1",
        "errors": [
          {
            "errorType": "1",
            "errorMsg": "Anti disassembly alarm of equipment(Device
Name:172.3.0.77)."
          }
        ]
      }
    ]
  }
}
```

3.16.5 Operating Sub-systems of the Alarm Control Panel in Batches

Brief description

- Operating sub-systems of the alarm control panel in batches

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/area/operate/batch

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
forceMode	Yes	string	Enforcement mode: 0 = No, 1 = Yes
operateType	Yes	string	Operation Type: 1 = arming, 2 = P1 arming, 3 = P2 arming, 5 = disarming
arealds	Yes	array	Sub-system ID

Request Example

```
POST /obms/api/v1.1/alarm-host/area/operate/batch HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: e32037bf02b54cf6aaa07a3087889c8f
```

```
{
  "forceMode": "0",
  "arealds": [
    "1000001$3$0$-1",
    "1000001$3$1$-1"
  ],
  "operateType": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Return result
- failedResults	array	Return result
- - deviceCode	string	Device code
- - deviceName	string	Device name

Parameter	Type	Description
- - supportForceMode	string	Enforcement mode: 0 = No, 1 = Yes
- - areas	array	Failed sub-system
- - - areald	string	Sub-system ID
- - - areaName	string	Sub-system name
- - errors	array	Error information
- - - errorType	string	Error type: 1 = Device, 2 = Module, 3 = Protection zone
- - - errorMsg	string	Reasons for error

Return Example

```
{
  "code": 1001,
  "desc": "Failed",
  "data": {
    "failedResults": [
      {
        "deviceCode": "1000001",
        "deviceName": "172.3.0.77",
        "supportForceMode": "1",
        "areas": [
          {
            "areald": "1000001$3$0$-1",
            "areaName": "AREA1"
          },
          {
            "areald": "1000001$3$1$-1",
            "areaName": "AREA2"
          }
        ],
        "errors": [
          {
            "errorType": "1",
            "errorMsg": "Anti disassembly alarm of equipment(Device
Name:172.3.0.77)."
          }
        ]
      }
    ]
  }
}
```

3.16.6 Operating the Protection Zones of the Alarm Control Panel in Batches

Brief description

- Operating the protection zones of the alarm control panel in batches

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/zone/operate/batch

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
operateType	Yes	string	Operation type: 7 = Recovery, 8 = Bypass, 9 = Isolation
zonelds	Yes	array	Protection zone ID

Request Example

```
POST /obms/api/v1.1/alarm-host/zone/operate/batch HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 2ec1e6ed676643488e762ef41833a0c1
```

```
{
  "operateType": "8",
  "zonelds": [
    "1161",
    "1163"
  ]
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Data

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

3.16.7 Get List of Alarm Output Channels of the Alarm Controller

Brief Description

- Get output channel list of alarm controller

Request Method

- GET

Request URL

- /obms/api/v1.1/alarm-host/alarm-out/channel/list?deviceCode={deviceCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
deviceCode	YES	string	Device code

Request Example

```
GET /obms/api/v1.1/alarm-host/alarm-out/channel/list?deviceCode=1000008 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

Return Parameter

Parameter	Type	Description
code	int	The error code
desc	string	Result description
data	object	None
- results	array	Return result
- - channelSeq	string	Channel number

Parameter	Type	Description
-- channelId	string	Channel number
-- channelName	string	The channel name
-- channelStatus	string	Channel status: 0=Disable; 1=Enable

Return Example

```
{
  "code": 1001,
  "desc": "Success",
  "data": {
    "results": [
      {
        "channelSeq": "1",
        "channelId": "1000008$4$0$0",
        "channelName": "111_1",
        "channelStatus": "0"
      }
    ]
  }
}
```

3.16.8 Get Alarm Channel List of Alarm Controller

Brief Description

- Get alarm channel list of alarm controller

Request Method

- GET

Request URL

- /obms/api/v1.1/alarm-host/alarm-signal/channel/list?deviceCode={deviceCode}

Request Parameter

- Path Parameters

Parameter	Required	Type	Description
deviceCode	Yes	string	Device code

Request Example

GET /obms/api/v1.1/alarm-host/alarm-signal/channel/list?deviceCode=1000008 HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

Return Parameter

Parameter	Type	Description
code	int	The error code
desc	string	Result description
data	object	None
- results	array	Return result
- - channelSeq	string	Channel sequence
- - channelId	string	Channel number
- - channelName	string	Channel name
- - channelStatus	string	Channel status: 0=Disable; 1=Enable

Return Example

```
{
  "code": 1001,
  "desc": "Success",
  "data": {
    "results": [
      {
        "channelSeq": "1",
        "channelId": "1000008$4$1$0",
        "channelName": "111_1",
        "channelStatus": "0"
      }
    ]
  }
}
```

3.16.9 Operating Output Channel of Alarm Controller in Batches

Brief Description

- Operate output channel of alarm controller in batches

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/alarm-out/channel/operate/batch

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
channelIds	Yes	array	Channel number
operateType	Yes	string	Operation Type: 0=Disable; 1=Enable

Request Example

```
POST /obms/api/v1.1/alarm-host/alarm-out/channel/operate/batch HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c
```

```
{
  "channelIds": [
    "1000008$4$0$0"
  ],
  "operateType": "1"
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code
desc	string	Result description
data	object	None
- failedChannelIds	object	Code of failed channel

Return Example

- Success

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

- Fail

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "failedChannelIds": ["1000001$4$0$0"]
  }
}
```

3.16.10 Operating Alarm Signal Channels of Alarm Controller in Batches

Brief Description

- Operate alarm signal channel of alarm controller in batches

Request Method

- POST

Request URL

- /obms/api/v1.1/alarm-host/alarm-signal/channel/operate/batch

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
channelIds	Yes	array	Channel number
operateType	Yes	string	Operation Type: 0=Disable; 1=Enable

Request Example

```
POST /obms/api/v1.1/alarm-host/alarm-signal/channel/operate/batch HTTP/1.1
Host: 192.168.1.1
Connection: close
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 5fad3e259cbc497fa3e2ece67676469c

{
  "channelIds": [
    "1000008$4$1$0"
  ],
  "operateType": "1"
}
```

```
}
```

Return Parameter

Parameter	Type	Description
code	int	The error code
desc	string	Result description
data	object	None
- failedChannelIds	object	Code of failed channel

Return Example

- Success

```
{
  "code": 1000,
  "desc": "Success",
  "data": null
}
```

- Fail

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "failedChannelIds": "1000001$22$0$0"
  }
}
```

3.17 MPT

3.17.1 Acquiring MPT Enforcement Records by Page

Brief Description

- Acquiring MPT enforcement records by page

Request Method

- POST

Request URL

- /brms/api/v1.1/law-enforcement/mpt/record/fetch/page

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
deviceCodes	Yes	array	Device code
fileType	No	string	File types: 0 = Unknown, 1 = Video, 2 = Audio, 3 = Picture
deviceName	No	string	Device name fuzzy query, case insensitive
userIds	No	array	MPT user ID
endTime	Yes	string	End time, time stamp (in seconds)
startTime	Yes	string	Start time, time stamp (in seconds)
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pageSize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
orderType	No	string	Sort type: 1 = by capture time
orderDirection	No	string	Sort direction: 0 = Ascending 1 = Descending
currentPage	No	string	The current page number.
splitTime	No	string	The timestamp of the first record when you scroll up a page or the timestamp of the last record when you scroll down a page, in seconds. This parameter is required when you scroll up or down a page.
splitId	No	string	The ID of the first record when you scroll up a page or the ID of the last record when you scroll down a page. This parameter is required when you scroll up or down a page.

Request Example

```
POST /brms/api/v1.1/law-enforcement/mpt/record/fetch/page HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Time-Zone: Asia/Shanghai
Content-Type: application/json;charset=UTF-8
X-Subject-Token: 1f212cb345fc401599112bf417c98a7c
```

```
{
  "deviceCodes": [
    "1000000"
  ],
  "startTime": "1632844800",
  "endTime": "1632931199",
  "userIds": [],
  "fileType": "",
  "deviceName": "",
  "page": "1",
  "pageSize": "20",
  "currentPage": 1,
  "orderType": "1",
  "orderDirection": 1,
  "splitTime": "",
  "splitId": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	None
- totalCount	string	Total number
- pageData	object	Page data
- - id	string	Record ID
- - deviceCode	string	Device code
- - deviceName	string	Device name
- - userId	string	MPT user ID
- - username	string	MPT user name
- - fileUrl	string	URL of images or videos
- - fileLength	string	File size (in bytes)
- - fileType	string	File type: 0 = Unknown, 1 = Video, 2 = Audio, 3 = Image
- - recordTime	string	File start time ,time stamp (in seconds)
- - fileTimeLength	string	File duration (in seconds)
- - sourceType	string	Source type: 1=MPT file supplement, 2=EEC device, 3=EEC file supplement, 4=MXVR file supplement
- - eecDeviceCode	string	Storage device code (EEC)

Parameter	Type	Description
- - sslId	string	SS service ID
- - orgCode	string	Organization code
- - orgName	string	Organization name
- - vkeKIds	array	VKEK Key ID
- - gps	array	GPS informations
- - - gpsX	string	Longitude
- - - gpsY	string	Latitude
- - - gpsTime	string	GPS reporting time, time stamp, unit: second

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "totalCount": "-1",
    "pageData": [
      {
        "id": "1",
        "deviceCode": "1000000",
        "deviceName": "MPT",
        "userId": "",
        "username": "",
        "fileUrl": "http://192.168.1.1:80/v1/s/192.168.1.1/9901/3369ad4d-1ffe-11ec-8064-02004c4f4f50/20210928/21/dsf_9c2bdbbc4-20cd-11ec-8346-02004c4f4f50_6827133_8017573.jpg",
        "fileLength": "1190440",
        "fileType": "3",
        "recordTime": "1632877343",
        "fileTimeLength": "0",
        "sourceType": "1",
        "eecDeviceCode": "",
        "sslId": "",
        "orgCode": "001001",
        "orgName": "org1",
        "vkeKIds": [],
        "gps": [
          {
            "gpsX": "120.123",
            "gpsY": "30.125",
            "gpsTime": "1632877343"
          }
        ]
      }
    ]
  }
}
```



```

    }
  ]
}
}

```

3.18 Video Intercom

3.18.1 Get Call Records by Device

Brief Description

- Get call records by device

Request Method

- POST

Request URL

- /obms/api/v1.0/vtalk/callLogList

Request Parameter

- Body Parameters

Parameter	Required	Type	Description
deviceCodes	Yes	array	Calling device code
callStartTime	Yes	string	Start time of call time period, timestamp (in seconds)
callEndTime	Yes	string	End time of call time period, timestamp (in seconds)
page	Yes	string	Page number, starting from 1, the maximum value is 2147483647
pagesize	Yes	string	Number of entries each page, starting from 1, the maximum value is 10000
currentPage	No	string	The current page number, used when flipping pages, helps to turn pages faster
splitId	No	string	Page splitting ID, when flipping up, it is the ID of the first record, and when flipping down, it is the ID of the last record. Used when flipping, it helps to flip pages faster
splitTime	No	string	Page splitting timestamp, when flipping up, it is the time of the first record, and when flipping down, it is the time of the last record. Timestamp (in seconds), used when flipping, helps to flip pages faster

Request Example

```
POST /obms/api/v1.0/vtalk/callLogList HTTP/1.1
Host: 192.168.1.1
Connection: keep-alive
Accept-Language: en
Content-Type: application/json;charset=UTF-8
X-Subject-Token: ca19ae28474040f192b9fc8cc7b9c69a
```

```
{
  "callEndTime": "1558627199",
  "callStartTime": "1556640000",
  "deviceCodes": [
    "1000059",
    "1000060"
  ],
  "page": 1,
  "pagesize": 20,
  "currentPage": 0,
  "splitId": 0,
  "splitTime": ""
}
```

Return Parameter

Parameter	Type	Description
code	int	Error code
desc	string	Result description
data	object	Response data
- totalCount	string	Total count
- pageData	array	Page data
- - id	string	Primary key ID
- - callNumber	string	Call number
- - callTime	string	Call duration, timestamp(in seconds)
- - talkTime	string	Talk duration, timestamp(in seconds)
- - endState	string	Final status: 0=not answered, 1=received, 2=not answered, other members answered, passively hung up, 3=not answered, actively rejected
- - callType	string	Call Type: 0=outbound, 1=inbound
- - formatTalkTime	string	Talk duration, for example 00:15
- - formatCallTime	string	Call duration, for example 00:15

Parameter	Type	Description
-- deviceName	string	Device name
-- callerNumber	string	Calling number
-- calledNumber	string	Called number

Return Example

```
{
  "code": 1000,
  "desc": "Success",
  "data": {
    "pageData": [
      {
        "id": "12",
        "callingParty": "2",
        "calledParty": "8888881000",
        "callNumber": "2",
        "callTime": "1557785967",
        "talkTime": "0",
        "endState": "0",
        "picturePath": "",
        "callType": "0",
        "formatCallTime": "2019-05-13 15:19:27",
        "formatTalkTime": "00:00",
        "deviceName": "4J03991PAZAD67B",
        "recordId": "123"
      }
    ],
    "nextPage": "-1",
    "totalCount": "2"
  }
}
```

4 MQ Notification Message

Public Message Protocol is received by all users, whereas Private Message Protocol is only sent to the specified users with the themes distinguished by the user ID.

4.1 Device

4.1.1 Notification of Device Online and Offline

Brief Description

- Notification of Device Online and Offline

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	The user ID.

Method

- vms.notifyDeviceStatus

Message Parameter

Parameter	Type	Description
method	string	The method name.
info	object	None
- deviceCode	string	The device code.
- onlineStatus	string	The status: 0 = Offline, 1 = Online.
- offlineReason	string	The offline reason.
- syncChannelStatus	string	Whether to synchronize the channel status: 0 = No, 1 = Yes. The default value is 1.
- updateTime	string	The change interval, in seconds.

Parameter	Type	Description
id	string	The message ID.

Message Example

```
{
  "method": "vms.notifyDeviceStatus",
  "info": {
    "deviceCode": "1000000",
    "onlineStatus": "1",
    "offlineReason": "0",
    "syncChannelStatus": "1",
    "updateTime": "1559913632"
  },
  "id": "16"
}
```

4.1.2 Notification of Channel Online and Offline

Brief Description

- Notification of Channel Online and Offline

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	The user ID.

Method

- vms.notifyDevChannelStatus

Message Parameter

Parameter	Type	Description
method	string	The method name.
info	array	None
- channelId	string	The channel code.
- onlineStatus	string	The status: 0 = Offline, 1 = Online.
- updateTime	string	The change interval, in seconds.

Parameter	Type	Description
id	string	The message ID.

Message Example

```
{
  "method": "vms.notifyDevChannelStatus",
  "info": [
    {
      "channelId": "1000010$1$0$4",
      "onlineStatus": "1",
      "updateTime": "1559913632"
    }
  ],
  "id": "25438"
}
```

4.2 Alarm(Event Center)

4.2.1 Alarm Messages

Brief Description

- Alarm Messages

Topic

- mq.alarm.msg.topic.{userId}
- mq.alarm.msg.group.topic.{userGroupId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID
userGroupId	Yes	string	User group ID

Method

- brms.notifyAlarms

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None

Parameter	Type	Description
- deviceCode	string	Device code
- channelSeq	string	Channel number
- unitSeq	string	Unit number
- unitType	string	Unit type
- nodeType	string	Node type
- nodeCode	string	Channel code
- alarmCode	string	Alarm code
- alarmStat	string	Alarm status
- alarmType	string	Alarm type, see [6.1.7 Alarm Type]
- alarmGrade	string	Alarm grade
- alarmPicture	string	Alarm snapshot
- alarmPictureSize	string	Alarm picture size
- alarmDate	string	Alarm time
- memo	string	Alarm plan remarks
- eventRemark	string	Alarm event remarks
- screenNum	string	Screen ID
- linkVideoChannels	array	Channel of associated video
- - channelId	string	Channel code
- - channelName	string	Channel name
- - streamType	string	Stream type
- - screenId	string	Window ID
- linkRecordChannels	array	Linkage video channel list
- - channelId	string	Channel code
- - alarmRecordPosition	string	Alarm video storage location: 1 = all, 2 = equipment, 3 = center
- - streamType	string	Stream type
- - recordTime	string	Recording time, in seconds
- - preRecordTime	string	Prerecorded time, in seconds
- userIds	string	ID of associated user
- alarmSourceName	string	Alarm source channel name
- deviceName	string	Device name
- linkedOutput	string	Linkage alarm output control: 0 = No, 1 = Yes
- linkedAudioLight	string	Audio light control linkage: 0 = No, 1 = Yes

Parameter	Type	Description
- linkedAudio	string	Whether audio control has been linked: 0 = No, 1 = Yes
- gpsX	number	Longitude
- gpsY	number	Latitude
- mapIds	array	Map ID
- height	number	Height
- angle	number	Direction angle
- speed	number	Speed
- ruleThreshold	number	Rule threshold value
- stayNumber	number	Number of stranded persons
- planTemplateId	string	Alarm plan template ID
- faceRecognitionInfo	object	Face recognition information
- - personId	string	Person ID
- - personName	string	Person name
- - repositoryId	string	Face library ID
- - repositoryName	string	Face Library name
- - personTypeId	string	Person type ID
- - personTypeName	string	Person type name
- - gender	string	Gender: 1 = male, 2 = female
- - birthday	string	Birthday, format: YYYY-MM-DD
- - nationality	string	Nationality
- - tel	string	Telephone
- - captureFaceImageUrl	string	Face capture image URL
- - personFaceImageUrl	string	The face image URL of the person in the library
- - similarity	string	Similarity
- linkedVoicePrompt	string	Whether the voice prompt has been linked: 0 = No, 1 = Yes
- voicePromptContent	string	Voice Content
- extData	string	Extended data: JSON format
- disarmUserIds	array	Block alarm user ID
- allowViewLinkInfo	string	Allow viewing of linkage information: 0 = No, 1 = Yes
id	string	Message ID

Message Example


```
{
  "method": "brms.notifyAlarms",
  "info": [
    {
      "deviceCode": "1000004",
      "channelSeq": 0,
      "unitSeq": 0,
      "unitType": 1,
      "nodeType": "2",
      "nodeCode": "1000004$1$0$0",
      "alarmCode": "{8C2C8056-D0A7-454B-845B-C566746D3B42}",
      "alarmStat": "1",
      "alarmType": "4",
      "alarmGrade": "1",
      "alarmPicture": "",
      "alarmPictureSize": null,
      "alarmDate": "1547014708",
      "memo": "9876",
      "eventRemark": "",
      "screenNum": null,
      "linkVideoChannels": [
        {
          "channelId": "1000003$1$0$0",
          "channelName": "test",
          "streamType": "1",
          "screenId": "0"
        }
      ],
      "linkRecordChannels": [
        {
          "channelId": "1000003$1$0$0",
          "alarmRecordPosition": "2",
          "streamType": "1",
          "recordTime": "10",
          "preRecordTime": "5"
        }
      ],
      "userIds": [
        1
      ],
      "alarmSourceName": "Channel0",
      "deviceName": "10",
      "gpsX": null,
      "gpsY": null,
      "mapIds": [
        "1"
      ],
      "height": null,
      "angle": null,
    }
  ]
}
```

```

    "speed": null,
    "ruleThreshold": 0,
    "stayNumber": 0,
    "planTemplateId": "1",
    "extData": "",
    "faceRecognitionInfo": {
      "personId": "31405411",
      "personName": "Jack",
      "repositoryId": "100",
      "repositoryName": "face",
      "personTypeId": "",
      "personTypeName": "",
      "gender": "1",
      "birthday": "",
      "nationality": "",
      "tel": "13012345678",
      "captureFaceImageUrl": "",
      "personFaceImageUrl": "",
      "similarity": "50"
    },
    "linkedVoicePrompt": "0",
    "voicePromptContent": "",
    "disarmUserIds": [],
    "allowViewLinkInfo": ""
  }
],
  "id": 1
}

```

4.2.2 Notification of Handling Alarms

Brief Description

- Notification of Handling Alarms

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- vms.handleAlarm

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None
- alarmCode	string	Alarm code
- alarmDate	string	Alarm time
- handleUser	string	Handler
- handleDate	string	Handling time
- handleStatus	string	Handling status
- handleMessage	string	Handling information
- mailReceivers	array	Email recipient
- alarmGrade	string	Alarm grade
- alarmPic	string	Alarm image
- alarmPicSize	string	Alarm image size
- alarmStat	string	Alarm status
- alarmType	string	Alarm type, see [6.1.7 Alarm Type]
- deviceCode	string	Device code
- deviceName	string	Device name
- unitType	int	Unit type
- channelSeq	int	Channel number
- nodeType	string	Channel type
- nodeCode	string	Channel code
- comment	string	Remarks
- alarmSourceName	string	Alarm channel name
- linkedVoicePrompt	string	Whether the voice prompt has been linked: 0 = No, 1 = Yes
- voicePromptContent	string	Voice Content
id	string	Message ID

Message Example

```
{
  "method": "vms.handleAlarm",
  "info": {
    "alarmCode": "{E44C5030-2192-4475-97B9-F99492084345}",
    "alarmDate": "1547122298",
    "handleUser": "system",
    "handleDate": "1547122305",
```

```

    "handleStatus": "1",
    "handleMessage": "",
    "mailReceivers": [],
    "alarmGrade": "1",
    "alarmPic": "",
    "alarmPicSize": "",
    "alarmStat": "1",
    "alarmType": "3",
    "deviceCode": "1000008",
    "deviceName": "dev8",
    "unitType": 1,
    "channelSeq": 0,
    "nodeType": "2",
    "nodeCode": "1000008$1$0$0",
    "comment": "",
    "alarmSourceName": "Channel0",
    "linkedVoicePrompt": "0",
    "voicePromptContent": ""
  },
  "id": "25704"
}

```

4.2.3 Notification of Alarm Images

Brief Description

- Notification of Alarm Images

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- vms.notifyAlarmPicture

Message Parameter

Parameter	Type	Description
method	string	Method name
info	string	None

Parameter	Type	Description
- alarmCode	string	Alarm code
- alarmDate	string	Alarm time, seconds
- picture	string	Picture URL, separated by semicolons. In sending request, ?token={credential} shall be appended to the URL for authentication
- pictureSize	string	Picture size
id	string	Message ID

Message Example

```
{
  "method": "vms.notifyAlarmPicture",
  "info": {
    "alarmCode": "{3B41B8FC-9312-4B2B-AD9E-7A2209510D4C}",
    "alarmDate": "1547455618",
    "picture": "https://192.168.1.1:9901/56602311-65f5-11eb-94b1-4cedfb428a36/20200101/1/dfs_cdaaefb1-6698-11eb-988d-4cedfb428a36_53486430_53797182.jpg",
    "pictureSize": ""
  },
  "id": "2848"
}
```

4.3 Access Control

4.3.1 Notification of Door Status

Brief Description

- Notification of Door Status

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- vms.doorAccessStatus

Message Parameter

Parameter	Type	Description
id	string	Message ID
info	object	Door status message
- channelCode	string	Channel code
- doorStatus	string	Door status: 0: Closed, 1: Open
- swipDate	string	Message generation time (seconds)
method	string	Method name

Message Example

```
{
  "id": "113",
  "info": {
    "channelCode": "1000007$7$0$2",
    "doorStatus": "2",
    "swipDate": "1562212982"
  },
  "method": "vms.doorAccessStatus"
}
```

4.3.2 Notification of Normally Open and Normally Closed Door Status

Brief Description

- Notification of Normally Open and Normally Closed Door Status

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- vms.doorAlwaysOpenCloseStatus

Message Parameter

Parameter	Type	Description
id	string	Message ID
info	object	None
- channelId	string	Access control channel code
- status	string	Door status: 0: Normal; 1: Normally closed; 2: Normally open
method	string	Method name

Message Example

```
{
  "id": "99",
  "info": {
    "channelId": "1000007$7$0$0",
    "status": "2"
  },
  "method": "vms.doorAlwaysOpenCloseStatus"
}
```

4.4 Person

4.4.1 Notification of Batch Adding Persons

Brief Description

- Notification of Batch Adding Persons

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.person.addPersons

Message Parameter

Parameter	Type	Description
method	string	Method name

Parameter	Type	Description
info	object	MQ message content
- personIds	array	Person ID
id	string	Message ID

Message Example

```
{
  "method": "obms.person.addPersons",
  "info": {
    "personIds": [
      "123"
    ]
  },
  "id": "12782"
}
```

4.4.2 Notification of Batch Updating Persons

Brief Description

- Notification of Batch Updating Persons

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.person.updatePersons

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	MQ message content
- personIds	array	Person ID
id	string	Message ID

Message Example

```
{
  "method": "obms.person.updatePersons",
  "info": {
    "personIds": [
      "123"
    ]
  },
  "id": "12782"
}
```

4.4.3 Notification of Batch Deleting Persons

Brief Description

- Notification of Batch Deleting Persons

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.person.deletePersons

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	MQ message content
- personIds	array	Person ID
id	string	Message ID

Message Example

```
{
  "method": "obms.person.deletePersons",
  "info": {
    "personIds": [
```

```
        "123"
      ],
      "id": "12782"
    }
  }
```

4.4.4 Notification of Batch Operating Access Control Records

Brief Description

- Notification of Batch Operating Access Control Records

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- brms.notifyAcsEvents

Message Parameter

Parameter	Type	Description
method	string	Method name
info	array	None
- channelCode	string	Channel code
- channelName	string	Channel name
- cardNum	string	Card number
- firstName	string	First name
- lastName	string	Last name
- tel	string	Phone number
- gender	string	Gender, 0 = Unknown; 1 = Male; 2 = Female, 0 by default
- idNum	string	ID number
- personId	string	ID of card swiping person
- stageId	string	Stage ID
- buildingId	string	Building ID

Parameter	Type	Description
- unitId	string	Unit ID
- roomId	string	Room number
- cardType	string	Card type for access control, 0 = General; 1=Blocklist; 2 = Visitor; 3 = Patrol; 4 = VIP; 5 = Other; 0 by default
- inOut	string	In and out: 0: In; 1: Out
- eventType	string	Event type. Dictionary type, see [6.1.7 Alarm Type]
- deviceType	string	Device type
- swipDate	string	Card swiping time: second
- pictureUrl	string	Face picture URL
- picture1	string	Face snapshot URL1
- picture2	string	Face snapshot URL2
- picture3	string	Face snapshot URL3
- picture4	string	Face snapshot URL4
- memo	string	Description
- alarmCode	string	Alarm code
- pointId	string	Passpoint ID
- pointName	string	Passpoint name
- pointType	string	Passpoint type: 1=access control, 2=elevator, 3=gate
- areaCode	string	Area code
- areaName	string	Area name
- orgInfos	array	Personnel groups
- - orgCode	string	Personnel group code
- - orgName	string	Personnel group name
- antiPassType	string	Anti-passback type: 0=local or non-anti-passback, 1=global
- operateType	string	Operation type: 0=other, 1=user, 2=room, 3=management machine
- operateInfo	string	Operational information
- temperature	string	Human body temperature
- temperatureUnit	string	Temperature units: 0=Celsius, 1=Fahrenheit, 2=Kelvin
id	string	Message ID

Message Example

```
{
  "method": "brms.notifyAcsEvents",
  "info": [
    {
      "channelCode": "1000748$7$0$0",
      "channelName": "Door1",
      "cardNum": "2BD7182A",
      "firstName": "zhangsan",
      "lastName": "",
      "tel": "",
      "gender": "0",
      "idNum": "",
      "personId": "1",
      "stageId": "",
      "buildingId": "",
      "unitId": "",
      "roomId": "1#1#1001",
      "cardType": "",
      "inOut": "0",
      "eventType": "51",
      "deviceType": "5",
      "swipDate": "1616385414",
      "pictureUrl": "https://192.168.1.1:8443/upload/obms/headPic/1.jpg",
      "picture1": "",
      "picture2": "",
      "picture3": "",
      "picture4": "",
      "memo": "",
      "alarmCode": "{75E566B8-00B9-482B-864E-4789510D6D8F}",
      "pointId": "1",
      "pointName": "point1",
      "pointType": "1",
      "areaCode": "001001",
      "areaName": "area1",
      "orgInfos": [
        {
          "orgCode": "001001",
          "orgName": "org1"
        }
      ],
      "antiPassType": "1",
      "operateType": "1",
      "operateInfo": "",
      "temperature": "37",
      "temperatureUnit": "0"
    }
  ],
  "id": "13"
}
```

4.5 Visitor

4.5.1 Notification of Adding a Visitor

Brief Description

- Notification of Adding a Visitor

Topic

- mq.common.msg.topic

Parameter

None

Method

- vms.addVisitor

Message Parameter

Parameter	Type	Description
method	string	Method name
info	string	None
- visitorId	string	Visitor ID
- status	string	Visitor status, 0 = Appointed; 1 = In visit
- source	string	Source of visitor, 1 = Client; 2 = VDPAPP
- registerType	string	Appointment registration method: 0-self-service reservation, 1-invitee invitation, 2-administrator registration, 3-administrator reservation, 4-passenger plane registration
- registerDetail	string	Appointment registration details
- createTime	string	Creation time, timestamp, unit: second
- visitorName	string	Visitor name
- visitorOrgName	string	Organization of visitor
- visitedName	string	Name of visited person
- visitedOrgName	string	Organization of visited person

Parameter	Type	Description
- idType	string	Certificate type, 0 = ID card; 1 = Officer ID card; 2 = Student ID card; 3 = Driver license; 4 = Passport; 5 = Tax number; 6 = Other
- idNum	string	ID number
- tel	string	Contact number
- email	string	Email address
- expectArrivalTime	string	Appointed time of visit, timestamp, unit: second
- arrivalTime	string	Actual time of visit, timestamp, unit: second
- expectLeaveTime	string	Appointed visitor leaving time, timestamp, unit: second
- leaveTime	string	Actual visitor leaving time, timestamp, unit: second
- plateNo	string	Plate number
- reason	string	Reason for visit
- remark	string	Remarks
- personId	string	ID of associated person
- authInfo	object	Certification information
- - sipId	string	Room number
- - cardNo	string	Card number
- - facePictures	array	Face picture (Url)
- - qrcode	string	QR code data
- - passportCardNo	string	Card number in pass certificate
- - idPicture	string	Photo
- rightInfo	object	Authorization information
- - acsChannelIds	array	Channel of access control devices
- - vtoChannelIds	array	Access control channel of VTO devices
- - positionIds	array	Entrance/exit ID
- - liftChannels	array	Lift control channels
- - - channelId	string	Lift control channel code
- - - floors	string	Lift control floors
id	string	Message ID

Message Example

```
{
  "method": "vms.addVisitor",
  "info": {
    "visitorId": "1",
    "status": "1",
    "source": "1",
    "registerType": "0",
    "registerDetail": "appointment",
    "createTime": "1572906543",
    "visitorName": "lisi",
    "visitorOrgName": "701",
    "visitedName": "zhangsan",
    "visitedOrgName": "009",
    "idType": "0",
    "idNum": "369552199004123354",
    "tel": "15500000001",
    "email": "duan1@12.cn",
    "expectArrivalTime": "1572916027",
    "arrivalTime": "1572916027",
    "expectLeaveTime": "1572923227",
    "leaveTime": "1572923227",
    "plateNo": "A12A26",
    "reason": "ju",
    "remark": "",
    "personId": "99999999",
    "authInfo": {
      "sipId": "1#1#1001",
      "cardNo": "AF123EB1",
      "facePictures": [
        "http://192.168.1.1:80/upload/obms/xxxx.jpg"
      ],
      "qrcode": "",
      "passportCardNo": "",
      "idPicture": ""
    },
    "rightInfo": {
      "acsChannelIds": [
        "1000001$7$0$0"
      ],
      "vtoChannelIds": [
        "1000002$7$0$0"
      ],
      "positionIds": [
        "1"
      ],
      "liftChannels": [
        {
          "channelId": "1000000$50$0$0",
          "floors": "1,2,3"
        }
      ]
    }
  }
}
```

```

    }
  ]
}
},
"id": "381"
}

```

4.5.2 Notification of Updating a Visitor

Brief Description

- Notification of Updating a Visitor

Topic

- mq.common.msg.topic

Parameter

None

Method

- vms.updateVisitor

Message Parameter

Parameter	Type	Description
method	string	Method name
info	string	None
- visitorId	string	Visitor ID
- status	string	Visitor status, 0 = Appointed; 1 = In visit
- source	string	Source of visitor, 1 = Client; 2 = VDPAPP
- registerType	string	Appointment registration method: 0-self-service reservation, 1-invitee invitation, 2-administrator registration, 3-administrator reservation, 4-passenger plane registration
- registerDetail	string	Appointment registration details
- createTime	string	Creation time, timestamp, unit: second
- visitorName	string	Visitor name
- visitorOrgName	string	Organization of visitor
- visitedName	string	Name of visited person

Parameter	Type	Description
- visitedOrgName	string	Organization of visited person
- idType	string	Certificate type, 0 = ID card; 1 = Officer ID card; 2 = Student ID card; 3 = Driver license; 4 = Passport; 5 = Tax number; 6 = Other
- idNum	string	ID number
- tel	string	Contact number
- email	string	Email address
- expectArrivalTime	string	Appointed time of visit, timestamp, unit: second
- arrivalTime	string	Actual time of visit, timestamp, unit: second
- expectLeaveTime	string	Appointed visitor leaving time, timestamp, unit: second
- leaveTime	string	Actual visitor leaving time, timestamp, unit: second
- plateNo	string	Plate number
- reason	string	Reason for visit
- remark	string	Remarks
- personId	string	ID of associated person
- authInfo	object	Certification information
- - sipId	string	Room number
- - cardNo	string	Card number
- - facePictures	array	Face picture (Url)
- - qrcode	string	QR code data
- - passportCardNo	string	Card number in pass certificate
- - idPicture	string	Photo
- rightInfo	object	Authorization information
- - acsChannelIds	array	Channel of access control devices
- - vtoChannelIds	array	Access control channel of VTO devices
- - positionIds	array	Entrance/exit ID
- - liftChannels	array	Lift control channels
- - - channelId	string	Lift control channel code
- - - floors	string	Lift control floors
id	string	Message ID

Message Example

```
{
  "method": "vms.updateVisitor",
  "info": {
    "visitorId": "1",
    "status": "1",
    "source": "1",
    "registerType": "0",
    "registerDetail": "appointment",
    "createTime": "1572906543",
    "visitorName": "lisi",
    "visitorOrgName": "701",
    "visitedName": "zhangsan",
    "visitedOrgName": "009",
    "idType": "0",
    "idNum": "369552199004123354",
    "tel": "15500000001",
    "email": "duan1@12.cn",
    "expectArrivalTime": "1572916027",
    "arrivalTime": "1572916027",
    "expectLeaveTime": "1572923227",
    "leaveTime": "1572923227",
    "plateNo": "A12A26",
    "reason": "ju",
    "remark": "",
    "personId": "99999999",
    "authInfo": {
      "sipId": "1#1#1001",
      "cardNo": "AF123EB1",
      "facePictures": ["http://192.168.1.1:80/upload/obms/xxxx.jpg"],
      "qrcode": "",
      "passportCardNo": "",
      "idPicture": ""
    },
    "rightInfo": {
      "acsChannelIds": [
        "1000001$7$0$0"
      ],
      "vtoChannelIds": [
        "1000002$7$0$0"
      ],
      "positionIds": [
        "1"
      ],
      "liftChannels": [
        {
          "channelId": "1000000$50$0$0",
          "floors": "1,2,3"
        }
      ]
    }
  }
}
```

```
}  
},  
"id": "381"  
}
```

4.5.3 Notification of Batch Deleting Visitors

Brief Description

- Notification of batch deleting visitors

Topic

- mq.common.msg.topic

Parameter

None

Method

- obms.visitor.deleteVisitors

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None
- visitorIds	array	Visitor ID
id	string	Message ID

Message Example

```
{  
  "method": "obms.visitor.deleteVisitors",  
  "info": {  
    "visitorIds": ["1"]  
  },  
  "id": "391"  
}
```

4.5.4 Notification of the Number of Appointments and On-site Visitors

Brief Description

- Notification of the number of appointments and on-site visitors. Due to business adjustments, the protocol will be obsolete in the next version.

Topic

- mq.common.msg.topic

Parameter

None

Method

- vms.visitorCountReport

Message Parameter

Parameter	Type	Description
method	string	Method name
info	string	None
- reserved	string	Number of appointments
- visiting	string	Number of online visitors
- approving	string	Number of visitors to be approved
- retention	string	Number of stranded visitors
id	string	Message ID

Message Example

```
{
  "method": "vms.visitorCountReport",
  "info": {
    "reserved": "1",
    "visiting": "1",
    "approving": "1",
    "retention": "1"
  },
  "id": "381"
}
```

4.6 Vehicle(Entrance and Exit)

4.6.1 Notification of Batch Add Vehicles

Brief Description

- Notification of batch add vehicles

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- ipms.vehicle.addVehicles

Message Parameter

Parameter	Type	Description
id	string	Message ID
method	string	Method name
info	object	MQ message content
- vehicleIds	array	Vehicle ID

Message Example

```
{
  "id": "168593",
  "method": "ipms.vehicle.addVehicles",
  "info": {
    "vehicleIds": ["1","2","3"]
  }
}
```

4.6.2 Notification of Batch Update Vehicles

Brief Description

- Notification of batch update vehicles

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- ipms.vehicle.updateVehicles

Message Parameter

Parameter	Type	Description
id	string	Message ID
method	string	Method name
info	object	MQ message content
- vehicleIds	array	Vehicle ID

Message Example

```
{
  "id": "168593",
  "method": "ipms.vehicle.updateVehicles",
  "info": {
    "vehicleIds": ["1","2","3"]
  }
}
```

4.6.3 Notification of Batch Delete Vehicles

Brief Description

- Notification of batch delete vehicles

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- ipms.vehicle.deleteVehicles

Message Parameter

Parameter	Type	Description
id	string	Message ID
method	string	Method name
info	object	MQ message content
- vehicleIds	array	Vehicle ID

Message Example

```
{
  "id": "168593",
  "method": "ipms.vehicle.deleteVehicles",
  "info": {
    "vehicleIds": ["1","2","3"]
  }
}
```

4.6.4 Notification of Snapshot Information of Vehicles at the Entrance and Exit

Brief Description

- Notification of Snapshot Information of Vehicles at the Entrance. When direct=9, the parking duration can be obtained through the parkingTime field.

Topic

- mq.common.msg.topic

Parameter

None

Method

- ipms.entrance.notifyVehicleCaptureInfo

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	MQ message content
- parkingLotId	string	Parking lot ID

Parameter	Type	Description
- parkingLotName	string	Parking lot name
- positionId	string	Position ID
- positionName	string	Position name
- pointId	string	Gate ID
- pointName	string	Gate name
- channelId	string	ID of snapshot channel
- channelName	string	Name of snapshot channel
- captureTime	string	Capture time (in seconds)
- bindingChannelIds	array	IDs of channels bound with resources
- plateNo	string	License plate number
- originalPlateNo	string	Original license plate number
- extPlateNo	string	Extended license plate number
- fuzzyMatched	string	Whether the license plate is fuzzy matching: 1=Yes, 0=No
- accessAuthority	string	Access authority: 1=allow access, 2=prohibit access, 3=verify access, 4=visitors are not in the validity period, 5=(site area) is insufficient, 6=(user) has no available parking space
- vehicleBrand	string	Vehicle brand, dictionary type. See [6.1.12 Brand of motor vehicles]
- vehicleColor	string	Vehicle color, dictionary type. See [6.1.10 Color of motor vehicles]
- inputVehicleBrand	string	Vehicle brand(input), dictionary type. See [6.1.12 Brand of motor vehicles]
- inputVehicleColor	string	Vehicle color(input), dictionary type. See [6.1.10 Color of motor vehicles]
- capturePicture	string	Snapshot image
- capturePlatePicture	string	License plate image
- direct	string	Driving direction: 8=In, 9=Out
- personId	string	Person ID
- firstName	string	First name of vehicle owner
- lastName	string	Last name of vehicle owner
- companyName	string	Company name
- orgCode	string	Person group code

Parameter	Type	Description
- orgName	string	Person group name
- facePicture	string	Face image URL
- passModel	string	Release mode: 1=Released, 2=Not released
- vehicleType	string	Vehicle type: 1=Visitor, 2=Allowlist, 4=Blocklist, 5=Temporary vehicle
- lastEnterRecordId	string	Last entry record ID
- parkingTime	string	Parking time, unit: second
- entranceGroups	array	Entrance and exit groups
- - groupId	string	ID of entrance and exit group
- - groupName	string	Name of entrance and exit group
- - defaultGroup	string	Default group: 1=Yes, 0=No
- - groupColor	string	Background color
- orgInfos	array	Person groups
- - orgCode	string	Person group code
- - orgName	string	Person group name
id	string	Message ID

Message Example

```
{
  "id": "12782",
  "method": "ipms.entrance.notifyVehicleCaptureInfo",
  "info": {
    "parkingLotId": "1",
    "parkingLotName": "test",
    "positionId": "2",
    "positionName": "e2",
    "pointId": "3",
    "pointName": "r4",
    "channelId": "100005$1$0$0",
    "channelName": "qwe",
    "captureTime": "1686569039",
    "bindingChannelIds": [],
    "plateNo": "cd",
    "originalPlateNo": "A15679",
    "extPlateNo": "ABC",
    "fuzzyMatched": "1",
    "accessAuthority": "1",
    "vehicleBrand": "1",
    "vehicleColor": "2",
    "inputVehicleBrand": "1",
```

```

    "inputVehicleColor": "2",
    "capturePicture": "",
    "capturePlatePicture": "",
    "direct": "8",
    "personId": "23",
    "firstName": "ca",
    "lastName": "t",
    "companyName": "qw",
    "orgCode": "",
    "orgName": "",
    "facePicture": "",
    "passModel": "1",
    "vehicleType": "1",
    "lastEnterRecordId": "1441683",
    "parkingTime": "24",
    "entranceGroups": [
      {
        "groupId": "1",
        "groupName": "vip",
        "defaultGroupName": "1",
        "groupColor": "1"
      }
    ],
    "orgInfos": [
      {
        "orgCode": "002",
        "orgName": "test"
      }
    ]
  }
}

```

4.6.5 Notification of Opening Results at the Entrance and Exit

Brief Description

- Notification of Opening Results at the Entrance and Exit

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- ipms.entrance.openSluice

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	MQ message content
- parkingLotId	string	Parking lot ID
- parkingLotName	string	Parking lot name
- positionId	string	Position ID
- positionName	string	Position name
- pointId	string	Gate ID
- pointName	string	Gate name
- channelId	string	Channel No.
- channelName	string	Channel name
- openSluiceResult	string	Gate opening result: 0 = Failure, 1 = Success
- openSluiceMode	string	Opening mode: 0=system opening, 1=remote opening, 2=card swiping / face fingerprint / QR code opening
- openSluiceUserId	string	Remote opening user ID
id	string	Message ID

Message Example

```
{
  "method": "ipms.entrance.openSluice",
  "info": {
    "parkingLotId": "1",
    "parkingLotName": "sc",
    "positionId": "2",
    "positionName": "da",
    "pointId": "2",
    "pointName": "dasd",
    "channelId": "10000$1$2$1",
    "channelName": "1",
    "openSluiceResult": "1",
    "openSluiceUserId": "",
    "openSluiceResult": "1"
  },
  "id": "12782"
```

```
}
```

4.7 Face

4.7.1 Notification of Sending Faces

Brief Description

- Notification of sending faces

Topic

- mq.cluster.common.msg.topic

Parameter

None

Method

- brms.notifyFaceAddRepo

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None
- data	object	None
- - repositoryId	int	Face comparison group ID
- - issuePerson	int	Indicate whether to send: 0=No, 1=Yes
- - deviceCodes	array	Device code
- serverCode	string	Server code

Message Example

```
{
  "info": {
    "data": {
      "deviceCodes": ["1003264"],
      "issuePerson": "1",
      "repositoryId": 3
    },
    "serverCode": "video#local"
  },
}
```

```
"method": "brms.notifyFaceAddRepo"
}
```

4.7.2 Notification of Face Snapshot/Recognition Messages

Brief Description

- Notification of face snapshot/recognition messages

Topic

- mq.common.msg.topic

Parameter

None

Method

- brms.notifyFaceInfos

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None
- channelId	string	Channel code
- faceImageUrl	string	Face picture
- pictureUrl	string	Scene picture
- appearTimes	string	Appearance times
- hited	string	Whether to recognize: 0: Capture; 1: Recognize
- recAge	string	Age
- recGender	string	Gender: 0=Unrecognized; 1=Male; 2=Female
- recFringe	string	Feature, fringe: 0=No; 1=Yes
- recEye	string	Feature, eye: 1=Unrecognized; 2=Closed; 3=Opened
- recMouth	string	Feature, mouth: 1=Unrecognized; 2=Closed; 3=Opened
- recMask	string	Feature, mask: 0=Unknown (SDK); 1=Unrecognized; 2=Without mask; 3=With mask

Parameter	Type	Description
- recBeard	string	Feature, beard: 0=Unknown (SDK); 1=Unrecognized; 2=Without beard; 3=With beard
- recGlasses	string	Feature, glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
- recEmotion	string	Feature, expressions: 0=Smile; 1=Angry; 2=Sad; 3=Disgusted; 4=Scared; 5=Surprised; 6=Normal; 7=Laugh; 8=Happy; 9=Confused; 10=Scream
- beginTime	string	Start time (second)
- endTime	string	End time (second)
- similarFaces	array	Similar faces
- - uid	string	Person ID
- - faceImageUrl	string	Face picture
- - name	string	Name
- - personTypeId	string	Person type ID
- - personTypeName	string	Person type name
- - personId	string	ID number, such as ID card, job number, etc.
- - gender	string	Gender: 0: Unrecognized; 1: Male; 2: Female
- - birthday	string	Date of birth
- - nationality	string	Nationality
- - memo	string	Remarks
- - repositoryId	string	Face database ID
- - repositoryName	string	Face database name
- - repositoryColor	string	Library color: 1: Gray; 2: Red; 3: Orange; 4: Yellow; 5: Green; 6: Cyan; 7: Blue; 8: Purple
- - similarity	string	Similarity
- - surveillance	string	Armed or not: 0: No; 1: Yes
id	string	Message ID

Message Example

```
{
```

```
"method": "brms.notifyFaceInfos",
"info": [{
  "channelId": "1000319$1$0$0",
  "faceImageUrl": "https://192.168.1.1:50001/afefb58d-5b66-11e9-8037-00000000
0000/20190612/1/dsf_739dc016-8d79-11e9-9f8a-000000000000_34998401_35001200.j
pg",
  "pictureUrl": "https://192.168.1.1:50001/afefb58d-5b66-11e9-8037-0000000000
000/20190612/1/dsf_739dc016-8d79-11e9-9f8a-000000000000_34782193_34998401.jpg
",
  "appearTimes": "0",
  "hited": "1",
  "recAge": "29",
  "recGender": "1",
  "recFringe": "0",
  "recEye": "3",
  "recMouth": "3",
  "recMask": "2",
  "recBeard": "2",
  "recGlasses": "1",
  "recEmotion": "6",
  "beginTime": "1560390144",
  "endTime": "1560390144",
  "serviceCode": "4001",
  "similarFaces": [
    {
      "uid": "24",
      "faceImageUrl": "https://192.168.1.1:50001/afefb58d-5b66-11e9-8037
-000000000000/20190612/1/dsf_739dc016-8d79-11e9-9f8a-000000000000_35001200_3
5021084.jpg",
      "name": "1212",
      "personType": "3",
      "personTypeName": "Staff",
      "personId": "14565456",
      "gender": "1",
      "birthday": "",
      "nationality": "9999",
      "memo": "",
      "repositoryId": "3",
      "repositoryName": "test",
      "repositoryColor": "3",
      "similarity": "38",
      "surveillance": "0"
    }
  ]
}],
"id": "1262006"
}
```

4.7.3 Notification of Face Search by Image Result Progress

Brief Description

- Notification of face search by image result progress

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- brms.face.searchDetectionByPictureProcess

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None
- session	string	Session mark
- process	string	Search progress. The value of 100 indicates that the search is completed.
- currentCount	string	The total number of results that have been found at present, with up to 1,000 records.
id	string	Message ID

Message Example

```
{
  "method": "brms.face.searchDetectionByPictureProcess",
  "info": {
    "session": "2cd252ce7ab9e0",
    "process": "13",
    "currentCount": "130"
  },
  "id": "196933"
}
```

4.8 Metadata

4.8.1 Notification of Human Body Snapshots

Brief Description

- Notification of human body snapshots

Topic

- mq.common.msg.topic

Parameter

None

Method

- brms.notifyHumanInfos

Message Parameter

Parameter	Type	Description
id	string	Message ID.
method	string	Method
info	object	None
- channelId	string	Channel number
- faceImageUrl	string	Face image
- humanImageUrl	string	Human body image
- pictureUrl	string	Scenarios
- gender	string	Gender
- coat	string	Tops
- coatColor	string	Top color
- trousers	string	Bottoms
- trousersColor	string	Bottom color
- hat	string	Hat
- bag	string	Bag
- emotion	string	Expression
- glasses	string	Glasses
- beard	string	Beard
- mask	string	Mask
- age	string	Age
- captureTime	string	Snapshot time

Parameter	Type	Description
- facelImageTop	string	Top coordinate of the face image
- facelImageLeft	string	Left coordinate of the face image
- facelImageBottom	string	Bottom coordinate of the face image
- facelImageRight	string	Right coordinate of the face image
- humanImageTop	string	Top coordinate of the human body image
- humanImageLeft	string	Left coordinate of the human body image
- humanImageBottom	string	Bottom coordinate of the human body image
- humanImageRight	string	Right coordinate of the human body image
- detectMode	string	The specific type of information contained in the detected object, 0: from attributes 1: from features

Message Example

```
{
  "id": 1,
  "method": "brms.notifyHumanInfos",
  "info": [
    {
      "channelId": "1000005$1$0$0",
      "facelImageUrl": "",
      "humanImageUrl":
        "https://192.168.1.1:443/upload/picture/body/20181206/11/14/2d6c344b-87d1-452b-833a-61ccf6463f16.jpg",
      "pictureUrl":
        "https://192.168.1.1:443/upload/picture/scenebody/20181206/11/14/6687ec54-cb82-45a2-ab55-fed465de3a8a.jpg",
      "gender": "0",
      "coat": "1",
      "coatColor": "5",
      "trousers": "1",
      "trousersColor": "4",
      "hat": "1",
      "bag": "1",
      "emotion": "",
      "glasses": "",
      "beard": "0",
    }
  ]
}
```

```

    "mask": "0",
    "age": "0",
    "captureTime": "1544252578",
    "faceImageTop": "",
    "faceImageLeft": "",
    "faceImageBottom": "",
    "faceImageRight": "",
    "humanImageTop": "3845",
    "humanImageLeft": "2394",
    "humanImageBottom": "5779",
    "humanImageRight": "2688",
    "detectMode": "0"
  }
}

```

4.8.2 Notification of Motor Vehicle Snapshots

Brief Description

- Notification of motor vehicle snapshots

Topic

- mq.common.msg.topic

Parameter

None

Method

- brms.notifyVehicleInfos

Message Parameter

Parameter	Type	Description
id	string	The message ID.
method	string	Method
info	object	None
- channelId	string	Channel number
- plateImageUrl	string	Plate image
- vehicleUrl	string	Vehicle image
- pictureUrl	string	Panoramic image
- plate	string	Plate number

Parameter	Type	Description
- plateColor	string	Plate color
- carType	string	The vehicle type, which is a dictionary. For details, see [6.1.11 Model of motor vehicles].
- carColor	string	The vehicle color, which is a dictionary. For details, see [6.1.10 Color of motor vehicles].
- carBrand	string	The vehicle trademark, which is a dictionary. For details, see [6.1.12 Brand of motor vehicles].
- captureTime	string	Snapshot time
- carImageTop	string	Top coordinate of the vehicle body image
- carImageLeft	string	Left coordinate of the vehicle body image
- carImageBottom	string	Bottom coordinate of the vehicle body image
- carImageRight	string	Right coordinate of the vehicle body image
- detectMode	string	The specific type of information contained in the detected object, 0: from attributes 1: from features

Message Example

```
{
  "id": 1,
  "method": "brms.notifyVehicleInfos",
  "info": [
    {
      "channelId": "1000009$1$0$0",
      "plateImageUrl":
        "https://192.168.1.1:443/upload/picture/vehicle/20181206/11/14/ce7bd226-6b9e-49c2-a380-fc5cb8a9004c.jpg",
      "vehicleUrl":
        "https://192.168.1.1:443/upload/picture/vehicle/20181206/11/14/ae7ac226-6b9e-47d2-a380-fc5fcsha0041.jpg",
      "pictureUrl":
        "https://192.168.1.1:443/upload/picture/scenecar/20181206/11/14/6e99e4e9-ff71-498a-bcfe-c8634fd10ac3.jpg",
      "plate": "",
      "plateColor": "3",
      "carType": "22",
      "carColor": "0",
      "carBrand": "0",
      "captureTime": "1544066039",
      "carImageTop": "2012",
      "carImageLeft": "3727",
      "carImageBottom": "2420",

```

```

        "carImageRight": "4312",
        "detectMode": "0"
    }
}

```

4.8.3 Notification of Non-motor Vehicle Snapshots

Brief Description

- Notification of non-motor vehicle snapshots

Topic

- mq.common.msg.topic

Parameter

None

Method

- brms.notifyNonVehicleInfos

Message Parameter

Parameter	Type	Description
id	string	Message ID
method	string	Method
info	string	None
- channelId	string	Channel number
- carImageUrl	string	Vehicle image
- pictureUrl	string	Panoramic image
- carType	string	The vehicle type, which is a dictionary. For details, see [6.1.14 Model of non motorized vehicles].
- carColor	string	The vehicle color, which is a dictionary. For details, see [6.1.13 Color of non motorized vehicles].
- riderNum	string	The number of riders.
- captureTime	string	Snapshot time
- carImageTop	string	Top coordinate of the vehicle image
- carImageLeft	string	Left coordinate of the vehicle image

Parameter	Type	Description
- carImageBottom	string	Bottom coordinate of the vehicle image
- carImageRight	string	Right coordinate of the vehicle image
- faceInfos	string	Face information
- - id	number	ID
- - faceId	string	Face related ID
- - faceOrder	number	Sort
- - faceUrl	string	Face images
- - bodyUrl	string	Human body image
- - captureTime	string	Snapshot time
- - sex	number	The gender: 0 = Unrecognized, 1 = Male, 2 = Female.
- - mask	number	Whether with Mask: 0 = unknown (SDK), 1 = unrecognized, 2 = no, 3 = yes
- - glass	number	Whether wearing glasses: -1=Unknown, 0=Not Wearing Glasses, 1=Regular Glasses, 2=Sunglasses, 3=Black-rimmed Glasses
- - mouth	number	MouthState: 1= unrecognized, 2 = closed, 3 = open
- - emotion	number	The emotion: -1 = N/A, 0 = Smiling, 1 = Angry, 2 = Sad, 3 = Disgusted, 4 = Feared, 5 = Surprised, 6 = Normal, 7 = Laughing, 8 = Happy, 9 = Confused, 10 = Screaming.
- - call	number	None
- - carrierBag	number	None
- - bag	number	None
- - umbrella	number	None
- - upClothes	number	None
- - downClothes	number	None
- - upperBodyColor	number	None
- - lowerBodyColor	number	None

Message Example

```
{
  "id": 1,
  "method": "brms.notifyNonVehicleInfos",
  "info": [
    {
      "channelId": "1000009$1$0$0",

```

```

        "carImageUrl":
        "https://192.168.1.1:443/upload/picture/nomotor/20181206/11/14/c5bc3861-4faf-42fa-
        8e7a-cf2a60af73f0.jpg",
        "pictureUrl":
        "https://192.168.1.1:443/upload/picture/scenenomotor/20181206/11/14/445594cf-
        54e2-4a85-9dcf-5f43bcd0b90d.jpg",
        "carType": "2",
        "carColor": "8",
        "riderNum": "1",
        "captureTime": "1544066042",
        "carImageTop": "2136",
        "carImageLeft": "3682",
        "carImageBottom": "2648",
        "carImageRight": "4184",
        "faceInfos": [
        {
        "id": null,
        "faceId": "126585xd56a55d",
        "faceOrder": 0,
        "faceUrl": "https://192.168.1.1:443/.../c5bc3861-4faf-42fa-8e7a-
        cf2a60af73f0.jpg",
        "bodyUrl": null,
        "captureTime": "",
        "sex": 1,
        "mask": 1,
        "glass": 1,
        "mouth": 1,
        "emotion": -1,
        "call": null,
        "carrierBag": null,
        "bag": null,
        "umbrella": null,
        "upClothes": null,
        "downClothes": null,
        "upperBodyColor": null,
        "lowerBodyColor": null
        }
        ]
    }
}

```

4.8.4 Notifying the Progress of Searching for Human Snapshots by Images

Brief Description

- Notification of progress of searching for human snapshots by images

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- brms.videoAnalyse.searchHumanByPictureProcess

Message Parameter

Parameter	Type	Description
method	string	Method name
info	object	None
- session	string	Session ID
- process	string	Search progress. When it is 100, indicating that the search is complete.
- currentCount	string	The total number of results currently searched. Maximum 1000 records can be searched.
id	string	Message ID

Message Example

```
{
  "method": "brms.videoAnalyse.searchHumanByPictureProcess",
  "info": {
    "session": "2cd252ce7ab9e0",
    "process": "13",
    "currentCount": "130"
  },
  "id": "196933"
}
```

4.9 Alarm Control Panel

4.9.1 Arming Notification of Alarm Control Panel Sub-systems

Brief Description

- Arming notification of alarm control panel sub-systems

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.alarmhost.notifyAreaArmStatus

Message Parameter

Parameter	Type	Description
method	string	Method
info	object	None
- areas	array	Sub-system
- - deviceCode	string	Device code
- - areald	string	Sub-system ID
- - armStatus	string	Arming status: 0=Disarmed, 1=Armed, 2=P1 armed, 3=P2 armed, 4=P1 + P2 armed.
id	string	Message ID

Message Example

```
{
  "method": "obms.alarmhost.notifyAreaArmStatus",
  "info": {
    "areas":
    [
      {
        "deviceCode": "1000001",
        "areald": "1000001$3$0$-1",
        "armStatus": "0"
      }
    ]
  },
  "id": "14766"
}
```

4.9.2 Bypass Notification of the Alarm Control Panel Protection Zone

Brief Description

- Bypass notification of the alarm control panel protection zone

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.alarmhost.notifyZoneBypassStatus

Message Parameter

Parameter	Type	Description
method	string	Method
info	object	None
- zones	array	Protection zone
- - deviceCode	string	Device code
- - zoneId	string	Protection zone ID
- - byPassStatus	string	Bypass status: 0 = Normal, 1 = Bypass, 2 = Isolated
id	string	Message ID

Message Example

```
{
  "method": "obms.alarmhost.notifyZoneBypassStatus",
  "info": {
    "zones":
    [
      {
        "deviceCode": "1000001",
        "zoneId": "1",
        "byPassStatus": "0"
      }
    ]
  }
}
```

```

    },
    "id": "14766"
  }

```

4.9.3 Fault Notification of the Alarm Control Panel Protection Zone

Brief Description

- Fault notification of the alarm control panel protection zone

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.alarmhost.notifyZoneFaultStatus

Message Parameter

Parameter	Type	Description
method	string	Method
info	object	None
- zones	array	Protection zone
- - deviceCode	string	Device code
- - zoneId	string	Protection zone ID
- - faultStatus	string	Fault status: 0 = Normal, 1 = Tampering, 2 = Short circuit, 3 = Anti-masking
id	string	Message ID

Message Example

```

{
  "method": "obms.alarmhost.notifyZoneFaultStatus",
  "info": {
    "zones":
    [
      {
        "deviceCode": "1000001",

```

```

        "zoneId": "1",
        "faultStatus": "0"
    }
]
},
"id": "14766"
}

```

4.9.4 Information Synchronization Notification of Alarm Control Panel Devices

Brief Description

- Information synchronization notification of alarm control panel devices

Topic

- mq.event.msg.topic.{userId}

Parameter

Parameter	Required	Type	Description
userId	Yes	string	User ID

Method

- obms.alarmHost.synchronizeDeviceInfo

Parameter

Parameter	Type	Description
method	string	Method
info	object	None
- deviceCode	string	Device code
- areas	array	Sub-system
- - armStatus	string	Arming status: 0 = Disarmed, 1 = Armed, 2 = P1 armed, 3 = P2 armed, 4 = P1 + P2 armed.
- - areaId	string	Sub-system ID
- - areaName	string	Sub-system name
- - partitions	array	Partition
- - - partitionId	string	Partition ID
- - - partitionName	string	Partition name

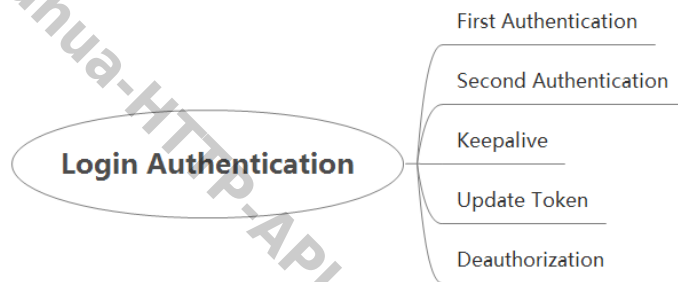
Parameter	Type	Description
id	string	Message ID

Message Example

```
{
  "method": "obms.alarmHost.synchronizeDeviceInfo",
  "info": {
    "deviceCode": "1000001",
    "areas": [
      {
        "armStatus": "0",
        "areaid": "1000001$3$0$-1",
        "areaName": "AREA1",
        "partitions": [
          {
            "partitionId": "1",
            "partitionName": "Partition_1"
          },
          {
            "partitionId": "2",
            "partitionName": "Partition_2"
          }
        ]
      }
    ]
  },
  "id": "14766"
}
```

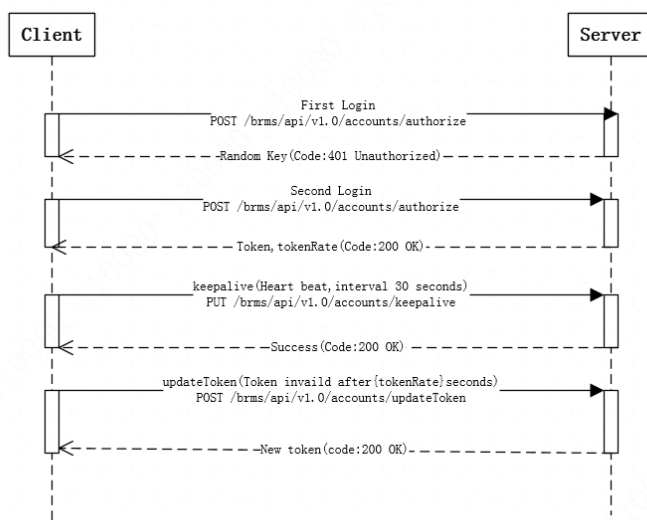
5 Best Practices

5.1 Login Authentication



Developers get the username and password of VMS and log in to the platform using HTTP Digest authentication. After successful login authentication, they can obtain the identity credential 'token' and the 'credential' information for accessing platform images or files. For security reasons, the token is valid for 30 seconds by default. Developers need to call the heartbeat keep-alive interface once within 30 seconds to keep the token alive. The validity period of the token is 30 minutes. Developers need to call the update token interface once within 30 minutes to refresh the Token. For security reasons, VMS separates the access API credential (Token) and the access image credential (Credential). Credential does not need to be kept alive or updated separately. When updating the token, VMS will synchronously associate and update the credential.

Login Authentication Process:



Docking Example:

See [6.3 Docking Example].

5.1.1 Get Authentication Token

Step1 First authentication For specific parameters, see the [3.1.1 Trigger Authentication] interface

For the first authentication request, carry the userName. The platform then returns Error Code 401 (meaning no permission), and contains encrypted information (including realm, randomKey, encryptType and publicKey) in the message body.

Step 2 Second authentication For specific parameters, see the [3.1.2 Submit Credentials] interface.

The requester generates a pair of AES secret key (32-bit string) and vector (16-bit string), and encrypts them using the RSA public key (publickey) returned by the first login. The second authentication request needs to use the encrypted AES secret key and vector, and use base64 encoding as parameters. Pseudocode examples:

```

aesSecretKey = UUID.replace("-", "");
aesSecretVector = UUID.replace("-", "").substring(16);
encryptedAesSecretKeyWithBase64 = Base64(encryptByPublicKey(aesSecretKey, Base64(firstLoginReturnPublicKey)));
encryptedAesSecretVectorWithBase64 = Base64(encryptByPublicKey(aesSecretVector, Base64(firstLoginReturnPublicKey)));

```

In the second authentication, user requests shall carry the signature. The signature is calculated with the userName, password, realm, randomKey in the following way:

Pseudocode examples:

```

temp1 = md5(password)
temp2 = md5(userName + temp1)
temp3 = md5(temp2)
temp4 = md5(userName + ":" + realm + ":" + temp3)
signature = md5(temp4 + ":" + randomKey)

```

As shown in the aforementioned pseudocodes, a total of 5 times of MD5 encryption is needed. The parameter values and results values are all character strings. The result value of MD5 encryption are all in lower case, and the length of MD5 encryption is 32 digits. The assumed values are as follows:

Parameter	Value
userName	"system"
password	"admin123"
realm	"VMS"
randomKey	"9c2b603650f54bcb"

Calculated values at each stage are as follows:

Times of encryption	Parameter/Result	Value
1	Parameter	"admin123"
1	Result	"0192023a7bbd73250516f069df18b500"
2	Parameter	"system0192023a7bbd73250516f069df18b500"
2	Result	"5a0fdbbe44b86807b5e5e127918bbc475"
3	Parameter	"5a0fdbbe44b86807b5e5e127918bbc475"
3	Result	"1e27fadce9af09e120ab5142a83a679e"
4	Parameter	"system:VMS:1e27fadce9af09e120ab5142a83a679e"
4	Result	"4b923d65cbbfd724285a164c3178b055"
5	Parameter	"4b923d65cbbfd724285a164c3178b055:9c2b603650f54bcb"
5	Result	"024a3dc397a3844bb31d24f22b4d6035"

In the case of a domain user, the signature algorithm is different. signature = RSA (password, platform public key). The platform public key is the same publickey returned in the first login.

Two times of login authentication shall be completed in 10 seconds. Otherwise, the randomKey returned from the first authentication interface becomes invalid, and the interface returns Error Code 1101 (The nonce is illegal.) Therefore, using tools like Postman to commission the login authentication interface one by one is not recommended. Instead, coding shall be used to quickly complete two login requests.

5.1.2 Token Keepalive

For specific parameters, see the [3.1.3 Heartbeat Keep-alive] interface.

After successful authentication, the heartbeat keeping interface shall be called for keep-alive during keep-alive period. The keep-alive interval is recommended to be 2/3 of the keep-alive lasting period (every 20 seconds). The keep-alive request needs to bring the current valid authentication token. The token value is put in the X-Subject-Token parameter of the request head, and null is passed to the request message body.

5.1.3 Update Token

For specific parameters, see the [3.1.4 Update Token] interface.

After successful authentication, although the token starts keeping alive, it remains invalid after tokenRate. Therefore, you need to call the update token interface to get a new token, and keep alive with it. It is recommended that the interval for updating token is 2/3 of tokenRate. If the tokenRate is 1,800, it means that the period of validity is 30 minutes, and the token update frequency is about 20 minutes. After token update, to ensure consistent request authentication, the old token will be deleted in one minute.

5.1.4 Cancel Token Authorization

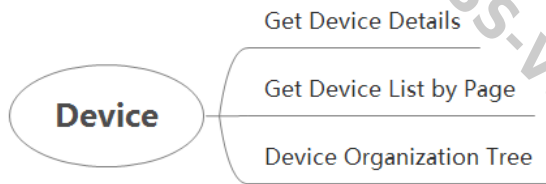
For specific parameters, see the [3.1.5 Logout] interface.

Logout on the client requires calling the authorization cancellation interface, and null is passed to the request message body.

Go to the following directory to view the detailed API interfaces

API Interface Directory	
API Interface	
++ Login Authentication	
+++ Trigger Authentication	
+++ Submit Credentials	
+++ Heartbeat Keep-alive	
+++ Update Token	
+++ Logout	

5.2 Device Management

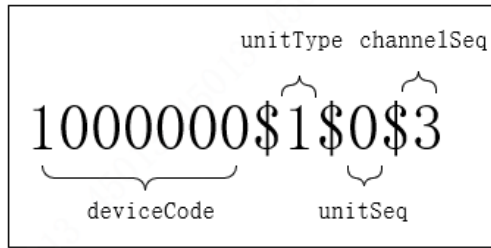


After the developer successfully logs in to the VMS platform, he can use the token to obtain resource information (organization, device, channel). Getting the device tree involves two protocols: Device organization tree and device list.

1. The device tree can get the relative relations of all organizations, devices, and channels.
2. The device list gets detailed information (such as device IP, port, and type) of devices, units, and channels.

Channel coding rules:

A channel code consists of the device code (deviceCode), unit type (unitType), unit number (unitSeq), and channel number (channelSeq), separated by \$. See the figure below:



Where The deviceCode is a 7-digit number. The device code of the newly added device on the platform increases one by one, starting from 1000000. The unitType see [6.1.3 Unit Type]. For detailed meanings, see the table below. The unitSeq represents the unit number that starts from 0. The channelSeq represents the channel number that starts from 0. In the figure above, 1000000\$1\$0\$3 means the 4th channel of the 1st unit of the coding unit of the device whose deviceCode is 1000000. If it is a subordinate device, the device code is like 1000000@001. This format indicates that the device comes from a cascaded domain with code 001, and the device code on the lower level platform is 1000000.

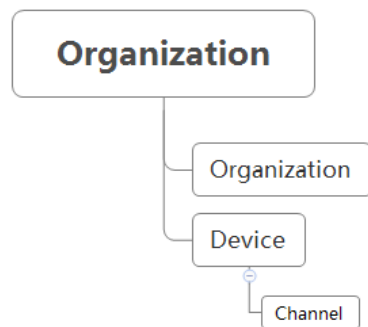
5.2.1 Getting Device Details

Search the device details on the platform through the device code.

5.2.2 Getting list of devices in pages

Get list of devices in pages.

5.2.3 Device Organization Tree (All)



Multiple organizations and devices can be loaded under an organization. Units of different types can be loaded under a device. Multiple channels can be loaded under a unit. The device tree interface supports conditional search by channel type (channelTypes) and organization (orgCode).

The organization tree established through the returned data by request. This organization tree has devices and channels, but no specific information of them. The specific information needs to be obtained from the device list.

5.2.4 Device Organization Tree (Graded)

Sometimes there are too many devices to be all displayed each time, so they can be displayed by grade. For this purpose, search child nodes (excluding grand child nodes) that meet conditions according to the parent node ID. The root node can be searched for null transmission. The subordinate level of an organization can be either organizations or devices. The subordinate level of devices can only be channels which have no subordinate levels.

Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Device
+++ Device Information
++++ Get Device Details
++++ Get the List of Devices in Pages
++++ Get the List of Device Types
++++ Search Offline Reasons of Devices
++++ Obtaining the Status of Devices and Channels in Batches
+++ Device Tree
++++ Getting Device List (all)
++++ Grading Device Organization Tree

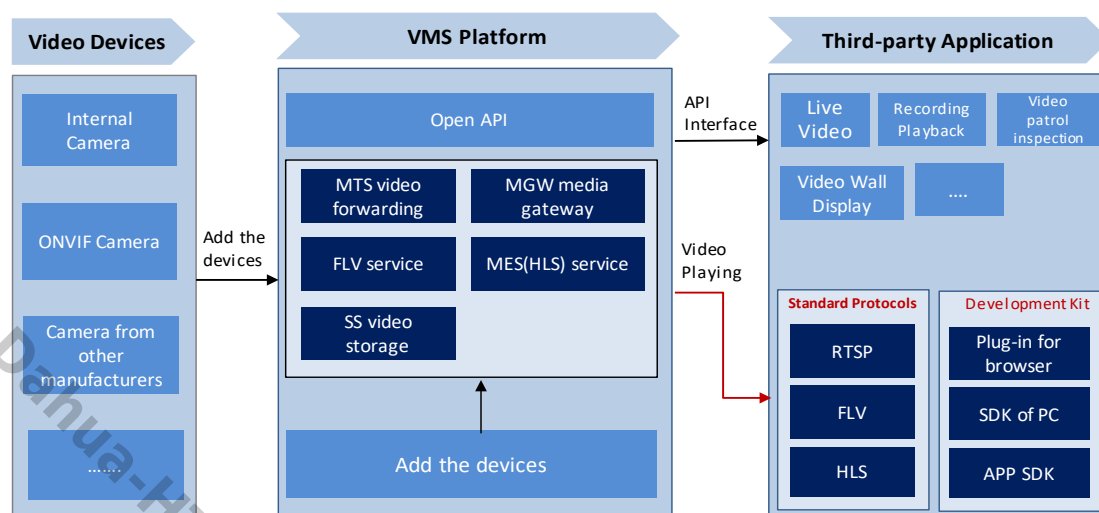
For specific MQ notifications, visit the following interfaces

MQ Notification Message
++ Device
+++ Notification of Device Online and Offline
+++ Notification of Channel Online and Offline

5.3 Video Services

5.3.1 Overall Solution of Video Playing

For complex video services, the VMS supports various playback protocols (RTSP/FLV/HLS) tailored to different application scenarios. To facilitate development, VMS also provides various development kits (browser plugins/PC SDK/App SDK), allowing developers to choose the appropriate integration method based on their specific use cases.



5.3.2 Application Scenarios and Recommended Solutions

- ✓ Recommended solutions for video playback in web browsers:

Integrate browser playback plugins or use FLV/HLS protocols for plugin-free video playback. Note that both FLV and HLS protocols do not support H.265 (HEVC) encoded streams by default.

- ✓ Recommended Solution for PC Client:

Integrate DPSDK or directly utilize RTSP/RTSPS protocols to enable video preview, recording playback, video downloads, and additional functionalities.

- ✓ Recommended Solution for App:

Integrate Android/iOS DPSDK or directly utilize RTSP/RTSPS protocols to enable video preview, recording playback, video downloads, and additional features.

- ✓ Recommended Solution for third-party player such as VLC or third-party pull stream service:

There is no need to login to the VMS platform; Use RTSP redirection to pull stream directly. In this mode, the RTSP URL remains valid indefinitely, eliminating token expiration issues. For third-party developers needing to persistently save stream addresses, we recommend utilizing RTSP redirection, directly pulling streams by redirecting to the platform's media gateway (MGW) on port 9090.

5.3.3 Different pulling stream protocols, their corresponding ports, and supported functionalities

Protocols	Redirection Port	Realtime Port	Playback Port	Supported Functions	Note
RTSP		9100	9320	Live video, playback, video download and Talk	When developing based on DSPDK, browser plugins, and the RTSP protocol, specific ports must be opened.
RTSP(S)		9102	9322		
RTSP Redirection	9090	9100	9320		
FLV		7896		Live video	H.264 decoding is supported, and stream encryption is not supported.
HLS		7096	7096	Live video and recordings playback	H.264/H.265 decoding is supported, and stream encryption is not supported.

5.3.4 Introduction of Pulling Stream Methods

✓ The playing process based on browser plugins:

VMS provides a web plug-in JavaScript SDK for browser users. This SDK encapsulates the RTSP over WebSocket pulling stream process, supports H.264/H.265 decoding, and allows for real-time playback and replay functions. The integration steps are as follows:

- 1) The developer contacts the VMS marketing representative to obtain the web plugin's JavaScript package.
- 2) The developer interacts with the VMS platform via the REST API to retrieve video channel information.
- 3) The developer integrates the web plugin package to play video or recordings from a specified video channel.

✓ The playing process based on DSPDK:

VMS provides the DSPDK dynamic library (DLL) for use in PC clients and apps. The DLL encapsulates the RTSP pulling stream process, enabling the implementation of complex video functionalities such as live video, playback, and intercom by integrating the DLL.

The steps are as follows:

- 1) The developer contacts the VMS marketing department representative to obtain the DSPDK dynamic library.

- 2) The developer interacts with the VMS platform through the REST API to obtain video channel information.
- 3) The developer calls the DPSDK dynamic library to play the video or recording of a specified video channel.

✓ **The playing process based on FLV/HLS:**

FLV/HLS are popular audio and video streaming protocols for browsers that enable plugin-free playing. VMS supports FLV/HLS pulling stream; however, these services are not included by default in the VMS baseline package. To enable FLV/HLS pulling stream, please contact the sales team to obtain the service. Once installed on the VMS platform, it will be supported.

The steps are:

- The developer contacts the VMS sales department to obtain the FLV/MES service package and installs it on the VMS platform.
- The developer interacts with the VMS platform through the REST API to obtain video channel resources.
- The developer uses the REST API to request the FLV/HLS pulling stream URL for the specified video resource from the VMS.
- The developer implements real-time video preview using FLV and video playback using HLS.

● **FLV URL format:**

`http://[service IP]:[service port]/live/cameraid/[camera ID]/substream/[stream type]/files.flv`

For example: `https://172.7.2.110:7896/live/cameraid/1000000%240/substream/1.flv`

● **HLS URL format:**

Central recording playback:

`http://[service IP]:[service port]/vod/center/cameraid/[camera ID]/substream/[stream type]/recordtype/recording type/totallength/[actual recording time]/begin time/[begin time]/endtime/[end time].m3u8`

Device recording playback:

`http://[ service IP]:[ service port]/vod/device/cameraid/[ camera ID D]/substream/[ stream type]/recordtype/recording type/totallength/[ actual recording time]/begin time/[start time]/endtime/[end time].m3u8`

● **Parameter description:**

[service IP]: The IP address of the server hosting the MES. If the MES is deployed

in a distributed manner, the IP is that of the server where the MESGW service is located.

[service port]: The port that the MES service listens on is 7086 by default. If the MES is deployed in a distributed manner, the port will be that of the MESGW service.

[camera ID]: such as 1000000%40001%240. 1000000 represents the device number, and 001 represents the cascade number. The two are separated by "%40," which represents "@". "%24" represents the symbol "\$", and the trailing "0" indicates channel 1 (starting from 0).

[stream type]: 1 represents the main stream and 2 represents the auxiliary stream.

[actual recording time]: The actual total recording time obtained from SS. The unit is second.

[recording type]: central recording: recordType=0 all, recordType=1 general recording, recordType=2 alarm recording, recordType=81

supplementary record, recordType=82 alarm pre-recording

device recording: recordType=0 all, recordType=2 alarm recording, recordType=3 motion detection, recordType=6 general recording

[begin time] and **[end time]:** The start and end times of the recording, represented as int. The unit is second.

✓ **The playing process based on RTSP redirection:**

Developers often do video streaming tests based on VLC tools. The advantage of using RTSP redirection is that they can pull streams without logging in to the VMS platform. Developers can assemble URLs according to the RTSP format required by VMS, and can directly assemble username and passwords into RTSP addresses. If they do not know the username and passwords, they can also open the RTSP allowlist on the VMS platform (System Configuration-> Security-> RTSP AllowList Configuration) to add the address of the pull stream to the allowlist. They can also pull the stream, and the URL of the pull stream is not valid for a long time, and there is no Token expiration problem.

● RTSP URL format of VMS pulling stream:

rtsp://username:password@platform

IP:9090/dss/monitor/params?cameraid=1000004%240&substream=1&username="username"

Note: Request the real-time video from the VMS for the 1000004\$0 channel (where 1000004 is the device code, %24 is the escape character for \$, and 0 is the channel number) for the main stream (substream: 1 for main stream, 2 for substream).

- RTSP URL format of VMS recording playback is as follows:

rtsp://username: password@platform

IP:9090/playback?cameraid=1000499%240&type=3&begintime=1602227183&endtime=1602227192&substream=1&recordType=1&username="username"

Note: Request the VMS media gateway for the 1000499\$0 channel (where %24 is the escape character for \$) for the main stream (substream: 1 for main stream, 2 for substream) of the ordinary recording (recordType: 1 for ordinary recording, 2 for alarm recording) with a start timestamp of 1602227183 and an end timestamp of 1602227192.

✓ **The playing process based on standard RTSP:**

After getting the RTSP address from the platform through the interface, the RTSP interaction process is supported. Starting from the OPTIONS signaling, the standard signaling process is adopted. A standard decoding library is used for decoding and playing. The platform only outputs streams. Functions like decoding, play, fast forward, and download are controlled by the play components, or to interface with the platform's DPSDK.

- Live View

Requesting for live video has two steps:

1. Get the playing URL and token through the Rest request.
2. Get stream on URL through RTSP.

Get URL

Get the URL and token through the [3.4.1 Start Real-time Video] interface. The returned URL and token field will be used in subsequent RTSP interaction process.

RTSP Interaction

When sending a video request, the token field shall be used as the URL parameter. Without the token, the video request might fail. **A token becomes invalid once it is used, or if it is not used after 30 seconds.** The format of URL with token is shown as follows:

rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=1190

Standard RTSP acquire stream is supported, and VLC can be used for verification. To do this set the network configuration of VLC to RTP over RTSP (for specific configuration steps, do an online search). Open the network stream, and enter the URL. The returned streams in this mode are standard streams. A standard decoding library can be selected on the client for decoding.

```
OPTIONS rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=1194 RTSP/1.0
CSeq: 2
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 2
Public: OPTIONS,DESCRIBE,SETUP,PLAY,PAUSE,TEARDOWN,SET_PARAMETER

DESCRIBE rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=1194 RTSP/1.0
CSeq: 3
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Accept: application/sdp

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 3
Content-Base: rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=1194
Content-Type: application/sdp
Content-Length: 485
Cache-Control: must-revalidate
x-Accept-Dynamic-Rate: 1

v=0
o=- 2209362008 2209362008 IN IP4 192.168.1.1
s=RTSP Session/2.0
c=IN IP4 192.168.1.1
t=0 0
a=control:*
a=range:npt=now-
m=video 0 RTP/AVP 96
a=control:trackID=0
a=framerate:25.000000
a=rtpmap:96 H264/90000
a=fmtp:96 packetization-mode=1;profile-level-id=640032;sprop-parameter-sets=J2QAMq2EBUViuKxUcQgKisVxWKjiECSFITk8nyfk/k/J8nm5s00IEkKQnJ5Pk/J/J+T5PNzZphcqAeAln5YQAAA+gAAMNQBA,KP4Jiw==
a=recvonly
m=audio 0 RTP/AVP 8
a=control:trackID=1
a=rtpmap:8 PCMA/16000
a=recvonly
SETUP rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=1194/trackID=0 RTSP/1.0
CSeq: 4
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Transport: RTP/AVP/TCP;unicast;interleaved=0-1
```

```
RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 4
Session: 450492010249;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=014CFFF3
x-dynamic-rate: 1

SETUP rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=
1194/trackID=1 RTSP/1.0
CSeq: 5
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Transport: RTP/AVP/TCP;unicast;interleaved=2-3
Session: 450492010249

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 5
Session: 450492010249;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=014CFFF5
x-dynamic-rate: 1

PLAY rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000002%240%26substream=1?token=1
194 RTSP/1.0
CSeq: 6
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Session: 450492010249
Range: npt=0.000-

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 6
Session: 450492010249
RTP-Info: url=trackID=0;seq=22015;rtptime=2659590554;ssrc=014CFFF3,url=trackID=1;seq=0;rtptime
=0;ssrc=014CFFF5
Range: npt=0.000000-
```

- Recording Playback

Recording Search

For the protocol format of recording search, see [3.5.1 Search Recording]. Based on different video storage sources, there are device recordings, and platform recordings (center recordings) which are stored in the storage space configured on the platform. Storage space can be configured for both the central platform and distributed platform. Center recordings produced during the period when the device is mounted to a platform are stored in the storage space of this platform. Device recordings are stored on the device, regardless of to which platform the device is mounted.

The API protocol does not support video download. It is recommended to use DPSDK docking, or the caller can obtain the RTSP address of the video obtained through the API and implement the download function themselves.

Platform Recording Search

Platform recordings are stored in the storage space configured by the SS service where the device is mounted. If the device ever changed the SS service where it is mounted, the platform recordings in the search period might be in storage space of different SS services. Therefore, when searching platform recordings, the center service searches recordings from all SS services. The search process ends after all SS services return the search results. The unit of the returned recordings is file. See the following:

```
{
  "channelId": "1000001$1$0$0",
  "recordSource": "3",
  "recordType": "6",
  "startTime": "1545840000",
  "endTime": "1545841800",
  "recordName": "{2db653e6-40d2-4bce-9df6-d2609b96d9ae}",
  "fileLength": "737280",
  "planId": "0",
  "ssId": "1001",
  "diskId": "1545840000-1545841800",
  "streamId": "18",
  "forgotten": "0"
}
```

Device Recording Search

When searching device recordings, the center service searches recordings from all SS services where the device is mounted. The unit of the returned recordings is file. See the following:

```
{
  "channelId": "1000012$1$0$0",
  "recordSource": "2",
  "recordType": "6",
  "startTime": "1545926402",
  "endTime": "1545926473",
  "recordName": "0-2-172108-12288-1545926402-1545926473",
  "fileLength": "12288",
  "planId": "0",
  "ssId": "1002",
  "diskId": "",
  "streamId": "0",
  "forgotten": "0"
}
```

To request for recording playback, first get the recording playback URL, and the recording stream through the RTSP.

Get Recording Playback URL

Playback by file is recommended for platform recordings, because recording files in a period might correspond to different SS services (if the device ever changed the SS service to which it is bound). Recording playback can only be controlled on the client. Play a recording first, and then request to play the next one. In this case, playing the recordings in a period requires multiple recording playback requests. Playback by time is

recommended for device recordings, because this mode requires only one recording playback request. For the request protocol format of recording playback by time, see [3.5.2 Video Playback by Time]. For the request protocol format of recording playback by file, see [3.5.3 Video Playback by File].

Video Request on the Playback URL through RTSP Interaction

Platform Recording

Standard RTSP acquire stream is supported, and VLC can be used for verification. To do this set the network configuration of VLC to RTP over RTSP (for specific configuration steps, do an online search). Open the network stream, and enter the URL. The returned streams in this mode are standard streams. A standard decoding library can be selected on the client for decoding. The URL format is:

```
rtsp://{IP}:{Port}/playback/center?streamID={streamID}&beginTime={beginTime}&endTime={endTime}
```

IP and Port correspond to the IP and Port in the recording playback URL obtained in the previous step. The streamID is the one in the record of requested recordings. The beginTime and endTime are the start time and end time of the recording to be played. The beginTime can be later than the start time but not the end time of the recording to be played.

The specific interaction procedure is as follows:

```
OPTIONS rtsp://192.168.1.1:9320/playback/center?streamID=18&beginTime=1545840000&endTime=1545841800 RTSP/1.0
CSeq: 2
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 2
Public: OPTIONS,DESCRIBE,SETUP,PLAY,PAUSE,TEARDOWN,SET_PARAMETER

DESCRIBE rtsp://192.168.1.1:9320/playback/center?streamID=18&beginTime=1545840000&endTime=1545841800 RTSP/1.0
CSeq: 3
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Accept: application/sdp

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 3
Content-Base: rtsp://192.168.1.1:9320/playback/center?streamID=18&beginTime=1545840000&endTime=1545841800
Content-Type: application/sdp
Content-Length: 419
Cache-Control: must-revalidate
x-Accept-Dynamic-Rate: 1

v=0
```

```
o=- 2209529590 2209529590 IN IP4 192.168.1.1
s=RTSP Session/2.0
c=IN IP4 192.168.1.1
t=0 0
a=control:*
a=range:npt=0-1800
m=video 0 RTP/AVP 96
a=control:trackID=0
a=framerate:25.000000
a=rtpmap:96 H264/90000
a=fmtp:96 packetization-mode=1;profile-level-id=4D4032;sprop-parameter-sets=J01AMouVAPAET8sl
AAAFQAAGGoQg,KP4JiA==
a=recvonly
m=audio 0 RTP/AVP 0
a=control:trackID=1
a=rtpmap:0 PCMU/16000
a=recvonly
SETUP rtsp://192.168.1.1:9320/playback/center?streamID=18&beginTime=1545840000&endTime=154
5841800/trackID=0 RTSP/1.0
CSeq: 4
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Transport: RTP/AVP/TCP;unicast;interleaved=0-1

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 4
Session: 540790241089;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=049DFFFD
x-dynamic-rate: 1

SETUP rtsp://192.168.1.1:9320/playback/center?streamID=18&beginTime=1545840000&endTime=154
5841800/trackID=1 RTSP/1.0
CSeq: 5
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Transport: RTP/AVP/TCP;unicast;interleaved=2-3
Session: 540790241089

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 5
Session: 540790241089;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=049DFFFD
x-dynamic-rate: 1

PLAY rtsp://192.168.1.1:9320/playback/center?streamID=18&beginTime=1545840000&endTime=1545
841800 RTSP/1.0
CSeq: 6
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Session: 540790241089
Range: npt=0.000-

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 6
Session: 540790241089
RTP-Info: url=trackID=0;seq=0;rtptime=20970;ssrc=049DFFFD,url=trackID=1;seq=0;rtptime=0;ssrc=04
9DFFFD
```

Range: npt=0.233000-1800.000000

Device Recording

Standard RTSP acquire stream is supported, and VLC can be used for verification. To do this set the network configuration of VLC to RTP over RTSP (for specific configuration steps, do an online search). Open the network stream, and enter the URL. The returned streams in this mode are standard streams. A standard decoding library can be selected on the client for decoding. The URL format is:

```
rtsp://{IP}:{Port}/vms/playback/pu?cameraid={cameraid}&begintime={begintime}&endtime={endtime}&substream={substream}
```

IP and Port correspond to the IP and Port in the recording playback URL obtained in the previous step. The cameraid consists of the device code and channel serial number in the channelId requested in the previous step. The escape character %24 of '\$' shall be added between the device code and channel code, for example, if the channelId is 1000012\$1\$2\$3, the cameraid will be 1000012%243. The begintime and endtime are the start time and end time of the recording to be played. The beginTime can be later than the start time but not the end time of the recording to be played. substream indicates main stream and sub stream. 1: Main stream; 2: Sub stream

The specific interaction procedure is as follows:

```
OPTIONS rtsp://192.168.1.1:9320/vms/playback/pu?cameraid=1000012%240&begintime=1545926402
&endtime=1546012799&substream=0 RTSP/1.0
CSeq: 2
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 2
Public: OPTIONS,DESCRIBE,SETUP,PLAY,PAUSE,TEARDOWN,SET_PARAMETER

DESCRIBE rtsp://192.168.1.1:9320/vms/playback/pu?cameraid=1000012%240&begintime=1545926402
&endtime=1546012799&substream=0 RTSP/1.0
CSeq: 3
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Accept: application/sdp

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 3
Content-Base: rtsp://192.168.1.1:9320/vms/playback/pu?cameraid=1000012%240&begintime=154592
6402&endtime=1546012799&substream=0
Content-Type: application/sdp
Content-Length: 416
Cache-Control: must-revalidate
x-Accept-Dynamic-Rate: 1

v=0
o=- 2209060076 2209060076 IN IP4 192.168.1.1
s=RTSP Session/2.0
```

```

c=IN IP4 192.168.1.1
t=0 0
a=control:*
a=range:npt=0-86397
m=video 0 RTP/AVP 96
a=control:trackID=0
a=framerate:25.000000
a=rtpmap:96 H264/90000
a=fmtp:96 packetization-mode=1;profile-level-id=4D0032;sprop-parameter-sets=Z00AMpY1QEABg03
AQEBAgA==,aO48gA==
a=recvonly
m=audio 0 RTP/AVP 8
a=control:trackID=1
a=rtpmap:8 PCMA/16000
a=recvonly
SETUP rtsp://192.168.1.1:9320/vms/playback/pu?cameraid=1000012%240&begintime=1545926402&e
ndtime=1546012799&substream=0/trackID=0 RTSP/1.0
CSeq: 4
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Transport: RTP/AVP/TCP;unicast;interleaved=0-1

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 4
Session: 71273842861;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=02250000
x-dynamic-rate: 1

SETUP rtsp://192.168.1.1:9320/vms/playback/pu?cameraid=1000012%240&begintime=1545926402&e
ndtime=1546012799&substream=0/trackID=1 RTSP/1.0
CSeq: 5
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Transport: RTP/AVP/TCP;unicast;interleaved=2-3
Session: 71273842861

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 5
Session: 71273842861;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=02250002
x-dynamic-rate: 1

PLAY rtsp://192.168.1.1:9320/vms/playback/pu?cameraid=1000012%240&begintime=1545926402&e
ndtime=1546012799&substream=0 RTSP/1.0
CSeq: 6
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
Session: 71273842861
Range: npt=0.000-

RTSP/1.0 200 OK
Server: VMS Rtsp Server/2.0
CSeq: 6
Session: 71273842861
RTP-Info: url=trackID=0;seq=0;rtptime=187200;ssrc=02250000,url=trackID=1;seq=0;rtptime=0;ssrc=0
2250002
Range: npt=2.080000-86397.000000

```

5.3.5 The Connection Procedure for Typical Applications: Video Playback

1. **Add a device installed with a video player** from [Basic Config>Device>Add Device] on the VMS client.

2. Start API connection. **Firstly, perform login authentication**, including first login, second login, heartbeat keepalive, and token update. For details, refer to [5.1 Login Authentication] or [6.3.1.1 Example of Login Authentication with Java] on Demo.

All subsequent API invocations need to carry the token field returned during logging in the X-Subject-Token request header.

3. **Get all devices** from API document through [3.3.2.1 Getting Device Tree (All)].

The API input parameter is simple enough to be unquestionable.

The output parameter is complicatedly structured and can be simplified as follows:

```
"devices": [
  {
    "code": "1000747",
    "name": "NVR",
    ...
    "units": [
      {
        "unitType": "1",
        ...
        "channels": [
          {
            "channelCode": "1000747$1$0$0",
            ...
          }
        ]
      }
    ]
  }
]
```

Wherein,

code: Indicates a unique device identifier, which can be displayed or ignored in the action bar of the client device list.

name: Indicates a device name. These two fields can be used to determine whether the device is the desired one.

unitType: Indicates a unit type. 1 indicates a video unit and other units are ignored.

channelCode: Indicates a unique channel identifier. All stream pull actions are based on channels.

4. **Get the video playback URL** from API document through [3.4.1 Start Real-time Video].

Focus on main parameters:

```
"data": {
  ...
  "channelId": "1000040$1$0$0",
  ...
}
```

Wherein,

channelId: Indicates a unique channel identifier, corresponding to the channelCode returned by the [3.3.2.1 Getting Device Tree (All)]API.

Check the return value of the API:

```
"data": {
  ...
  "url": "rtsp:// 192.168.1.1:9100/vms/monitor/param/cameraid=1000040%240%26substream=1",
  ...
  "token": "2204",
  ...
}
```

Wherein,

url: Indicates the RTSP address of the pull stream. Your access will be denied when this address is directly used for pulling stream.

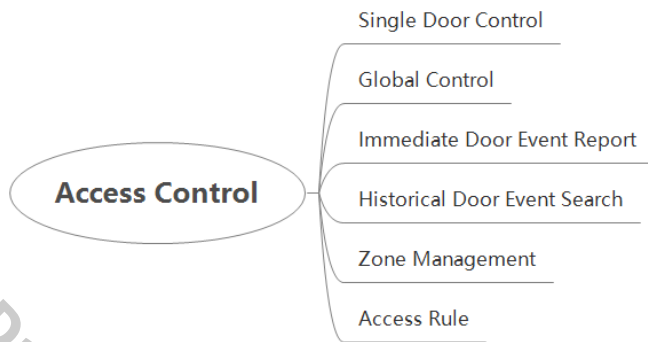
token: Indicates a video service authentication token. The token different from that of the platform becomes invalid after being used once or not used for more than 30 seconds.

5. Assemble the return value of the [3.4.1 Start Real-time Video] to make a complete RTSP stream
 pull URL: rtsp://
 192.168.1.1:9100/vms/monitor/param/cameraid=1000040%240%26substream=1?token=
 2204, and use VLC or other standard decoding libraries to play videos.

Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Intercom Preview
+++ Start Real-time Video
+++ Start Talk
+++ Stop Talk
+++ Acquiring Streaming Media URL for Live View(FLV)
+++ Acquiring Streaming Media URL for Live View(HLS)
++ Record Playback
+++ Search Recording
+++ Video Playback by Time
+++ Video Playback by File
+++ Acquiring Stream Media URL for Video Playback(HLS)

5.4 Access Control

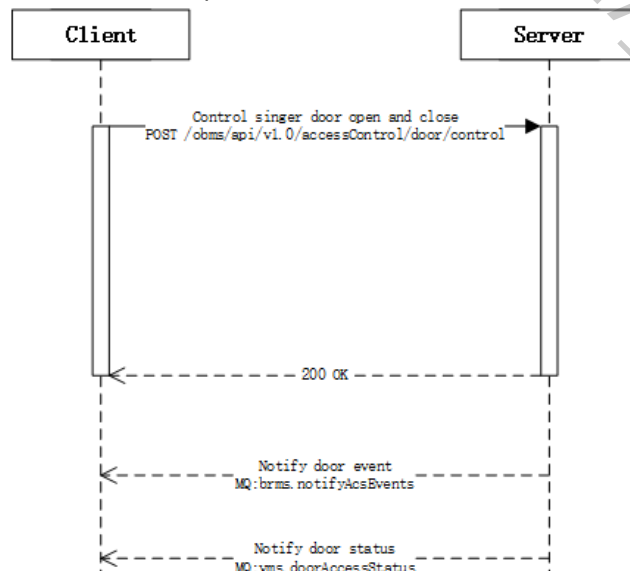


An access control module can unlock and lock a single door, implement global control of normally open and normally closed doors, and zone management and access rule. Zone management and access rule configuration are not supported in API documentation

5.4.1 Unlock and Lock a Single Door

On the platform, click the remote unlock and remote lock buttons of the specified door channel to unlock and lock the access control channel. The interface [3.7.1 Unlock and Lock a Single Door] is called. After operations of the access control channel is successful, the platform reports the operation record notification of the access control channel through MQ (notification of batch access control records: brms.notifyAcsEvents), and reports the door status notification through MQ (notification of door status: vms.doorAccessStatus).

The interaction process is as follows:



5.4.2 Global Control of Normally Open and Normally Closed Doors

This function is to select access control channels in batch, and then perform operations on the channels. For example, keep them normally open, normally closed and restore them.

5.4.3 Zone Management

The platform supports operations such as adding, updating, deleting, querying lists, and querying details of zones. Access points are automatically generated based on access control/elevator control channels, and each zone can be associated with one or more access points. It supports configuring access rules based on zones and access points for authorization issuance.

5.4.4 Access Rule

The platform supports the addition, update, deletion, list query, and detail query operations of traffic rules. The passage rules are divided into ordinary rules and advanced rules. Ordinary rules support time templates and holiday plans, and can be associated with one or more zones and passage points, bound to one or more personnel, vehicle organizations, and personnel for authorization issuance. There are many types of advanced rules that support time templates and authorization distribution based on the dimensions of access points and personnel.

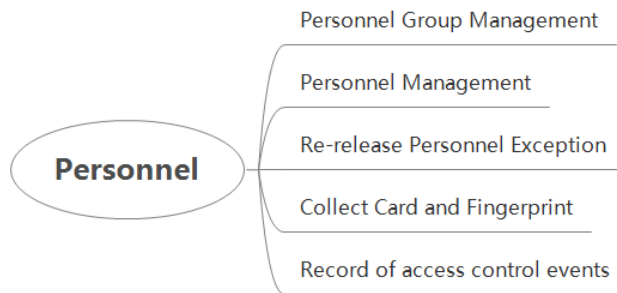
Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Access Control
+++ Unlock and Lock of a Single Door
+++ Configuring the Global Status of Access Control and Elevator Control for Access Control Zones
+++ Obtaining a List of Access Rules that Were Not Configured for Resources
+++ Get Access Control Area List
+++ Get the Access Point List of the Specified Area of Access Control by Paging
+++ Get the Access Control Point Access Status

Go to the following directory to view the detailed MQ notifications

MQ Notice Directory
MQ Notification Message
++ Access Control
+++ Notification of Door Status
+++ Notification of Normally Open and Normally Closed Door Status

5.5 Personnel Management



5.5.1 Configure Personnel Group

A person group is an organization to which persons belong, and is required when you add or update persons. By default, the platform has a person group named "All Persons and Vehicles" (Organization code 001, cannot be deleted).

Call the [3.8.1.4 Get the List of Person Groups] interface to get list of person groups.

Call the [3.8.1.1 Add a Person Group] interface to add person groups.

Call the [3.8.1.2 Update a Person Group] interface to update person groups.

Call the [3.8.1.3 Delete a Person Group] interface to delete person groups.

5.5.2 Configure Person

Call the [3.8.2.4 Get the Details of a Person] interface to get details of individual persons.

Call the [3.8.2.5 Get the List of Persons in Pages] interface to get list of persons in pages.

Call the [3.8.2.6 Get the List of Persons in Pages (Brief Information)] interface to get list of persons (brief information).

Call the [3.8.2.1 Add a Person] interface to add person information. After successfully adding person information, the platform sends a notice through MQ ([4.4.1 Notification of Batch Adding Persons]) for successfully adding person information.

Call the [3.8.2.2 Update a Person] interface to modify person information. After successfully modifying person information, the platform sends a notice through MQ ([4.4.2 Notification of Batch Updating Persons]) for successfully modifying person information.

Call the [3.8.2.3 Batch Delete Persons of a Specified Persons Group] interface to delete persons by group. After successfully deleting person information, the platform sends a notice through MQ ([4.4.3 Notification of Batch Deleting Persons]) for successfully deleting persons.

5.5.3 Exceptions in Person and Search Re-sending

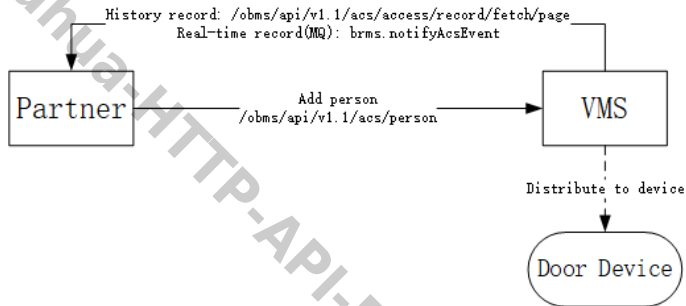
There are three types of exception information: Access control, face comparison, entrance and exit.

After calling the [3.8.2.5 Get the List of Persons in Pages] interface, call the [3.8.2.7 Get the List of Exceptions in Sending Data of Persons] interface to obtain the exception information of a batch of personnel. Then call the [3.8.2.8 Resend the Persons with Exceptions in Sending Data] interface to resend the exception information of a batch of personnel.

5.5.4 Access Control Event

When the user triggers unlocking by card, fingerprint, password, and face recognition on an access control device, the platform pushes these real-time access control events through MQ (brms.notifyAcsEvents) to the client.

The platform supports searching for the access control records by specified criteria. Call the interface [3.8.3.1 Obtaining Access Control Records by Page] to search for the access control history records by period and person ID.



5.5.5 The Connection Procedure for Typical Applications: Add a Person

The entire step is quite complex, and various associated information needs to be configured on the VMS client first. However, these are optional and there is no related business to omit. You can explain them in two scenarios.

Scene 1: Only configure personnel basic information

Just focus on baseInfo and extensionInfo, and assign values according to default values for others. The following is an example of the most concise parameters, without the need to issue any data, only the personId and firstName have been set:

```

{
  "baseInfo": {
    "personId": "18069001",
    "lastName": "",
    "firstName": "Jack",
    "gender": "0",
    "orgCode": "001",
    "orgCodes": [
      "001"
    ],
  },
  "email": "",
  "tel": "",
  "remark": "",
  "source": "0",
  "facePictures": []
},
  "extensionInfo": {
    "nickName": "",
    "address": "",
    "idType": "6",
  }
}
  
```

```

        "idNo": "",
        "nationalityId": "9999",
        "birthday": "",
        "companyName": "",
        "department": "",
        "position": ""
    },
    "userDefineFields": [],
    "residentInfo": {
        "houseHolder": "0",
        "sipId": "",
        "vdpUser": "0"
    },
    "authenticationInfo": {
        "combinationPassword": "",
        "cards": [],
        "fingerprints": [],
        "palmVeins": [],
        "startTime": "1751385600",
        "endTime": "2067004799"
    },
    "accessInfo": {
        "accessType": "0",
        "guestUseTimes": "200",
        "passageRuleIds": []
    },
    "faceComparisonInfo": {
        "enableFaceComparisonGroup": "0",
        "faceComparisonGroupIds": []
    },
    "entranceInfo": {
        "parkingSpaceQuotas": [],
        "vehicles": []
    }
}

```

Scene 2: Require authorization for face, vehicle, and other business operations

If it is necessary to configure associated services for authorization, relevant configuration information needs to be queried through the API, and the [3.8.2.1 Add a Person] interface needs to be used.

1. **Add access control/face recognition/entrance & exit device** from [Basic Config>Device>Add Device] on the VMS client.
2. **Add person groups** from [Applications Config>Person and Vehicle Info>Person List] on the VMS client or skip this step. The default group ID is 001.
3. **Add access rule** from [Applications Config>Person and Vehicle Info>Person List], on the VMS client. Configure zones and access points.
4. **Add face comparison** from [Applications Config>Watch List>Face Watch List] on the VMS client, and distribute the comparison group to the facial device.
5. **Add parking lot** from [Applications Config> Parking Lot>Parking Lot Basic Config] on the VMS client.

6. **Add vehicle** from [Applications Config> Person and Vehicle Info>Vehicle List] on the VMS client, and associated parking lot vehicle group.

7. Start API connection. **Firstly, perform login authentication**, including first login, second login, heartbeat keepalive, and token update. For details, refer to [5.1 Login Authentication] or [6.3.1.1 Example of Login Authentication with Java] on Demo.

All subsequent API invocations need to carry the token field returned during logging in the X-Subject-Token request header.

8. **Get the list of person groups** from API document through [3.8.1.4 Get the List of Person Groups]. This is used when you add a person, corresponding to the baseInfo module of the API.

Check the return value of the API:

```
"results": [
  {
    "parentOrgCode": "001",
    "orgCode": "001001",
    "orgName": "test",
    "remark": "",
    "childNum": "0"
  }
]
```

Wherein,

orgCode: Indicates a person group ID.

9. **Get access rule** from API document through [3.7.3 Obtaining a List of Access Rules that Were Not Configured for Resources]. This is used when you add a person, corresponding to the accessInfo module of the API.

Check the return value of the API:

```
"results": [
  "pageData": [
    {
      "id": "10",
      "ruleName": "VVV",
      "ruleType": "1",
      "pointType": "1",
      "timeTemplateId": "1",
      "timeTemplateName": "time.template.name.all",
      "holidayPlanId": "",
      "holidayPlanName": null,
      "personCount": 1,
      "pointCount": 1
    }
  ],
  "nextPage": "-1",
  "totalCount": "1"
]
```

Wherein,

id: Indicates a access rule id.

10. **Get face comparison groups by pages** from API document through [3.12.1.4 Get Face Comparison Group List in pages]. This is used when you add a person, corresponding to the faceComparisonInfo module of the API.

Check the return value of the API:

```
"pageData": [  
  {  
    "repositoryId": "1",  
    "repositoryName": "7788",  
    ...  
  }  
]
```

Wherein,

repositoryId: Indicates a face comparison group ID.

repositoryName: Indicates a face comparison group name.

11. **Get parking lot list** from API document through [3.11.4.1 Get Parking Resource Tree]. This is used when you add a person, corresponding to the entranceInfo module of the API.

Check the return value of the API:

```
"results": [  
  {  
    "id": "5",  
    "parkingLotName": "parkingLot3",  
    ...  
  }  
]
```

Wherein,

id: Parking lot ID

parkingLotName: Parking lot name

12. **Get vehicle groups** from API document through [3.11.1.4 Get the List of Entrance and Exit Groups of Vehicles]. This is used when you add a person, corresponding to the entranceInfo module of the API.

Check the return value of the API:

```
"pageData": [  
  {  
    "groupId": "2",  
    "groupName": "General",  
    ...  
  }  
]
```

Wherein,

groupId: Indicates a vehicle group ID.

groupName: Indicates a vehicle group name.

13. **Add a person** from API document through [3.8.2.1 Add a Person].

The following shows an example of a full parameter:

```
{  
  "baseInfo": {  
    "personId": "00002126",  
    "lastName": "",  
    "firstName": "Jerry Green",  
  }  
}
```



```
"gender": "1",
"orgCode": "001",
"orgCodes": [
    "001"
],
"email": "abc@163.com",
"tel": "13812345678",
"remark": "Excellent staff",
"source": "0",
"facePictures": [
    "/9j/4AAQSk...ZCi8KKKAP/9k=",
    "/9j/4AAQSk...moJETadRs//Z"
]
},
"extensionInfo": {
    "nickName": "Jerry",
    "address": "Green Road",
    "idType": "0",
    "idNo": "D123456",
    "nationalityId": "248",
    "birthday": "2000-08-02",
    "companyName": "Internet company",
    "department": "",
    "position": "Software engineer"
},
"userDefineFields": [],
"residentInfo": {
    "houseHolder": "1",
    "sipId": "1#1#1",
    "vdpUser": "1",
    "inheritHouseHolder": "0"
},
"authenticationInfo": {
    "combinationPassword": "6cf9d67210ec9590a2f688d76c41b19b",
    "cards": [
        {
            "cardNo": "8B1C3E2A",
            "duressFlag": "0"
        },
        {
            "cardNo": "DC1AD20B",
            "duressFlag": "1"
        }
    ],
    "fingerprints": [
        {
            "fingerprint": "E6D8F09...F97FL/D0A91B",
            "name": "..... 1",
            "duressFlag": "0"
        }
    ],
    "palmVeins": [],
    "startTime": "1719331200",
    "endTime": "2034950399"
},
"accessInfo": {
    "accessType": "0",
```

```

    "guestUseTimes": "200",
    "passageRuleIds": [
        "35"
    ]
},
"faceComparisonInfo": {
    "enableFaceComparisonGroup": "1",
    "faceComparisonGroupIds": [
        "6"
    ]
},
"entranceInfo": {
    "parkingSpaceQuotas": [
        {
            "parkingLotId": "1",
            "parkingSpaceQuota": 10
        }
    ],
    "vehicles": [
        {
            "id": "",
            "plateNo": "RJM1601",
            "vehicleColor": "8",
            "vehicleBrand": "26",
            "bindingDeviceCode": "",
            "companyTelephone": "12345678",
            "companyOwner": "Jack",
            "affiliatedCompany": "Test Company",
            "remark": "Family car",
            "entranceGroups": [
                {
                    "plateNo": "RJM1601",
                    "parkingLotId": "1",
                    "entranceLongTerm": "1",
                    "entranceStartTime": "-1",
                    "entranceEndTime": "-1",
                    "entranceGroupIds": [
                        "5"
                    ]
                }
            ]
        }
    ]
}
}

```

Roughly, the following modules are involved. Please focus on main parameters:

baseInfo: { Basic information }

personId: Indicates a unique person identifier which can be customized and contain up to 30 alphanumeric characters.

orgCode: Indicates a person group ID, corresponding to the orgCode returned by the [3.8.1.4 Get the List of Person Groups] API.

facePictures: Indicates face pictures. A maximum of two pictures can be added. For requirements, refer to the description about picture frame when you add a person on the VMS client. The list displays Base64-encoded picture files. The face recognition camera and some access control devices deliver only the first picture. Try to keep the first picture in

high definition. The second picture is for assistance and can be a picture with a person wearing a hat or mask. The second picture is only delivered by some devices.

extensionInfo: { Extended information }

nationalityId: Indicates a region ID. Default set: Unknown=9999.

residentInfo: { Residence information }

houseHolder: If video call service is unavailable, set: 0=no house owner.

sipId: If video call service is unavailable, set: ""=null.

userDefineFields: { User defined information }

Generally, setting an empty array is sufficient.

authenticationInfo: { Access control authentication information }

combinationPassword: Indicates a combined unlock password. This can be used in combination with card and fingerprint according to door configurations and the password needs to be AES encrypted. For details, refer to [3.3.5 Example of Encryption mode].

cardNo: Indicates a card number., up to 16 characters.

fingerprints: Fingerprint data is generally collected directly from the device, which is relatively complex. If you do not have it, you can set it: [].

palmVeins: Palm vein data is generally collected directly from the device, which is relatively complex. If you do not have it, you can set it: [].

accessInfo: { Access control information }

accessType: Indicates a person type. General setting: 0=General guestUseTimes: Indicates an allowed unlock attempt count for a guest. For non guests, set: 0.

passageRuleIds: Indicates the list of access rule IDs to be authorized, corresponding to the id returned by the [3.7.3 Obtaining a List of Access Rules that Were Not Configured for Resources] API.

faceComparisonInfo: { Face comparison information }

faceComparisonGroupId: Indicates the list of face comparison group IDs, corresponding to the repositoryId returned by the [3.12.1.4 Get Face Comparison Group List in pages] API.

entranceInfo: { Vehicle entrance/exit information }

vehicles: Indicates the list of vehicles.

parkingSpaceQuotas: Parking available parking array

parkingLotId: Parking lot ID, corresponding to the ID returned by the [3.11.4.1 Get Parking Resource Tree] interface

entranceGroups: Parking vehicle group array

entranceGroupIds: Indicates the list of vehicle entrance/exit group IDs, corresponding to the groupId returned by the [3.11.1.4 Get the List of Entrance and Exit Groups of Vehicles] API.

14. After a person is successfully added, as long as the setting information is complete, card No. and other information will be automatically sent to an access control device, face pictures will be sent to the face recognition camera, and the license plate will be sent to the entrance camera.

15. For access control services, unlock events will be reported to the platform. To get unlock events in real time, **monitor the messaging component named ActiveMQ**. For operation methods, refer to [4 MQ Notification Message] or [6.3.2.1 Example of MQ Connection with Java] on Demo.

16. Check the details about unlock event messages from [4.4.4 Notification of Batch Operating Access Control Records]. **Focus on the method=brms.notifyAcsEvents message in the mq.event.msg. userId topic.**

Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Person
+++ Person Group
++++ Add a Person Group
++++ Update a Person Group
++++ Delete a Person Group
++++ Get the List of Person Groups
+++ Person Information
++++ Add a Person
++++ Update a Person
++++ Batch Delete Persons
++++ Get the Details of a Person
++++ Get the List of Persons in Pages
++++ Get the List of Persons in Pages (Brief Information)
++++ Get the List of Exceptions in Sending Data of Persons
++++ Resend the Persons with Exceptions in Sending Data
+++ Access Control Record
++++ Obtaining Access Control Records by Page
++++ Obtaining the Details of Access Control Records

Go to the following directory to view the detailed MQ notifications

MQ Notice Directory
MQ Notification Message
++ Person
+++ Notification of Batch Adding Persons
+++ Notification of Batch Updating Persons
+++ Notification of Batch Deleting Persons
+++ Notification of Batch Operating Access Control Records

5.6 Visitor Management



5.6.1 Visitor Manage

✓ **Generate the QR Code Data and Card Number in the Visitor Pass Permit**

Generate the QR code data and card number in the visitor pass permit. See [3.9.5 Generate the QR Code Data and Card Number in the Visitor Pass Permit].

✓ **New Visitors' Visit Registration and Appointment Registration**

Used to add a new visitor. See [3.9.3 Add a New Visitor]. The visitor status is visit after visit registration and appointment after appointment registration. Here are the specific steps:

Step 1: Add user information.

Step 2: Add visitor information.

Step 3: Send a notification of adding the visitor. See [4.5.1 Notification of Adding a Visitor].

✓ **Appointed Visit Registration, On-site Visitor's Leaving Registration, Cancellation of Appointment, and Update of On-site Visitors/Appointments**

Update the visitors' information and status. See [3.9.4 Update a Visitor]. Here are the specific steps:

Step 1: Search for the visitor by visitor ID.

Step 2: Delete the visitor who have left or canceled the appointment. Otherwise, update the visitor information.

Step 3: If the visitor is deleted in Step 2, the corresponding person information and visitor information will be deleted too.

Step 4: After Step 3 is completed, send a notification of deleting the visitor. See [4.5.3 Notification of Batch Deleting Visitors].

Step 5: If the visitor is updated in Step 2, the corresponding person information and visitor information will be updated too.

Step 6: After Step 5 is completed, send a notification of updating the visitor. See [4.5.2 Notification of Updating a Visitor].

✓ **Sign off Expired Visitors**

Sign off the expired visitors immediately. See [3.9.8 Sign off Expired Visitors].

5.6.2 Visitor Search

✓ **Search for Visitor (Appointment and Visit Status)**

Search for the information about a specified visitor by visitor ID. See [3.9.2 Search for a Specified Visitor].

✓ **Search for the List of Visitors in Pages (Appointment and Visit Status)**

Search for the list of visitors in pages. See [3.9.1 Search the List of Visitors in Pages]. Search the information about visitors according to searching criteria in pages.

✓ **Search for the History Records of Visitors in Pages**

Search for the list of history records of visitors in pages. See [3.9.7 Get the List of Visitor History Records in Pages]. Search for the access records of visitors according to searching criteria in pages.

✓ **Get the Number of Appointments and On-site Visitors**

Get the number of appointments and on-site visitors. See [3.9.6 Get the Number of Appointments and On-site Visitors].

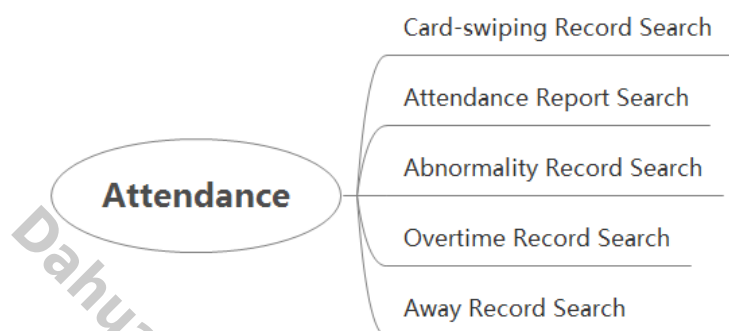
Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Visitor
+++ Add a New Visitor
+++ Update a Visitor
+++ Search for a Specified Visitor
+++ Search for the List of Visitors in Pages
+++ Search for the List of History Records of Visitors in Pages
+++ Sign off Expired Visitors
+++ Get the Number of Appointments and On-site Visitors
+++ Generate the QR Code Data and Card Number in the Visitor Pass Permit

For specific MQ notifications, visit the following interfaces

MQ Notice Directory
MQ Notification Message
++ Visitor
+++ Notification of Adding a Visitor
+++ Notification of Updating a Visitor
+++ Notification of Deleting a Visitor
+++ Notification of the Number of Appointments and On-site Visitors

5.7 Attendance Management



5.7.1 Description of Unified Attendance Configuration Rules

Before using the attendance function, you need to configure the following rules on the platform client first:

- Attendance check point
Add an access control device and configure the access control channel as the attendance check point.
- Attendance periods
Add corresponding fixed and unfixed attendance periods as required.
- Attendance check shifts
Add corresponding attendance shifts as required.
- Attendance check holiday plan
Configure the attendance check holiday plan as required.
- Shift management
Shifts are required during non-holiday periods for check-in/out to be recorded into attendance check history. You can use department shift, person shift and temporary shift as required. The person shift and department shift use the overwriting mechanism, and the last configuration should prevail.
- Description of holiday and shift priority
When there are multiple shifts or holidays on the same day, the priority is as follows: Temporary shift > Holiday shift > Person/Department shift.

5.7.2 Description of Record Search

Person ID uses precise search and name uses fuzzy search.

Card Swiping Records

- The generation conditions are as follows:
 1. Swiping card within the valid check-in range of the fixed attendance is check-in, and swiping card within the valid check-out range is check-out. Other events are null.
 2. All events where the day is a day off or holiday are null.

3. Unfixed attendance check does not enable rounding half to even. The first card swiping is check in, and the subsequent card swipings before the latest check-out time are all check-out.
 4. Unfixed attendance check enables rounding half to even. The first card swiping is check-in, the second valid card swiping is outing, and the third valid card swiping is returning; this is a loop. Events during card swiping intervals are null.
- Search in pages
Search card swiping records in pages.

Attendance Report

- The generation conditions are as follows:
 1. Fixed attendance check: There are card swiping records within the check-in or check-out range.
 2. There are card swiping records on days off or holidays.
 3. Unfixed attendance check: There is one or multiple card swiping records.
 4. If there is no card swiping record, for shifts that do not cross day, the records will be generated by the system on the second day; for shifts that cross day, the records will be generated by the system on the third day.
- Search in pages
Search for attendance reports in pages.

Exception Records

- The generation conditions are as follows:
 1. Fixed attendance: According to the configured rules, the conditions include late, leave early, unauthorized absence, forget to check in/out, late and forget to check in, late and leave early, forget to check out and leave early.
 2. Unfixed attendance check: According to the configured rules, the conditions include late, forget to check in/out, late and forget to check in.
 3. Fixed attendance check: If you did not check in/out within the valid range, the system will generate unauthorized absence records. For shifts that do not cross day, the records will be generated on the second day; for shifts that cross day, the records will be generated on the third day.
- Search in pages
Search exception records in pages.

Overtime Records

- The generation conditions are as follows:
 1. Fixed attendance check: Checking out N minutes after the configured time is considered as overtime.
 2. Days off or holidays are considered as overtime.
 3. Unfixed attendance: Exceeding the configured minimum overtime duration is considered as overtime.
- Search in pages
Search overtime records in pages.

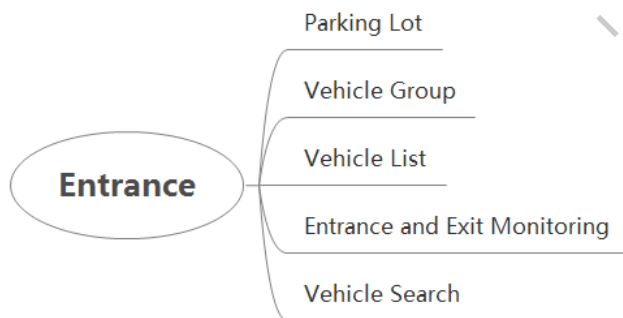
Outing Records

- The generation conditions are as follows:
Unfixed attendance enables rounding half to even. The first card swiping is check-in, the second valid card swiping is outing, and the third valid card swiping is returning; this is a loop.
- Search in pages
Search outing records in pages.

Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Attendance
+++ Get Card Swiping Records in Pages
+++ Search Attendance Reports in Pages
+++ Search Exception Records in Pages
+++ Search Overtime Records in Pages
+++ Get Outing Records in Pages

5.8 Entrance and Exit



[Parking lot]: The unique name of a specified parking lot.

[Position]: As a part of the parking lot, it can be understood as a gate of the parking lot, such as the north gate. The position name must be unique in one parking lot.

[Gate]: As a part of the position, it can be understood as a barrier arm. It is the minimum unit of the parking lot, including an entrance and an exit. One gate consists of the following devices: main camera, auxiliary camera (which can be configured in case of dual cameras), VTO, and general screen. The main camera is the core attribute and indispensable, so the main camera is a required field. The configuration of other devices is not required, such as the auxiliary camera, VTO, and general screen.

5.8.1 Configure Parking Lots

After the parking lot is configured, the system will determine and process the passing records of bound devices according to the barrier-opening logics.

[Get the List of Parking Lots] Through [3.11.4.1 Get Parking Resource Tree], you can search for the basic information about each parking lot. Check whether the configured parking lot has empty parking spots, and whether the blocklist and allowlist are sent, as well as the number of positions, entering gates, and exiting gates.

[Add a Parking Lot] Through [3.11.4.2 Add a Parking Lot], add the information about a specified parking lot, and configure the barrier-opening release rules of vehicle groups and positions.

[Update a Parking Lot] Through [3.11.4.3 Update a Parking Lot], modify the information about a specified parking lot, and modify the release rules.

[Batch Delete Parking Lots] Through [3.11.4.4 Batch Delete Parking Lots], delete the parking lots. The snapshot requests received by the deleted parking lots will not be processed.

5.8.2 Configure the List of Vehicles

A vehicle can be added on the platform. When the platform recognizes its license plate, the platform can automatically match the information about persons and vehicle groups that are bound with the vehicle.

[Get the List of Vehicles] Through [3.11.2.3 Get the List of Vehicles in Pages], search for the information about persons and groups that are bound with a specified vehicle.

[Update Vehicle Information] Through the interface [3.11.2.1 Batch Save Vehicle Information], add or modify the vehicle information.

[Batch Delete Vehicles] Through [3.11.2.2 Delete Vehicle Information], delete the vehicle information by vehicle Ids from the platform.

5.8.3 Configure Vehicle Groups

Add vehicles to a specified vehicle group, and bind the group with position release rules. Also, batch send the blocklist and allowlist through the vehicle group.

[Get the List of Entrance and Exit Groups of Vehicles] Through [3.11.1.4 Get the List of Entrance and Exit Groups of Vehicles], search for the basic information about group color and vehicles.

[Add a Vehicle Entrance and Exit Group] Through [3.11.1.1 Add a Vehicle Entrance and Exit Group], add a group, and bind the parking lot position with group rules. The added rules will be synchronized to the parking lot.

[Update a Vehicle Entrance and Exit Group] Through [3.11.1.2 Update a Vehicle Entrance and Exit Group], modify the basic information about a group, as well as the binding of the parking lot position with the group. The modified rules will be synchronized to the parking lot.

[Batch Delete Vehicle Entrance and Exit Groups] Through [3.11.1.3 Delete a Vehicle Entrance and Exit Group], delete a group. The deleted rules will be synchronized to the parking lot.

5.8.4 Configure Reserved Parking Space Group

Through the vehicle group or vehicle two ways to add, configure the exclusive parking space group bound parking lot parking space, can be configured to alert the parking space occupancy. Generally used for special parking spaces (such as private parking spaces that have been sold), parking non-special vehicles, remind the administrator to proceed to the next step.

5.8.5 Configure Available Parking Space

By adding vehicle groups or personnel, available parking spaces are configured, and the passage of personnel and vehicles can be limited by the number of available parking spaces. Generally, it is used in combination with the rules of card swiping and release to check the number of parking spaces parked under personnel or vehicle groups.

5.8.6 License Plate Recognition

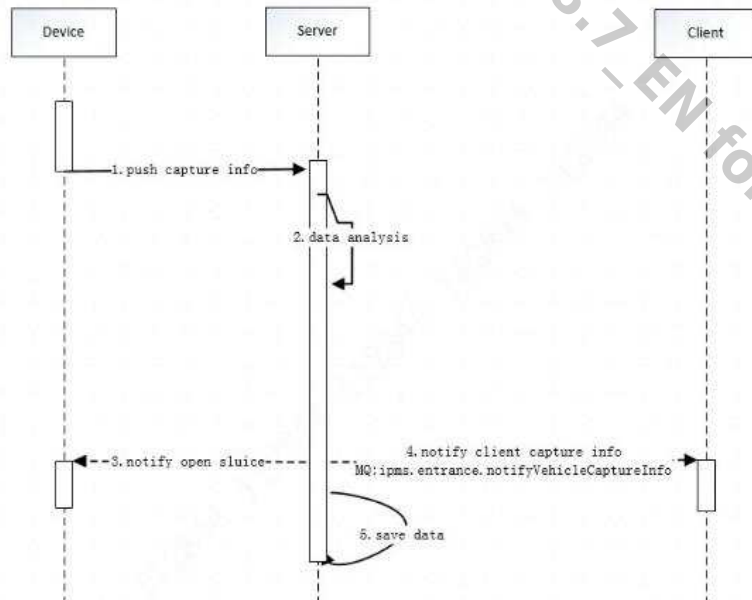
[Step 1] The device (ITC/NVR) is bound with a parking lot, takes snapshot images, and reports them to the platform.

[Step 2] The platform receives the snapshot information, and analyzes according to the configured parking lot rules.

[Step 3] If a vehicle has the access permission, the platform notifies the device to open the barrier automatically.

[Step 4] MQ (ipms.entrance.notifyVehicleCaptureInfo) notifies the client of the vehicle entry.

[Step 5] Save the data of snapshot, entry and exit, as well as empty parking spots.



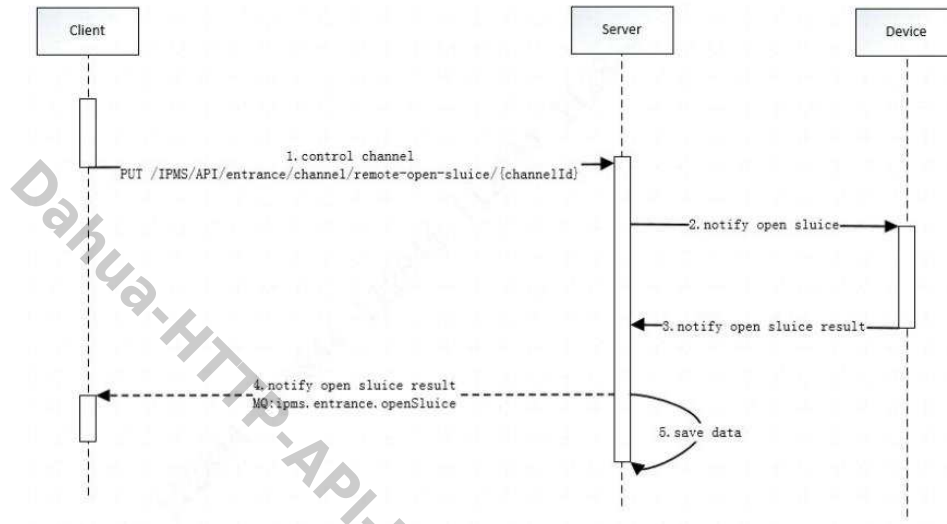
5.8.7 Remote Opening

When a vehicle without the access permission shall be released, the barrier can be opened remotely, manually, or through VTO calls. The specific procedure is as follows:

[Step 1] Call the command [3.11.4.6 Remote Opening] to open the barrier.

[Step 2] The platform sends a barrier-opening command to the device.

- [Step 3] The device returns the barrier-opening result to the platform synchronously.
- [Step 4] Through MQ (ipms.entrance.openSluice), the platform notifies the client of the barrier-opening result.
- [Step 5] Save the data of snapshot, entry and exit, as well as empty parking spots.



5.8.8 Information Search

1. After the device bound with the parking lot reports the license plate number, the platform will record the snapshot information. The snapshot information can be queried in the snapshot records of information search.
2. After logic processing of the snapshot information, the platform determines whether it enters or leaves the parking lot. If the vehicle has the access permission, an entry record will be created in the entry records, and the status is "Not Exited".
3. When the snapshot information is determined to be exiting, search for the latest "Not Exited" record of the vehicle with the same license plate number, update the entry record to be "Exited" (no update is performed if the matching record is not found), and generate an exit record.
4. If an error occurs during exit recognition, one entry record is not matched for a long time, but actually the vehicle is not in the parking lot already. Call the forced exiting interface to update this record to be exited, and generate a record in the forced exiting table.

Go to the following directory to view the detailed API interfaces

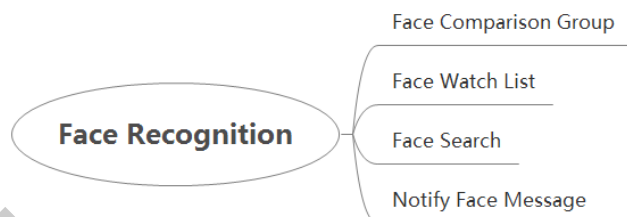
API Interface Directory
API Interface
++ Vehicle
+++ Vehicle Group
++++ Get the List of Entrance and Exit Groups of Vehicles
++++ Add a Vehicle Entrance and Exit Group

API Interface Directory
++++ Update a Vehicle Entrance and Exit Group
++++ Delete a Vehicle Entrance and Exit Group
++++ Obtain the List of Vehicle Survey Group List
+++ Vehicle Information
++++ Get the List of Vehicles in Pages
++++ Batch Save Vehicle Information
++++ Delete Vehicle Information
++++ Obtain Vehicle Details by License Plate Number
+++ Vehicle Search
++++ Get the Records of Entering Vehicles at the Entrance and Exit in Pages
++++ Get the Record Details of Entering Vehicles at the Entrance and Exit
++++ Get the Records of Exiting Vehicles at the Entrance and Exit in Pages
++++ Get the Record Details of Exiting Vehicles at the Entrance and Exit
++++ Get the Snapshot Records of Vehicles at the Entrance and Exit in Pages
++++ Obtaining the Parking Records at the Vehicle Entrance/Exit by Page
++++ Obtaining the Parking Records at the Vehicle Entrance/Exit in Batches
+++ Parking Lot
++++ Get Parking Resource Tree
++++ Add a Parking Lot
++++ Update a Parking Lot
++++ Batch Delete Parking Lots
++++ Get Details of the Parking Lot
++++ Remote Opening

Go to the following directory to view the detailed MQ notifications

MQ Notice Directory
MQ Message
++ Vehicle
+++ Notification of Batch Add Vehicles
+++ Notification of Batch Update Vehicles
+++ Notification of Batch Delete Vehicles
+++ Notification of Snapshot Information of Vehicles at the Entrance and Exit
+++ Notification of Opening Results at the Entrance and Exit

5.9 Face Recognition



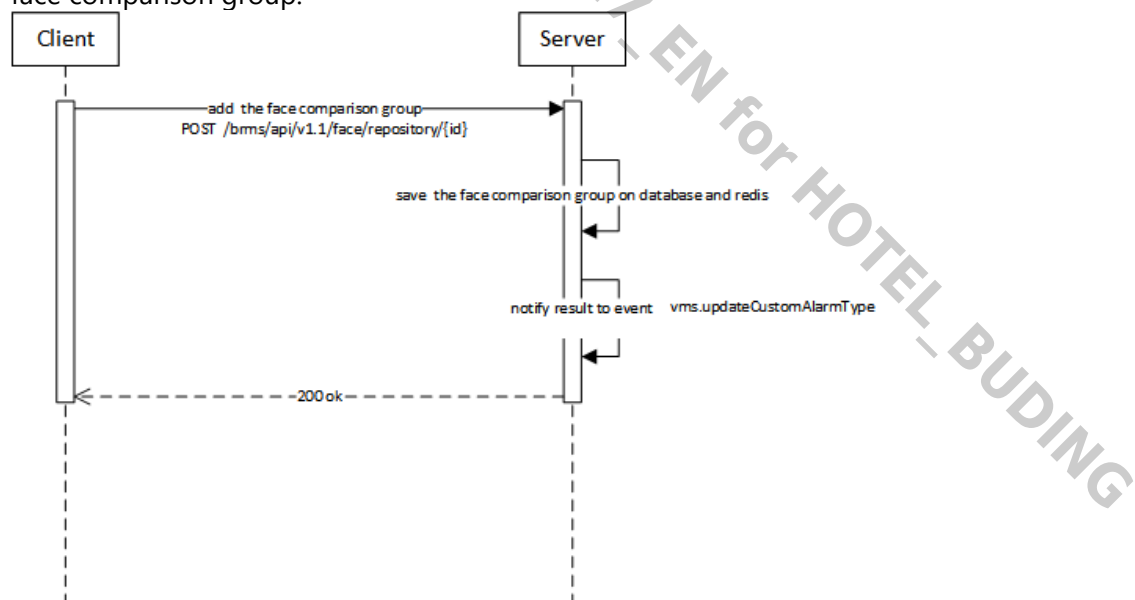
The main functions of the face recognition module include: Face comparison group (adding, editing and deletion), Face database arming, face search and face snapshot message reporting.

5.9.1 Face Comparison Group

On the platform, you can add, edit and delete face comparison groups, add person information to the face comparison group, arm and issue face comparison groups to the face device. You can create up to 50 face comparison groups and add 50,000 persons to a single face comparison group. The main face devices include IVSS, IPC and NVR.

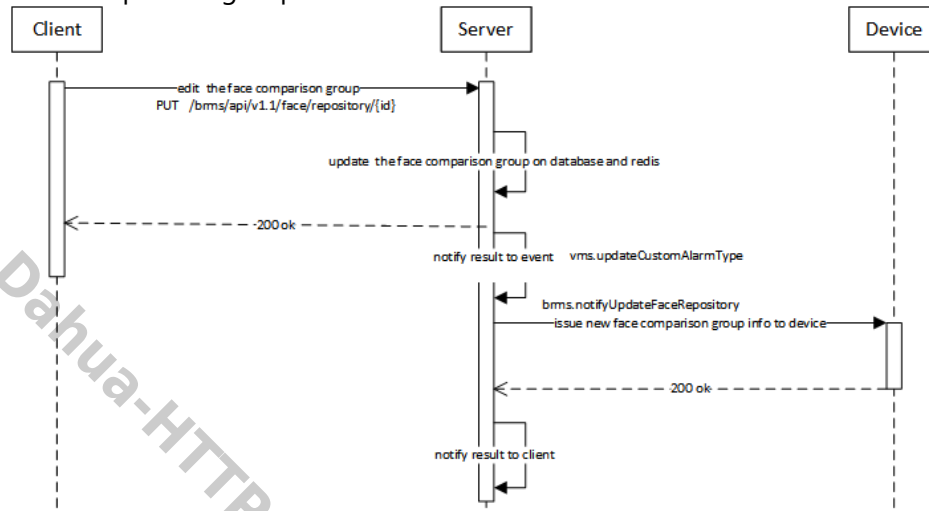
Add Face Comparison group

Select [3.12.1.1 Add a Face Comparison Group], and then call the interface to add a new face comparison group:



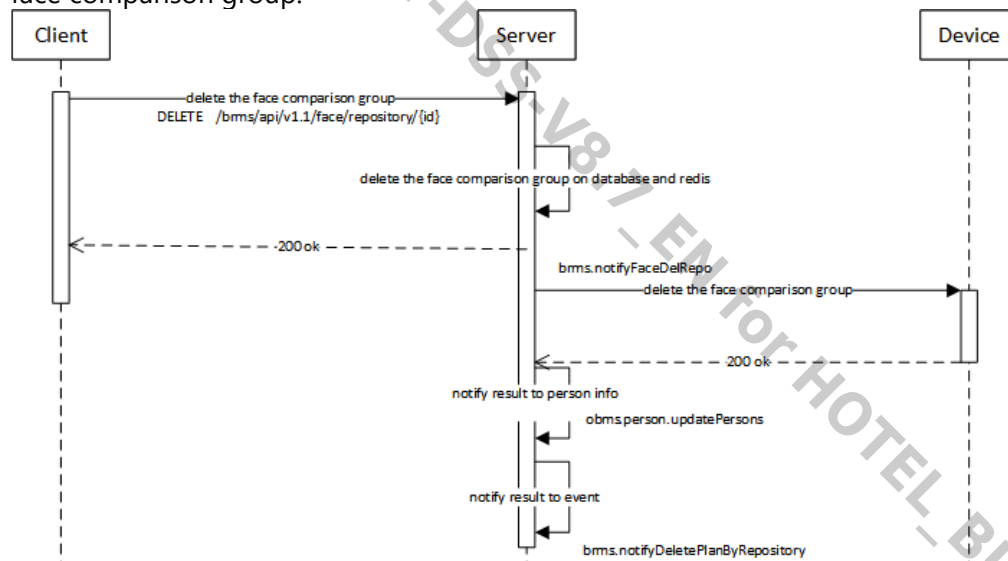
Modify Face Comparison Group

Select [3.12.1.2 Update a Face Comparison Group], and then call the interface to edit the face comparison group:



Delete Face Comparison Group

Select [3.12.1.3 Delete a Face Comparison Group], and then call the interface to delete the face comparison group:

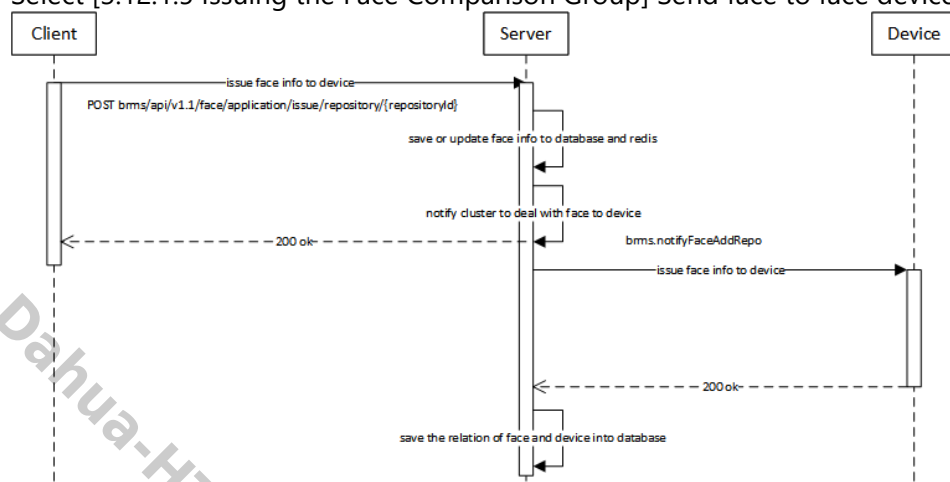


5.9.2 Face Database Arming

On the platform, you can arm the persons added to the face comparison group to the specified face device, or disarm them from the specified face device. The platform allows the maximum 10 million of face records to be distributed, mainly including face distribution, face deletion, channel distribution and channel withdrawal.

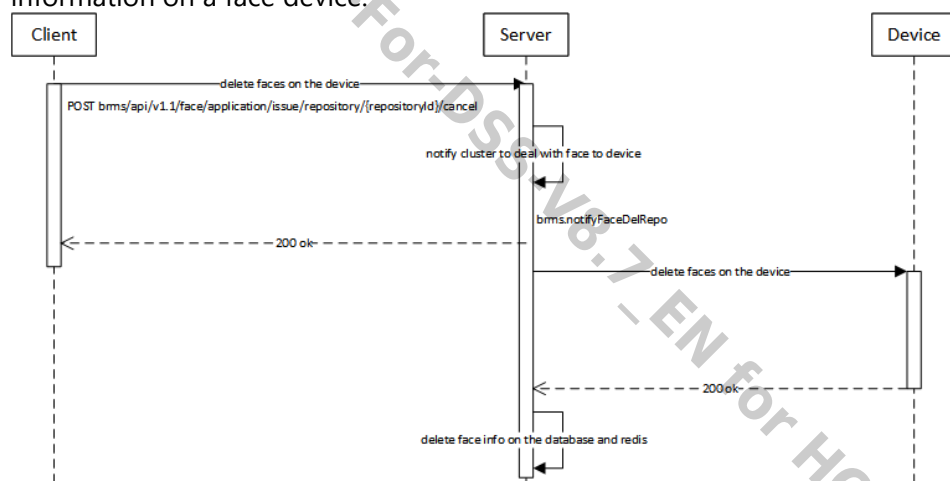
Face distribution

Select [3.12.1.5 Issuing the Face Comparison Group] Send face to face device:



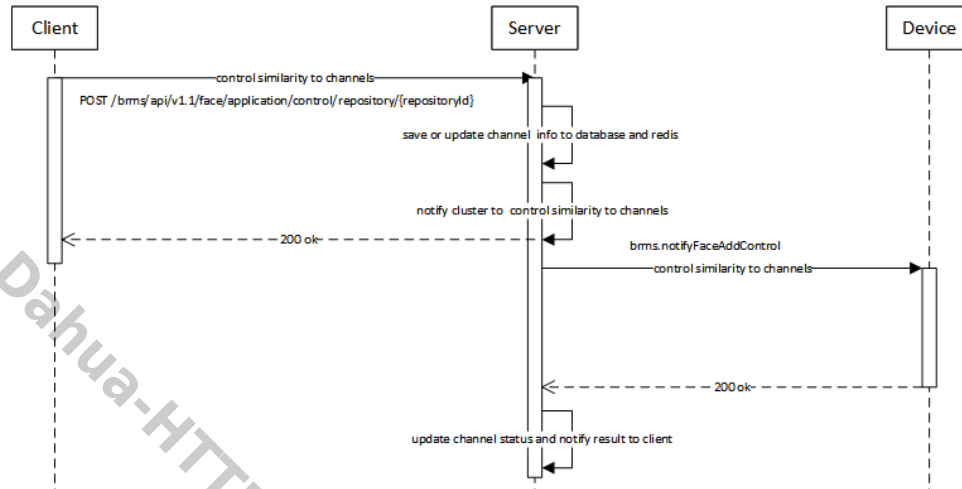
Face deletion and delivery

Select [3.12.1.6 Cancel the Issue of the Face Comparison Group] To delete face information on a face device:



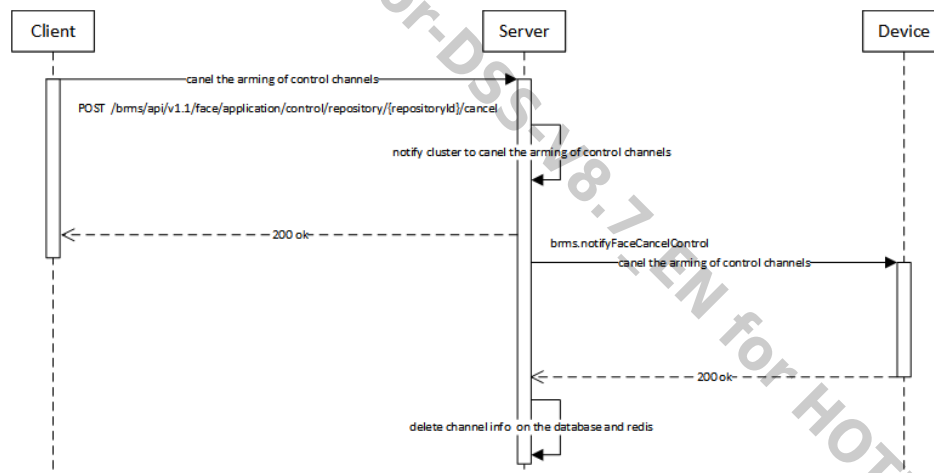
Arming

Select [3.12.1.7 Arming the Face Comparison Group], and then call the interface to arm faces to the face device:



Disarm

Select [3.12.1.8 Canceling the Arming of the Face Comparison Group].



5.9.3 Face search

On the platform, you can search the historical information of face snapshots based on conditions (search by face feature in the platform DB, and search by image on the device), and search for the historical information of face recognition by face feature and face image (search by face feature in the platform DB).

Face Snapshot Record Search

Select [3.12.2.1 Starting Search the Face Snapshot Records Asynchronously by Searching by Image], and then call the interface to get face snapshot or recognition records, and the interface returns session for the search of the next interface. Based on the session returned by the first interface, select [3.12.2.3 Search the Face Snapshot Records in Pages by Searching by Image], and then call the interface to get face snapshot or recognition records.

Select [3.12.2.5 Search the Face Snapshot Records in Pages Based on Features], and then call the interface to get face snapshot records based on the feature values.

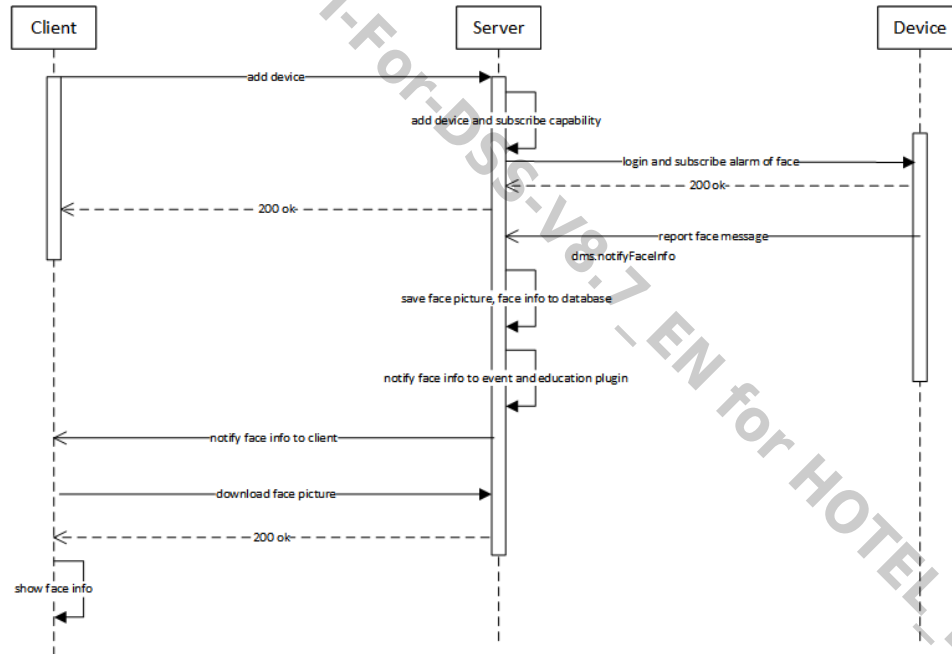
Face Recognition Record Search

Select [3.12.2.6 Search the Face Recognition Records in Pages Based on Features], and then call the interface to get face recognition records.

Face Snapshot Message Reporting

The face snapshot function module means that the camera channel of the device reports the captured face information to the platform, which then parses, processes and displays the event.

The platform parses and processes the information by receiving MQ messages in [4.7.2 Notification of Face Snapshot/Recognition Messages], and then notifies the alarm module and client.



Go to the following directory to view the detailed API interfaces

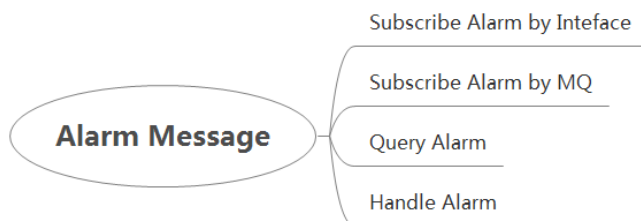
API Interface Directory
API Interface
++ Face
+++ Face Comparison Group
++++ Add a Face Comparison Group
++++ Update a Face Comparison Group
++++ Delete a Face Comparison Group

API Interface Directory
++++ Issuing the Face Comparison Group
++++ Cancel the Issue of the Face Comparison Group
++++ Arming the Face Comparison Group
++++ Canceling the Arming of the Face Comparison Group
++++ Get Face Comparison Group List in pages
+++ Face search
++++ Starting Search the Face Snapshot Records Asynchronously by Searching by Image
++++ Stopping Asynchronously Querying Face Capture Records in Reverse Image Search
++++ Search the Face Snapshot Records Asynchronously by Searching by Image
++++ Search the Face Snapshot Records in Pages by Searching by Image
++++ Face Snapshot Records — Search by Image (Download Image)
++++ Search the Face Snapshot Records in Pages Based on Features
++++ Search the Face Recognition Records in Pages Based on Features

Go to the following directory to view the detailed MQ notifications

MQ Notification Directory
MQ Notification Message
++ Face
+++ Notification of Face Snapshot/Recognition Messages
+++ Notification of Sending Faces
+++ Notification of Face Search by Image Result Progress

5.10 Alarm Messages



The platform supports two methods to send external alarms:

- Call the alarm port, and then push the alarm through the callback URL.
- Connect with the MQ, subscribe to the alarm topic, and then receive real-time alarm messages.

If you only connect to real-time alarms, the first method is recommended, because it is simpler. If you need to connect to other real-time information, such as real-time card swiping records and passed vehicle records, the second method is recommended, because its functions are more complete. Only alarm messages support the interface callback method now.

5.10.1 Interface Subscription

After completing login authentication, call the alarm port. See [3.6.1 Subscribe to Alarms]. That is, the specified logged in user subscribes to alarm messages from the platform. When the platform client configures the alarm scheme, a linked user is designated. If the linked user has subscribed to the alarm, when reporting the specified type of alarm, the callback interface provided during subscription is called to push an alarm message.

Notes:

1. The linked user of the alarm scheme must match the user who has subscribed to the alarm to receive alarms.
2. The URL and return value of the callback interface can be customized, while the interface must be a standard REST interface. The request method and input parameters must be in the specified formats.
3. The signature field of the input parameter is optional. The requester can freely define a string of no more than 255 characters, which is used for matching during callback to make a simple verification. You can repeatedly call the interface and modify the signature field to update the signature.
4. Subscription and unsubscription use the same interface and are distinguished by the action field.

5.10.2 MQ Subscription

Get MQ Address

Receive alarm messages from the platform through MQ. See [4 MQ Notification Message] to obtain the MQ address.

Alarm Subscription

The client gets alarm messages by subscribing to the Topic. The format of the alarm topic is as follows: **mq.alarm.msg.topic.{UserID}**. UserID is user ID, which is returned in the second login. If the user ID of the system is 1, after the system user logs in to the platform, the subscribed topic is: **mq.alarm.msg.topic.1**. See [6.3.2 MQ Connection Examples] for subscription connection.

5.10.3 Search for Alarm

Real-time Alarm Message

After the platform receives the alarm, if the alarm in the alarm scheme is configured with a linked user, the alarm will be sent to the corresponding online user. See [4.2.1 Alarm Messages] for alarm messages. Subscription by user group is also supported.

Historical Alarm Query

Search for alarm records in pages. Some alarm messages might carry images. If there is no image or you need to view the alarm details, you need to call the details viewing interface, which varies with the alarm type.

Process Alarm

Process alarm information and record processing comments.

5.10.4 The Connection Procedure for Typical Applications: Receive an Alarm

1. **Add a device that can trigger an alarm** from [Basic Config>Device>Add Device] on the VMS client.
2. **Configure an alarm scheme** from [Applications Config>Event] on the VMS client and set alarm type, alarm source, linkage rules, and intended users.
3. Start API connection. **Firstly, perform login authentication**, including first login, second login, heartbeat keepalive, and token update. For details, refer to [5.1 Login Authentication] or [6.3.1.1 Example of Login Authentication with Java] on Demo. All subsequent API invocations need to carry the token field returned during logging in the X-Subject-Token request header.
4. Alarms can be received by either ActiveMQ monitoring or subscription callback. Only the subscription callback method with lower connection difficulty is described here. For details, refer to [5.10 Alarm Messages].
5. **Subscribe to alarms** from API document through [3.6.1 Subscribe to Alarms]. Focus on all parameters:

```
{
  "callbackUrl": "https://192.168.1.1:443/brms/api/v1.1/push-data/alarm/push",
  "action": "1",
  "signature": "c743b433407be68ee62c46ede8fb5a0c"
}
```

Wherein,

callbackUrl: Indicates a third-party custom alarm receiving API, which must be defined in the specified callback format.

action: Indicates a subscription or unsubscription action. The token carried in this API is only effective for the subscriber. A repeated subscription action will overwrite the last subscription. Note: The subscriber must be the notified subscriber specified in the alarm scheme.

signature: Indicates a random string. If this field is not null, the platform will carry this string while calling back an alarm. The subscriber will accept this alarm only when the signature is valid.

6. A device triggers an alarm and sends it to the VMS platform. The VMS platform pushes the alarm to the VMS client while invoking the specified subscription callback API according to the notified subscriber specified in the alarm scheme, pushing key information including alarm time, alarm source, and alarm type to a third party platform.

Go to the following directory to view the detailed API interfaces

API Interface Directory
API Interface
++ Alarm
+++ Subscribe to Alarms
+++ Get Alarm Messages in Pages
+++ Handle Alarms
+++ Get Alarm Related Vehicle Blocklist Information — Vehicle database arming
+++ Get Alarm Related Vehicle Violation Information
+++ Get Alarm Details of Timeout Parking at the Entrance and Exit
+++ Get Alarm Details of Vehicles without Entering and Exiting Records at the Entrance and Exit
+++ Get Blocklist Information of Alarm Related Vehicle at the Entrance and Exit
+++ Get Alarm Related Face Snapshot Information
+++ Get Alarm Related Face Recognition Informationm
+++ Get Alarm Related Access Control Information
+++ Get Details of Parking over Line Alarm at the Entrance and Exit
+++ Get Details of Vehicle Arming Alarm
+++ Obtain the Alarm Details of the Entrance and Exit
+++ Obtain a list of alarm types

Go to the following directory to view the detailed MQ notifications

MQ Notification Directory
MQ Message
++ Alarm
+++ Alarm Messages
+++ Notification of Handling Alarms
+++ Notification of Alarm Images

Dahua-HTTP-API-For-DSS-V8.7_EN for HOTEL_BUDING

6 Appendix

6.1 Data Dictionary

6.1.1 Manufacturer Type

Value	Type
1	Dahua
2	Hikvision
4	AXIS
5	UNIVIEW
6	BOSCH
185	JIUZHOU
246	Other

6.1.2 Device Type

Device Category		Device Type	
Value	Type	Value	Type
1	Encoder	1	DVR/XVR
		2	IPC
		3	NVS
		5	MDVR/MXVR/MNVR
		6	NVR
		9	MPT
		10	EVS
		26	Thermal Camera
		43	IVSS/IVD
3	Alarm Controller	97	EEC
		1	Alarm Controller
5	ANPR Device	1	ANPR Device
		7	Indoor Parking Space Detector
		8	Access ANPR Camera
		19	Edge Storage Terminal
		20	Speedometer
		25	Outdoor Parking Space Detector

7	Intelligent Device	43	IVS
8	Access Control	1	Access Controller
		16	Access Control Recognition Terminal
		99	Third-party Access Control Device
15	UVSS	2	UVSS
16	Display Device	5	Indoor Parking Space Available Display
		6	Outdoor Parking Space Available Display
		7	Outdoor Entrance Parking Space Available Display
21	Video Intercom	2	Unit VTO
		3	VTH
		5	Fence VTO
		7	2nd VTO
		8	VTs
30	Visitor Terminal	16	Standing Visitor Terminal
		17	Desktop Visitor Terminal
34	Emergency	1	Emergency Phone Tower
		2	Emergency Phone Box
43	Radar Device	3	Radar Device
45	Video Wall Control	2	NVD
		4	DsCON
		6	VMP
46	Security Device	37	Security Screening Machine
		38	Walk-through Metal Detector
		42	EAS
50	Lift Control	1	Lift Controller
51	Network Device	1	Network Switch
53	IP Speaker	1	IP Speaker

6.1.3 Unit Type

Value	Type
1	Video Channel
2	Decode Channel
3	Alarm Input Channel
4	Alarm Output Channel
7	Access Control Channel
11	POS Channel
12	Radar Channel
15	LED Channel
16	HDCVI External Alarm Channel

Value	Type
17	Alarm Box Channel
18	Audio Light Alarm Channel
19	Display Settings
20	Parking Space Info
21	EAS Alarm Channel
22	Siren Output Channel
23	Audio Channel
24	UVSS Channel
26	Alarm Box Output Channel
50	Lift Control Channel

6.1.4 Camera Type

Value	Type
1	Fixed Camera
2	Speed Dome
3	Dome Camera

6.1.5 Encoding Channel Capability Set

Decimal is converted to 32-bit binary, each bit representing a set of capabilities: 0 = no, 1 = yes. According to the enumeration value description, from right to left, the value of each bit is determined by determining whether it is 1 to confirm whether there is a corresponding capability set. For example, the capability set returned by the interface to a certain channel is 9, and the conversion to binary is 1001, that is to say, starting from the right, the first and fourth bits have capability sets, corresponding to "Intelligent Alarm" and "IR Temperature Analysis" respectively.

Value	Type
1	Intelligent Alarm
2	Fisheye
3	Electric Focus
4	IR Temperature Analysis
5	General Heat Map
8	Face Detection
9	Face Recognition
10	Metadata Attribute Report Capability
11	Main Sub Track
12	Entrance ANPR
17	People Counting
18	Privacy Protection

Value	Type
19	Advanced Heat Map
20	Road ANPR
21	Face & Body Detection
24	Indoor Parking Space Detector
27	Outdoor Parking Space Detector
28	Metadata Feature Report Capability
29	AR
32	Smart Traffic
33	Crowd Distribution
34	Vehicle Density

6.1.6 Gender

Value	Type
0	Unknown
1	Male
2	Female

6.1.7 Alarm Type

Event Source Type		Alarm Category		Alarm Sub-category	
Value	Type	Value	Type	Value	Type
1	Device	101	Common Device Alarm	16	Device Disconnected
		1	Encoder	5	Disk Full
				6	Disk Error
				225	Low Battery
				1916	No Disk
				4204	Fall Alarm
				19068	Fire Sensor Fault Alarm
				19069	Fire Equipment Tamper Alarm
				19093	Device Self-check
		20	Emergency	1907	SOS
				13010	Tamper
		25	Access Control Device	600004	Device Tamper Alarm
		31	Video Intercom Device	900009	Call from Video Intercom
		22	EVS/NVR Device	1917	Software Version Abnormal

				1918	Cluster Switch
				1924	Storage Pool Error
				1925	Raid Alarm
				1926	Share Service Exception
				1960	Main and Sub Board Switch
				4402	Disk Check
				19024	Disk Health Abnormal
		107	Walk-through Metal Detector	15623	Metal Alarm
		108	Security Screening Machine	847	Contraband object alarm
				19071	Control board failure. Communication exception
				19072	Interface board failure. Communication exception
				19073	The emergency stop button is pressed
				19074	X-ray Detector box interlock was opened
				19075	The security screening machine housing was opened
				19076	Keyboard failure. Communication exception
				19079	X-ray generator failure
		113	EAS	19111	AC Frequency Alarm
		8	Alarm Controller	84	Device Low Battery
				85	Device Main Power Failure
				1649	Device Storage Battery Missing
				1651	PSTN Offline
				22020	Fire Alarm
				22021	Medical Alarm
				22022	Panic Alarm
				22023	Hold-up Alarm
				22025	Duress Alarm
				22028	2G or 4G Module Missing
				22029	Expansion Module Not Connected
				22034	SIM Card Missing
				22037	Device Tamper
				22038	Siren Tamper
				22039	Keypad Locked
				22086	Device Main Power Restored
				22087	Device Battery Restored
				22088	Device Storage Battery Restored
				22089	Device Tamper Resolved
				701609	Keypad Tamper

				701671	Keypad Offline
2	Video Channel	102	Operation and Maintenance Alarm	1	Video Loss
				4	Video Tampering
				13	Channel Disconnected
				665	Scene Changing
		111	Video Quality Analytics	951	Image Freeze
				955	High Brightness
				956	Color Cast
				17054	Blurry Video
				17055	Low Contrast
				17056	Video Jitter
				17061	Interference Fringe
				17062	Static Noise
		4	Intelligent Event	3	Motion Detection
				302	Tripwire
				303	Intrusion
				305	Abandoned Object
				306	Stay Detection
				307	Loitering Detection
				309	Fast Moving Detect
				310	Tailing Detection
				311	Crowd Gathering
				312	Flame Event
				313	Smoke Alarm
				314	Violence Detection
				564	Illegal Parking
				574	Running Detection
				575	Wrong-way Driving Detection
				577	Smart Missing Object
				578	Parking Detection
				587	Crossing Virtual Fence
				597	Leave Post Detection
				675	Fall Detection
				825	Stand Detection
				886	People Approach Detection
				901	Abnormal Face
				902	Adjacent Face
				961	Tripwire by Animal
				962	Tripwire(Human)

				963	Tripwire(Vehicle)
				964	Intrusion(Human)
				965	Intrusion(Vehicle)
				966	Intrusion by Animal
				980	Fast Moving(Human)
				981	Fast Moving(Vehicle)
				982	Animal Moving Fast
				15591	Illegal Parking (Motor Vehicle)
				15592	Illegal Parking (Non-motor Vehicle)
				15599	Floating Debris Detection
				15602	Call Detection
				15607	Garbage Exposure
				15608	Full Garbage
				15618	Classroom Behavior Analysis
				15631	Shelf Replenishment
				15666	Walking Detection
				15667	Back Detection
				15668	Blackboard Writing Detection
				15670	Sleeping while on Duty
				15768	Using Mobile Phone
				15804	Blocked Fire Exit
				15908	Exam Behavior Analysis
				15917	Crowd Detection
				16030	Head-up Detection
				16235	Crowd Status
				16236	Vehicle Status
				16238	Abandoned Shopping Cart Items
				16239	Shopping Cart Detection
				16994	Low Water Level Alarm
				16995	High Water Level Alarm
				16998	Crowd Map(Area)
				16999	Crowd Map(Global)
				17105	Boat Tripwire
				18133	Boat Intrusion
				18288	Theft Detection
				19010	SMD(Human)
				19011	SMD(Vehicle)
				19022	Congestion Detection
				19023	Parking Upper Limit

				900002	Smart Object Abandoned(Non-motor Vehicle)
				900003	Smart Object Abandoned(Luggage/Bag/Box)
				900004	Smart Missing Object (Non-motor Vehicle)
				900005	Smart Missing Object (Luggage/Bag/Box)
		127	Open AI Event	16000000-16999999	Open AI Event Name
		103	Flow Event	316	People Counting
				826	Area People Counting
				15610	Queue Stay Time
				15611	Abnormal Number of People in Queue
		11	Thermal	5120	Temperature Alarm
				5121	Flashpoint Alarm
				5123	Hot Spot
				5124	Cold Spot
				5125	Temperature Contrast
				5128	Low Temperature Abnormal
				5129	High Temperature Abnormal
				5130	Blackbody Abnormal
				15603	Smoking Detection
				17508	Drone Tripwire
				17509	Drone Intrusion
		109	PPE Detection	15622	PPE Detection
		104	Face Feature Detection	967	Not Wearing Safety Hat
				17029	Not Wearing Reflective Vest
				17044	Wearing Face Mask
				17045	Wearing Glasses
				17046	Wearing Hat
				17047	Wearing Helmet
				17048	Wearing Safety Helmet
				17050	Wearing Bag
				17051	Not Wearing Glasses Alarm
				299999	Not Wearing Face Mask
		105	Face Arming	100000-199999	Face Arming Group Name
				200000	Stranger Alarm
		122	Vehicle Detection	300101	Speeding
				400000	Unlicensed Vehicle
		106	Vehicle Arming	400001	High Risk Vehicle

				400002-400099	Vehicle Arming Group Name
		24	Audio Detection	1909	Audio Exception
				1998	Intensity Change
3	Alarm Input Channel	3	Alarm Input Channel	2	External Alarm
				28	Remote External Alarm
		30	Custom Alarm Type	19	Infrared Detect
				4300	Door Sensor
				4301	PIR
				4302	Gas Sensor
				4303	Smoke Sensor
				4304	Water Leak Detection
				4305	Active Infrared Sensor
				4306	Glasses Sensor
				4321	Emergency Button
				4322	Stolen Alarm
				4323	Perimeter
				4324	Preventer Move
				4331	Fire Alarm
				4339	Wireless IR Curtain
				4349	Siren
				4352	Duress Password
				12000000-12999999	Custom type name
18	Subsystem	124	Change Status	22072	Arm
				22073	Disarm
16	Zones	118	Zone Alarm	81	Zone Alarm
				82	Flame Alarm
				1604	Medical Alarm
				1666	Burglary Alarm
				22066	High Temperature Alarm
				22067	Low Temperature Alarm
				22076	Medical Alarm Restored
				22077	Flame Alarm Restored
				22078	Burglary Alarm Restored
				22079	High Temperature Alarm Restored
				22080	Low Temperature Alarm Restored
				12200019	Infrared Detect
				12202201	PIR
				12202604	Zone Disarm

4	EAS Alarm Channel			12204302	Gas Sensor
				12204303	Smoke Sensor
				12204306	Glasses Sensor
				12204323	Perimeter
		119	Zone Fault Alarm	1635	Peripheral Short Circuit
				1636	Peripheral Tamper
				22002	Peripheral Masking
				22026	Peripheral Low Battery
				22027	Peripheral Not Connected
				22032	Peripheral Main Power Failure
				22035	Peripheral Storage Battery Missing
				22036	External Device Triggered
				22081	Peripheral Tamper Resolved
				22082	Peripheral Main Power Restored
				22083	Peripheral Battery Restored
				22084	Peripheral Storage Battery Restored
				22085	Peripheral Not Connected Resolved
		125	Change Status	22074	Arm
				22075	Disarm
12	EAS Alarm Channel	114	EAS Alarms	19107	EAS Alarm
				19108	TX Closed Alarm
				19109	RX Closed Alarm
				19110	Tag Too Close Alarm
				19112	Channel Disconnected
4	Access Control Channel	16	Alarm Event	41	Duress Alarm
				72	Door Open Timeout Alarm
				1433	Blocklist Alarm
				1446	Malicious Alarm
				13105	Anti-pass Back Alarm
				13110	Intrusion Alarm
				13130	Global Interlock (Device Offline)
				600003	Reader Tamper Alarm
				701020	High Temperature
				702001	Incorrect Password Alarm
		15	Abnormal Access	43	Stranger (Password)
				46	Stranger (Fingerprint)
				52	Stranger (Card)
				62	Stranger (Face)
				13100	No Permission

				13103	Unlock Method Verification Failed
				13104	Validity Error
				13107	Door NC Unlock
				13108	Multi-door Interlock (Locked Door)
				13111	Period Error
				13112	Unlock Time Error During Holiday
				13113	Non-first Card (daily)
				13114	Card Correct but Password Error
				13116	Card Correct but Fingerprint Error
				13118	Fingerprint Correct but Password Error
				13120	Incorrect Sequence of Multi-person Combination Unlock
				13127	The user is not in the authorized group
				13129	Incorrect Sequence of Single-person Combination Unlock
				13140	Repeated entering
				13142	Identity of the same user was verified more than once during group combination unlock
				13143	Frozen Person Permissions
				13144	The visiting times reached the upper limit
				600012	Card Correct but Face Error
		5	Normal Access	42	Valid Password Unlock
				45	Valid Fingerprint Unlock
				48	Platform Remote Open
				49	Normal Button Unlock
				51	Valid Swipe
				13109	Patrol User
				13121	Unlock Require Verification
				13123	Card and Password Unlock
				13124	Card and Fingerprint Unlock
				13125	Remote Verification
				13126	Group combination unlock is approved
				13316	Unlock by Palm Veins
				13317	Unlock by Palm Veins and Password
				13318	Unlock by Face and Palm Veins
				13319	Unlock by Card and Palm Veins
				13323	Unlock by Face, Palm Veins and Password
				13324	Unlock by Card, Face and Palm Veins
				13325	Unlock by Card, Palm Veins and Password

				13329	Unlock by Card, Face, Palm Veins and Password
				600005	Valid Face Unlock
				600013	Card and Face Unlock
				700000	Fingerprint and Password Unlock
				700001	Fingerprint and Face Unlock
				700002	Face and Password Unlock
				700003	Card+Fingerprint+Password Unlock
				700004	Card+Fingerprint+Face Unlock
				700005	FingerPrint+Face+Password Unlock
				700006	Card+Face+Password Unlock
				700007	Card+Fingerprint+Face+Password Unlock
				900000	VTH Remote Open
				900001	VTX Remote Open
6	Radar	21	Radar Alarm	15661	Radar Alarm
7	Parking Lot	23	Parking Lot Alarm	10701	Blocklist Alarm
				10702	Overtime Parking in Parking Lot
				10703	No Entry and Exit Record
				10704	Reserved Parking Space Occupied
				10705	Parking Overline
				10706	Overtime Parking in Parking Space
				10707	Parking Lot Full
				10708	Strange Vehicle
				10709	Small Vehicle Parked in Large Vehicle's Space
				10710	Unauthorized Occupation
17	People Counting Group	123	People Counting Group Alarm	14000000	Overlimit Threshold Alarm
				14000001	Crowd Threshold Alarm
9	Lift Control Channel	27	Alarm Event	1003	VTM Offline Alarm
		28	Normal Access	1394	Authentication on VTO
				1395	Authentication on Platform
				1396	Authentication on Lift Controller
				13128	Authentication on Access Control Device
21	Mobile Device	132	Face Arming	1100000-1199999	Face Arming Group Name
		131	Vehicle Arming	1200000-1299999	Vehicle Arming Group Name
10	Soft Trigger	110	Soft Trigger Alarm	11000001	Soft Trigger_1
				11000002-12000000	Soft Trigger Name
11	POS Alarms	112	POS Alarms	8600	POS Original Price Alarm

				8601	POS Discount Alarm
				8602	POS Refund Alarm
13	Combined Event	115	Combined Event	500000-599999	Combined Event Name
14	Custom Alarm	116	DHOP Event	13000001	DHOP Event
		117	Extended Private Event	13000002-13999999	Custom Alarm Name
		121	SIA Event	13000002-13999999	SIA Event Name
19	Generic Event	126	Generic Event	15000000	Generic Event

6.1.8 Offline Reasons of Devices

Value	Type
Empty	Online
-1	High security login is not supported by the device
0	Online
1	Incorrect username or password
2	Incorrect username or password
3	Login timed out
4	Account was logged in to
5	Account locked
6	The account is in the blocklist
7	Insufficient resources. System is busy
8	Abnormal network
9	Abnormal network
10	Exceeds max connections
11	Unknown error
12	No USB keys were plugged in the device or the USB key information is incorrect
13	The IP address does not have login permission
18	The device account was not initialized
19	Login prohibited
20	The device only supports high security login
21	The device password expired
90	P2P service communication failed
96	SIP No. repeated.
97	Building enable not matched.
98	Unit enable not matched.
99	Unknown error
100	Phase enable not matched.

Value	Type
101	The residence scenario is not configured.
102	2nd VTO is not support.
103	Service runs abnormally
104	The device was not registered.
105	Failed to receive the heartbeat.
106	Device Logout
108	Failed to subscribe for the event. Another platform might be using the device.
2000	Unknown error
2011	Failed to authenticate P2P service
2012	Failed to log in to the P2P service
2013	P2P service is offline
2014	Failed to log in to the P2P service
2015	Failed to log in to the P2P service
2016	Failed to log in to the P2P service
2017	Failed to authenticate P2P service
2018	Unknown error
2019	The device is not registered with P2P
2020	Failed to log in to the P2P service
2021	Failed to log in to the P2P service
2022	Failed to log in to the P2P service
2023	Failed to log in to the P2P service
3001	The DMS service is running abnormally
3002	The PCPS service is running abnormally
3003	The ARS service is running abnormally
3004	The DMS and PCPS services are running abnormally
3005	The DMS and ARS services are running abnormally
3006	The ARCMS service is running abnormally
3007	The ARCMS and ARS services are running abnormally
3008	The ACDG service is running abnormally
3009	The ACDG and ARS services are running abnormally
3010	The VMS service is running abnormally

6.1.9 Plate Color

Value	Type
0	Blue
1	Yellow
2	White
3	Black

Value	Type
4	Green
5	Shadow Green
99	Unrecognized
100	Other colors

6.1.10 Color of motor vehicles

Value	Type
0	White
1	Black
2	Red
3	Yellow
4	Gray
5	Blue
6	Green
8	Purple
10	Pink
11	Brown
12	Silver
18	Dark Orange
99	Unrecognized
100	Other

6.1.11 Model of motor vehicles

Value	Type
-1	Other
0	Unrecognized
17	Public Bus
18	Motorcycle
19	Bus
20	Large Truck
21	Medium Truck
22	Sedan
23	Van
24	Small Truck
26	Medium Bus
27	SUV
28	MPV

Value	Type
29	Pickup
32	Mini Sedan

6.1.12 Brand of motor vehicles

The dictionary type can be obtained through [3.2.5 Get a List of Data Dictionaries], parameter types=2016.

6.1.13 Color of non motorized vehicles

Value	Type
0	White
1	Black
2	Red
3	Yellow
4	Gray
5	Blue
6	Green
8	Purple
10	Pink
11	Brown
17	Sliver
21	Dark Orange
99	Unrecognized
100	Other

6.1.14 Model of non motorized vehicles

Value	Type
-1	Other
0	Unrecognized
1	Tricycle
4	Bicycle
14	Van Cargo Tricycle
15	Open Tricycle with Passenger
16	Open Tricycle without Passenger

6.1.15 Region

The dictionary type can be obtained through [3.2.5 Get a List of Data Dictionaries], parameter types=9046.

6.2 Common Error Codes

6.2.1 HTTP Status Codes

Error code	Error explanation
400	Bad request, please check whether the request information is correct.
401	Authentication failed. The possible cause is that no token authentication information is carried or the token has become invalid. After successful login, a token is returned. This returned token shall be carried in subsequent requests. Also, the token has a valid time. It needs a keep-alive within 30 seconds, and an update within 30 minutes.
403	The request URL is illegal. Please check the correctness of the URL.
415	The Content-type type is incorrect. The input parameters of our most POST/PUT interfaces are in standard JSON format. The request header of Content-type must be set as application/json.
429	The single business request is too frequent. Please reduce the request frequency and try again later.
503	The server is busy, please pause your request and try again later.

6.2.2 Business Error Codes

Error code	Error explanation
1001	Internal server error. Check the parameters or contact the maintenance personnel to view the log for problem analysis.
1004	Invalid request parameter. It may be that a required parameter is missing or the parameter value format is incorrect.
1005	The name already exists. Modify the name and try again.
1006	The maximum number limit is exceeded. The limit varies with the function module.
1007	The operation data does not exist. It is recommended to restart the client and try again.
1008	The current task is not completed. Do not repeat the operation. Try again later.
1101	Login error. The username userName is null or the random value randomKey is invalid. Check the login interface parameters. Long interval between two login operations results in invalid randomKey. Complete two login operations within 10 seconds.
1103	The current user does not have operation permission for this channel or device. Contact the administrator for authorization.
1105	Service offline. Try again later, or contact the maintenance personnel to check the platform.
2001	Username or password error. Check and try again.
2002	The user is locked. Try again later, or contact the administrator to unlock.
7000	Authentication failed. The token might be null or invalid.

6.3 Docking Example

6.3.1 API Connection Examples

6.3.1.1 Example of Login Authentication with Java

Introduce Maven Dependency

```
<dependencies>
  <dependency>
    <groupId>org.apache.httpcomponents</groupId>
    <artifactId>httpclient</artifactId>
    <version>4.3.6</version>
  </dependency>
  <dependency>
    <groupId>com.alibaba</groupId>
    <artifactId>fastjson</artifactId>
    <version>1.2.58</version>
  </dependency>
</dependencies>
```

Example Code

```
import com.alibaba.fastjson.JSON;
import com.alibaba.fastjson.JSONObject;
import org.apache.commons.codec.digest.DigestUtils;
import org.apache.http.HttpEntity;
import org.apache.http.client.methods.CloseableHttpResponse;
import org.apache.http.client.methods.HttpEntityEnclosingRequestBase;
import org.apache.http.client.methods.HttpPost;
import org.apache.http.client.methods.HttpPut;
import org.apache.http.conn.ssl.SSLConnectionSocketFactory;
import org.apache.http.entity.StringEntity;
import org.apache.http.impl.client.CloseableHttpClient;
import org.apache.http.impl.client.HttpClients;
import org.apache.http.util.EntityUtils;

import javax.crypto.Cipher;
import javax.net.ssl.SSLContext;
import javax.net.ssl.TrustManager;
import javax.net.ssl.X509TrustManager;
import java.io.IOException;
import java.nio.charset.StandardCharsets;
import java.security.*;
import java.security.cert.X509Certificate;
import java.security.spec.X509EncodedKeySpec;
import java.util.*;

/**
 * This is a LoginExample.
 */
public class LoginExample {

    static final String AUTH_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/authorize";
    static final String KEEP_ALIVE_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/keepalive";
    ;
```

```

static final String UPDATE_TOKEN_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/updateToken";

static final String SUCCESS_CODE = "1000";
static String SIGNATURE_MD5_TEMP4 = null;
static String TOKEN_VALUE = null;
static final String TOKEN = "token";
static final String POST = "post";
static final String PUT = "put";
static int HEART_COUNT = 0;

/**
 * main method
 */
@param args args
@throws Exception exception
*/
public static void main(String[] args) throws Exception {

    //Step1: Try to login first time.
    Map<String, Object> firstLoginParams = new HashMap<>(3);
    firstLoginParams.put("userName", "system");
    firstLoginParams.put("ipAddress", "");
    firstLoginParams.put("clientType", "WINPC_V2");
    String firstResponseString = sendPostOrPut(AUTH_URL, firstLoginParams, POST);
    JSONObject firstLoginResponse = JSONObject.parseObject(firstResponseString);

    //Step2: Try to login second time.
    String userName = "system";
    String passWord = "admin1234567";
    String realm = firstLoginResponse.getString("realm");
    String randomKey = firstLoginResponse.getString("randomKey");
    String platformPublicKeyWithBase64 = firstLoginResponse.getString("publickey");
    String signature = generateSignature(userName, passWord, realm, randomKey);
    //Generate the AES secret key and AES secret vector, encrypted by RSA public key.
    String aesSecretKey = UUID.randomUUID().toString().replace("-", "");
    String aesSecretVector = UUID.randomUUID().toString().replace("-", "").substring(16);
    String encryptedAesSecretKeyWithBase64 = Base64.getEncoder().encodeToString(encryptBy
    PublicKey(aesSecretKey, Base64.getDecoder().decode(platformPublicKeyWithBase64)));
    String encryptedAesSecretVectorWithBase64 = Base64.getEncoder().encodeToString(encrypt
    ByPublicKey(aesSecretVector, Base64.getDecoder().decode(platformPublicKeyWithBase64)));

    Map<String, Object> secondLoginParams = new HashMap<>(9);
    secondLoginParams.put("mac", "2C:F0:5D:4D:5E:DB");
    secondLoginParams.put("signature", signature);
    secondLoginParams.put("userName", userName);
    secondLoginParams.put("randomKey", randomKey);
    secondLoginParams.put("publicKey", "");
    secondLoginParams.put("encryptType", "MD5");
    secondLoginParams.put("ipAddress", "");
    secondLoginParams.put("clientType", "WINPC_V2");
    secondLoginParams.put("userType", "0");
    secondLoginParams.put("secretKey", encryptedAesSecretKeyWithBase64);
    secondLoginParams.put("secretVector", encryptedAesSecretVectorWithBase64);
    String secondResponseString = sendPostOrPut(AUTH_URL, secondLoginParams, POST);
    JSONObject secondResponse = JSONObject.parseObject(secondResponseString);
    TOKEN_VALUE = secondResponse.getString(TOKEN);
}

```

```

        System.out.println(String.format("Your AES key {%s} and vector {%s}, which you can use to
        o decrypt encrypted data returned by other interfaces", aesSecretKey, aesSecretVector));
        System.out.println(String.format("The token is {%s}", TOKEN_VALUE));
        System.out.println(String.format("The credential is {%s}", secondResponse.getString("credential")));
        System.out.println(String.format("The userId is {%s}", secondResponse.getString("userId")));

        //Step3: Keep alive and Update the token.
        while (true) {
            //Send a heart per 20 seconds.
            Thread.sleep(20000);
            Map<String, Object> keepAliveParamMap = new HashMap<>(1);
            keepAliveParamMap.put("token", TOKEN_VALUE);
            JSONObject heartResponse = JSONObject.parseObject(sendPostOrPut(KEEP_ALIVE_URL,
            keepAliveParamMap, PUT));
            if (SUCCESS_CODE.equals(heartResponse.getString("code"))) {
                System.out.println("Heart success.");
            }
            HEART_COUNT++;
            //Update the token per 20 minutes.
            if (HEART_COUNT % 60 == 0) {
                //Restore the heartCount to zero.
                HEART_COUNT = 0;
                Map<String, Object> updateTokenParamMap = new HashMap<>();
                JSONObject updateTokenResponse = JSONObject.parseObject(sendPostOrPut(UPDATE_TOKEN_URL,
                updateTokenParamMap, POST));
                if (SUCCESS_CODE.equals(updateTokenResponse.getString("code"))) {
                    String newTokenValue = updateTokenResponse.getJSONObject("data").getString(TOKEN);
                    System.out.println(String.format("Update token success. New token: %s", newTokenValue));
                    TOKEN_VALUE = newTokenValue;
                }
            }
        }

        //Login is over, do your own things with the token...
    }

    /**
     * Send post or put request method with HTTP.
     * @param url url
     * @param params params
     * @param requestMode POST or PUT or GET or DELETE
     * @return {@link String}
     * @throws IOException ioexception
     */
    static String sendPostOrPut(String url, Map<String, Object> params, String requestMode) throws IOException {
        if (!requestMode.equals(POST) && !requestMode.equals(PUT)) {
            return null;
        }
        // Create a trust manager that does not validate certificate chains
        TrustManager[] trustAllCerts = new TrustManager[] {
            new X509TrustManager() {
                @Override
                public X509Certificate[] getAcceptedIssuers() {

```

```

        return new X509Certificate[0];
    }
    @Override
    public void checkClientTrusted(X509Certificate[] certs, String authType) {
    }
    @Override
    public void checkServerTrusted(X509Certificate[] certs, String authType) {
    }
}
};
// Install the all-trusting trust manager
try {
    SSLContext sslContext = SSLContext.getInstance("SSL");
    sslContext.init(null, trustAllCerts, new SecureRandom());
    // Create an ssl socket factory with our all-trusting manager
    SSLConnectionSocketFactory sslSocketFactory = new SSLConnectionSocketFactory(sslContext, (hostname, session) -> true);

    // Create a custom HttpClient with the SSL socket factory
    CloseableHttpClient httpClient = HttpClients.custom()
        .setSSLSocketFactory(sslSocketFactory)
        .build();
    HttpEntityEnclosingRequestBase httpRequest = null;
    if (requestMode.equals(POST)) {
        httpRequest = new HttpPost(url);
    } else if (requestMode.equals(PUT)) {
        httpRequest = new HttpPut(url);
    }
    StringEntity entity = new StringEntity(JSON.toJSONString(params), "UTF-8");
    httpRequest.setEntity(entity);
    httpRequest.setHeader("Content-Type", "application/json;charset=UTF-8");
    httpRequest.setHeader("X-Subject-Token", TOKEN_VALUE);

    CloseableHttpResponse response = httpClient.execute(httpRequest);
    HttpEntity responseEntity = response.getEntity();
    //In order to avoid messy code, encode the response data in UTF-8.
    String reply = EntityUtils.toString(responseEntity, "UTF-8");
    //Release resources finally.
    if (httpClient != null) {
        httpClient.close();
    }
    if (response != null) {
        response.close();
    }
    return reply;
} catch (NoSuchAlgorithmException | KeyManagementException e) {
    throw new RuntimeException(e);
}
}

/**
 * Generate signature with MD5-transformation five times.
 * @param userName login username
 * @param password login password
 * @param realm the 'realm' returned from the first login
 * @param randomKey the 'randomKey' returned from the first login
 * @return signature

```

```

    */
    static String generateSignature(String userName, String password, String realm, String randomKey) {
        String temp1 = DigestUtils.md5Hex(password);
        String temp2 = DigestUtils.md5Hex(userName + temp1);
        String temp3 = DigestUtils.md5Hex(temp2);
        String temp4 = DigestUtils.md5Hex(userName + ":" + realm + ":" + temp3);
        //Retain the temp4, in order to calculate a signature for update token later.
        SIGNATURE_MD5_TEMP4 = temp4;
        String signature = DigestUtils.md5Hex(temp4 + ":" + randomKey);
        return signature;
    }

    /**
     * RSA encryption. Public key encryption, private key decryption
     * @param text data
     * @param publicKeyBytes public key string array
     * @return string after encryption
     * @throws Exception
     */
    static byte[] encryptByPublicKey(String text, byte[] publicKeyBytes) throws Exception {
        byte[] result = null;
        int rsaBytesLenEncrypt = 245;

        //Get public key object
        X509EncodedKeySpec x509KeySpec = new X509EncodedKeySpec(publicKeyBytes);
        KeyFactory keyFactory = KeyFactory.getInstance("RSA");
        Key publicKeyObject = keyFactory.generatePublic(x509KeySpec);

        //Encrypt data
        byte[] dataBytes = text.getBytes(StandardCharsets.UTF_8);
        Cipher cipher = Cipher.getInstance(keyFactory.getAlgorithm());
        cipher.init(Cipher.ENCRYPT_MODE, publicKeyObject);
        for (int i = 0; i < dataBytes.length; i += rsaBytesLenEncrypt) {
            int to = Math.min(i + rsaBytesLenEncrypt, dataBytes.length);
            byte[] temp = cipher.doFinal(Arrays.copyOfRange(dataBytes, i, to));
            result = sumBytes(result, temp);
        }
        return result;
    }

    /**
     * Sum 2 bytes
     * @param bytes1 byte 1
     * @param bytes2 byte 2
     * @return the new array formed by merging two arrays
     */
    static byte[] sumBytes(byte[] bytes1, byte[] bytes2) {
        byte[] result = null;
        int len1 = 0;
        int len2 = 0;
        if (null != bytes1) {
            len1 = bytes1.length;
        }
        if (null != bytes2) {
            len2 = bytes2.length;
        }
    }

```

```
        if (len1 + len2 > 0) {
            result = new byte[len1 + len2];
        }
        if (len1 > 0) {
            System.arraycopy(bytes1, 0, result, 0, len1);
        }
        if (len2 > 0) {
            System.arraycopy(bytes2, 0, result, len1, len2);
        }
        return result;
    }
}
```

6.3.1.2 Example of Add Person with Java

Introduce Maven Dependency

```
<dependencies>
  <dependency>
    <groupId>org.apache.httpcomponents</groupId>
    <artifactId>httpclient</artifactId>
    <version>4.3.6</version>
  </dependency>
  <dependency>
    <groupId>com.alibaba</groupId>
    <artifactId>fastjson</artifactId>
    <version>1.2.58</version>
  </dependency>
</dependencies>
```

Example Code

```
import com.alibaba.fastjson.JSON;
import com.alibaba.fastjson.JSONObject;
import org.apache.http.HttpEntity;
import org.apache.http.client.methods.CloseableHttpResponse;
import org.apache.http.client.methods.HttpEntityEnclosingRequestBase;
import org.apache.http.client.methods.HttpGet;
import org.apache.http.client.methods.HttpPost;
import org.apache.http.conn.ssl.SSLConnectionSocketFactory;
import org.apache.http.entity.StringEntity;
import org.apache.http.impl.client.CloseableHttpClient;
import org.apache.http.impl.client.HttpClients;
import org.apache.http.util.EntityUtils;
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.net.ssl.SSLContext;
import javax.net.ssl.TrustManager;
import javax.net.ssl.X509TrustManager;
import java.io.File;
import java.io.IOException;
import java.nio.file.Files;
import java.security.*;
```

```

import java.security.cert.X509Certificate;
import java.util.*;
import java.util.stream.Collectors;

/**
 * This is a PersonAddExample.
 */
public class PersonAddExample {

    static final String PERSON_ADD_URL = "/obms/api/v1.1/acs/person";
    static final String PERSON_GROUP_TREE_URL = "/obms/api/v1.1/acs/person-group/list";
    static final String PARKING_LOT_URL = "/ipms/api/v1.1/parking-lot/summary/list?parkingLotId=
&mode=";
    static final String ENTRANCE_GROUP_URL = "/ipms/api/v1.1/entrance-group/list?parkingLotId=
{parkingLotId}&keyword=";
    static final String FACE_REPOSITORY_URL = "/brms/api/v1.1/face/application/control/repository
/page?orderType=0&orderDirection=0&keyword=&page=1&pageSize=1";
    static final String PASSAGE_RULE_URL = "/obms/api/v1.1/acp/passage/rule/exclude/page?page
=1&pageSize=20&keyword=&resourceType=4&resourceCode=";
    static final String POST = "post";
    static final String GET = "get";
    static final String RESULT = "results";
    static final String PAGE_DATA = "pageData";
    static final Integer SUCCESS_CODE = 1000;
    static final Integer RETURN_CODE_DATA_NOT_EXIST = 1007;
    //Attention: Replace it with your own IP and port.
    static final String URL_PREFIX ="https://192.168.1.1:443";
    //Attention: Tokens are updated every 20 minutes,Please use the latest token,otherwise,the in
terface will respond 7000.
    static final String TOKEN_VALUE = "8379186362.....ac859679";
    static final String SECRET_KEY = "fa5a43ee33.....4e73c6";
    static final String SECRET_VECTOR = "a9840.....0e07a51";

    public static void main(String[] args) throws Exception {

        //Before start, read relevant API first, and this example is long, be patient please.
        //Assign values to parameters.This step has 7 modules.
        //You can optionally assign values to your parameters, because some parameters can b
e empty.
        Map<String, Object> param = new HashMap<>(8);

        /**Module 1: Basic Info*/
        Map<String, Object> baseInfoParams = new HashMap<>(10);
        //Person source, 0 = People management list, 1 = Access control group, 2 = Face com
parison group, 3 = Entrance and exit vehicle group; 0 by default
        baseInfoParams.put("source", "0");
        //general 8 random number
        baseInfoParams.put("personId", randomNumber(8));
        baseInfoParams.put("firstName", "Andy");
        baseInfoParams.put("lastName", "Lau");
        //Gender: 0 = Unknown, 1 = Male, 2 = Female; 0 by default.
        baseInfoParams.put("gender", "1");
        //Person Group : We got it through Get the List of Person Groups interface.
        List<TreeNode> treeNodeList = deserializeData(sendPostOrGet(PERSON_GROUP_TREE_URL,
null, GET), TreeNode.class,RESULT,"Person Group");
        //Select your own personnel group node,we only use the first here.
        baseInfoParams.put("orgCode", treeNodeList.get(0).getOrgCode());
    }
}

```

```

baselInfoParams.put("orgCodes", Collections.singletonList(treeNodeList.get(0).getOrgCode()));
baselInfoParams.put("email", "1234@1234.com");
baselInfoParams.put("tel", "1234");
baselInfoParams.put("remark", "");
//Attention:The picture information is encoded by Base64,some characters are omitted here.
baselInfoParams.put("facePictures", Arrays.asList(imageToBase64("C:/face.jpg")));
//Data source: 1 = platform, 2 = domain, 3 = third-party
baselInfoParams.put("sourceType", "1");
//Data source id: If third-party is selected as sourceType(means sourceType = 3), the sourceId is required.
baselInfoParams.put("sourceId", "1");
//The unique id in third-party system, which is used to associate third-party system.
baselInfoParams.put("associateId", "1");
param.put("baselInfo", baselInfoParams);

/**Module 2: Additional Info*/
Map<String, Object> extensionInfoParams = new HashMap<>(8);
extensionInfoParams.put("nickName", "Andy");
extensionInfoParams.put("address", "No.1000, JX Road");
extensionInfoParams.put("idType", "0");
extensionInfoParams.put("idNo", "100101198012255310");
extensionInfoParams.put("birthday", "1980-12-25");
//Region: Reference the API.
extensionInfoParams.put("nationalityId", "51");
extensionInfoParams.put("companyName", "Dandelion Technology");
extensionInfoParams.put("position", "Engineer");
extensionInfoParams.put("department", "HR department");
param.put("extensionInfo", extensionInfoParams);

/**Module 3: Residence Info*/
Map<String, Object> residentInfoParams = new HashMap<>(2);
residentInfoParams.put("sipld", "02#1#1002");
//Video intercom householder or not, 0 = No, 1 = Yes; 0 by default.
residentInfoParams.put("houseHolder", "1");
residentInfoParams.put("vdpUser", "0");
residentInfoParams.put("inheritHouseHolder", "0");
param.put("residentInfo", residentInfoParams);

/**Module 4: Authentication Info*/
Map<String, Object> authenticationInfoParams = new HashMap<>(9);
//Combination unlocking password, AES encryption.
authenticationInfoParams.put("combinationPassword", encryptPassWordWithAES("123456"));
//Start time and end time
authenticationInfoParams.put("endTime", "1924963200");
authenticationInfoParams.put("startTime", "1609430400");
//Card list
List<Map<String, Object>> cardList = new ArrayList<>(1);
Map<String, Object> cardMap = new HashMap<>(3);
cardMap.put("cardNo", "87E307E5");
cardMap.put("duressFlag", "0");
cardList.add(cardMap);
authenticationInfoParams.put("cards", cardList);
//Palm vein list. Feature values can be obtained by access control equipment with palm vein extraction capability.
List<Map<String, Object>> palmVeinList = new ArrayList<>(2);
Map<String, Object> palmVeinMap1 = new HashMap<>(5);

```



```

palmVeinMap1.put("palmType", "1");
palmVeinMap1.put("style", "1");
palmVeinMap1.put("feature", "0/LoRPVr5DTwjj4q9QEJHvSSTuL.....vVea6rw=cr0==");
palmVeinMap1.put("picture", imageToBase64("C:/palmVein1.jpg"));
palmVeinMap1.put("version", "131072");
palmVeinList.add(palmVeinMap1);
Map<String, Object> palmVeinMap2 = new HashMap<>(5);
palmVeinMap2.put("palmType", "2");
palmVeinMap2.put("style", "1");
palmVeinMap2.put("feature", "Eo0uPWp0JbyJ54W9VZOK.....i86Lw=6rw=cr0==");
palmVeinMap2.put("picture", imageToBase64("C:/palmVein2.jpg"));
palmVeinMap2.put("version", "131072");
palmVeinList.add(palmVeinMap2);
authenticationInfoParams.put("palmVeins", palmVeinList);
//Fingerprint list, limit three.
List<Map<String, Object>> fingerPritList = new ArrayList<>(1);
Map<String, Object> fingerprintMap = new HashMap<>(3);
//Attention:The fingerprint information is encoded by Base64,some characters are omitt
d here.
fingerprintMap.put("fingerprint", "xR9pAOCgm+UQSbjE.....AAAdR4QDxsAAOva");
fingerprintMap.put("name", "fingerprint01");
//Duress fingerprint or not, 0 = No, 1 = Yes
fingerprintMap.put("duressFlag", "0");
fingerPritList.add(fingerprintMap);
authenticationInfoParams.put("fingerprints", fingerPritList);
param.put("authenticationInfo", authenticationInfoParams);

/**Module 5: Access Control Permissions*/
Map<String, Object> accessInfoParams = new HashMap<>(5);
// get passageRuleIds
List<ResourceExcludeRuleVo> passageRuleList = deserializeData(sendPostOrGet(PASSAGE_R
ULE_URL, null, GET), ResourceExcludeRuleVo.class,PAGE_DATA,"passage rule info");
if (null != passageRuleList && passageRuleList.size() > 0) {
    accessInfoParams.put("passageRuleIds", passageRuleList.stream().map(ResourceExclueR
uleVo::getId).collect(Collectors.toList()));
} else {
    accessInfoParams.put("passageRuleIds", null);
}
//Number of times visitors can unlock; 200 by default.
accessInfoParams.put("guestUseTimes", "200");
//Access control person type.
accessInfoParams.put("accessType", "0");
param.put("accessInfo", accessInfoParams);

/**Module 6: Face Comparison*/
Map<String, Object> faceComparisonInfoParams = new HashMap<>(2);
//Enable face comparison group, 0 = No, 1 = Yes.
faceComparisonInfoParams.put("enableFaceComparisonGroup", "1");
//Face comparison group ID: We got it through [Get Face Watch List in pages] interfac
e.
List<FaceRepositoryGroup> faceRepositoryGroupList = deserializeData(sendPostOrGet(FACE
_REPOSITORY_URL, null, GET), FaceRepositoryGroup.class,PAGE_DATA,"Face Repository Group");
//Select your own face repository group,we only use the first here.
if (null != faceRepositoryGroupList && faceRepositoryGroupList.size() > 0) {
    faceComparisonInfoParams.put("faceComparisonGroupIds", faceRepositoryGroupList.stre
am().map(e -> e.getRepositoryId()).collect(Collectors.toList()).subList(0, 1));
} else {

```

```

        faceComparisonInfoParams.put("faceComparisonGroupIds", null);
    }
    param.put("faceComparisonInfo", faceComparisonInfoParams);

    /**Module 7: Vehicle Info & Entrance and Exit Vehicle Group*/
    List<ParkingLot> parkingLotList = deserializeData(sendPostOrGet(PARKING_LOT_URL, null,
    GET), ParkingLot.class, RESULT, "PARKING LOT");
    String parkingLotId = "";
    if (null != parkingLotList && parkingLotList.size() > 0) {
        parkingLotId = parkingLotList.get(0).getId();
    }
    Map<String, Object> entranceInfoParams = new HashMap<>(4);
    List<Map<String, Object>> vehicles = new ArrayList<>(1);
    Map<String, Object> vehicleMap = new HashMap<>(8);
    vehicleMap.put("plateNo", "ZHE132");
    //Vehicle color: Reference the API.
    vehicleMap.put("vehicleColor", "1");
    //Vehicle entrance and exit group ID: We got it through [Get the List of Entrance and
    Exit Groups of Vehicles in Pages] interface.
    List<EntranceGroup> entranceGroupList = deserializeData(sendPostOrGet(ENTRANCE_GRO
    UP_URL.replaceAll("\\{parkingLotId\\}", parkingLotId), null, GET), EntranceGroup.class, RESULT, "Entran
    ce and Exit Group");
    List<Map<String, Object>> entranceGroups = new ArrayList<>(1);
    //Select your own entrance group, we only use the first here.
    if (null != entranceGroupList && entranceGroupList.size() > 0) {
        Map<String, Object> entranceGroup = new HashMap<>(5);
        entranceGroup.put("parkingLotId", parkingLotId);
        entranceGroup.put("entranceStartTime", "-1");
        entranceGroup.put("entranceEndTime", "-1");
        entranceGroup.put("entranceLongTerm", "1");
        entranceGroup.put("entranceGroupIds", entranceGroupList.stream().map(e -> e.getGrou
        pld()).collect(Collectors.toList()).subList(0, 1));
        entranceGroups.add(entranceGroup);
        vehicleMap.put("entranceGroups", entranceGroups);
    } else {
        vehicleMap.put("entranceGroups", null);
    }
    vehicleMap.put("remark", "");
    //Vehicle brand: Reference the API.
    vehicleMap.put("vehicleBrand", "47");
    vehicleMap.put("bindingDeviceCode", "");
    vehicleMap.put("companyTelephone", "12345678");
    vehicleMap.put("companyOwner", "Jack");
    vehicleMap.put("affiliatedCompany", "Test Company");
    vehicles.add(vehicleMap);
    entranceInfoParams.put("vehicles", vehicles);
    List<Map<String, Object>> parkingSpaceQuotas = null ;
    if (null != parkingLotList && parkingLotList.size() > 0) {
        parkingSpaceQuotas = new ArrayList<>(1);
        Map<String, Object> parkingSpaceQuota = new HashMap<>(2);
        parkingSpaceQuota.put("parkingLotId", parkingLotList.get(0).getId());
        parkingSpaceQuota.put("parkingSpaceQuota", "10");
        parkingSpaceQuotas.add(parkingSpaceQuota);
    }
    entranceInfoParams.put("parkingSpaceQuotas", parkingSpaceQuotas);
    param.put("entranceInfo", entranceInfoParams);

```

```

        //Last: Send a POST request with the param.If you get the return code of 1000, it means you succeed.
        String responseString = sendPostOrGet(PERSON_ADD_URL, param, POST);
        Integer responseCode = (Integer) JSONObject.parseObject(responseString).get("code");
        if (responseCode.compareTo(SUCCESS_CODE) == 0) {
            System.out.println(responseString);
        }else{
            throw new Exception(responseString);
        }
    }

    /**
     * Obtain 8 random digits
     * @param count
     * @return
     */
    public static String randomNumber(int count) {
        Random random = new Random();
        StringBuilder val = new StringBuilder();
        for(int i = 0; i < count; ++i) {
            val.append(random.nextInt(10));
        }
        return val.toString();
    }

    /**
     * Send post or get request method with HTTP.
     * @param url url
     * @param params params
     * @param requestMode POST or GET
     * @return reply
     */
    static String sendPostOrGet(String url, Map<String, Object> params, String requestMode) throws IOException {
        String realUrl = URL_PREFIX + url;
        if (!requestMode.equals(POST) && !requestMode.equals(GET)) {
            return null;
        }
        // Create a trust manager that does not validate certificate chains
        TrustManager[] trustAllCerts = new TrustManager[]{
            new X509TrustManager() {
                @Override
                public X509Certificate[] getAcceptedIssuers() {
                    return new X509Certificate[0];
                }
                @Override
                public void checkClientTrusted(X509Certificate[] certs, String authType) {
                }
                @Override
                public void checkServerTrusted(X509Certificate[] certs, String authType) {
                }
            }
        };
        // Install the all-trusting trust manager
        try {
            SSLContext sslContext = SSLContext.getInstance("SSL");
            sslContext.init(null, trustAllCerts, new SecureRandom());

```

```

        // Create an ssl socket factory with our all-trusting manager
        SSLConnectionSocketFactory sslSocketFactory = new SSLConnectionSocketFactory(sslContext, (hostname, session) -> true);

        // Create a custom HttpClient with the SSL socket factory
        CloseableHttpClient httpClient = HttpClients.custom()
            .setSSLSocketFactory(sslSocketFactory)
            .build();
        CloseableHttpResponse httpResponse = null;
        if (requestMode.equals(POST)) {
            HttpEntityEnclosingRequestBase httpRequest = new HttpPost(realUrl);
            StringEntity entity = new StringEntity(JSON.toJSONString(params), "UTF-8");
            httpRequest.setEntity(entity);
            httpRequest.setHeader("Content-Type", "application/json;charset=UTF-8");
            httpRequest.setHeader("X-Subject-Token", TOKEN_VALUE);
            httpResponse = httpClient.execute(httpRequest);
        } else if (requestMode.equals(GET)) {
           HttpGet httpGet = new HttpGet(realUrl);
            httpGet.setHeader("X-Subject-Token", TOKEN_VALUE);
            httpGet.setHeader("Content-Type", "application/json;charset=UTF-8");
            httpResponse = httpClient.execute(httpGet);
        }
        HttpEntity responseEntity = httpResponse.getEntity();
        //In order to avoid messy code,encode the response data in UTF-8.
        String reply = EntityUtils.toString(responseEntity, "UTF-8");
        if (!reply.contains("1000")) {
            System.out.println("Request \"" + url + "\" failed, response: " + reply);
        }
        //Release resources finally.
        if (httpClient != null) {
            httpClient.close();
        }
        if (httpResponse != null) {
            httpResponse.close();
        }
        return reply;
    } catch (NoSuchAlgorithmException | KeyManagementException e) {
        throw new RuntimeException(e);
    }
}

/**
 * AES encryption
 * @param password
 * @return
 */
static String encryptPassWordWithAES(String password){
    //This three params you can obtain from the LoginExample or your client.
    try {
        return encryptWithAES7(password,SECRET_KEY,SECRET_VECTOR);
    } catch (Exception e) {
        e.printStackTrace();
    }
    return null;
}

/**

```

```

    * AES encryption mode is CBC, and the filling mode is pkcs7padding
    * @param text
    * @param aesKey
    * @param aesVector
    * @return
    * @throws Exception
    */
    static String encryptWithAES7(String text, String aesKey, String aesVector) throws Exception {
        SecretKey keySpec = new SecretKeySpec(aesKey.getBytes("UTF-8"), "AES");
        //If your program run with an exception : "Cannot find any provider supporting AES/CBC
        /PKCS7Padding",you can replace "PKCS7Padding" with "PKCS5Padding".
        Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
        IvParameterSpec iv = new IvParameterSpec(aesVector.getBytes("UTF-8"));
        cipher.init(Cipher.ENCRYPT_MODE, keySpec, iv);
        byte[] encrypted = cipher.doFinal(text.getBytes("UTF-8"));
        return parseByte2HexStr(encrypted);
    }

    /**
     * Image to base64
     * @param imagePath
     * @return
     */
    public static String imageToBase64(String imagePath) {
        String base64Image = "";
        try {
            byte[] imageBytes = Files.readAllBytes(new File(imagePath).toPath());
            base64Image = org.apache.commons.codec.binary.Base64.encodeBase64String(imageBy
tes);
        } catch (IOException e) {
            e.printStackTrace();
        }
        return base64Image;
    }

    /**
     * Byte array to 16 bit string
     * @param buf
     * @return
     */
    static String parseByte2HexStr(byte buf[]) {
        StringBuffer sb = new StringBuffer();
        for (int i = 0; i < buf.length; i++) {
            String hex = Integer.toHexString(buf[i] & 0xFF);
            if (hex.length() == 1) {
                hex = '0' + hex;
            }
            sb.append(hex.toLowerCase());
        }
        return sb.toString();
    }

    /**
     * Deserialize data
     * @param responseString
     * @param clazz
     * @param fieldName

```

```

    * @param businessName
    * @param <T>
    * @return
    * @throws Exception
    */
    static <T> List<T> deserializeData(String responseString, Class<T> clazz,String fieldName,String
    businessName) throws Exception {
        JSONObject jsonObject=JSONObject.parseObject(responseString);
        Integer code = (Integer) jsonObject.get("code");
        if(code.compareTo(SUCCESS_CODE) == 0){
            Object dataObject = jsonObject.getJSONObject("data").get(fieldName);
            if (dataObject != null) {
                List<T> resultList=JSONObject.parseArray(dataObject.toString(), clazz);
                if (resultList == null || resultList.size() == 0) {
                    throw new Exception("If you want to add a " + businessName + ", please
                    add it in client first.");
                }
                return JSONObject.parseArray(dataObject.toString(), clazz);
            }
        } else if (code.compareTo(RETURN_CODE_DATA_NOT_EXIST) == 0){
            throw new Exception("If you want to add a " + businessName + ", please add it in
            client first.");
        } else {
            throw new Exception(responseString);
        }
        return null;
    }

    /**
     * Tree node
     */
    static class TreeNode {
        private String orgCode;
        private String parentOrgCode;
        private String orgName;
        private String remark;
        private List<TreeNode> children;
        private String childNum;
        public String getOrgCode() {
            return orgCode;
        }
        public void setOrgCode(String orgCode) {
            this.orgCode = orgCode;
        }
        public String getParentOrgCode() {
            return parentOrgCode;
        }
        public void setParentOrgCode(String parentOrgCode) {
            this.parentOrgCode = parentOrgCode;
        }
        public String getOrgName() {
            return orgName;
        }
        public void setOrgName(String orgName) {
            this.orgName = orgName;
        }
        public String getRemark() {

```

```
        return remark;
    }
    public void setRemark(String remark) {
        this.remark = remark;
    }
    public List<TreeNode> getChildren() {
        return children;
    }
    public void setChildren(List<TreeNode> children) {
        this.children = children;
    }
    public String getChildNum() {
        return childNum;
    }
    public void setChildNum(String childNum) {
        this.childNum = childNum;
    }
}

/**
 * Resource exclude rule
 */
static class ResourceExcludeRuleVo{
    private String id;
    private String ruleName;
    private String ruleType;
    private String pointType;
    private String timeTemplateId;
    private String timeTemplateName;
    private String holidayPlanId;
    private String holidayPlanName;
    private int personCount;
    private int pointCount;
    public String getRuleType() {
        return ruleType;
    }
    public void setRuleType(String ruleType) {
        this.ruleType = ruleType;
    }
    public int getPersonCount() {
        return personCount;
    }
    public void setPersonCount(int personCount) {
        this.personCount = personCount;
    }
    public int getPointCount() {
        return pointCount;
    }
    public void setPointCount(int pointCount) {
        this.pointCount = pointCount;
    }
    public String getId() {
        return id;
    }
    public void setId(String id) {
        this.id = id;
    }
}
```

```
public String getRuleName() {
    return ruleName;
}
public void setRuleName(String ruleName) {
    this.ruleName = ruleName;
}
public String getPointType() {
    return pointType;
}
public void setPointType(String pointType) {
    this.pointType = pointType;
}
public String getTimeTemplateId() {
    return timeTemplateId;
}
public void setTimeTemplateId(String timeTemplateId) {
    this.timeTemplateId = timeTemplateId;
}
public String getTimeTemplateName() {
    return timeTemplateName;
}
public void setTimeTemplateName(String timeTemplateName) {
    this.timeTemplateName = timeTemplateName;
}
public String getHolidayPlanId() {
    return holidayPlanId;
}
public void setHolidayPlanId(String holidayPlanId) {
    this.holidayPlanId = holidayPlanId;
}
public String getHolidayPlanName() {
    return holidayPlanName;
}
public void setHolidayPlanName(String holidayPlanName) {
    this.holidayPlanName = holidayPlanName;
}
}

/**
 * Parking lot
 */
static class ParkingLot{
    private String id;
    private String parkingLotName;
    public String getId() {
        return id;
    }
    public void setId(String id) {
        this.id = id;
    }
    public String getParkingLotName() {
        return parkingLotName;
    }
    public void setParkingLotName(String parkingLotName) {
        this.parkingLotName = parkingLotName;
    }
}
```



```
/**
 * Entrance group
 */
static class EntranceGroup {
    private String groupId;
    private String groupName;
    private String groupColor;
    private String remark;
    private String defaultGroup;
    private String vehicleCount;
    public String getGroupId() {
        return groupId;
    }
    public void setGroupId(String groupId) {
        this.groupId = groupId;
    }
    public String getGroupName() {
        return groupName;
    }
    public void setGroupName(String groupName) {
        this.groupName = groupName;
    }
    public String getGroupColor() {
        return groupColor;
    }
    public void setGroupColor(String groupColor) {
        this.groupColor = groupColor;
    }
    public String getRemark() {
        return remark;
    }
    public void setRemark(String remark) {
        this.remark = remark;
    }
    public String getDefaultGroup() {
        return defaultGroup;
    }
    public void setDefaultGroup(String defaultGroup) {
        this.defaultGroup = defaultGroup;
    }
    public String getVehicleCount() {
        return vehicleCount;
    }
    public void setVehicleCount(String vehicleCount) {
        this.vehicleCount = vehicleCount;
    }
}

/**
 * Face repository group
 */
static class FaceRepositoryGroup{
    private String repositoryId;
    private String repositoryName;
    private String color;
    private String colorName;
```

```
private String memo;
private String createTime;
private String updateTime;
private String facePersonCount;
private String isFailed;
private String thumbnail;
public String getRepositoryId() {
    return repositoryId;
}
public void setRepositoryId(String repositoryId) {
    this.repositoryId = repositoryId;
}
public String getRepositoryName() {
    return repositoryName;
}
public void setRepositoryName(String repositoryName) {
    this.repositoryName = repositoryName;
}
public String getColor() {
    return color;
}
public void setColor(String color) {
    this.color = color;
}
public String getColorName() {
    return colorName;
}
public void setColorName(String colorName) {
    this.colorName = colorName;
}
public String getMemo() {
    return memo;
}
public void setMemo(String memo) {
    this.memo = memo;
}
public String getCreateTime() {
    return createTime;
}
public void setCreateTime(String createTime) {
    this.createTime = createTime;
}
public String getUpdateTime() {
    return updateTime;
}
public void setUpdateTime(String updateTime) {
    this.updateTime = updateTime;
}
public String getFacePersonCount() {
    return facePersonCount;
}
public void setFacePersonCount(String facePersonCount) {
    this.facePersonCount = facePersonCount;
}
public String getIsFailed() {
    return isFailed;
}
}
```

```
public void setIsFailed(String isFailed) {
    this.isFailed = isFailed;
}
public String getThumbnail() {
    return thumbnail;
}
public void setThumbnail(String thumbnail) {
    this.thumbnail = thumbnail;
}
}
```

6.3.1.3 Example of Login Authentication with JS

Description

Import the plug-ins of axios and crypto-js/md5.

Install NodeJS first and install the plug-ins with npm command.

```
npm install axios crypto-js/md5
```

Or import the plug-ins with the <script> tag.

```
<script src="https://unpkg.com/axios/dist/axios.min.js"></script>
<script type="text/javascript" src="path-to/bower_components/crypto-js/crypto-js.js"></script>
```

Only partial functions of the plug-ins are mentioned in the text. For more functions and details, go to the websites:

[axios.js - official document](#)

[npm/crypto-js](#)

[npm/uuid](#)

Example Code

Run the codes in ES6 environment. Use Chrome 76 and later versions or use front-end packaging tools such as webpack to assist running.

```
const axios = require('axios');
const CryptoJS = require('crypto-js');
const { v4: uuidv4 } = require('uuid');
const crypto = require('crypto');
const https = require('https');
const AUTH_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/authorize";
const KEEP_ALIVE_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/keepalive";
const UPDATE_TOKEN_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/updateToken";
const SUCCESS_CODE = 1000;
```

```

let SIGNATURE_MD5_TEMP4 = null;
let TOKEN_VALUE = null;
let HEART_COUNT = 0;

async function main() {
  try {
    //Step1: Try to login first time.
    const firstLoginParams = {
      userName: "system",
      ipAddress: "",
      clientType: "WINPC_V2"
    };
    const firstResponse = await sendPostOrPut(AUTH_URL, firstLoginParams, 'POST');
    const firstLoginResponse = firstResponse.data;

    if (!firstLoginResponse || !firstLoginResponse.realm) {
      console.error("First login failed or response is invalid:", firstLoginResponse);
      return;
    }

    //Step2: Try to login second time.
    const userName = "system";
    const passWord = "admin1234567";
    const realm = firstLoginResponse.realm;
    const randomKey = firstLoginResponse.randomKey;
    const platformPublicKeyWithBase64 = firstLoginResponse.publickey;
    const signature = generateSignature(userName, passWord, realm, randomKey);

    //Generate the AES secret key and AES secret vector, encrypted by RSA public key.
    const aesSecretKey = uuidv4().replace(/-/g, "");
    const aesSecretVector = uuidv4().replace(/-/g, "").substring(16);
    const encryptedAesSecretKeyWithBase64 = encryptByPublicKey(aesSecretKey, platformPublicKeyWithBase64);
    const encryptedAesSecretVectorWithBase64 = encryptByPublicKey(aesSecretVector, platformPublicKeyWithBase64);

    const secondLoginParams = {
      mac: "2C:F0:5D:4D:5E:DB",
      signature,
      userName,
      randomKey,
      publicKey: "",
      encryptType: "MD5",
      ipAddress: "",
      clientType: "WINPC_V2",
      userType: "0",
      secretKey: encryptedAesSecretKeyWithBase64,
      secretVector: encryptedAesSecretVectorWithBase64
    };
    const secondResponse = await sendPostOrPut(AUTH_URL, secondLoginParams, 'POST');
    const secondResponseData = secondResponse.data;
    TOKEN_VALUE = secondResponseData.token;
    console.log(`Your AES key ${aesSecretKey} and vector ${aesSecretVector}, which you can use to decrypt encrypted data returned by other interfaces`);
    console.log(`The token is ${TOKEN_VALUE}`);
    console.log(`The credential is ${secondResponseData.credential}`);
    console.log(`The userId is ${secondResponseData.userId}`);
  }
}

```

```

//Step3: Keep alive and Update the token.
while (true) {
    //Send a heart per 20 seconds.
    await new Promise(resolve => setTimeout(resolve, 20000));
    const keepAliveParamMap = {
        token: TOKEN_VALUE
    };
    const heartResponse = await sendPostOrPut(KEEP_ALIVE_URL, keepAliveParamMap, 'PUT');
    if (SUCCESS_CODE === heartResponse.data.code) {
        console.log("Heart success.");
    }
    HEART_COUNT++;
    //Update the token per 20 minutes.
    if (HEART_COUNT % 60 === 0) {
        // Restore the heartCount to zero.
        HEART_COUNT = 0;
        const updateTokenParamMap = {};
        const updateTokenResponse = await sendPostOrPut(UPDATE_TOKEN_URL, updateTokenParamMap, 'POST');
        if (SUCCESS_CODE === updateTokenResponse.data.code) {
            const newTokenValue = updateTokenResponse.data.data.token;
            console.log('Update token success. New token: ${newTokenValue}');
            TOKEN_VALUE = newTokenValue;
        }
    }
}

//Login is over, do your own things with the token...
} catch (error) {
    console.error("Error:", error);
}
}

/**
 * Send a POST or PUT request.
 * @param url
 * @param params
 * @param requestMode
 * @returns {Promise<*>|null}
 */
async function sendPostOrPut(url, params, requestMode) {
    if (requestMode !== 'POST' && requestMode !== 'PUT') {
        return null;
    }
    try {
        const config = {
            headers: {
                'Content-Type': 'application/json;charset=UTF-8',
                'X-Subject-Token': TOKEN_VALUE
            },
            data: JSON.stringify(params),
            httpsAgent: new https.Agent({
                rejectUnauthorized: false
            })
        };
    }
}

```

```

        return await axios[requestMode.toLowerCase()](url, config.data, config);
    } catch (error) {
        if (error.status === 401) {
            return error.response;
        }

        throw new Error('HTTP request failed: ${error.message}');
    }
}

/**
 * Generate the signature.
 * @param userName
 * @param password
 * @param realm
 * @param randomKey
 * @returns {*}
 */
function generateSignature(userName, password, realm, randomKey) {
    const temp1 = CryptoJS.MD5(password).toString();
    const temp2 = CryptoJS.MD5(userName + temp1).toString();
    const temp3 = CryptoJS.MD5(temp2).toString();
    const temp4 = CryptoJS.MD5(userName + ':' + realm + ':' + temp3).toString();
    // Retain the temp4, in order to calculate a signature for update token later.
    SIGNATURE_MD5_TEMP4 = temp4;
    return CryptoJS.MD5(temp4 + ':' + randomKey).toString();
}

/**
 * Encrypt the text by public key.
 * @param text
 * @param publicKeyBase64
 * @returns {string}
 */
function encryptByPublicKey(text, publicKeyBase64) {
    try {
        // Convert public key from base64 to PEM format
        const publicKeyPem = `-----BEGIN PUBLIC KEY-----\n${publicKeyBase64}\n-----END PUBLIC KEY-----`;

        // Encrypt the data using PKCS1_v1_5 padding
        const encryptedData = crypto.publicEncrypt({
            key: publicKeyPem,
            padding: crypto.constants.RSA_PKCS1_PADDING
        }, Buffer.from(text));

        // Convert the encrypted data to base64
        const encryptedDataBase64 = encryptedData.toString('base64');

        return encryptedDataBase64;
    } catch (error) {
        console.error('Encryption failed:', error);
        throw new Error('Encryption failed: ${error.message}');
    }
}

main();

```

6.3.1.4 Example of Login Authentication with C#

Example Code

```
using System;
using System.Net.Http;
using System.Security.Cryptography;
using System.Text;
using System.Text.Json;
using Org.BouncyCastle.Crypto.Parameters;
using Org.BouncyCastle.Security;

/**
 * This is a LoginExample.
 */
namespace LoginApplication
{
    public class LoginExample
    {
        public const string AUTH_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/authorize";
        public const string KEEP_ALIVE_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/keepalive";
        public const string UPDATE_TOKEN_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/updateToken";

        public const string SUCCESS_CODE = "1000";
        private static string? SIGNATURE_MD5_TEMP4 = null;
        private static string? TOKEN_VALUE;
        public const string TOKEN = "token";
        public const string POST = "post";
        public const string PUT = "put";
        private static int HEART_COUNT = 0;

        /**
         * Main method
         * @param args args
         */
        static void Main(string[] args)
        {
            //Step1: Try to login first time.
            var firstLoginParams = new Dictionary<string, object>
            {
                { "userName", "system" },
                { "ipAddress", "" },
                { "clientType", "WINPC_V2" }
            };
            string? firstResponseString = SendPostOrPut(AUTH_URL, firstLoginParams, POST);
            Dictionary<string, string>? firstLoginResponse = JsonSerializer.Deserialize<Dictionary<string, string>>(firstResponseString!);
            //Step2: Try to login second time.
            String userName = "system";
            String passWord = "admin1234567";
            String? realm = firstLoginResponse?["realm"];
            String? randomKey = firstLoginResponse?["randomKey"];
            String? platformPublicKeyWithBase64 = firstLoginResponse?["publickey"];
```

```

String signature = GenerateSignature(userName, passWord, realm, randomKey);

//Generate the AES secret key and AES secret vector, encrypted by RSA public key.
string aesSecretKey = Guid.NewGuid().ToString("N");
string aesSecretVector = Guid.NewGuid().ToString("N").Substring(16);
string encryptedAesSecretKeyWithBase64 = Convert.ToBase64String(EncryptByPublicKe
y(aesSecretKey, Convert.FromBase64String(platformPublicKeyWithBase64)));
string encryptedAesSecretVectorWithBase64 = Convert.ToBase64String(EncryptByPublic
Key(aesSecretVector, Convert.FromBase64String(platformPublicKeyWithBase64)));

Dictionary<string, object> secondLoginParams = new()
{
    { "mac", "2C:F0:5D:4D:5E:DB" },
    { "signature", signature },
    { "userName", userName },
    { "randomKey", randomKey },
    { "publicKey", "" },
    { "encryptType", "MD5" },
    { "ipAddress", "" },
    { "clientType", "WINPC_V2" },
    { "userType", "0" },
    { "secretKey", encrypt.....WithBase64 },
    { "secretVector", encrypted.....hBase64 }
};

String? secondResponseString = SendPostOrPut(AUTH_URL, secondLoginParams, POS
T);

Dictionary<string, object>? secondResponse = JsonSerializer.Deserialize<Dictionary<st
ring, object>>(secondResponseString!);

if (secondResponse!.TryGetValue(TOKEN, out object? token))
{
    TOKEN_VALUE = Convert.ToString(token);
}

Console.WriteLine($"Your AES key {{ {aesSecretKey} }} and vector {{ {aesSecretVector}
}}, which you can use to decrypt encrypted data returned by other interfaces");
Console.WriteLine($"The token is {{ {TOKEN_VALUE} }}");
Console.WriteLine($"The credential is {{ {secondResponse["credential"]} }}");
Console.WriteLine($"The userId is {{ {secondResponse["userId"]} }}");

//Step3: Keep alive and Update the token.
while (true)
{
    //Send a heart per 20 seconds.
    Thread.Sleep(20000);
    Dictionary<String, Object> keepAliveParamMap = new()
    {
        {"token", TOKEN_VALUE!}
    };
    Dictionary<string, object>? heartResponse = JsonSerializer.Deserialize<Dictionary
<string, object>>(SendPostOrPut(KEEP_ALIVE_URL, keepAliveParamMap, PUT)!);
    if (SUCCESS_CODE.Equals(Convert.ToString(heartResponse?["code"])))
    {
        Console.WriteLine("Heart success.");
    }
    HEART_COUNT++;
}

```



```

        //Update the token per 20 minutes.
        if (HEART_COUNT % 60 == 0)
        {
            //Restore the heartCount to zero.
            HEART_COUNT = 0;
            Dictionary<String, Object> updateTokenParamMap = new({});
            Dictionary<string, object>? udapteTokenResponse = JsonSerializer.Deserialize
            <Dictionary<string, object>>(SendPostOrPut(UPDATE_TOKEN_URL, updateTokenParamMap, POST!));
            if (SUCCESS_CODE.Equals(Convert.ToString(udapteTokenResponse?["code"])))
            {
                object data = udapteTokenResponse!["data"];
                Dictionary<string, object>? newToken = JsonSerializer.Deserialize<Dictio
                nary<string, object>>(JsonSerializer.Serialize(data));
                String newTokenValue = Convert.ToString(newToken?[TOKEN]);
                Console.WriteLine("Update token success. New token: " + newTokenVal
                ue);
                TOKEN_VALUE = newTokenValue;
            }
        }

        //Login is over, do your own things with the token...
    }

    /**
     * Send post or put request method with HTTP.
     * @param url
     * @param params
     * @return reply
     */
    public static string? SendPostOrPut(string url, Dictionary<string, object> param, string re
    questMode)
    {
        if (!requestMode.Equals(POST) && !requestMode.Equals(PUT))
        {
            return null;
        }

        HttpRequestMessage httpRequestMessage = new(HttpMethod.Post, url);

        if (requestMode.Equals(PUT))
        {
            httpRequestMessage = new HttpRequestMessage(HttpMethod.Put, url);
        }
        StringContent stringContent = new(JsonSerializer.Serialize(param), Encoding.UTF8, "ap
        plication/json");

        httpRequestMessage.Content = stringContent;
        httpRequestMessage.Headers.Add("X-Subject-Token", TOKEN_VALUE);

        return SendAsync(httpRequestMessage).Result;
    }

    /**
     * Sending http requests asynchronously.
     * @param httpRequestMessage
     * @return response

```

```

    */
    public static async Task<string> SendAsync(HttpRequestMessage httpRequestMessage)
    {
        // Create a custom HttpClientHandler and ignore certificate validation
        HttpClientHandler handler = new HttpClientHandler
        {
            ServerCertificateCustomValidationCallback = (message, cert, chain, sslPolicyErrors)
=> true
        };
        HttpClient httpClient = new HttpClient(handler);
        HttpResponseMessage response = await httpClient.SendAsync(httpRequestMessage);

        //In order to avoid messy code,encode the response data in UTF-8.
        string reply = await response.Content.ReadAsStringAsync();
        return reply;
    }

    /**
     * Generate signature with MD5-transformation five times.
     * @param userName
     * @param password
     * @param realm
     * @param randomKey
     * @return signature
     */
    public static string GenerateSignature(string userName, string password, string? realm, string? randomKey)
    {
        string temp1 = Md5Hex(password);
        string temp2 = Md5Hex(userName + temp1);
        string temp3 = Md5Hex(temp2);
        string temp4 = Md5Hex(userName + ":" + realm + ":" + temp3);
        //Retain the temp4,in order to calculate a signature for update token later.
        SIGNATURE_MD5_TEMP4 = temp4;
        String signature = Md5Hex(temp4 + ":" + randomKey);
        return signature;
    }

    /**
     * First MD5 encryption, then convert to hexadecimal string.
     * @param data
     * @return
     */
    public static string Md5Hex(string data)
    {
        byte[] bytes = Encoding.UTF8.GetBytes(data);
        byte[] result = MD5.HashData(bytes);
        var stringBuilder = new StringBuilder();
        for (int i = 0; i < result.Length; i++)
        {
            stringBuilder.Append(result[i].ToString("x2"));
        }
        return stringBuilder.ToString();
    }

    /**
     * RSA public key encryption.

```

```
* @param text
* @param publicKeyBytes
* @return
*/
public static byte[] EncryptByPublicKey(string text, byte[] publicKeyBytes)
{
    var publicKey = (RsaKeyParameters)PublicKeyFactory.CreateKey(publicKeyBytes);
    var cipher = CipherUtilities.GetCipher("RSA/NONE/PKCS1Padding");
    cipher.Init(true, publicKey);

    byte[] dataBytes = Encoding.UTF8.GetBytes(text);
    int rsaBytesLenEncrypt = 245;
    List<byte[]> encryptedParts = new List<byte[]>();

    for (int i = 0; i < dataBytes.Length; i += rsaBytesLenEncrypt)
    {
        int to = Math.Min(i + rsaBytesLenEncrypt, dataBytes.Length);
        byte[] temp = cipher.DoFinal(dataBytes, i, to - i);
        encryptedParts.Add(temp);
    }

    return encryptedParts.SelectMany(bytes => bytes).ToArray();
}
}
```

6.3.1.5 Example of Login Authentication with Python

Example Code

```
import requests
import json
import hashlib
import base64
import uuid
import time
from Crypto.Cipher import AES, PKCS1_v1_5
from Crypto.PublicKey import RSA
from Crypto.Util.Padding import pad, unpad
import urllib3

AUTH_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/authorize"
KEEP_ALIVE_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/keepalive"
UPDATE_TOKEN_URL = "https://192.168.1.1:443/brms/api/v1.0/accounts/updateToken"
SUCCESS_CODE = 1000
TOKEN = "token"
POST = "post"
PUT = "put"
HEART_COUNT = 0
SIGNATURE_MD5_TEMP4 = None
TOKEN_VALUE = None

"""
Send a POST or PUT request.
```

This function sends an HTTP request to the specified URL with the given parameters and request mode (POST or PUT), including necessary headers, and returns the response.

Parameters:

- url (str): The target URL for the request.
- params (dict): The parameters for the request, which will be sent as JSON.
- request_mode (str): The request mode, supporting "POST" or "PUT".

Returns:

- str: The response text from the request.

```
"""
def send_post_or_put(url, params, request_mode):
    if params is None:
        params = {}
    # Disable SSL verification and suppress InsecureRequestWarning
    urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)
    headers = {
        "Content-Type": "application/json;charset=UTF-8",
        "X-Subject-Token": TOKEN_VALUE
    }
    response = requests.request(request_mode, url, headers=headers, data=json.dumps(params), verify=False)
    return response.text
"""
```

Generate a signature for user authentication.

This function creates an MD5 signature based on the provided username, password, realm, and random key.

The process involves multiple hashing steps to ensure the security and integrity of the generated signature.

Parameters:

- username (str): The username of the user.
- password (str): The password of the user.
- realm (str): The authentication realm.
- random_key (str): A random key used in the signature generation.

Returns:

- str: The generated MD5 signature.

```
"""
def generate_signature(username, password, realm, random_key):
    temp1 = hashlib.md5(password.encode()).hexdigest()
    temp2 = hashlib.md5((username + temp1).encode()).hexdigest()
    temp3 = hashlib.md5(temp2.encode()).hexdigest()
    temp4 = hashlib.md5((username + ":" + realm + ":" + temp3).encode()).hexdigest()
    global SIGNATURE_MD5_TEMP4
    SIGNATURE_MD5_TEMP4 = temp4
    signature = hashlib.md5((temp4 + ":" + random_key).encode()).hexdigest()
    return signature
"""
```

Encrypts the given text using the provided public key.

Parameters:

- text (str): The text to be encrypted.

- public_key_bytes (bytes): The public key in bytes format.

Returns:

- bytes: The base64 encoded encrypted data.

"""

```
def encrypt_by_public_key(text, public_key_bytes):
    public_key = RSA.import_key(public_key_bytes)
    cipher = PKCS1_v1_5.new(public_key)
    encrypted_data = cipher.encrypt(text.encode())
    return base64.b64encode(encrypted_data)
```

"""

Concatenates two byte sequences.

This function takes two byte sequences as input, checks if they are None, and if so, assigns them an empty byte sequence.

Then it concatenates these two byte sequences and returns the result.

Parameters:

- bytes1: The first byte sequence, can be None.

- bytes2: The second byte sequence, can be None.

Returns:

- The concatenated byte sequence.

"""

```
def sum_bytes(bytes1, bytes2):
    if bytes1 is None:
        bytes1 = b''
    if bytes2 is None:
        bytes2 = b''
    return bytes1 + bytes2
```

def main():

Step1: Try to login first time.

```
first_login_params = {
    "userName": "system",
    "ipAddress": "",
    "clientType": "WINPC_V2"
}
```

```
first_response_string = send_post_or_put(AUTH_URL, first_login_params, POST)
```

```
first_login_response = json.loads(first_response_string)
```

Step2: Try to login second time.

```
username = "system"
```

```
password = "admin1234567"
```

```
realm = first_login_response["realm"]
```

```
random_key = first_login_response["randomKey"]
```

```
platform_public_key_with_base64 = first_login_response["publickey"]
```

```
signature = generate_signature(username, password, realm, random_key)
```

Generate the AES secret key and AES secret vector, encrypted by RSA public key.

```
aes_secret_key = str(uuid.uuid4()).replace("-", "")
```

```
aes_secret_vector = str(uuid.uuid4()).replace("-", "")[-16:]
```

```
encrypted_aes_secret_key_with_base64 = encrypt_by_public_key(aes_secret_key, base64.b64decode(platform_public_key_with_base64))
```

```
encrypted_aes_secret_vector_with_base64 = encrypt_by_public_key(aes_secret_vector, base64.b64decode(platform_public_key_with_base64))
```

```

second_login_params = {
    "mac": "2C:F0:5D:4D:5E:DB",
    "signature": signature,
    "userName": username,
    "randomKey": random_key,
    "publicKey": "",
    "encryptType": "MD5",
    "ipAddress": "",
    "clientType": "WINPC_V2",
    "userType": "0",
    "secretKey": encrypted_aes_secret_key_with_base64.decode(),
    "secretVector": encrypted_aes_secret_vector_with_base64.decode()
}
second_response_string = send_post_or_put(AUTH_URL, second_login_params, POST)
second_response = json.loads(second_response_string)
global TOKEN_VALUE
TOKEN_VALUE = second_response[TOKEN]
print(f"Your AES key {aes_secret_key} and vector {aes_secret_vector}, which you can use to d
encrypt encrypted data returned by other interfaces")
print(f"The token is {TOKEN_VALUE}")
print(f"The credential is {second_response['credential']}")
print(f"The userId is {second_response['userId']}")

# Step3: Keep alive and Update the token.
while True:
    # Send a heart per 20 seconds.
    time.sleep(20)
    keep_alive_param_map = {
        "token": TOKEN_VALUE
    }
    heart_response = json.loads(send_post_or_put(KEEP_ALIVE_URL, keep_alive_param_map, PU
T))
    if heart_response["code"] == SUCCESS_CODE:
        print("Heart success.")
    global HEART_COUNT
    HEART_COUNT += 1
    # Update the token per 20 minutes.
    if HEART_COUNT % 60 == 0:
        # Restore the heartCount to zero.
        HEART_COUNT = 0
        update_token_param_map = {}
        update_token_response = json.loads(send_post_or_put(UPDATE_TOKEN_URL, update_to
ken_param_map, POST))
        if update_token_response["code"] == SUCCESS_CODE:
            new_token_value = update_token_response["data"][TOKEN]
            print(f"Update token success. New token: {new_token_value}")
            TOKEN_VALUE = new_token_value

    # Login is over, do your own things with the token...

if __name__ == "__main__":
    main()

```

6.3.2 MQ Connection Examples

Get MQ Configuration

- MQ password is AES-encrypted. First, complete login authentication. The second login interface returns AES-decrypted secret key and vector. The secret key and vector have been encrypted with the RSA public key of the requester, and shall be decrypted by RSA private key of the requester.
- See [3.2.1 Get MQ Configuration] to get server MQ configuration data.
- Get the AES secret key and vector of the decrypted plain text. Use AES decryption method to decrypt the MQ password into plain text.
- The client connects to the server through SSL.
- When there may be different dependency packages listening to MQ, the naming format of the topic will be different, which is not our common "." sign separation. Take the alarm topic mq.alarm.msg.topic.{userId} as an example:

Environment: Browser
Dependent package: mqtt
Protocol: ws/wss
Topic format: mq/alarm/msg/topic/{userId}

Environment: Node
Dependent package: mqtt
Protocol: mqtt/mqtts
Topic format: mq/alarm/msg/topic/{userId}

Environment: Node
Dependent package: stompit
Protocol: stomp
Topic format: /topic/mq.alarm.msg.topic.{userId}

6.3.2.1 Example of MQ Connection with Java

Introduce Maven Dependency

```
<dependencies>
  <dependency>
    <groupId>org.apache.httpcomponents</groupId>
    <artifactId>httpclient</artifactId>
    <version>4.3.6</version>
  </dependency>
  <dependency>
    <groupId>com.alibaba</groupId>
    <artifactId>fastjson</artifactId>
    <version>1.2.58</version>
  </dependency>
  <dependency>
    <groupId>org.apache.activemq</groupId>
    <artifactId>activemq-client</artifactId>
    <version>5.15.12</version>
  </dependency>
  <dependency>
    <groupId>org.bouncycastle</groupId>
    <artifactId>bcprov-jdk15on</artifactId>
    <version>1.61</version>
  </dependency>
</dependencies>
```

Example Code

```
import com.alibaba.fastjson.JSON;
import com.alibaba.fastjson.JSONObject;
import org.apache.activemq.ActiveMQSslConnectionFactory;
import org.apache.activemq.command.ActiveMQQueue;
import org.apache.activemq.command.ActiveMQTopic;
import org.apache.http.HttpEntity;
import org.apache.http.client.methods.CloseableHttpResponse;
import org.apache.http.client.methods.HttpEntityEnclosingRequestBase;
import org.apache.http.client.methods.HttpPost;
import org.apache.http.client.methods.HttpPut;
import org.apache.http.conn.ssl.SSLConnectionSocketFactory;
import org.apache.http.entity.StringEntity;
import org.apache.http.impl.client.CloseableHttpClient;
import org.apache.http.impl.client.HttpClients;
import org.apache.http.util.EntityUtils;
import org.bouncycastle.jce.provider.BouncyCastleProvider;

import javax.crypto.Cipher;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.jms.*;
import javax.net.ssl.*;
import java.net.Socket;
import java.nio.charset.StandardCharsets;
import java.security.KeyManagementException;
import java.security.NoSuchAlgorithmException;
import java.security.SecureRandom;
import java.security.Security;
import java.security.cert.X509Certificate;
import java.util.HashMap;
import java.util.Map;

/**
 * This is a MqConnectionExample.
 */
public class MqConnectionExample {

    static final String MQ_URL_PREFIX = "ssl://%s";

    static final String MQ_CONFIG_URL = "https://192.168.1.1:443/brms/api/v1.0/BRM/Config/GetMqConfig";

    static String TOKEN_VALUE = null;

    static final String TOPIC = "topic";

    static final String QUEUE = "queue";

    public static void main(String[] args) throws Exception {

        //Step1: Call the "Get the MQ configuration" interface, in order to obtain the MQ connection info.
        //You can obtain the value of token from "LoginExample".
        TOKEN_VALUE = "F2094B224A774771b65494be5e3a871d";
    }
}
```



```

        String userId = "1";
        String mqConfigResponseString = sendPost(MQ_CONFIG_URL, null);
        JSONObject mqConfigResponse = JSONObject.parseObject(mqConfigResponseString).getJS
ONObject("data");
        String userName = mqConfigResponse.getString("userName");
        String userPasswordWithAes = mqConfigResponse.getString("password");
        if (null == userPasswordWithAes) {
            System.out.println("Password encryption exception, please check the second login pa
rameters 'secretKey' and 'secretVector'.");
        }
        String mqUrlAddr = mqConfigResponse.getString("addr");

        //Step1: Decrypt the "password".
        //Secret key and secret vector are generated by the caller himself, as a parameter of th
e second login interface.
        String secretKey = "ddb623.....18d0bf6bd59";
        String secretVector = "8780ae.....450046b";
        String userPassword = decryptByAesCbc(userPasswordWithAes, secretKey, secretVector);

        //Step3: Initialize the MQ connection information, and start the message listening.
        //Replace it with a queueName or a topicName which you would listened.
        String messageName = "mq.event.msg.topic." + userId;
        listenMqMessage(String.format(MQ_URL_PRFIX, mqUrlAddr), userName, userPassword, mes
sageName, TOPIC);

        //The connection is over, do your own things here...
    }

    /**
     * Listen a queue or topic message.
     * @param mqUrl MQ URL string
     * @param userName MQ username
     * @param passWord MQ password
     * @param messageName Queue name or a topic name
     * @param messageType Message type, fixed string 'topic' or 'queue'
     * @throws Exception
     */
    static void listenMqMessage(String mqUrl, String userName, String passWord, String message
Name, String messageType) throws Exception {
        if (!messageType.equals(TOPIC) && !messageType.equals(Queue)) {
            return ;
        }
        CustomerMqFactory factory = new CustomerMqFactory();
        factory.setUserName(userName);
        factory.setTrustStore("");
        factory.setPassword(passWord);
        factory.setBrokerURL(mqUrl);
        Connection connection = factory.createConnection();
        connection.start();
        Session session = connection.createSession(false, Session.AUTO_ACKNOWLEDGE);
        MessageConsumer consumer;
        if(messageType.equals(TOPIC)){
            consumer = session.createConsumer(new ActiveMQTopic(messageName));
        }else{
            consumer = session.createConsumer(new ActiveMQQueue(messageName));
        }
        System.out.println("Start listening...");
    }

```

```

        consumer.setMessageListener((message) -> {
            TextMessage textMessage = (TextMessage) message;
            try {
                String text = textMessage.getText();
                //Print the listening message.
                System.out.println(text);
            } catch (JMSException e) {
                e.printStackTrace();
            }
        });
    }

    /**
     * Send post request method with HTTP.
     * @param url URL string
     * @param params interface params
     * @return response data
     */
    static String sendPost(String url, Map<String, Object> params) throws Exception {
        if (params == null) {
            params = new HashMap<>();
        }
        //Create a trust manager that does not validate certificate chains
        TrustManager[] trustAllCerts = new TrustManager[] {
            new X509TrustManager() {
                @Override
                public X509Certificate[] getAcceptedIssuers() {
                    return new X509Certificate[0];
                }
                @Override
                public void checkClientTrusted(X509Certificate[] certs, String authType) {
                }
                @Override
                public void checkServerTrusted(X509Certificate[] certs, String authType) {
                }
            }
        };
        //Install the all-trusting trust manager
        try {
            SSLContext sslContext = SSLContext.getInstance("SSL");
            sslContext.init(null, trustAllCerts, new SecureRandom());
            //Create an ssl socket factory with our all-trusting manager
            SSLConnectionSocketFactory sslSocketFactory = new SSLConnectionSocketFactory(sslContext, (hostname, session) -> true);

            //Create a custom HttpClient with the SSL socket factory
            CloseableHttpClient httpClient = HttpClients.custom()
                .setSSLSocketFactory(sslSocketFactory)
                .build();
            HttpEntityEnclosingRequestBase httpRequest = new HttpPost(url);
            StringEntity entity = new StringEntity(JSON.toJSONString(params), "UTF-8");
            httpRequest.setEntity(entity);
            httpRequest.setHeader("Content-Type", "application/json;charset=UTF-8");
            httpRequest.setHeader("X-Subject-Token", TOKEN_VALUE);

            CloseableHttpResponse response = httpClient.execute(httpRequest);
            HttpEntity responseEntity = response.getEntity();

```

```

        //In order to avoid messy code,encode the response data in UTF-8.
        String reply = EntityUtils.toString(responseEntity, "UTF-8");
        //Release resources finally.
        if (httpClient != null) {
            httpClient.close();
        }
        if (response != null) {
            response.close();
        }
        return reply;
    } catch (NoSuchAlgorithmException | KeyManagementException e) {
        throw new RuntimeException(e);
    }
}

/**
 * AES decryption, the mode is CBC, and the filling mode is pkcs7padding
 * @param text data
 * @param aesKey AES key
 * @param aesVector AES vector
 * @return string after decryption
 * @throws Exception
 */
static String decryptByAesCbc(String text, String aesKey, String aesVector) throws Exception {
    Security.addProvider(new BouncyCastleProvider());
    byte[] raw = aesKey.getBytes(StandardCharsets.UTF_8);
    SecretKeySpec skeySpec = new SecretKeySpec(raw, "AES");
    Cipher cipher = Cipher.getInstance("AES/CBC/PKCS7Padding");
    IvParameterSpec iv = new IvParameterSpec(aesVector.getBytes(StandardCharsets.UTF_8));
    cipher.init(Cipher.DECRYPT_MODE, skeySpec, iv);
    byte[] encrypted = parseHexStr2Byte(text);
    byte[] originalPassByte = cipher.doFinal(encrypted);
    return new String(originalPassByte, StandardCharsets.UTF_8);
}

/**
 * Hex string to byte array
 * @param hexStr hex string
 * @return byte array
 */
static byte[] parseHexStr2Byte(String hexStr) {
    byte[] result = new byte[hexStr.length() / 2];
    for (int i = 0; i < hexStr.length() / 2; i++) {
        int high = Integer.parseInt(hexStr.substring(i * 2, i * 2 + 1), 16);
        int low = Integer.parseInt(hexStr.substring(i * 2 + 1, i * 2 + 2), 16);
        result[i] = (byte) (high * 16 + low);
    }
    return result;
}

/**
 * CustomerMqFactory, used by SSL connection
 */
static class CustomerMqFactory extends ActiveMQSslConnectionFactory {
    @Override
    protected TrustManager[] createTrustManager() {
        return new TrustManager[]{

```

```

        new X509ExtendedTrustManager() {
            @Override
            public void checkClientTrusted(X509Certificate[] x509Certificates, String s, Socket socket) {
            }
            @Override
            public void checkServerTrusted(X509Certificate[] x509Certificates, String s, Socket socket) {
            }
            @Override
            public void checkClientTrusted(X509Certificate[] x509Certificates, String s, SSLEngine sslEngine) {
            }
            @Override
            public void checkServerTrusted(X509Certificate[] x509Certificates, String s, SSLEngine sslEngine) {
            }
            @Override
            public void checkClientTrusted(X509Certificate[] x509Certificates, String s) {
            }
            @Override
            public void checkServerTrusted(X509Certificate[] x509Certificates, String s) {
            }
            @Override
            public X509Certificate[] getAcceptedIssuers() {
                return new X509Certificate[]{};
            }
        }
    };
}
}
}

```

6.3.2.2 Example of MQ Connection with JS

Description

Use plug-ins to run utf8, hex, base64, AES, mqtt and import the plug-ins with ES6 or the <script> tag.

```

// axios
<script src="https://unpkg.com/axios/dist/axios.min.js"> </script>
// crypto-js
<script type="text/javascript" src="path-to/bower_components/crypto-js/crypto-js.js"> </script>
// mqttjs
<script src="https://cdn.bootcdn.net/ajax/libs/mqtt/4.2.8/mqtt.min.js"> </script>

```

For detailed instructions and description, go to the websites:

[axios.js - official document](#)

[npm/crypto-js](#)

[npm/mqtt](#)

Example Code

Run the code in ES6 environment. Adapt to browser scenarios through WSS protocol docking.

```
const axios = require('axios');
const CryptoJS = require('crypto-js');
const { v4: uuidv4 } = require('uuid');
const crypto = require('crypto');
const utf8 = require('crypto-js/enc-utf8');
const hex = require('crypto-js/enc-hex');
const base64 = require('crypto-js/enc-base64');
const AES = require('crypto-js/AES');
const mqtt = require('mqtt');
const https = require('https');

const MQ_URL_PREFIX = "wss://%s/mqtt";
const MQ_CONFIG_URL = "https://192.168.1.1:443/brms/api/v1.0/BRM/Config/GetMqConfig";
const TOPIC = "topic";
const QUEUE = "queue";
let TOKEN_VALUE = null;

async function main() {
  try {
    //Step1: Call the "Get the MQ configuration" interface, in order to obtain the MQ connection info.
    //You can obtain the value of token from "LoginExample".
    TOKEN_VALUE = "F2094B22.....4be5e3a871d";
    const userId = "1";
    const mqConfigResponse = await sendPost(MQ_CONFIG_URL, null);
    const mqConfig = mqConfigResponse.data;
    const userName = mqConfig.userName;
    const userPasswordWithAes = mqConfig.password;
    const mqttUrlAddr = mqConfig.wss;

    if (userPasswordWithAes === null) {
      console.error("Password encryption exception, please check the second login parameters 'secretKey' and 'secretVector'.");
      return;
    }

    //Step1: Decrypt the "password".
    //Secret key and secret vector are generated by the caller himself, as a parameter of the second login interface.
    const secretKey = "ddb6237.....d0bf6bd59";
    const secretVector = "8780a.....0046b";
    const userPassword = decryptByAesCbc(userPasswordWithAes, secretKey, secretVector);
    //Step3: Initialize the MQ connection information, and start the message listening.
    //Replace it with a queueName or a topicName which you would listened.
    const messageName = `mq/event/msg/topic/${userId}`;
    listenMqMessage(MQ_URL_PREFIX.replace("%s", mqttUrlAddr), userName, userPassword, messageName, TOPIC);

    //The connection is over, do your own things here...
  } catch (error) {
    console.error("Error:", error);
  }
}
```

```
    }  
  }  
  
  /**  
   * Send a POST request to the specified URL with the given parameters.  
   * @param url  
   * @param params  
   * @returns {Promise<any>}  
   */  
  async function sendPost(url, params) {  
    if (params === null) {  
      params = {};  
    }  
  
    try {  
      const response = await axios.post(url, params, {  
        headers: {  
          'Content-Type': 'application/json;charset=UTF-8',  
          'X-Subject-Token': TOKEN_VALUE  
        },  
        httpsAgent: new https.Agent({  
          rejectUnauthorized: false  
        })  
      });  
      //In order to avoid messy code, encode the response data in UTF-8.  
      return response.data;  
    } catch (error) {  
      throw error;  
    }  
  }  
}  
  
/**  
 * Listen to the MQ message.  
 * @param mqUrl  
 * @param userName  
 * @param userPassword  
 * @param messageName  
 * @param messageType  
 */  
function listenMqMessage(mqUrl, userName, userPassword, messageName, messageType) {  
  console.log(mqUrl)  
  const client = mqtt.connect(mqUrl,{  
    clientId: 'client-1',  
    username: userName,  
    password: userPassword,  
    rejectUnauthorized: false,  
  });  
  
  client.on('connect', function () {  
    const topicOrQueue = messageType === TOPIC ? messageName : messageName;  
  
    client.subscribe(topicOrQueue, (err, granted) => {  
      if (err) {  
        console.error('Subscription error:', err);  
      } else {  
        console.log('Subscribed to the topic successfully', granted);  
      }  
    });  
  });  
}
```

```
    })
    client.on('message', (topic, message) => {
        console.log('Received message on ${topic}: ${message.toString()}');
    });
    client.on('reconnect', () => {
        console.log('Reconnecting to MQ');
    });

    client.on('offline', () => {
        console.log('Offline from MQ');
    });
    client.on('close', () => {
        console.log('Connection closed');
    });
}

/**
 * AES encryption mode is CBC, and the filling mode is pkcs7padding.
 * @param text
 * @param secretKey
 * @param secretVector
 * @returns {string}
 */
function decryptByAesCbc(text, secretKey, secretVector) {
    const key = utf8.parse(secretKey);
    const iv = utf8.parse(secretVector);
    const encryptedHexWord = hex.parse(text);
    const srcs = base64.stringify(encryptedHexWord);
    const decrypt = AES.decrypt(srcs, key, { iv });
    const decryptedWord = decrypt.toString(utf8);
    return decryptedWord.toString();
}

main();
```

6.3.2.3 Example of MQ Connection with C#

Example Code

```
using System;
using System.Security.Cryptography;
using System.Text;
using System.Text.Json;
//https://activemq.apache.org/components/nms/providers/activemq/downloads/
using Apache.NMS.ActiveMQ;
using Apache.NMS;
using ISession = Apache.NMS.ISession;
using Apache.NMS.ActiveMQ.Commands;
using System.Net.Security;
using System.Security.Cryptography.X509Certificates;

/**
 * This is a MqConnectionExample.
 * using Apache.NMS.ActiveMQ.Transport.Tcp.Ssl
```

```

*/
namespace MqConnectionApplication
{
    public class MqConnectionExample
    {
        public const string MQ_URL_PREFIX = "ssl://";

        public const string MQ_CONFIG_URL = "https://192.168.1.1:443/brms/api/v1.0/BRM/Config/GetMqConfig";

        public const string TOPIC = "topic";

        public const string QUEUE = "queue";

        private static string? TOKEN_VALUE;

        private static readonly object _lock = new object();

        /**
         * Main method
         * @param args args
         */
        static void Main(string[] args)
        {
            System.Net.ServicePointManager.ServerCertificateValidationCallback = (sender, certificate, chain, sslPolicyErrors) => true;
            System.Net.ServicePointManager.SecurityProtocol = System.Net.SecurityProtocolType.Tls12;

            //Step1: Call the "Get the MQ configuration" interface, in order to obtain the MQ connection info.
            //You can obtain the value of token from "LoginExample".
            TOKEN_VALUE = "2D6784.....5405081c60c2b";
            String userId = "1";
            Dictionary<string, object>? mqConfimqConfigResponse = JsonSerializer.Deserialize<Dictionary<string, object>>(SendPost(MQ_CONFIG_URL, null)!);
            Dictionary<string, object>? mqConfigResponse = JsonSerializer.Deserialize<Dictionary<string, object>>(JsonSerializer.Serialize(mqConfimqConfigResponse?["data"]));

            String userName = Convert.ToString(mqConfigResponse?["userName"]);
            String userPasswordWithAes = Convert.ToString(mqConfigResponse?["password"]);
            String mqUrlAddr = Convert.ToString(mqConfigResponse?["addr"]);

            if (string.IsNullOrEmpty(userPasswordWithAes))
            {
                Console.WriteLine("Password encryption exception, please check the second login parameters 'secretKey' and 'secretVector'.");
            }
            //Step2: Decrypt the "password".
            //Secret key and secret vector are generated by the caller himself, as a parameter of the second login interface.
            string secretKey = "2c379be579.....99655d60d9cc";
            string secretVector = "8b7d.....4147";

            //Decrypt the "Password".
            String userPassword = DecryptWithAES7(userPasswordWithAes, secretKey, secretVector);

            Console.WriteLine($"The userPassword is {{ {userPassword} }}");
        }
    }
}

```



```

        Console.WriteLine($"The userName is {{ {userName} }}");
        Console.WriteLine($"The mqUrlAddr is {{ {mqUrlAddr} }}");

        //Step3: Initialize the MQ connection information,and start the message listening.
        //Replace it with a queueName or a topicName which you would listened.
        String messageName = "mq.event.msg.topic." + userId;
        ListenMqMessage(MQ_URL_PREFIX + mqUrlAddr, userName, userPassword, messageName, TOPIC);

        //The connection is over, do your own things here...
    }

    /**
     * Send post request method with HTTP.
     * @param url
     * @param params
     * @return reply
     */
    static String SendPost(String url, Dictionary<String, Object>? param)
    {
        param ??= new Dictionary<string, object>();
        HttpRequestMessage httpRequestMessage = new(HttpMethod.Post, url);

        StringContent stringContent = new(JsonSerializer.Serialize(param), Encoding.UTF8, "application/json");

        httpRequestMessage.Content = stringContent;
        httpRequestMessage.Headers.Add("X-Subject-Token", TOKEN_VALUE);

        //In order to avoid messy code,encode the response data in UTF-8.
        return SendAsync(httpRequestMessage).Result;
    }

    /**
     * Sending http requests asynchronously.
     * @param httpRequestMessage
     * @return response
     */
    public static async Task<string> SendAsync(HttpRequestMessage httpRequestMessage)
    {
        // Create a custom HttpClientHandler and ignore certificate validation
        HttpClientHandler handler = new HttpClientHandler
        {
            ServerCertificateCustomValidationCallback = (message, cert, chain, sslPolicyErrors)
=> true
        };
        HttpClient httpClient = new HttpClient(handler);
        HttpResponseMessage response = await httpClient.SendAsync(httpRequestMessage);

        //In order to avoid messy code,encode the response data in UTF-8.
        string reply = await response.Content.ReadAsStringAsync();
        return reply;
    }

    /**
     * AES encryption mode is CBC, and the filling mode is pkcs7padding.
     * @param text

```

```

    * @param aesKey
    * @param aesVector
    * @return
    * @throws Exception
    */
    public static String DecryptWithAES7(String? text, String aesKey, String aesVector)
    {
        byte[] raw = Encoding.UTF8.GetBytes(aesKey);
        using (Aes aesAlg = Aes.Create())
        {
            aesAlg.Key = raw;
            aesAlg.IV = Encoding.UTF8.GetBytes(aesVector);
            aesAlg.Mode = CipherMode.CBC;
            aesAlg.Padding = PaddingMode.PKCS7;

            ICryptoTransform decryptor = aesAlg.CreateDecryptor(aesAlg.Key, aesAlg.IV);
            byte[] encrypted = ParseHexStr2Byte(text);
            byte[] originalPassByte = decryptor.TransformFinalBlock(encrypted, 0, encrypted.Length);

            return Encoding.UTF8.GetString(originalPassByte);
        }
    }

    /**
     * Parse hex string to byte array.
     *
     * @param hexStr
     * @return
     */
    public static byte[] ParseHexStr2Byte(string hexStr)
    {
        byte[] result = new byte[hexStr.Length / 2];
        for (int i = 0; i < hexStr.Length / 2; i++)
        {
            int high = Convert.ToInt32(hexStr.Substring(i * 2, 1), 16);
            int low = Convert.ToInt32(hexStr.Substring(i * 2 + 1, 1), 16);
            result[i] = (byte)(high * 16 + low);
        }
        return result;
    }

    /**
     * Listen a queue or topic message.
     * @param mqUrl
     * @param userName
     * @param passWord
     * @param messageName
     * @throws Exception
     */
    public static void ListenMqMessage(String mqUrl, String? userName, String? passWord, String? messageName, String messageType)
    {
        if (!messageType.Equals(TOPIC) && !messageType.Equals(QUEUE))
        {
            return;
        }
    }

```

```

        Uri connecturi = new(mqUrl + "?transport.SslProtocol=Tls12&transport.ClientCertificate=
me=&transport.AcceptInvalidBrokerCertificate=true");
        // Accept all certificates
        AppContext.SetSwitch("NMS-SSL_CERTIFICATE_VALIDATION", true);
        System.Net.ServicePointManager.ServerCertificateValidationCallback =
            (sender, certificate, chain, sslPolicyErrors) => true;

        lock (_lock)
        {
            ConnectionFactory connectionFactory = new ConnectionFactory
            {
                UserName = userName,
                Password = passWord,
                BrokerUri = connecturi
            };

            // Create a Connection and Session with topic
            IConnection connection = connectionFactory.CreateConnection();
            connection.Start();
            ISession session = connection.CreateSession(AcknowledgementMode.AutoAcknowl
edge);

            IMessageConsumer consumer;

            if (messageType.Equals(TOPIC))
            {
                consumer = session.CreateConsumer(new ActiveMQTopic(messageName));
            }
            else
            {
                consumer = session.CreateConsumer(new ActiveMQQueue(messageName));
            }
            Console.WriteLine("Start listening...");
            consumer.Listener += new MessageListener(ConsumerListener);
            while (true)
            {
                Thread.Sleep(1000);
            }
        }
    }

    /**
     * The callback function of the message listener.
     * @param message
     */
    static void ConsumerListener(IMessage message)
    {
        ITextMessage textMessage = (ITextMessage)message;
        String text = textMessage.Text;
        //Print the listening message.
        Console.WriteLine(text);
    }
}

```

6.3.2.4 Example of MQ Connection with Python

Example Code

```
import json
import requests
from stomp import PrintingListener
import stomp

from Crypto.Cipher import AES
from Crypto.Util.Padding import unpad
import base64
from binascii import unhexlify
import ssl
import time
import urllib3

MQ_CONFIG_URL = "https://192.168.1.1:443/brms/api/v1.0/BRM/Config/GetMqConfig"
TOPIC = "topic"
QUEUE = "queue"
TOKEN_VALUE = None

def main():
    # Step1: Call the "Get the MQ configuration" interface, in order to obtain the MQ connection info.
    global TOKEN_VALUE
    TOKEN_VALUE = "148C4EC1.....79bf61c1f2b5"
    user_id = "1"
    mq_config_response_string = send_post(MQ_CONFIG_URL, None)
    mq_config_response = json.loads(mq_config_response_string)["data"]
    user_name = mq_config_response["userName"]
    user_password_with_aes = mq_config_response["password"]

    if not user_password_with_aes:
        print("Password encryption exception, please check the second login parameters 'secretKey' and 'secretVector'.")
        return

    mq_url_addr = mq_config_response.get("stomp")

    # Step2: Decrypt the "password".
    secret_key = "9c9d026.....ae8a231a8d6"
    secret_vector = "b6d3.....91cc"
    user_password = decrypt_by_aes_cbc(user_password_with_aes, secret_key, secret_vector)

    # Step3: Initialize the MQ connection information, and start the message listening.
    message_name = f"mq.event.msg.topic.{user_id}"
    print(user_password)
    listen_mq_message_stomp(mq_url_addr, user_name, user_password, message_name, TOPIC)

"""
Sends a POST request to the specified URL with optional parameters.

This function is used to interact with an HTTP server by sending a POST request.

Parameters:
- url (str): The target URL to which the request will be sent.
```

- params (dict, optional): A dictionary of parameters to be sent with the request. Defaults to an empty dictionary if not provided.

Returns:

- str: The response text from the server.

"""

```
def send_post(url, params=None):
    if params is None:
        params = {}
    headers = {
        "Content-Type": "application/json;charset=UTF-8",
        "X-Subject-Token": TOKEN_VALUE
    }
    # Disable SSL verification and suppress InsecureRequestWarning
    urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)
    print(headers)
    response = requests.post(url, headers=headers, json=params, verify=False)
    print(response.text)
    response.raise_for_status()
    return response.text
```

"""

Connects to a Message Queue (MQ) using the STOMP protocol and listens for messages.

Parameters:

- mq_url (str): The URL of the MQ server in the format "host:port".
- user_name (str): The username for connecting to the MQ.
- password (str): The password for connecting to the MQ.
- message_name (str): The name of the topic or queue to subscribe to.
- message_type (str): The type of message destination, either 'topic' or 'queue'.

Returns:

- None: If the `message\_type` is neither 'topic' nor 'queue', the function returns immediately.

Summary:

This function establishes an SSL-enabled connection to the MQ server using STOMP. It subscribes to either a topic or a queue based on the `message\_type` parameter and starts listening for incoming messages. A non-blocking loop prevents the program from exiting prematurely.

"""

```
def listen_mq_message_stomp(mq_url, user_name, password, message_name, message_type):
    if message_type not in [TOPIC, QUEUE]:
        return
```

```
    # Use Connection with SSL support
```

```
    conn = stomp.Connection12(host_and_ports=[(mq_url.split(":")[0], int(mq_url.split(":")[1]))])
```

```
    conn.set_listener("", PrintingListener())
```

```
    # Configure SSL manually
```

```
    ssl_params = {
```

```
        'key_file': None,
```

```
        'cert_file': None,
```

```
        'ca_certs': None,
```

```
        'ssl_version': ssl.PROTOCOL_TLSv1_2
```

```
    }
```

```
    conn.transport.set_ssl([(mq_url.split(":")[0], int(mq_url.split(":")[1]))], **ssl_params)
```

```
    conn.connect(user_name, password, wait=True)
```

```
    if message_type == TOPIC:
```

```
        conn.subscribe(destination=f"/topic/{message_name}", id=1, ack='auto')
```

```

else:
    conn.subscribe(destination=f"/queue/{message_name}", id=1, ack='auto')

print("Start listening...")

# The method of listening to MQ is non blocking, while loop prevents program exit
while True:
    time.sleep(10)

"""
Decrypts the given text using AES-CBC mode.
:param text: The encrypted text, which should be a hexadecimal encoded string.
:param aes_key: The AES key used for decryption.
:param aes_vector: The initialization vector (IV) used for decryption.
:return: The decrypted text as a UTF-8 string.
"""
def decrypt_by_aes_cbc(text, aes_key, aes_vector):
    key = bytes(aes_key, 'utf-8')
    iv = bytes(aes_vector, 'utf-8')
    cipher = AES.new(key, AES.MODE_CBC, iv)
    encrypted = unhexlify(text)
    decrypted = unpad(cipher.decrypt(encrypted), AES.block_size)
    return decrypted.decode('utf-8')

if __name__ == "__main__":
    main()

```

6.4 Encryption and Decryption Example

The VMS platform uses encryption algorithms such as MD5, RSA, and AES for encryption and decryption. The development language used in the example is JAVA, which has been verified in JDK1.8.0_381. The client and platform interact with each other, involving encryption and decryption fields, using the AES algorithm. The RSA algorithm is only used to encrypt AES key and vector, and is not used for encryption of other business fields.

MD5 Encryption

```

String text = "0123456789ABCDEF";
//Encryption
String encryptTextByMd5 = encryptByMd5(text);

/**
 * MD5 encryption
 * @param text data
 * @return string after encryption
 * @throws Exception
 */
public static String encryptByMd5(String text) throws Exception {
    MessageDigest md = MessageDigest.getInstance("MD5");
    byte[] messageDigest = md.digest(text.getBytes());
    StringBuilder buffer = new StringBuilder();
    for (byte b : messageDigest) {
        buffer.append(String.format("%02x", b));
    }
}

```

```
}  
    return buffer.toString();  
}
```

RSA Encryption and Decryption

```
String text = "0123456789ABCDEF";  
//Generate key pairs  
KeyPair keyPair = getRsaKeys();  
PublicKey publicKey = keyPair.getPublic();  
PrivateKey privateKey = keyPair.getPrivate();  
//Encryption  
byte[] encryptByteByRsaPublicKey = encryptByPublicKey(text, publicKey.getEncoded());  
String encryptText = parseByte2HexStr(encryptByteByRsaPublicKey.toUpperCase());  
//Decryption  
byte[] decryptByteByRsaPrivateKey = decryptByPrivateKey(encryptText, privateKey.getEncoded());  
String decryptText = new String(decryptByteByRsaPrivateKey, StandardCharsets.UTF_8);  
  
/**  
 * Generate RSA public key and private key. The private key is in pkcs8 format, and the public key is in X509 format  
 * @return key pair  
 * @throws Exception  
 */  
public static KeyPair getRsaKeys() throws Exception {  
    Provider provider = Security.getProvider("SunRsaSign");  
    KeyPairGenerator keyPairGen = KeyPairGenerator.getInstance("RSA", provider);  
    keyPairGen.initialize(2048, new SecureRandom());  
    KeyPair keyPair = keyPairGen.generateKeyPair();  
    return keyPair;  
}  
  
/**  
 * RSA encryption. Public key encryption, private key decryption  
 * @param text data  
 * @param publicKeyBytes public key string array  
 * @return string after encryption  
 * @throws Exception  
 */  
public static byte[] encryptByPublicKey(String text, byte[] publicKeyBytes) throws Exception {  
    byte[] result = null;  
    int rsaBytesLenEncrypt = 245;  
  
    //Get public key object  
    X509EncodedKeySpec x509KeySpec = new X509EncodedKeySpec(publicKeyBytes);  
    KeyFactory keyFactory = KeyFactory.getInstance("RSA");  
    Key publicKeyObject = keyFactory.generatePublic(x509KeySpec);  
  
    //Encrypt data  
    byte[] dataBytes = text.getBytes(StandardCharsets.UTF_8);  
    Cipher cipher = Cipher.getInstance(keyFactory.getAlgorithm());  
    cipher.init(Cipher.ENCRYPT_MODE, publicKeyObject);  
    for (int i = 0; i < dataBytes.length; i += rsaBytesLenEncrypt) {  
        int to = Math.min(i + rsaBytesLenEncrypt, dataBytes.length);  
        byte[] temp = cipher.doFinal(Arrays.copyOfRange(dataBytes, i, to));  
        result = sumBytes(result, temp);  
    }  
}
```

```

    }
    return result;
}

/**
 * RSA decryption. Public key encryption, private key decryption
 * @param text data
 * @param privateKeyBytes private key string array
 * @return string after decryption
 * @throws Exception
 */
public static byte[] decryptByPrivateKey(String text, byte[] privateKeyBytes) throws Exception {
    byte[] result = null;
    int rsaBytesLenDecrypt = 256;
    byte[] dataBytes = parseHexStr2Byte(text);

    //Get private key object
    PKCS8EncodedKeySpec pkcs8KeySpec = new PKCS8EncodedKeySpec(privateKeyBytes);
    KeyFactory keyFactory = KeyFactory.getInstance("RSA");
    Key privateKey = keyFactory.generatePrivate(pkcs8KeySpec);

    //Decrypt data
    Cipher cipher = Cipher.getInstance(keyFactory.getAlgorithm());
    cipher.init(Cipher.DECRYPT_MODE, privateKey);

    for (int i = 0; i < dataBytes.length; i += rsaBytesLenDecrypt) {
        int to = (i + rsaBytesLenDecrypt) < dataBytes.length ? (i + rsaBytesLenDecrypt) : dataBytes.length;
        byte[] temp = cipher.doFinal(Arrays.copyOfRange(dataBytes, i, to));
        result = sumBytes(result, temp);
    }

    return result;
}

/**
 * Hex string to byte array
 * @param hexStr hex string
 * @return byte array
 */
private static byte[] parseHexStr2Byte(String hexStr) {
    byte[] result = new byte[hexStr.length() / 2];
    for (int i = 0; i < hexStr.length() / 2; i++) {
        int high = Integer.parseInt(hexStr.substring(i * 2, i * 2 + 1), 16);
        int low = Integer.parseInt(hexStr.substring(i * 2 + 1, i * 2 + 2), 16);
        result[i] = (byte) (high * 16 + low);
    }
    return result;
}

/**
 * Sum 2 bytes
 * @param bytes1 byte 1
 * @param bytes2 byte 2
 * @return the new array formed by merging two arrays
 */
private static byte[] sumBytes(byte[] bytes1, byte[] bytes2) {

```



```

byte[] result = null;
int len1 = 0;
int len2 = 0;
if (null != bytes1) {
    len1 = bytes1.length;
}
if (null != bytes2) {
    len2 = bytes2.length;
}
if (len1 + len2 > 0) {
    result = new byte[len1 + len2];
}
if (len1 > 0) {
    System.arraycopy(bytes1, 0, result, 0, len1);
}
if (len2 > 0) {
    System.arraycopy(bytes2, 0, result, len1, len2);
}
return result;
}

```

AES Encryption and Decryption

```

String text = "0123456789ABCDEF";
//Generate key and vector
String aesKey = UUID.randomUUID().toString().replace("-", "");
String aesVector = UUID.randomUUID().toString().replace("-", "").substring(16);
//Encryption
String encryptTextByAesCbc = encryptByAesCbc(text, aesKey, aesVector);
//Decryption
String decryptTextByAesCbc = decryptByAesCbc(encryptTextByAesCbc, aesKey, aesVector);

/**
 * AES encryption, the mode is CBC, and the filling mode is pkcs7padding
 * If you cannot provide BouncyCastleProvider(), you can comment out this code "Security.addPro
vider(new BouncyCastleProvider())", and change "AES/CCB/PKCS7Padding" to "AES/CCB/PKCS5Paddi
ng" in the following code
 * @param text data
 * @param aesKey AES key
 * @param aesVector AES vector
 * @return string after encryption
 * @throws Exception
 */
public static String encryptByAesCbc(String text, String aesKey, String aesVector) throws Exception
{
    //If you cannot provide BouncyCastleProvider(), you can comment out this code and change
"AES/CCB/PKCS7Padding" to "AES/CCB/PKCS5Padding" in the following code.
    Security.addProvider(new BouncyCastleProvider());
    byte[] raw = aesKey.getBytes(StandardCharsets.UTF_8);
    SecretKeySpec keySpec = new SecretKeySpec(raw, "AES");
    Cipher cipher = Cipher.getInstance("AES/CBC/PKCS7Padding");
    IvParameterSpec iv = new IvParameterSpec(aesVector.getBytes(StandardCharsets.UTF_8));
    cipher.init(Cipher.ENCRYPT_MODE, keySpec, iv);
    byte[] encrypted = cipher.doFinal(text.getBytes(StandardCharsets.UTF_8));
    return parseByte2HexStr(encrypted).toLowerCase();
}

```

```
/**
 * AES decryption, the mode is CBC, and the filling mode is pkcs7padding
 * If you cannot provide BouncyCastleProvider(), you can comment out this code "Security.addPro
 * vider(new BouncyCastleProvider())", and change "AES/CCB/PKCS7Padding" to "AES/CCB/PKCS5Paddi
 * ng" in the following code
 * @param text data
 * @param aesKey AES key
 * @param aesVector AES vector
 * @return string after decryption
 * @throws Exception
 */
public static String decryptByAesCbc(String text, String aesKey, String aesVector) throws Exception
{
    Security.addProvider(new BouncyCastleProvider());
    byte[] raw = aesKey.getBytes(StandardCharsets.UTF_8);
    SecretKeySpec skeySpec = new SecretKeySpec(raw, "AES");
    Cipher cipher = Cipher.getInstance("AES/CBC/PKCS7Padding");
    IvParameterSpec iv = new IvParameterSpec(aesVector.getBytes(StandardCharsets.UTF_8));
    cipher.init(Cipher.DECRYPT_MODE, skeySpec, iv);
    byte[] encrypted = parseHexStr2Byte(text);
    byte[] originalPassByte = cipher.doFinal(encrypted);
    return new String(originalPassByte, StandardCharsets.UTF_8);
}

/**
 * Byte array to hex string
 * @param bytes byte array
 * @return hex string
 */
private static String parseByte2HexStr(byte[] bytes) {
    StringBuffer buffer = new StringBuffer();
    for (int i = 0; i < bytes.length; i++) {
        String hex = Integer.toHexString(bytes[i] & 0xFF);
        if (hex.length() == 1) {
            hex = '0' + hex;
        }
        buffer.append(hex);
    }
    return buffer.toString();
}

/**
 * Hex string to byte array
 * @param hexStr hex string
 * @return byte array
 */
private static byte[] parseHexStr2Byte(String hexStr) {
    byte[] result = new byte[hexStr.length() / 2];
    for (int i = 0; i < hexStr.length() / 2; i++) {
        int high = Integer.parseInt(hexStr.substring(i * 2, i * 2 + 1), 16);
        int low = Integer.parseInt(hexStr.substring(i * 2 + 1, i * 2 + 2), 16);
        result[i] = (byte) (high * 16 + low);
    }
    return result;
}
```

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7.1 Common Errors

7.1.1 Image URL cannot be opened

Q: The interface return value or MQ message includes an image URL as follows.

```
https://192.168.1.1:50001/afefb58d-5b66-11e9-8037-000000000000/20190612/1/dsf_739dc016-8d79-11e9-9f8a-000000000000_34782193_34998401.jpg
```

After copying the URL to the browser bar, the image cannot be opened.

A: To get image resources, you need to add the authentication token as follows.

```
https://192.168.1.1:50001/afefb58d-5b66-11e9-8037-000000000000/20190612/1/dsf_739dc016-8d79-11e9-9f8a-000000000000_34782193_34998401.jpg?token=122dc1ecda074d1ea6b5e8865ef028bb
```

This token is different from the token sent in the interface request header X-Subject-Token. Instead, it comes from the return value credential of the [3.1.2 Submit Credentials] interface, but also follows the same expiration mechanism: Without heartbeat keep-alive, the credential expires after 30 seconds. With heartbeat keep-alive, the credential expires after 30 minutes. So you must add a valid credential to the end of the image URL.

If the image still cannot be opened, open it on the VMS client. If the image cannot be opened on the VMS client, the image might not have been stored for the lack of image disk. If the image can be opened on the VMS client, the token might have been expired.

Another case is that the image URL includes "?" as follows:

```
https://192.168.1.1:8082/1000054$1$0$0/20210206/23/5457-3196639-0.jpg?streamID:1260@capTime:1612684797
```

In this case, you must use "&" to add the token.

```
https://192.168.1.1:8082/1000054$1$0$0/20210206/23/5457-3196639-0.jpg?streamID:1260@capTime:1612684797&token=122dc1ecda074d1ea6b5e8865ef028bb
```

7.1.2 RTSP URL cannot pull stream

Q: While playing videos on VLC after getting the live video RTSP URL from the interface, users still need to enter the username and password or the platform reports error.

A: We recommend you call [3.4.1 Start Real-time Video] to get the live video RTSP URL, and then pull stream with standard decoding tools (such as VLC) or DPSDK.

Return data of the interface:

```
{
  "data": {
    "minRate": null,
    "protocol": null,
    "ip": null,
    "port": null,
    "stunEnable": null,
    "stunPort": null,
    "url": "rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000040%240%26substream=1",
    "url2": "",
    "connectType": null,
    "session": "2204",
    "token": "2204",
    "trackId": ""
  }
}
```

When requesting a video, use the token field as a URL parameter. Otherwise, the video request will fail.

This token is different from the token in X-Subject-Token and is returned by MTS. Each token expires after being used once, or not used within 30 seconds. The token for continuous stream pulling does not need to be updated.

The format of an URL with token is as follows:

```
rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000040%240%26substream=1?
token=2204
```

In addition, there is another situation where in the internal and external network environment, the platform may return multiple streaming addresses, separated by "|", namely the intranet address and the external network address. When using it, you need to separate it into multiple addresses according to the network environment of the current login platform, and finally splice the token at the end. For example, the address returned by the platform is:

```
rtsp://192.168.1.1:9100/vms/monitor/param/cameraid=1000040%240%26substream=1|r
tsp://10.10.10.10:9100/vms/monitor/param/cameraid=1000040%240%26substream=1
```

Among them, 192.168.1.1 is the intranet IP, and 10.10.10.10 is the external IP. If you are currently docking in an external network environment, the final streaming address is

```
rtsp://10.10.10.10:9100/vms/monitor/param/cameraid=1000040%240%26substream=1?
token=2204
```

7.1.3 Failed to get real-time alarms after listening to MQ

Q: After listening to a MQ alarm topic, users cannot receive alarms.

A: Multiple reasons might cause the problem.

1 Failed to configure linked users. For users to receive alarms from the platform, you must configure linked users while configuring the alarm protocols. Check whether the VMS client can receive real-time alarms. If yes, configurations are normal. If no, contact the technical support.

2 Listening to wrong MQ alarm topics. To receive alarms, you need to listen to `mq.alarm.msg.topic.{userId}`. UserID is the ID of the login user, it can be obtained through the [3.1.2 Submit Credentials] interface. For example, if the userID of the login user is 100, listen to `mq.alarm.msg.topic.100`.

3 Users are offline. Alarms, events and other messages on the platform are only sent to online users by default (System users are excluded). After logging in to the platform through the API, you need to call [3.1.3 Heartbeat Keep-alive] and [3.1.4 Update Token] interfaces to keep online status. Otherwise, alarm messages will not be sent to the related MQ topics. During test, keep the user online on the VMS client to check whether the user can receive alarm messages. After the test, keep the user online through API interface.

7.1.4 No image information in the alarm messages

Q: No image information is included in the alarm messages.

A: Listen to `mq.alarm.msg.topic.{userId}` to get alarm events. See [4.2.1 Alarm Messages].

Only few intelligent alarms support automatic capture. Get the snapshots with `alarmPicture`. Snapshots of other linked cameras are reported asynchronously. Listen to `mq.event.msg.topic.{userId}` to link snapshots. See [4.2.3 Notification of Alarm Images]. Alarm messages and images are interconnected through `alarmCode`.

7.1.5 No person information in the alarm messages

Q: No person information is included in the access control alarm messages.

A: Listen to `mq.alarm.msg.topic.{userId}` to get alarm events. See [4.2.1 Alarm Messages]. The messages only include the basic information about an event. For linked information such as persons, faces, entrances and exits and traffic, call other interfaces. For example, to obtain personnel information of access control alarms, refer to [3.6.10 Get Alarm Related Access Control Information] and associate through the `alarmCode` field.